



**GOVERNMENT OF INDIA
CENTRAL PUBLIC WORKS DEPARTMENT**

**GENERAL SPECIFICATIONS FOR
ELECTRICAL WORKS**

**PART-II
EXTERNAL**

2023



**PUBLISHED UNDER THE AUTHORITY
OF
DIRECTOR GENERAL, CPWD, NEW DELHI**

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A GOVERNMENT OF INDIA PUBLICATION

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FOREWORD

This revised and enlarged edition of "CPWD General Specifications for Electrical Works Part-II External 2023" is of immense importance in view of the need to update the long pending External EI specifications. The updated specifications include the energy efficiency requirements in road/ external lighting, execution of external electrical work, selection/ designing of power distribution cables and General guidelines to ensure safe, efficient, reliable and economical use of electricity.

Earlier editions were brought out in the year 1949, 1960, 1974 & 1994 and now the present edition has been finalized in 2023. This revision reflects the green initiatives and technological developments in the field of external electrical installations, including lighting aspects of ECBC / CPWD Green Rating compliant buildings/ projects.

These revised specifications incorporate the relevant provisions of NBC 2016, ECBC 2017, NEC 2023, Central Electricity Authority (Measures related to Safety and Electric Supply) Regulations 2010 (amended in 2015, 2018, 2019) etc. covering different aspects related to Cable work, Overhead line work and others. New chapters have been added covering trenchless Cable Work, Aerial Bunched Cable Work, Road Lighting and Control Work, other External lighting and control, Poles and High Masts etc. The IS codes mentioned in the specifications were also revised wherever these have undergone change.

I acknowledge the hard work put in by Sh. Dharmesh Chandra Goel, ADG(Tech), Sh. Vikas Rana, Chief Engineer, CSQ (E), Sh. Vimal Kumar, Chief Engineer (E), Sh. Ramayan Prasad Gupta, SE(E) TAS, Sh. Ashok Kumar Meena, EE (E)TAS, Sh. Sandeep Kumar Das, AE(E)TAS, and other- staff of CSQ(E) whose names are not mentioned here for the sake of brevity, towards revision and finalisation of General Specifications for Electrical Works Part-II External 2023.

The contribution of Sh. Ujjwal Mitra, ADG (Training & Research) Ghaziabad, Sh. AK Kain, CE (BFLZ), Sh. S.S. Garg, CE, North Zone, PWD GNCTD, Delhi, Sh. Ujjwal Kumar, CE NDZ-III, Sh. Rahul Sitaram Kamble SE(E), CVPEC, Sh. R. P. Gupta SE, CSQ(E) towards reviewing and finalizing the CPWD General Specifications for Electrical Works Part-II External 2023 is acknowledged with appreciation.

Suggestions for any modifications, errors and omissions may be sent to SE (E) TAS, office of Chief Engineer (E) CSQ, CPWD, Nirman Bhawan, New Delhi.

In case of any discrepancy between English and Hindi versions, the English version shall be held valid.

Place: New Delhi
Dated: 03.07.2023


(Rajesh Kumar Kaushal)
Director General, CPWD



Dharmesh Chandra Goel
Addl. Director General (Tech)



भारत सरकार
Government of India



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PREFACE

CPWD envisages a lead role for itself in the execution, maintenance and standardization of the built environment in India, while continuing to play the role of a government department in facilitating the implementation of policies for sustainable development and transparency in governance along with assimilation of knowledge and experience. CPWD strives to educate its clients to aspire for green buildings and develops norms for the same. Its Vision is to create and maintain a sustainable and inclusive built environment within the available resources while ensuring world class quality.

The Central Public Works Department (CPWD) is a 168 years old institution and is the principal agency of the Government of India responsible for creating assets and providing comprehensive services including planning, designing, construction and maintenance of office and residential buildings as well as other infrastructures of various ministries, departments of Government of India, autonomous bodies and public sector enterprises. Its activities are spread throughout the country.

Therefore, the following is the sequence of various editions of CPWD General Specifications for Electrical Works (External).

1974

1994

2023 (Present edition)

This edition aims to incorporate latest technological trends and green building norms in central government buildings being constructed and maintained by CPWD.

The highlight of the present edition are:-

- Green Building initiatives in the field of External electrical installations for CPWD Green Rating/ ECBC/ GRIHA compliant buildings under construction by CPWD.
- Incorporates the relevant provisions of NBC 2016, ECBC 2017, NEC 2023, Central Electricity Authority (Measures related to Safety and Electric Supply) Regulations 2010 (amended in 2015, 2018, 2019) etc. covering different aspects related to External lighting design, lighting control, safety measures.

- Recommended Illuminance levels for External lighting.
- Addition of New chapters for:
- Trenchless Cable Work,
- Aerial Bunched Cables Work,
- Road Lighting and Control Work,
- Other External lighting.
- Poles and High Masts.
- Important Indian standards are also updated.

I am grateful to Shri Rajesh Kumar Kaushal, Director General, CPWD, Shri Shailendra Sharma, Director General (Former), CPWD for reposing trust in our team to undertake this work. I acknowledge the efforts put in by Chairman and members of the specification committee in drafting the present specification which is technically updated, modern & user friendly. I express my deep appreciation for Sh. Ujjwal Mitra, ADG (Training & Research) Ghaziabad, Chairman of the committee and members Sh. AK Kain, CE (BFLZ), Sh. S.S. Garg, CE, North Zone, PWD GNCTD, Delhi, Sh. Ujjwal Kumar, CE NDZ-III, Sh. Rahul Sitaram Kamble SE(E), CVPEC and Sh. R. P. Gupta SE, CSQ(E) for their valuable contributions and inputs.

I complement Sh. Vikas Rana, Chief Engineer, CSQ (E), Sh. Vimal Kumar CE (E), Sh. Ramayan Prasad Gupta, SE (E) TAS, CSQ. Sh. Ashok Kumar Meena, EE(E) TAS, Sh. Sandeep Kumar Das, AE(E) TAS, who made their sincere efforts to update the specifications and making the publication available in very short time.

Errors or omissions, and suggestions for improvement, if any, may kindly be brought to the notice of the Superintending Engineer (E) TAS, in the Office of Chief Engineer (E) CSQ. CPWD, New Delhi-110011 (Tel No. 01123061418; email: delseetas.cpwd@nic.in, deleetas.cpwd@gov.in).

Place: New Delhi
Dated: 04.07.2023


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CHAPTER 1

GENERAL

1.1 SCOPE

These General Specifications for External electrical works aims to lay down General guidelines to ensure safe, efficient, reliable and economical use of electricity and execution of external electrical work:

- 1.1.1 These specifications generally shall not be deviated. However the appropriate technical sanctioning authority may depart from these specifications to meet the particular requirements of any work or for other technical reasons with recorded reason and justification for deviation from these specifications.
- 1.1.2 These General Specifications are subject to revision from time to time.
- 1.1.3 This Chapter covers the general requirements applicable to works contracts for such External Electrical Installation works.

1.2 RELATED DOCUMENTS

Each work has its own particular requirements. Therefore, in addition to the General Specifications, governing BIS, Central Electricity Authority (Measures relating to Safety and Electricity Supply) Regulations, 2010, National building code 2016, National Electrical Code 2023, Energy Conservation Building Code of India 2017 (ECBC 2017) all as amended upto date, Standard condition of contract Conditions etc., there would be necessity of Additional conditions/ Specifications for a particular work. The Deviation from this specification in Additional conditions/ Specifications for a particular work shall be duly justified and recorded by NIT/Technical Sanctioning Authority.

In the case of discrepancy between the Schedule of Quantities, the Specifications and/ or the Drawings, the following order of preference shall be observed:

- (i) Description of Schedule of Quantities.
- (ii) Particular Specification and Special Condition, if any.
- (iii) NIT Drawings.
- (iv) CPWD Specifications.
- (v) Indian Standard Specifications of B.I.S.
- (vi) Central Electricity Authority (Measures relating to Safety and Electricity Supply) Regulations, 2010
- (vii) National Electrical Code 2023 as amended up to date and
- (viii) National building code 2016 as amended up to date and Energy Conservation Building Code of India 2017 (ECBC 2017) as amended up to date.

1.2 TERMINOLOGY

The definition of terms shall be in accordance with relevant Bureau of Indian Standards of related items, Central Electricity Authority (Measures relating to Safety and Electricity Supply) Regulations, 2010, National building code 2016, National Electrical Code 2023. Some of the commonly used terms are indicated in Appendix A.

1.4 SUBMISSION OF TENDERS

- 1.4.1 The tender shall be submitted complete with the following: -
 - 1.4.1.1 Complete tender documents as purchased/downloaded from CPWD/website duly filled in and submitted. The price part of the tender shall be indicated only on the tender schedule of work as per NIT.

- 1.4.1.2 Earnest Money deposit in one of the specified forms as per laid down rules issued from time to time.
- 1.4.1.3 Any other supplementary details required for the evaluation of the tenders such as drawings, technical literature/ catalogues, data etc.
- 1.4.2 Where one/ two-part tendering system is proposed to be adopted in any particular work (depending upon composite/ EPC or individual External EI work), the procedure for submission and opening of tenders shall be indicated in tender documents for that work.

1.5 RATES

- 1.5.1 The work shall be treated as on works contract basis and the rates tendered shall be for complete items of work (except the materials, if any, stipulated for supply by the department) inclusive of all taxes, GST (including works contract tax, if any), duties, and levies etc. and all charges for items contingent to the work, such as, packing, forwarding, insurance, freight and delivery at site for the materials to be supplied by the contractor, watch and ward of all materials (including those, if any, supplied by the department) for the work at site etc.
- 1.5.2 Prices quoted shall be firm. Price adjustments shall however be governed by Clause 10C, 10CC the Conditions of Contract given in form CPWD 7 or 8 of the tender documents, for works executed under these forms, as applicable. All relevant documents shall be produced by the contractor to the Engineer-in-charge, whenever called upon by him to do so, for working out such adjustments in rates.

1.6 TAXES AND DUTIES

- 1.6.1 Being an indivisible works contract, GST, custom duty, import duty, Octroi and any other taxes and duties etc. are not payable separately.
- 1.6.2 The works contract tax shall be deducted from the bills of the contractor as applicable in the State in which the work is carried out, at the time of payments.

1.7 MOBILISATION ADVANCE

No mobilization advance shall be paid for the work, unless otherwise stipulated in tender papers for any individual works. In case of EPC contracts, the conditions as per NIT shall be applicable.

1.8 COMPLETENESS OF TENDER

All sundry fittings, assemblies, accessories, hardware items, foundation nut & bolts, termination lugs for electrical connections, levelling and dressing of earth, cleaning as required, and all other sundry items which are useful and necessary for proper assembly and efficient working of the various components of the work shall be deemed to have been included in the tender, irrespective of the fact whether such items are specifically mentioned in the tender documents or not.

1.9 WORKS TO BE ARRANGED BY THE DEPARTMENT

Unless and otherwise specified in the tender documents, the following works shall be arranged by the Department:

- 1.9.1 Suitable space for storage of all equipment, components and materials pertaining to the work shall be provided by the Department free of cost to the contractor but the required arrangement/set up for storage of materials in the spared space including locking arrangement shall be provided by the agency. Also, the watch and ward of the stores and their safe custody

shall be the responsibility of the contractor till the final taking over of the installation by the Department.

Note: In case of EPC contracts, the firm shall be responsible for above as a whole since the complete building/structure to be built shall be within his scope and hence the above items shall also be in firm's scope, which may defined in NIT accordingly.

1.9.2 Supply of materials to the contractor, if any, stipulated in the tender documents.

1.10 WORKS TO BE DONE BY THE CONTRACTOR

Unless and otherwise mentioned in the tender documents, the following works shall be done by the contractor, and therefore their cost shall be deemed to be included in their tendered cost: -

- 1.10.1 Foundation for equipment, poles, supports and brackets (other than for mounting of light fitting) and components wherever required, including foundation bolts.
- 1.10.2 Suspenders and brackets for suspending/supporting cables, as required.
- 1.10.3 Suspenders for cable trays for laying the cables, where required.
- 1.10.4 Excavation and refilling of trenches in soil wherever the pipes/ cables are to be laid directly in ground, including necessary base treatment and supports for pipes as specified.
- 1.10.5 Cutting and making good all damages caused during installation and restoring the same to their original finish.
- 1.10.6 Sealing of all floor openings provided by him for pipes, earthing, rising mains and cables, from fire safety and water ingress point of view, after laying, of the same.
- 1.10.7 Painting at site of all exposed metal surfaces of the installation other than pre- painted items like fittings, poles, pipes, switchgear/ distribution gear items, cubicle switchboard etc. Damages to finished surfaces of these items while handling and erection, shall however be rectified to the satisfaction of the Engineer-in-charge.
- 1.10.8 Testing and commissioning of completed installation.

1.11 STORAGE AND CUSTODY OF MATERIALS

The storage space, if available shall be provided by the Department free of cost to-the contractor subject to provision as contained in para 1.9.1

1.12 ELECTRIC POWER SUPPLY

- 1.12.1 Unless otherwise specified, 3 phase, 415 Volts, 50 Hz power supply shall be provided by the department free of charge to the contractor at one point. Termination switchgear however, shall be provided by the contractor. Further extension if required shall be done by the contractor and nothing extra shall be paid on this account.
 - 1.12.1.1 Unless otherwise specified in the contract, further power distribution to the various equipment shall be done by the contractor.
 - 1.12.1.2 Where the power supply has to be arranged by the Department at more than one point as per the terms of the contract, the termination of all such power feeders in the incomer of respective control panels (provided by the contractor) shall be the responsibility of the contractor at his own cost.
 - 1.12.1.3 The contractor shall not use the power supply for any other purpose than that for which it is intended for. No major fabrication work shall be done at site. The power supply shall be disconnected in case of such default and the contractor shall then have to arrange the required power supply at his cost.
 - 1.12.1.4 Contractor may have to install their DG Set for construction activity. The department do not guarantee for continuous power supply for the work to be carried out.

Note: In case of EPC contracts, the firm shall be responsible for above as a whole since the complete building/structure to be built shall be within his scope and hence the above items are also in firm's scope, which may be defined in NIT accordingly.

1.13 Tools, Machinery and Equipment for Handling, Erection Testing and Commissioning

All tools, Machinery, Equipment and tackles required for handling of equipment and materials at site of work as well as for their assembly and erection and also necessary for testing and commissioning shall be the responsibility of the contractor.

1.14 PAYMENT TERMS

- 1.14.1 Unless otherwise specified in the conditions of the contract, the payment shall be made as per the relevant clauses of form PWD 7/8 forming a part of the tender documents.
- 1.14.2 Security deposit shall be deducted in such form and as per provisions contained in CPWD form 7/8 as amended from time to time. The security deposit shall be released on the expiry of guarantee/ maintenance period stipulated in the contract.

1.15 COORDINATION WITH OTHER AGENCIES

The contractor shall coordinate with all other agencies involved in the site/ building work so that the works of other agencies are not hampered due to delay in his work, security or other reasons, which directly affect the progress of building work, should be given priority.

1.16 CARE OF BUILDINGS

- 1.16.1 Care shall be taken by the contractor to avoid damage to the building during execution of his part of the work. He shall be responsible for repairing of all damages and restoring the same to their original finish at his cost. He shall also remove at his cost all unwanted and waste materials arising out of his work from the site.
- 1.16.2 As far as possible, cutting of roads, lawns etc. should be avoided. Where it becomes inescapable to cut them, these shall be repaired immediately.

1.17 STRUCTURAL ALTERATIONS TO BUILDINGS

- 1.17.1 No structural member in the building shall be damaged/altered, without prior approval from the competent authority through the Engineer-in-charge.
- 1.17.2 Structural provisions like openings, pipes, if any, provided by the department for the work, shall be used. Where these require modifications, such contingent works shall be carried out by the contractor, at his cost.
- 1.17.3 All cut out openings in floors provided by the Department shall be closed, after installing the cables, in accordance with the item therefore in the Schedule of work.
- 1.17.4 All cuttings made by the contractor in connection with the works shall be filled by him at his cost to the original finish.
- 1.17.5 In case of EPC tender, all above provisions/requirements shall be included in the scope of the work and nothing extra shall be payable on this account.

1.18 ADDITION TO AN INSTALLATION

An addition, temporary or permanent, shall not be made to the authorized load of an existing installation until it has been definitely ascertained that the current carrying capacity and the

condition of the existing accessories, conductors, switches etc. affected, including those of the Supply Authorities are adequate for the increased load and with Approval of Engineer-in-charge.

1.19 WORK IN OCCUPIED BUILDINGS

- 1.19.1 When work is executed in occupied buildings/complex, there should be minimum inconvenience to the occupants. The work shall be programmed in consultation with the Engineer-in-charge and the occupying department. If required, the work may have to be done even before and after office hours.
- 1.19.2 The contractor shall be responsible to abide by the Regulations or restrictions set in regard to entry into and movement within the premises of the site of work.
- 1.19.3 The contractor shall not tamper with any of the existing installations, including their switching operations or connections there to, without specific approval from the Engineer-in-charge.

1.20 DRAWINGS

- 1.20.1 The work shall be carried out in accordance with the drawing(s) if any, enclosed with the tender and also in accordance with modification(s) if any thereto from time to time approved by the Engineer-in-charge, and also instructions from him in the course of execution of the works.
- 1.20.2 All schematics, layout diagrams etc. shall be deemed to be 'Drawings' within the meaning of the terms as used in Clause 11 of the Conditions of Contract (PWD 7 or PWD 8).
- 1.20.3 All circuits, poles, feeder pillars etc. shall be indicated and numbered in the layout diagram(s).

1.21 CONFORMITY TO IE ACT, IE RULES, AND STANDARDS

- 1.21.1 All Electrical works, shall be carried out in accordance with the provisions of Indian Electricity Amendment Act, 2007 as amended upto date and Indian Electricity Rules 2005 As the Electricity (Amendment rule 2022 as amended up to date, Central Electricity Authority (Measure related to Safety and Electric Supply) Regulation, 2010 as amended upto date, National Electric Code, 2023 as amended upto date, National Building Code, 2016 as amended upto date, BIS standards as amended upto date, (Last Date of submission of tender or extended date of submission of tender unless specified otherwise).
- 1.21.2 The works shall also conform to relevant Bureau of Indian Standard Codes of Practice (COP) for the type of work involved. (See Appendix B for the list of such Codes).
- 1.21.3 Materials to be used in work shall be ISI marked, whenever such ISI marked materials are available.
- 1.21.4 In all electrical installation works, relevant, safety codes of practices shall be followed. Guidelines on safety procedure as outlined in the General Specification for Electrical works-part I (Internal), 2023 shall be followed.

1.22 GENERAL REQUIREMENTS OF COMPONENTS

1.22.1 Quality of materials

All materials and equipment supplied by the contractor shall be new i.e. the date of manufacturing of material shall not be more than six months from date of delivery at site. They shall be of such design, size and material as to satisfactorily function under the rated conditions of operation and to withstand the environmental conditions at site. Only quality materials of reputed make as specified in the contract will be used in work.

1.22.2 INSPECTION OF MATERIALS AND EQUIPMENT

- 1.22.2.1 Materials and equipment to be used in the work shall be inspected by the Departmental officers. Such inspection shall be of following categories:

- a) Inspection of materials/equipment to be witnessed at the Manufacturer's premises in accordance with relevant BIS/ Agreement Inspection Procedure.
 - b) To receive materials at site with Manufacturer's Test Certificate(s).
 - c) To inspect materials at the Authorized Dealer's Godowns to ensure delivery of genuine materials at site.
 - d) To receive materials after physical inspection at site.
- 1.22.2.2 The Departmental officers will take adequate care to ensure that only tested and genuine materials of proper quality are used in work.
- 1.22.2.3 Similarly, for fabricated equipment, the contractor will first submit dimensional detailed drawings for approval before fabrication is taken up in the factory. Suitable stage inspection at factory also will be made to ensure proper use of materials, workmanship and quality control.
- 1.22.2.4 The tender specifications will stipulate the inspection requirements or their waiver for various materials/ equipment including norms of inspection in specific cases.
- 1.22.2.5 All material shall be accompanied with manufacturer test certificate, acceptance test certificate, guarantee/warranty certificate, installation, testing and commissioning manual etc.
- 1.22.2.6 The material shall also be accompanied with other certificates as mentioned in the contract i/c Third party certification certificates.

1.22.3 Ratings of components

All current carrying components in an installation shall be of appropriate ratings of voltage, current, and frequency, as required at the respective sections of the electrical installation in which they are used, without their respective ratings being exceeded.

1.22.4 Conformity to Standards

- 1.22.4.1 All components shall conform to relevant Indian Standard Specifications, wherever existing. Materials with ISI certification mark shall be preferred.
- 1.22.4.2 A broad list of relevant Indian Standards is given in Appendix B. These Indian Standards, including amendments or revisions thereof up to the date of tender acceptance, shall be applicable in the respective contracts.

1.22.5 Interchangeability

Similar parts of all switches, lamp holders, distribution fuse boards, switch gears, ceiling roses, brackets, pendants, fans and all other fittings of the same type shall be interchangeable in each installation.

1.23 WORKMANSHIP

- 1.23.1 Good workmanship is an essential requirement to be complied with. The entire work of manufacture/ fabrication, assembly and installation shall conform to sound engineering practice.
- 1.23.2 Proper Supervision/Skilled Workmen
The contractor shall be a licensed electrical contractor of appropriate class suitable for execution of the electrical work. He shall engage suitably licensed and skilled workmen of various categories (in compliance to Clause-3 of Central electricity Authority (Measure related to Safety and Electric Supply) Regulation 2010 as amended upto dated for execution of work supervised by supervisors / Engineer of appropriate qualification and experience to ensure proper execution of work. They will carry out instructions of Engineer-in-charge and other senior officers of the Department during the progress of work.

1.23.3 Fabrication of panels :

Switch boards and LT panels shall be fabricated in a factory/ workshop having modern facilities like quality fabrication, seven tank process, powder/epoxy paint plant, proper testing facilities, manned by qualified technical personnel. The Panel manufacturer shall be minimum CPRI approved.

The tender shall specify some quality makes of fabricators with modern facilities of design, fabrication and testing capable of delivering high quality LT panels and switch boards after testing as per relevant specifications.

1.24 TESTING

All tests prescribed in these General Specifications & Standards of Bureau Indian Standards to be done before, during and after installation i/c testing of material, shall be carried out, and the test results shall be submitted to the Engineer-in-charge in prescribed proforma, forming part of the Completion Certificate. (See Appendix C).

1.25 COMMISSIONING ON COMPLETION

After the work is completed, it shall be ensured that the installation is tested and commissioned.

1.26 COMPLETION PLAN AND COMPLETION CERTIFICATE

1.26.1 For all works costing more than Rs. 1,00,000/- completion certificate as given in Appendix C including the proforma for test results shall be submitted to the Engineer-in-charge, after completion of work.

1.26.2 Completion plan drawn to a suitable scale indicating the following along with Soft copy and three print copies in A1 paper size duly signed and laminated shall also be submitted:

1.26.2.1 General layout of the site showing therein routes of cables, poles, Panels, tapping Point, Earthing, Controls and overhead lines.

1.26.2.2 Schedule of lengths, types and sizes of cables and overhead conductors.

1.26.2.3 Positions of all cable joints type wise, supports, stays, struts, lightning arresters, feeder pillars and pipes or closed ducts.

1.26.2.4 Positions of cable route markers and joint markers with respect to permanent land marks available at site.

1.26.2.5 Types of street light fittings, Lighting Control.

1.26.2.6 Name of work, job number, accepted tender reference, actual date of completion, names of Division/Sub-Division, and name of the firm who executed the work with their signature(s).

1.26.3 In the case of works costing less than Rs.1,00,000/- the completion plan shall be prepared by the department and signed by the contractor before final payment is made.

1.27 Guarantee

The installation will be handed over to the Department after necessary testing and commissioning. The materials supplied by the contractor will be guaranteed against any manufacturing defect, inferior quality. The guarantee period will be for a period of 12 months from the date of handing over to the Department. Installation/ equipment or components thereof shall be rectified/ repaired at the discretion of the Engineer-in-Charge.

Note: Conditions for LED fittings to be added.

1.28 MAINTENANCE

The periodicity of important maintenance activities is indicated in the Appendix D for guidance.

1.29 Voltage Classifications — Nomenclature of commonly used short names of voltages (A.C.):

1.29.1 ELV (extra low voltage) — A system with a nominal voltage $U_n \leq 50 \text{ V a.c.}$

1.29.2 LV (low voltage) — The voltage which normally exceeds 50 V but does not normally exceed 250 V.

1.29.3 MV (medium voltage) — The voltage which normally exceeds 250 V but does not exceed 650 V.

1.29.4 HV (high voltage) — The voltage which normally exceeds 650 V but less than or equal to 33 kV.

1.30 Permissible Voltage Variation:

1.30.1 The voltage at any point of the system under normal conditions shall not depart from the declared voltage by more than the values given below:

- (a) 6 percent in the case of low or medium voltage;
- (b) 6 percent on the higher side or 9 percent on the lower side in the case of high voltage
- (c) 12.5 percent in the case of extra high voltage.

1.31 License, Certificate of Competency, Permit for Electrical Work

Electrical installations shall be carried out under the supervision of a person holding a certificate of competency issued by the appropriate government. The workmen shall also hold the appropriate certificate of competency issued by appropriate government. The Contractor shall also have electrical license from appropriate Government for appropriate class and voltage.

Helpers/trainees may assist licensed person on the installation work strictly under supervision of licensed person who shall allow him to work within the limited area, which has not been connected to source of electrical supply.

CHAPTER 2

CABLE WORK

2.1 SCOPE

This chapter covers the requirements for the selection, installation and jointing of power cables for low, medium and high-voltage applications up to and including 33KV. For details not covered in these Specifications, IS:1255-1983 (Reaffirmed 1996) as amended up to date shall be referred to. All references to BIS- Specifications and codes are for codes with amendments issued up to date i.e. till the date of call of tender.

2.2 TYPES OF CABLES

2.2.1 The cables for applications upto and including 1.1kV supply shall be one of the following :-

- (a) PVC insulated and PVC sheathed, conforming to IS: 1554 (Part-1) – 1988 amended upto date.
- (b) Cross linked polyethylene insulated, PVC sheathed (XLPE), conforming to IS: 7098 (Part-1) – 1988 amended upto date.
- (c) Halogen Free Flame Retardant (HFFR) cable conforming to IS: 17048: 2018 amended upto date.

2.2.2 The cables for applications for high voltage (from 3.3 KV but upto and including 11KV supply) supply shall be one of the following :-

- (a) PVC insulated and PVC sheathed, conforming to IS: 1554 (Part-2) 1988 amended upto date.
- (b) Cross linked polyethylene (XLPE) insulated, PVC sheathed conforming to IS: 7098 (Part-2) – 1985, 2011 (Reaffirmed 2021) amended upto date.

2.2.3 The cables for applications above 11kV but upto. and including 33KV supply shall be one of the following :-

Cross linked, polyethylene insulated (XLPE) conforming to IS: 7098 (Part-2) – 1985, 2011 (Reaffirmed 2021).

2.2.4 The cables shall be with solid or stranded aluminium conductors, as specified. Copper conductors may be used, only in special applications, where use of aluminium conductors is not technically acceptable.

2.3 ARMOURING AND SERVING

2.3.1 All multi core cables liable for mechanical damage and all HV cables (irrespective of the situation of installation) shall be armoured. Where armouring is unavoidable in single core cables, either the armour should be made of nonmagnetic material, or it should be ensured that, the armouring is not shorted at terminations, thus preventing the flow of circulating currents therein.

2.3.2 Short runs of cables laid in pipes, closed masonry trenches and similar protected or secured enclosures need not be armoured.

2.3.3 PVC and XLPE cables, when armoured, shall have galvanized steel wires (flat or round) for armouring.

2.4 SELECTION OF CABLE SIZES

2.4.1 (a) LV (upto 1kV) (distribution) cables : The cable of sizes shall be selected by considering the voltage drop and current carrying capacity. The higher size among size as derived based on mentioned parameter shall be selected.

(b) HV cables: The HV cables shall be selected based on short circuit current rating and current carrying capacity. The higher size among size as derived based on mentioned parameter shall be selected.

2.4.2 While deciding upon the cable sizes, de-rating factors for the type of cable and depth of laying, grouping, ambient temperature, ground temperature, and soil resistivity shall be taken into account.

2.4.3 Guidance for the selection of cables shall be derived from relevant Indian Standards such as IS:3961 (Part-2)-2017 for PVC insulated and PVC sheathed heavy duty cables and IS:3961 (Part-6)-2017 for XLPE insulated and PVC sheathed heavy duty cables, IS:5819-1970 (Reaffirmed in November 2021) for recommended short circuit ratings of high voltage PVC cables, [IS:1255-1983 on code of practice for installation and maintenance of power cables up to and including 33KV rating etc. [See Table No. I (A) to I(D) and Table II to Table XVI for M.V. H.V. selection].

2.4.4 Voltage Drop in Consumers Installation:

The voltage drop between origin of an installation and any load point shall have limitations as given in the Table -XX.

2.4.5 The cable size shall be selected after due consideration of above clauses 2.4.3 and 2.4.4.

2.5 STORAGE AND HANDLING

2.5.1 Storage

- (a) The cable drums shall be stored on a well-drained, hard surface. so that the drums do not sink in the ground causing rot and damage to the cable drums. Paved surface is preferred, particularly for long term storage.
- (b) The drums shall always be stored on their flanges, and not on their flat sides.
- (c) Both ends of the cables should be properly sealed to prevent ingress by the insulation during storage.
- (d) Protection from rain and sun is preferable for long term storage for all types of cables. There should also ventilation between cable drums.
- (e) During storage, periodical rolling of drums is recommended (i.e. once in 3 months through 90 degrees). Rolling shall be in the direction of the arrow marked on the drum.
- (f) Damaged battens of drums etc. should be replaced, as may be necessary.
- (g) Cable drum protective closure should be maintained until Cable is utilized.
- (h) Observe Fire Precaution rules during storage.
- (i) If drums are expected to be stored for long time, it should be specially treated by use of Pesticides at regular intervals in the storage area to avoid termite and rodent attack on wooden drums.
- (j) In hot climate area the drums should be stored under a shade and should be protected from direct sunlight.

2.5.2 Handling:

2.5.2.1

- (a) When the cable drums have to be moved over short distances, they should be rolled in the direction of arrow marked on the drum.
- (b) For manual transportation over long distances, the drum should be mounted on cable drum wheels, strong enough to carry the weight of the drum, and pulled by means of ropes. Alternatively, they may be mounted on a trailer or on a suitable mechanical transport.
- (c) For loading into and unloading from vehicles, a crane or a suitable lifting tackle should be used. Small sized cable drums can also be rolled down carefully suitable ramp or rails, for unloading, provided damage is likely to be caused to the cable or to the drum.

2.5.2.2

- (a) If forklift or crane used for lifting purpose of cable, then correct slings and spindle should be taken with checking their conditions.
- (b) If a crane is to be used, ensure that a spreader is incorporated to prevent damage to drum flanges.
- (c) Ensure the drum bolts are tight.
- (d) Ensure that the truck surface is clear of obstructions, nails, etc.
- (e) Do not drop drums onto truck loading bed.
- (f) Do not lay cables drums flat.
- (g) When rolling a cable drum to change the direction, use two steel plates with grease between them and by standing one flange on these plates. Then the cable drum may be swivelled in the desired direction.
- (h) There are two methods of rolling drums offloading beds if cranes are not available. (See below).

Method 1: Hole excavated (Maximum slope 1 in 10) to receive truck.



Method 2: Ramp construction (Maximum slope 1 in 4).



Drum stop

2.6 INSTALLATION

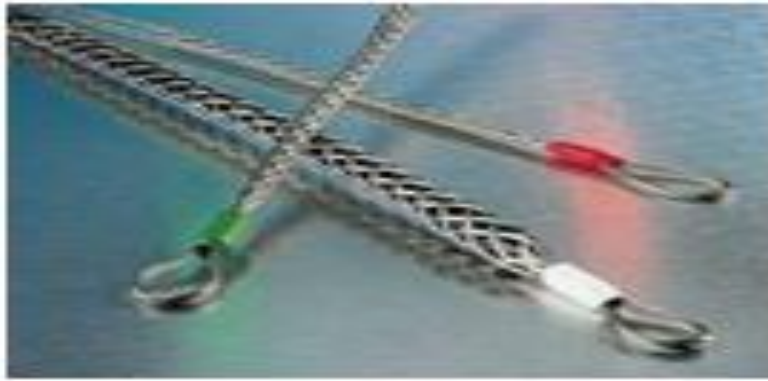
2.6.1 General

- (a) Cables with kinks, straightened kinks or any other apparent defects like defective armouring etc. shall not be installed.
- (b) Cables shall not be bent sharp to a small radius either while handling or in installation. The minimum safe bending radius for PVC/XLPE (MV) cables shall be 12 times the overall diameter of the cable. The minimum safe bending radius for XLPE (HV) cables shall be as given in Table-II. At joints and terminations, the bending radius of individual cores of a multi core cable of any type shall not be less than 15 times its overall diameter.
- (c) In case of PVC cables, suitable sealing compound/tape shall be used for this purpose, if likely exposed to rain in transit storage. Suitable heat shrinkable caps may also be used for the purpose.
- (d) Continuity of protective conductor shall not depend on its intermediate connections with enclosures/ system of wiring enclosures, equipment or armouring.
- (e) Mechanical Forces on Cables during Installation

All cables have a maximum pulling force which should not be exceeded during installation. The cable construction imposes the limitation on the pull-in force (F). When a cable stocking (Pulling Sock) is used the maximum force can be related to the overall cable diameter (D) in mm as follows:-

Steel Wire Armoured cables $F = 9 \times D^2$ (in Newtons)

Steel Tape Armoured or Unarmoured Cables $F = 5 \times D^2$ (in Newtons)



Limit the pulling force required to a minimum to avoid stretching in the outer layers of the cable which is particularly relevant when installing cables at high ambient temperatures. The thermoplastic bedding and sheathing materials may be softer than usual, and thus having reduced tensile properties.

2.6.2 Route

Before the cable laying work is undertaken, the route of the cable shall be decided by the Engineer-in-Charge considering the following:

- (a) While the shortest practicable route should be preferred, the cable route shall generally follow fixed developments such as roads, foot paths etc. with proper offsets so that future maintenance, identification etc. are rendered easy. Cross country run merely to shorten the route length shall not be adopted.
- (b) Cable route shall be planned away from drains and near the property, especially in the case of LV/MV cables, subject to any special local requirements that may have to be necessarily complied with.
- (c) As far as possible, the alignment of the cable route shall be decided after taking into consideration the present and likely future requirements of other services including cables enroute, possibility of widening of roads/lanes etc.
- (d) Corrosive soils, ground surrounding sewage effluent etc. shall be avoided for the routes.
- (e) Route of cables of different voltages:
 - (i) Whenever cables are laid along well demarcated or established roads, the LV/MV cables shall be laid far from HV cables.
 - (ii) Cables of different voltages, and also power and control cables shall be kept in different trenches with adequate separation. Where available space is restrict- such that this requirement cannot be met, LV/MV cables shall be laid above HV cables.
 - (iii) Where cables cross one another, the cable of higher voltage shall be laid at a lower level than the cable of lower voltage.

2.6.3 Proximity to communication cables

Power and communication cables shall as far as possible cross each other at right angles. The horizontal and vertical clearances between them shall not be less than 60cm.

2.6.4 Railway crossing

Cables under railway tracks shall be laid in spun reinforced concrete, or cast iron or steel pipes at such depths as may be specified by the railway authorities, but not less than 1m, measured from the bottom of the sleepers to the top of the pipe. Inside railway station limits, pipes shall be laid up to the point of the railway boundary or to a point to be decided by the railway authorities.

Outside the railway station limits, pipes shall be laid up to a minimum distance of 3m from the centre of the nearest track on either side.

2.6.5 Way Leave

Way leave for the cable route shall be obtained as necessary, from the appropriate authorities, such as, Municipal authorities, Department of telecommunication, Gas Works, Railways, Civil Aviation authorities, Owners of properties etc. In case of private property, Section 12/51 of the Indian Electricity Act shall be complied with.

2.6.6 Methods of laying

The cables shall be laid direct in ground, pipe, closed or open ducts, cable trays or on surface of wall. The method (s) of laying required shall be specified in the tender/ schedule of work.

2.6.7 Laying direct in ground

This method shall be adopted where the cable route is through open ground, along roads/ lanes, etc., and where no frequent excavations are likely to be encountered and where re-excavation is easily possible without affecting other services. This method shall be adopted where the cable route is through open ground, along roads/lanes, etc, and where no frequent excavation is likely to be encountered and where re-excavations easily possible without affecting other services.

2.6.8 Trenching

2.6.8.1 Width of trench

The width of the trench shall first be determined on the following basis (Refer figure 1).

- (a) The minimum width of the trench for laying a single cable shall be 35cm.
- (b) Where more than one cable is to be laid in the same trench in horizontal formation, the width of the trench shall be increased such that the inter-axial spacing between the cables, except where otherwise specified, shall be at least 20cm.
- (c) There shall be a clearance of at least 15cm between axis of the end cables and the sides of the trench.

2.6.8.2 Depth of trench

The depth of the trench shall be determined on the following basis (Refer figure 1)

- (a) Where the cables are laid in a single tier formation, the total depth of trench shall not be less than 75cm for cables upto 1.1KV and 1.2m for cables above 1.1KV.
- (b) When more than one tier of cables is unavoidable and vertical formation of laying is adopted, the depth of the trench in (a) above shall be increased by 30cm for each additional tier to be formed.
- (c) Where no sand cushioning and protective covering are provided for the cables below, the depth of the trench as per 2.6.8.2(a) & (b) above shall be increased by 25cm.

2.6.8.3 Excavation of trenches

- (a) The trenches shall be excavated in reasonably straight lines. Wherever there is a change in the direction, a suitable curvature shall be adopted complying with the requirements of clause 2.6.1.b.
- (b) Where gradients and changes in depth are able, these shall be gradual.
- (c) The bottom of the trench shall be level from stones, brick bats etc.
- (d) The excavation should be done by suitable means manual or mechanical. The excavated soil shall be stacked firmly by the side of the trench such that it may not fall back into the trench.
- (e) Adequate precautions should be taken not to damage any existing cable (s), pipes or any other such installations in the route during excavation. Wherever bricks, tiles or protective covers or

bare cables are encountered, further excavation shall not be carried out without the approval of the Engineer-in-Charge.

- (f) Existing property, if any, exposed during trenching shall be temporarily supported adequately as directed by the Engineer-in-Charge. The trenching in such cases shall be done in short lengths, necessary pipes laid for passing cables therein and the trench refilled in accordance with clause 2.6.8.5.
- (g) If there is any danger of a trench collapsing or endangering adjacent structures, the sides should be well shored up with sheeting as the excavation proceeds. Where necessary, these may even be left in place when back filling the trench.
- (h) Excavation through lawns shall be done in consultation with the Department concerned.

2.6.8.4 Laying of cable in trench

(a) Sand cushioning

- (i) The trench shall then be provided with a layer of clean, dry, sand cushion of not less than 8cm in depth, before laying the cables therein.
- (ii) However, sand cushioning as per (a) above need not be provided for MV cables, where there is no possibility of any mechanical damage to the cables due to heavy or shock loading on the soil above. Such stretches shall be clearly specified in the tender documents.
- (iii) Sand cushioning as per (a) above shall however be invariably provided in the case of HV cables.

(b) Testing before laying

At the time of issue of cables for laying, the cables shall be tested for continuity and insulation resistance (See also clause 2.8.1).

- (c) The cable drum shall be properly mounted on jacks, on a cable wheel at a suitable location, making sure that the spindle, jack etc. are strong enough to carry the weight of the drum without failure, and that the spindle is horizontal in the bearings so as to prevent the drum creeping to one side while rotating.
- (d) The cable shall be pulled over on rollers in the trench steadily and uniformly without jerks and strain. The entire cable length shall as far as possible be laid off in one stretch. PVC/XLPE cables less than 120 sq mm size may be removed by "Flaking" i.e. by making one long loop in the reverse direction.

Note: - For short runs and sizes up to 50sq cables, any other suitable method of direct handling and laying can be adopted without strain or excess bending of the cables.

- (e) After the cable has been so uncoiled, it shall be lifted slightly over the rollers beginning from one end by helpers standing about 10m apart and drawn straight. The cable shall then be lifted off the rollers and laid in a reasonably straight line.

(f) Testing before covering

The cables shall be tested for continuity of cores insulation resistance (Refer clause 2.8.2) and the cable length shall be measured, before closing the trench. The cable end shall be sealed/covered.

(g) sand covering

- (i) Cables laid in trenches in a single tier formation shall have a covering of dry sand of not less than 17cm above the base cushion of sand before the protective cover is laid.
- (ii) In the case of vertical multi-tier formation, after the first cable has been laid, a sand cushion of 30cm shall be provided over the base cushion before the second tier is laid. If additional tiers are formed each of the subsequent tiers also shall have a sand cushion 30cm as stated above. Cables in the top most tier shall have a final sand covering not less than 17cm before the protective cover is laid.

Sand covering as per g(i) and g(ii) above need not be provided for MV cables where a decision is taken by the Engineer-in-charge as per sub-clause (g) (i) above," but the inter tier spacing should be maintained as in g(ii) above with soft soil instead of sand between tiers for covering.

Sand cushioning as per g(i) and g(ii) above shall however be invariably provided in the case of HV cables.

(h) **Extra loop cable**

At the time of original installation approximately 3m of surplus cable shall be left on each terminal end of the cable and on each side of the underground joints. The surplus cable shall be left in the form of a loop. Where there are long runs of cables such loose cable may be left at suitable intervals as specified by the Engineer- in-Charge.

Where it may not be practically possible to provide separation between cables when forming loops of a number of cables as in the case of cables emanating from a substation, measurement shall be made only to the extent of actual volume of excavation, sand filling etc. and paid for accordingly.

(i) **Mechanical protection over the covering**

- (a) Mechanical protection to cables shall be laid over the covering in accordance with (b) and (c) below provide warning to future excavators of the presence of the cable and also to protect the cable against accidental mechanical damage by pickaxe blows etc.
- (b) Unless otherwise specified, the cables, shall be protected by second class brick of nominal size 22cm x 11.4cm x 7cm or locally available size, placed on top of the sand (or, soil as the case may be). The bricks shall be placed breadth-wise for the full length of the cable. Where more than one cable is to be laid in the same trench, this protective covering shall cover all the cables and project at least 5cm over sides of the end cables.
- (c) Where bricks are not easily available, or comparatively costly, there is no objection to use locally available material such as tiles or slates or stone/cement concrete slabs or precast concrete cable protection tiles (as per IS 5820). Where such an alternative is acceptable, the same shall be clearly specified in the tender specifications.
- (d) Protective covering as per (b) and (c) above need not be provided only for MV cables, in exceptional cases where there is normally no possibility of subsequent excavation. Such cases shall be particularly specified in the Tender specifications.
- (e) The protective covering as per (b) and (c) above shall, however invariably be provided in the case of HV cables.

2.6.8.5 Back filling

- (a) The trenches shall be then back-filled with excavated earth, free from stones or other sharp edged debris and shall be rammed and watered, if necessary in successive layers not exceeding 30 cm depth.
- (b) Unless otherwise specified, a crown of earth not less than 50mm and not exceeding 100mm in the centre and tapering towards the sides of the trench shall be to allow for subsidence. The crown of the however, should not exceed 10 cms. so as not to be a hazard to vehicular traffic.
- (c) The temporary reinstatements of roadways should be inspected at regular intervals, particularly during wet weather and settlements should be made good by further filling as may be required.
- (d) After the subsidence has ceased, trenches cut through roadways or other paved areas shall be restored to the same density and materials as the surrounding area and repaved in accordance with the relevant building specifications to the satisfaction of the Engineer-in- Charge.
- (e) Where road berms or lawns have been cut out of necessity, or kerb stones displaced, the same shall be repaired and made good, except for turfing/asphalting, to the satisfaction of the Engineer-in-Charge, and all the surplus earth or rock shall be removed to places as specified.

2.6.8.6 Laying of single core cables

- (a) Three single core cables forming one three phase circuit shall normally be laid in close trefoil formation and shall be bound together at intervals of approximately 1m.
- (b) The relative position of the three cables shall be changed at each joint at the time of original installation, complete transposition being effected in every three consecutive cable lengths.

2.6.8.7 Route markers

(a) **Location**

Route markers shall be provided along the cables at locations approved by the Engineer-in-Charge and generally at intervals not exceeding 100m. Markers shall also be provided to identify change in the direction of the cable route and at locations of underground joints.

(b) **Plate type marker**

Route markers shall be made out of 100mm x 5mm GI/aluminum plate welded/bolted on 35mm x 35mm x 6mm angle iron, 60cm long. Such plate markers shall mounted parallel to and at about 0.5m away from edge of the trench.

(c) **CC marker**

Alternatively, cement concrete 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate of 20mm in size) as shown in figure 2 shall be laid flat and centered over the cable. The concrete markers, unless otherwise instructed by the Engineer-in-Charge, shall project over the surrounding surface So as to make the cable route easily identifiable.

(d) **Inscription**

The words 'CPWD-MV/HV CABLE' as the case may be inscribed on the marker.

2.6.9 Laying in pipes / closed ducts

2.6.9.1 In locations such as road crossing, entry in to buildings, paved areas etc., and cables shall be laid in pipes or closed ducts. Metallic pipe shall be used as protection pipe for cables fixed on poles of overhead lines.

2.6.9.2

- (a) Stone ware pipes, GI, CI or spun, reinforced concrete pipes, DWC pipes shall be used for cables in general; however only GI pipe shall be used as protection pipe on poles.
- (b) The size of the pipe shall not be less than 10cm in diameter for a single cable and not less than 15 cm for more than one cable.
- (c) Where steel pipes are employed for protection single core cable feeding AC load, the pipe should be large enough to contain both cables in the case of single phase system and all cables in the case of poly phase system.
- (d) Pipes for LV, MV and HV cables shall be independent ones.

2.6.9.3

- (a) In the case of new construction, pipes as required (including for anticipated future requirements) shall be laid along with the civil works and jointed according to the CPWD Building Specifications.
- (b) Pipes shall be continuous and clear of debris or concrete before cables are drawn. Sharp edges if any at ends shall be smoothened to prevent damage to cable sheathing.
- (c) These pipes shall be laid directly in ground without any special bed except for SW pipe which shall be laid over 10cm thick cement concrete 1:5:10 (1 cement: S coarse sand: 10 graded stone aggregate of 40mm nominal size) bed. No sand cushioning or tiles need be used in such situations.

2.6.9.4 In certain instances when long runs in pipes or ducts are encountered it may be beneficial to grease the cable with petroleum jelly or some other non-aggressive compound to facilitate pulling-in.

Considerable damage can be done to cable servings at the mouth of a pipe and precautions should be exercised at such points. (The potential for damage is increased when a cable is installed in hot conditions, as materials may soften due to the heat and the tensile strength of the serving material is thus reduced). Fitting of a rubber grommet, or a Bell mouth, at the mouth to the pipe will reduce the potential for serving damage.

A Bell mouth (in separable hinged halves) will guide the cable into the duct, avoiding potential damage at the ducts entrance edges.

2.6.9.5 Road crossings

- (a) The top surface of pipes shall be at a minimum depth of 1m from the pavement level when laid under roads, pavements etc.
- (b) The pipes shall be laid preferably askew to reduce the angle of bend as the cable enters and leaves the crossing. This is particularly important for HV cables.
- (c) When pipes are laid cutting an existing road, care shall be taken so that the soil filled up after laying the pipes is rammed well in layers with watering required to ensure proper compaction. A crown of earth not exceeding 10cm should be left at the top.
- (d) The temporary reinstatements of roadways should be inspected at regular intervals, particularly after a rain, and any settlement should be made good by further filling as may be required.
- (e) After the subsidence has ceased, the top of the filled up trenches in roadways or other paved areas shall be restored to the same density and material as the surrounding area in accordance with the relevant CPWD Building Specifications to the satisfaction of the Engineer-in-charge.

2.6.9.6 Manholes shall be provided to facilitate feeding/ drawing in of cables with sufficient working space for the purpose. They shall be covered by suitable manhole covers. Sizes and other details shall be indicated in the Schedule of work.

2.6.9.7 Cable entry into the building

Pipes for cable entries to the building shall slope downwards from the building. The pipes at the building end shall be suitably sealed to avoid entry of water, after the cables are laid.

2.6.9.8 Cable-grip / draw-wires, winches etc. may be employed for drawing cables through pipes / closed ducts.

2.6.9.9 Measurement for drawing/ laying cables in pipes/ closed duct shall be on the basis of the actual length of the pipe / duct for each run of the cable, irrespective of the length of cable drawn through.

2.6.10 Laying in open ducts

- (a) Open ducts with suitable removable covers (RCC slabs or chequered plates) are generally provided in sub- stations, switch rooms, plant rooms, workshops etc, for taking the cables. The cable ducts should be of suitable dimensions for the number of cables involved.
- (b) Laying of cables with different voltage ratings in the same duct shall be avoided. Where it is inescapable to take HV & MV cables same trench, they shall be laid with a barrier between them or alternatively, one of the two (HV/MV) cables may be taken through pipe (s).
- (c) Splices or joints of any type shall not be permitted inside the ducts.
- (d) The cables shall be laid directly in the duct such that unnecessary crossing of cables is avoided.

- (e) Where specified, cables may be fixed with clamps on the walls of the duct or taken in hooks/ brackets/ troughs in ducts.
- (f) Where specified, ducts may be filled with dry sand after the cables are laid and covered as above, or finished with cement plaster, especially in high voltage applications.

2.6.11 Laying on surface

- 2.6.11.1 This method may be adopted in places like switch rooms, workshops, tunnels, rising (distribution) mains in buildings etc. This may also be necessitated in the works of additions and/or alterations to the existing installation, where other methods of laying may not be feasible.
- 2.6.11.2 Cables may be laid in surface by any of the following methods as specified:
 - (a) Directly clamped by saddles or clamps,
 - (b) Supported on cradles,
 - (c) Laid on troughs/trays duly clamped.
- 2.6.11.3
 - (a) The saddles and clamps used for fixing the cables on surface shall comply with the requirements given in Table-III.
 - (b) Saddles shall be secured with screws to suitable approved plugs. Clamps shall be secured with nuts on to the bolts, grouted in the supporting structure in an approved manner.
 - (c) In the case of single core cables, the clamps shall be of non-magnetic material. A suitable non-corrosive packing shall be used for clamping unarmoured cables to prevent damage to the cable sheath.
 - (d) Cables shall be fixed neatly without undue sag or kinks.
- 2.6.11.4 The arrangement of laying the cables in cradles permitted only in the case of cables of 1.1KV grade of size exceeding 120sqmm. In such cases, the cables may be suspended on MS flat cradles of size 50mm x 5mm which in turn shall be fixed on the wall by bolts grouted into the wall in an approved manner at spacing of not less than 60 cm.
- 2.6.11.5 All MS components used in fixing the cables shall be either galvanised or given a coat of red oxide and finished with 2 coats of approved paint.

2.6.12 Laying on cable tray

- 2.6.12.1 This method may be adopted in places like indoor stations, air-conditioning plant rooms, generator etc., or where long horizontal runs of cables required within the building and where it is not convenient to carry the cable in open ducts. This method is preferred where heavy sized cables or a number of cables are required to be laid. The cable trays may be either of perforated sheet type or of ladder type.

2.6.12.2 Perforated type cable tray

- (a) The cable tray shall be fabricated perforated MS double bended out of slotted/ sheets as channel sections, single or The channel sections shall be supplied in convenient lengths and assembled at site to the desired lengths. These may be galvanised or painted as specified. Alternatively, where specified, the cable tray may be fabricated by two angle irons of 50mm x 50mm x 5mm as two longitudinal members, with cross bracings between them by 50mm x 5mm flats welded/bolted to the angles at 1 m spacing. 2mm thick MS perforated sheet shall be suitably welded/bolted to the base as well as on the two sides.
- (b) Typically, the dimensions, fabrication details etc. are shown in figure 3A, B, and C.
- (c) The jointing between the sections shall be made with coupler plates of the same material and thickness as the channel section. Two coupler plates, each of mini- mum 200mm length, shall be bolted on each of the two sides of the channel section with 8mm dia round headed bolts, nuts and washers. In order to maintain proper earth continuity bond, the paint on the contact faces between the coupler plates and cable tray be scraped and removed before the installation.

- (d) The maximum permissible uniformly distributed load various sizes of cables trays and for different ported span are given in Table IV. the sizes shall specified considering the same.
- (e) The width of the cable tray shall be chosen so as to accommodate all the cables in one tier, plus 30 to 50% additional width for future expansion. This additional width shall be minimum 100mm. The overall width of one cable tray shall be limited to 800mm.
- (f) Factory fabricated bends, reducers, tee/cross junctions, etc. shall be provided as per good engineering practice. The radius of bends, junctions etc. shall not be less than the minimum permissible radius of bending of the largest size of cable to be carried by the cable tray.
- (g) The cable tray shall be suspended from the ceiling slab with the help of 10mm dia MS rounds or 25mm x 5mm flats at specified spacing (based on Table III). Flat type suspenders may be used for channels upto 450mm width bolted to cable trays. Round suspenders shall be threaded and bolted to the cable trays or to independent support angles 50mm x 50mm x 5mm at the bottom end as specified. These shall be grouted to the ceiling slab at the other end through an effective means, as approved by the Engineer-in-Charge, to take the weight of the cable tray with the cables.
- (h) The entire tray (except in the case of galvanised type) and the suspenders shall be painted with two coats of red oxide primer paint after removing the dirt and rust, and finished with two coats of spray paint of approved make synthetic enamel paint.
- (i) The cable tray shall be bonded to the earth terminal of the switch bonds at both ends.
- (j) The cable trays shall be measured on unit length basis, along the centre line of the cable tray, including bends, reducers, tees, cross joints, etc, and paid for accordingly.

2.6.12.3 Ladder type cable tray

- (a) The ladder type of cable tray shall be fabricated of double bended channel section longitudinal members with single bended channel section rungs of cross members welded to the base of the longitudinal members at a centre to centre spacing of 250cm.
- (b) Alternatively, where specified, ladder type cable trays may be fabricated out of 50mm x 50mm x 6mm (minimum) angle iron for longitudinal members, and 30mm x 6mm flat For rungs.
- (c) Typical details of fabrication and dimensions of both the types of trays are shown in figure 4A, B, C and D.
- (d) The maximum permissible loading, jointing of channel sections, width of the cable tray, provision of elbows, bends, reducers, horizontal tee/ cross junctions suspension of cable tray from the ceiling slab, painting and measurement of the cable tray shall be as per clause 2.6.12.2, except that the overall width of one cable tray may be limited to 800mm.

2.6.12.4 Cables laid on cable trays shall be clamped on to the tray at suitable intervals as per Table-III.

2.6.13 Cable identification tags

Whenever more than one cable is laid / run, side by side, marker tags as approved, inscribed with cable identification details shall be permanently attached to all the cables in the manholes / pull pits joint pits/ entry, points in buildings/ open ducts etc. shall also be attached to cables laid direct in at specified intervals, before the trenches are filled.

2.7 JOINTING

2.7.1 Location

- 2.7.1.1 Before laying a cable, proper locations for the proposed cable Joints, if any, shall be decided, so that when the cable is actually laid, the joints are made in the most suitable places. As far as possible, water logged locations, carriage ways, pavements, proximity to telephone cables, gas or water mains, inaccessible places, ducts, pipes, racks etc. shall be avoided for locating the cable joints.
- 2.7.1.2 Joints shall be staggered by 2m to 3m when joints are to be done for two or more cables laid together in the same trench.

2.7.2 Joints pits

- 2.7.2.1 Joint pits shall be of sufficient dimensions as to allow easy and comfortable working. The sides of the pit shall be well protected from loose earth falling into it. It shall also be covered by a tarpaulin to prevent dust and other foreign matter being blown on the exposed joints and jointing materials.
- 2.7.2.2 Sufficient ventilation shall be provided during jointing operation in order to disperse fumes given out by fluxing.

2.7.3 Safety precaution

- 2.7.3.1 A caution board indicating "CAUTION - CABLE JOINTING WORK IN PROGRESS" shall be displayed to warn the public and traffic where necessary.
- 2.7.3.2 Before jointing is commenced, all safety precautions like isolation, discharging, earthing, display of caution board on the controlling switchgear etc. shall be taken to ensure that the cable would not be inadvertently charged from live supply. Metallic armour and external metallic bonding shall be connected to where "Permit to work system is in vogue, procedures prescribed shall be complied with.

2.7.4 Jointing materials

- 2.7.4.1 Jointing materials and accessories like conductor ferrules, solder, flux, insulating and protective: tapes, filling compound, jointing boxes, heat shrinking joint kit, cold shrinkable joint kit, cast resin based joint kit etc. of right quality and correct sizes; conforming to relevant Indian Standards, wherever they exist, shall be used.
- 2.7.4.2 The design of the joint box and the composition of the filling compound shall be such as to provide an effective sealing against entry of moisture in addition to affording proper electrical characteristic to joints.
- 2.7.4.3 Where special type of splicing connector kits or epoxy resin spliced joints or heat shrinkable jointing kits or cold shrinkable jointing kits, cast resin based jointing kits are specified, materials approved for such application shall be used. Storing as well as jointing instructions of the manufacturer of such materials shall be strictly followed.

2.7.5 Joints

Jointing work shall be carried out by a licensed/experienced (where there is no licensing system for jointers) cable jointer.

2.7.6 Cable work with joints

- 2.7.6.1 About 3m long surplus cable shall be left on each side of joints as laid down in clause 2.6.8.4(h).
- 2.7.6.2 Insulation resistance of cables to be jointed shall be tested as per clause 2.8.1. Unless the insulation resistance values are satisfactory, jointing shall not be done.
- 2.7.6.3 Cores of the cables must be properly identified before jointing.
- 2.7.6.4 Where a cable is to be jointed with the existing cable, the sequence should be so arranged as to avoid crossing of cores while jointing.
- 2.7.6.5 Whenever the aluminium conductor is exposed to outside atmosphere a highly tenacious oxide film is formed which makes the soldering of aluminium conductor difficult. This oxide film should be removed by using appropriate type of flux.
- 2.7.6.6 The clamps for the armour shall be clean and tight.

2.7.7 Joint types

The type of joint shall be suitable for the type of cable as per Table V and shall be specified in the Tender Schedule of Work.

2.7.8 Jointing procedure

While it would be necessary to follow strictly the instructions for jointing furnished by the manufacturers of cables and joint kits, a brief on the jointing procedures is given for general guidance in Appendix E.

2.8 TESTING

2.8.1 Testing before laying

All cables, before laying, shall be tested with a megger for cables of 1.1KV grade, or with a 2500/5000V megger for cables of higher voltage. The cable cores shall be tested for continuity, absence of crossing, insulation resistance from conductors to earth /armour and between conductors.

2.8.2 Testing before backfilling

All cables shall be subjected to the above mentioned tests, before covering the cables by protective covers and back filling and also before taking up any jointing operation.

2.8.3 Testing after laying

2.8.3.1 After laying and jointing, the cable shall be subjected to a 15 minutes pressure test. The test pressure shall be as given in Table VI. DC pressure testing may normally be preferred to AC pressure testing.

2.8.3.2 In the absence of facilities for pressure testing as above, it is sufficient to test for one minute with 1,000V megger for cables of 1.1KV grade and with 2,500/5,000v megger for cables of higher voltages.

2.8.4 Mandatory Testing

The acceptance tests as mentioned in the applicable BIS codes of cables as mentioned in the clause 2.2.1 and 2.2.2 shall be treated as mandatory tests for respective cables. The Engineer in charge can carry out additional tests as required.

TYPICAL EXAMPLE FOR CABLE SELECTION

Step-I:- Assumptions

- Now total connected load (P) = 150 KW
- Supply Voltage – 415 Volt
- Power factor as 0.9 (As per NBC)
- Distance of the Load panel from the Main LT Panel is 100 mtrs
- Cable is directly buried in ground in trench at the depth of 0.75 meter
- Ground temperature is approximate 35°C
- Number of cable per trench is 1
- Thermal resistivity of soil is 1.5 K.m/W and considering Permissible Voltage Drop is 2%

Note 1:- Permissible Voltage drop may be taken as specified by concerned State Electricity board /NEC 2023 /NBC 2016 and ECBC, whichever is on lower side.

Note 2:- Power factor should be according to energy efficiency and Electrical Supply Agency requirements.

Step II: Application of cable Correction / de-rating factor

Connected Load in KVA	= KW/P.F. = 150/0.9 = 166.67 KVA
Full Load Current	= (KVA x 1000) / (1.732 x Voltage) = (166 x 1000) / (1.732 x 415) = 230 Amp.
Temperature correction factor (K1) when Cable in the Ambient Air and Cable group rating factor (K6) when cable laid in open air (K6) are not considered as Cable is in ground	
Ground temperature correction factor (K2)	= 0.96 (Refer Table XII)
Thermal Resistance Correction factor for Soil (Thermal resistance of soil is 1.5 Km/W) (K3)	= 1 (Refer Table XIV)
Cable depth correction factor (K4)	= 1.0 (Refer Table XIII)
Cable Grouping rating factor (K6)	= 0.84 (Refer Table XVI)
Total de-rating effect	= K2 x K3 x K4 x K6 = 0.96 x 1 x 1 x 0.84 = 0.8064
Net Load current after applying cable de-rating effect	= 230/0.8064 = 285 Amp
From Table IX Corresponding current rating, cable size may be selected 240 sq.mm	

Step III : From the Electrical Parameters taken from manufacturer specification/BIS

$$\begin{aligned}\text{Resistance R [Refer Table1(D)]} &= 0.163 \, \Omega/\text{km and} \\ \text{Reactance X [Refer Table1(D)]} &= 0.071 \, \Omega/\text{Km} \\ \text{Cos } \phi \text{ given is 0.9 so} & \\ \text{Sin } \phi &= 0.436, \\ \text{From the Formula } \sin^2\phi + \cos^2\phi &= 1 \\ \text{\% Voltage Drop of Cable} &= 1.732 \times \text{load Current} \times (\text{Rcos}\phi + \text{X}_L\sin\phi) \times \\ &\quad \text{Cable Length} \times 100 / (\text{line Voltage} \times \text{No. of run} \times 1000) \\ &= (1.732 \times 285 \times (0.163 \times 0.9 + 0.071 \times 0.436) \times \\ &\quad 100 \times 100) / (415 \times 1 \times 1000) \\ &= 2.11 \% \text{ (Voltage drop of cable} = 2.11\%) \end{aligned}$$

Here Voltage drop in 240 Sq.mm Cable (2.11%) is higher than considered Permissible Limit i.e. 2%. So, we will have to go to next higher size of Cable given in the current rating Table IX.

Attempt-2

Now, we take Next size of cable as 300 Sq.mm

$$\begin{aligned}\text{For 300 Amp load current, as per above formula, \% Voltage Drop of Cable} & \\ &= (1.732 \times 300 \times (0.132 \times 0.9 + 0.071 \times 0.436) \times 100 \times 100) / (415 \times 1 \times 1000) \\ &= 1.87\% \text{ Voltage drop of cable} \\ &= 1.87\% \text{ which is within permissible limit.} \end{aligned}$$

As such, we can select 3.5 x 300 sq mm Aluminium conductor XLPE Insulated, armoured cable for above assumed parameters.

CHAPTER 3

TRENCHLESS CABLE WORK

3.1 Introduction

Horizontal directional drilling (HDD) is a steerable trenchless method of installing underground pipes, conduits and cables in a shallow along a prescribed bore path by using a surface launched drilling rig with minimum impact on the surrounding area. Directional boring is used when trenching or excavation is not practical. Installation of a pipeline string by HDD is generally accomplished in following stages: -

(a) **Geo-technical investigation:**

The purpose of the geo tech investigation is not only to determine the feasibility of HDD crossing, but also to establish the most efficient way to accomplish it. On the geo tech information governs the determination of best crossing route along with selection of drilling tools and execution methodology.

(b) **Drill Path Design:**

One of the key considerations in the design of the drill path is creating as large a radius of curvature as possible within the limits of the right of way. Small radius of curvature induces bending stress and increases the pullback load. Curvature requirement are dependent on site geometry (crossing length, required depth to provide safe cover etc.) But, the degree of curvature is limited by the bending radius of the drilling rod and the minimum elastic bending radius of the pipe.

(c) **Pilot Hole Drilling:**

Pilot hole drilling typically is considered the most challenging and time consuming step. This is performed by mechanical action of fluid assisted cutting head.

(d) **Boring/reaming operation:**

Once the drilling of the pilot hole is completed, the reaming operation is to be started. The reaming operation consists of using an appropriate tool to open the pilot hole to a slightly larger diameter than the carrier pipeline. The percentage oversize depends on many variables including soil type, soil stability, depth, drilling mud, borehole hydrostatic pressure etc. Normal oversize may be from 1.4 to 1.5 times the diameter of the carrier pipe.

(e) **Pipe pull back and installation:**

The pullback operation involves pulling the entire pipeline string in one segment back through the drilling mud along the reamed hole pathway. The pulling equipment is attached to the leading end of the drill pipes string, and the prepared pipe string is fed gently into the bored hole.

3.2 Materials

(a) **Pipe**

Pipe used in this method should be smooth, flexible, and have sufficient strength to resist tension, bending, and external installation pressure loads. This method requires structurally strong joints that resist elongation or cross section reduction.

High-density polyethylene (HDPE) or steel pipes are normally used for this method. However, recently other pipe materials such as fusible PVC, restrained joint PVC, and ductile iron pipe have become available for this method.

HDPE pipes shall conform to the current ASTM 01248-05 standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable", & ASTM 0335006 •standard Specification for Polyethylene Plastics Pipe and Fittings Materials", and ASTM F714-06a "Standard Specification for Polyethylene (PE) Plastic Pipe (SDRPR) Based on Outside Diameter".

Steel pipe shall conform to the current ASTM A53-06a "Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded And Seamless", and ASTM A139/A 139M-04, "Standard Specification for Electric Fusion (Arc)-Welded Steel Pipe"

Ductile iron pipe shall confirm to the current ASTM 716-03 "Standard Specification for Ductile Iron Culvert Pipe" and ASTM 746-03 "Standard Specification for Ductile Iron Gravity Sewer Pipe"

PVC pipe shall confirm to the current ASTM F1962-05MStandard Guide for Use of Maxi-Horizontal Directional Drilling for Placement of Polyethylene Pipe or Conduit Under Obstacles, Including River Crossings" and ASTM 02321-05 "Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications" Or the current issues of the respective and equivalent Indian Standards as released by Bureau of Indian Standards.

3.3 Pipe dimensions

- (a) HDPE pipe shall have a minimum SDR of 11.
- (b) Pipe shall be round. Steel pipe shall have a roundness tolerance, so that the difference between the major and minor outside diameters shall not exceed 1% of the specified nominal outside diameter, or 6 mm, whichever is less. Likewise, HDPE, ductile iron, and PVC pipe shall have similar roundness tolerances.
- (c) Pipe shall have square and machine bevelled ends. The pipe end maximum out-of square tolerance shall be 1 mm, (measured across the diameter).
- (d) Pipe shall be straight in most cases. The maximum allowable straightness deviation over any 3.3 m length of steel casing pipe is 3 mm. Likewise, ductile iron, and PVC pipe shall have similar straightness tolerances. HDPE pipe does not need to be straight.
- (e) Pipe shall be without any significant dimensional or surface deformities. All pipes shall be free of visible cracks, holes, foreign material, foreign inclusions, blisters, or other deleterious or injurious faults or defects. Any section of the pipe with a gash, blister, abrasion, nick, scar, or other deleterious fault greater in depth than ten percent (10%) of the wall thickness, shall not be used and shall be immediately removed from the site.
- (f) Any of the following defects warrants pipe rejection:
 - (i) Concentrated ridges, discoloration, excessive spot roughness, and pitting.
 - (ii) Insufficient or variable wall thickness.
 - (iii) Pipe damage from bending, crushing, stretching or other stress.
 - (iv) Pipe damage that impacts the pipe strength, the intended use, the internal diameter of the pipe, and internal roughness characteristics.
 - (v) Any other defect of manufacturing or handling.

3.4 Protective Coatings (Steel Pipe)

The product pipe may be exposed to significant abrasion during pullback. Therefore, a coating to provide a corrosion barrier as well as an abrasion barrier is required. The coating shall be bonded well to the pipe and have a hard-smooth surface to resist soil stresses and reduce friction.

3.5 Drilling Fluids

All fluids used in a borehole during the drilling or reaming operations are termed as drilling fluids. Generally the drilling fluids are prepared by mixing additives like clays, bentonite, polymers and other thickening or thinning materials with water. Such additives provide the desired properties like Viscosity, Gel Strength, Filtration, and Lubricity to the drilling fluids to perform the above tasks.

These fluids are circulated in the borehole to minimize the intrinsic difficulties of horizontal drilling. The fluids should be able to perform the following tasks:

- (a) Stabilize the borehole;
- (b) Build wall cake and control filter loss;
- (c) Control groundwater pressures;
- (d) Provide gel strength for soil modification. (e) Remove cuttings from the borehole;
- (f) Provide steering capability by jet cutting;
- (g) Reduce torque on drill pipe;
- (h) Cool and lubricate the cutting bit and transmitter devices;
- (i) Reduce wear of drilling equipment;
- (j) Reduce friction encountered on pipe during Pulling Operations

Due to non-homogeneity of soils, the relative importance of above tasks change. The composition of drilling fluids, therefore, has to be location or site specific and designed accordingly.

The migratory drilling fluid leaving the drilling cycle, or spent drilling fluids or the fluid remaining after the work poses environmental threat and should not be left unattended but should be contained adequately or recycled back to the drilling system or disposed in suitable manners so as to avoid environmental pollution, clogging the drains, or carrying channels, or other underground appurtenances. The process should be in conformity to the local environmental requirements as in force at the time of works.

In Case 'Polymers' are being used along with Bentonite, 'Spoils/Cuttings' are to be collected in a collection pit, & stabilizing materials like bleaching-powder or other chemical as per the characteristics of the polymers should be added with the collected fluid to neutralize effects of 'Polymers' before disposing spoils from site.

3.6 CONSTRUCTION & SAFETY

(a) Construction

Construction plan

A construction plan shall be prepared before commencement of construction to assist in the control of the work. The plan shall contain a description of the construction; a health, safety and environment plan; and the quality requirements. The description of the construction should include methods, personnel and equipment required for the construction and working procedures.

The health, safety and environment plan (HSE Plan) should describe requirements and measures for the protection of the health and safety of the public, personnel involved in the construction and the environment. It should contain the applicable statutory and regulatory requirements,

identification of hazards and measures required for their control and procedures for handling emergency situations.

Some of the important components of the HSE plan are Safety plan and check lists, Identification of the responsible persons including Engineer-in-Charge, Operators, Support personnel. Utility location plans, Traffic Control Plans, Workmen Safety Plans, Drilling Reaming & Installation Precautions, and Emergency Preparedness for Electricity, Gas, Telecom/Optic Fibre network, Water sewer and storm water strikes.

All facilities, which may include existing roads and railways, rivers/canals, footpaths, pipelines, cables and buildings; that may be affected by the construction of the pipelines shall be identified prior to the beginning of the work. Temporary provisions and safety measures necessary to protect the identified facilities during construction should be established. A brief illustrative description of this planning is as under:

The Contractor shall prepare and submit a detailed execution methodology package, including design, engineering drawings and installation procedures etc. for review and approval for trenchless installation of pipeline before the project initiation.

(b) **Execution Methodology**

The execution methodology package shall include, but not limited to, the following:

- (i) Construction plan
- (ii) Site layout plan
- (iii) Operation schedule
- (iv) Communication plan
- (v) Emergency procedures
- (vi) Contingencies plan and drilling fluid management plan.
- (vii) Bore Path Profile.

(c) **Construction Plan**

The Contractor shall, as a minimum, include the following in the construction plan:

- (i) Location of entry and exit pits.
- (ii) Working areas and their approximate size.
- (iii) Proposed pipe fabrication and layout areas.
- (iv) State right-of-way lines, property lines.
- (v) Other utility right-of way and easement lines.
- (vi) Pipe material and wall thickness.
- (vii) Location of test pits or boreholes undertaken during the soil investigation.
- (viii) Proposed bore / drilling alignment (both plan & profile view) from entry to exit.
- (ix) All grades and curvature radii.
- (x) All utilities (both horizontal and vertical).
- (xi) Method of installation, covering all the stages of construction.
- (xii) Structures with their clearances from the proposed drill alignment.
- (xiii) Confirm minimum clearance requirements of affected utilities & structures
- (xiv) Required minimum clearances from existing utilities and structures.
- (xv) Diameter of pilot hole, number and size of pre-reams / back-reams.
- (xvi) Access requirements to site
- (xvii) Crew experience.
- (xviii) Equipment to be deployed for preparation of pipe string & installation of crossing.

- (xix) Tracking, monitoring & control equipment(s).
 - (xx) Pipeline string preparation details (hauling, stringing & welding etc.)
 - (xxi) Hydrostatic test procedure (Pre & post installation).
 - (xxii) Time schedule for construction.
- (d) **Construction Drawings:**
The Contractor shall, based on the result of design and engineering carried out by him, prepare construction drawings for the crossing. The construction drawings shall indicate the pipeline profile with levels furnished at sufficient intervals for proper control during construction. Other relevant details viz. entry and exit angles, radius of bends etc. shall also be indicated.
- (e) **Equipment Setup and Site Layout:**
(i) Required clear span shall be made available on the rig side to safely set up and operate the equipment.
(ii) Sufficient space shall be allocated to fabricate the pipeline into one string, thus enabling the pull back to be conducted in a single continuous operation.
The Contractor shall make arrangement for the work area(s) required for fabrication, construction, material storage & camp office/workshop etc.
- (f) **Communication Plan:**
The communication plan shall include the following:
(i) The phone numbers for communication with the Company's representative on the site.
(ii) Identification of all key personnel which will be responsible for ensuring that the communications plan is followed.
(iii) Means of communication between units/ teams involved in the crossing work.
- (g) **Contingency Plans:**
Contractor shall prepare contingency plans to handle any emergent situation during the operation. The Contingency plan shall besides other aspects also address the following:
(i) Inadvertent boring into a natural gas line, live power cable, water main, sewer lines, or a fibre-optic cable etc.
(ii) Containment and removal, of an inadvertent return or spill (e.g., drilling fluids, and hydraulic fluids).
- (h) **Drilling Fluids Management Plan**
The Contractor shall provide the following information as part of the drilling fluid management plan:
(i) Identify source of fresh water for mixing the drilling mud
(ii) Method of slurry containment
(iii) Method of recycling drilling fluid and spoils (if applicable)
(iv) Method of transporting drilling fluids and spoils off site
- (i) **Precautions to be observed during installation**
While formulating the installation procedures, following aspects should be given due consideration:
(i) Width of crossing, minimum requirements of cover, limitations of deployed equipment and additional drilling lengths as may be necessary for project execution.

- (ii) Access to work site.
 - (iii) Avoidance of inconvenience to public.
 - (iv) Undue interference with the normal use of the land and watercourses.
- (j) **Pre-Construction Survey & Site Evaluation**
- Prior to commencement of the execution of works, following actions in order to ensure safe and efficient construction with minimum interruption of normal, everyday activities at the site must be taken:
- (i) Inspect the site for evidence of substructures such as manhole covers, valve box covers, meter boxes, electrical transformers, conduits or drop lines from utility poles, and pavement patches etc.
 - (ii) Notify owners of subsurface utilities along and on either side of the proposed path of the impending work. All utilities along and on either side of the proposed path shall be located.
 - (iii) Obtain all necessary permits or authorizations to carry construction activities near or across all such buried obstructions.
 - (iv) Construction schedule should be arranged so as to minimize disruption (e.g. drilling under railroad beds, major highways, and/or river crossings).
 - (v) The proposed path should be determined and documented, including its horizontal and vertical alignments and the location of buried utilities and substructures along the path.
 - (vi) Ensure availability of sufficient clear span at the site for:
 - Entrance and exit pits
 - Stationing of equipment and its safe unimpeded operation
 - Support vehicles
 - Fusion machines
 - (vii) Stringing out the pipe to be pulled back in a single continuous Operation.
 - (viii) Suitability of soil conditions.
 - (ix) Interference Check to get the level of Active Interference in the desired path.

3.7 Crossings

(a) General

As far as possible, crossings shall be made at right angles. The surveyor shall record the angles of crossing for all fences, property lines, utilities, roads, railways, canals, streams, etc. that are crossed. In addition, the true bearings of the center-line of the road, railway, canal, as well as that of the pipe centre-line shall be recorded. Turning Points (TPS) provided near crossings shall be located, at least 50 m from the boundary of the crossings, in stable and firm ground.

(b) Railway crossings

The angles for all railway crossings shall be as close to 90 degrees as possible, but in no case less than 85 degrees to the centre-line of the railway tracks.

(c) Road crossings

The angle of crossings for primary roads i.e. national and state highways shall be as close to 90 degrees as possible, but in no case less than 80 degrees to the center-line of the road. The angles of crossing for secondary roads shall be as close to 90 degrees as possible, but in no case less than 45 degrees to the centre-line of the roads. All seasonal roads, unpaved village roads, cart-tracks, etc. come under this category.

(d) Major rivers crossings

These crossings shall be established as close as possible to the locations shown on the route map. Crossings shall be located in a comparatively straight reach of the river, where the banks are

stable and there is sufficient area for construction. Angle of crossing (Centre line of the pipeline to the flow direction) shall be as close to 90 degrees as possible. Wherever there is an evidence of bank erosion and the slope is less than 45 degree, application of necessary bank protection measures should be explored. In case of difficulty in their implementations, an alternate location should be selected.

(e) **Canals/Drainage/Ditch/Nala/Stream crossings**

The angle of crossing shall be as close to 90 degrees as possible but in no case less than 60 degrees to the centre-line of the canal/drainage ditch. Crossings shall be located where there is no evidence of slumping or erosion of banks or bed.

(f) **Existing Utility crossings**

The specific requirements of owner/operator of existing utility shall generally be followed. Unless specified otherwise, there shall be a minimum clearance of 300 mm between the existing Utilities and pipelines.

3.8 Alignment

The bore layout and the final pipe alignment must meet the alignment requirements with respect to minimum and maximum depths and minimum clearances as stipulated hereafter:

(a) **Minimum Allowable Depths**

The minimum allowable installation depth of cover of a HDD installed pipe under the ground at different locations is correlated to the locations in as given below as per requirements of local/statutory authorities, whichever is higher.

Minimum Allowable Depth		
Location	Minimum Allowable Installation Depth	Measurement to be made from
Under Road/Highways	1.5 m	The lowest point of the road/highway camber
Under Railways tracks	2.0 m	The soffit of the track
Under Lined Canal	2.5 m	The bottom of lining
Under Unlined Canal	4.0 m	Below Scour Level
Under River	10.0 m	Below Scour Level
Under Water bodies	4.0 m	Below Bed Level

To help with future locating of installed pipes, installation of a trace wire on plastic pipes and submission of an as-built (both plan and profile) for all installations are required.

In locations where the road surface is super-elevated, the minimum depth of the bore shall be measured from the lowest side of the pavement surface.

(b) **Maximum Allowable Depths**

The Maximum allowable depths for gravity flow networks where invert levels play a major role in fixing the pipeline levels, the maximum depths any no location shall exceed the maximum depths stipulated/designed at the installation locations. For other networks where the limits depths

are not critical, the maximum depth shall be maintained within the stipulated region to retain the maintainability.

(c) **Minimum Clearance**

A minimum clearance as given below shall be provided between a new HDD crossing and other underground structure not used in conjunction with the new pipeline. Wherever such clearance cannot be provided, adequate precautions to protect the pipeline by means of casing, bridging, or insulating material shall be ensured.

Minimum Clearance		
SI. No.	Description	Minimum Clearance
1	General Structures	500 mm
2	Gas or Petrochemical product structures	2000 mm
3	Water Sewer or drainage lines	1000 mm
4	Electrical cables	500 mm

3.9 Equipment

Equipment used for this method varies greatly. However, the basic operations of boring and pulling the pipe into position are essential. In addition, deployment of suitable mud pumps and motors for critical crossings may be important. A proper equipment deployment schedule including the anticipated spares based on the site requirements should be prepared and approved prior to the work initiation as a part of the construction plan.

For equipment sizing, reference may be made to the specific equipment operator's manual for more information.

Any equipment deployed or proposed to be deployed must be capable to perform the intended duties. For older equipment, or where the equipment capacity is suspect, authentication of the equipment capacity should be done prior to deployment.

Detailed hereafter is an illustrative list of some of the equipment considered essential for HDD works:

(a) **Drilling Rigs**

- (i) Drilling rigs must have a hydraulically powered system to rotate, push, and pull hollow drill pipe into the ground at a variable angle while delivering a pressurized fluid mixture to a trackable and guidable drill head.
- (ii) Drilling rigs must have electrical grounding capabilities for use during drilling and pull-back operations.
- (iii) Any significant mechanical deficiencies identified during inspection on a previous project must be repaired prior to starting a new project.

(b) **Tracking/Guidance Systems**

The guidance system should be of a proven, accurate type with an interface meeting the following requirements:

- (i) Must provide a continuous and accurate determination of the location of the drill head during the drilling operation.

- (ii) Must be capable of tracking at all depths expected in the project in any soil condition (including hard rock).
- (iii) Sonde must provide immediate information on the tool face, azimuth (horizontal direction indicator ball in box) and inclination/pitch (vertical direction) and must be accurate to $\pm 5\%$ of the vertical depth of the bore hole.
- (iv) System must be setup and operated by personnel trained and experienced. Operator must be aware of any geo-magnetic anomalies and must consider such influences in the operation of the guidance system if using a magnetic system.

(c) **Drilling Fluid Mixing Systems**

Self-contained, closed drilling fluid mixing system should be of sufficient size to mix and deliver drilling fluid composed of bentonite clay, potable water and appropriate additives.

The mixing system must be able to molecularly shear individual bentonite particles and polymers from the dry powder to avoid clumping and ensure thorough mixing.

The system must be able to continually agitate the drilling fluid during drilling operations.

Any significant mechanical deficiencies identified by inspector on a previous project must be repaired prior to starting a new project.

(d) **Pipe Rollers**

Sufficient number of pipe rollers or skids should be used to prevent excess sagging of the pipe and/or dragging of the pipe on the ground.

(e) **Reamers**

An appropriate reamer type (e.g., barrel, spiral, fluted, etc.) that is designed for the specific soil type that it will encounter should be used.

(f) **Testing Equipment**

Following testing equipment needed to document the physical and chemical properties of the drilling fluid may also be used:

- (i) Slurry test kit (to test viscosity and density)
- (ii) pH test strips (to test the make-up water)
- (iii) Hardness test strips (to test the make-up water quality)

3.10 Method

A pre-determined bore path plan should be prepared, using the approved bore planning software. Bore path plans and related requirements should be approved by the engineer-in-charge and subsequently discussed and agreed with the contractor during the project pre-construction meeting.

(a) **Staking/Field Marking**

The planned entry point, exit point, and other points of interest along the bore path, as required, to clearly communicate the desired path should be field marked. Wooden stakes with yellow ribbon, or white paint on the ground should be used along with the defined station number, where determined necessary by the engineer.

(b) **Separation from Other Utilities**

- (i) Separation from Other Utilities Bore paths should be designed with a minimum separation as stipulated in 3.8 (c) from other utilities (measured from centreline of bore path to outside of adjacent utility) to account for the $\pm 300\text{mm}$ allowable variance of the drill head in relation to the planned bore path, the radius of the reamer, and to provide a minimal error margin.

- (ii) If additional utilities are discovered in the field, or if potholing reveals that certain utilities are at significantly different depths than was assumed when the bore path was created, the Engineer-in-Charge or Inspector must be notified so that a revised bore path plan can be generated and approved prior to the start of the drill.
- (c) **Crossing Profile & Other Parameters**
 Basic profile drawings of rivers and other obstacles should be provided in the Bidding Document. Contractor shall specify proposed profile & other parameters and submit it to Engineer-in-Charge for approval the acceptance as final profile and other parameters and their limited value that define the location & configuration of the bore path. These should include, but not limited, to the following:
 - (i) Locations of entry and exit point.
 - (ii) Equipment and pipe layout areas.
 - (iii) Proposed drill path alignment (both plan & profile view).
 - (iv) Location, elevations and proposed clearances of all utility crossings and structures.
 - (v) Proposed depth of cover.
 - (vi) Soil analysis.
 - (vii) Product material, length, diameter-wall thickness, reamer diameter.
 - (viii) Detailed pipe calculations, confirming ability of product pipe to withstand installation loads and long term operational loads.
 - (ix) Proposed composition of drilling fluid (based on soil analysis), viscosity and density.
 - (x) Drilling fluid pumping capacity, pressures and flow rates proposed.
 - (xi) State right-of-way lines, property, and other utility right-of-way or easement lines.
 - (xii) Elevations.
 - (xiii) Type of tracking method/ system with Log While Drilling Facility for machine recording as built in the Tracking device itself during locating process.
 Survey grid establishment for monitoring ground surface movement (settlement or heave) due to the drilling operation.
- (d) **Bore Path Geometry**
 - (i) When generating a bore path plan, the drill entry angle will be limited by the drilling machine, but should generally be designed between 10°-12° from horizontal if drilling into existing grade. If drill entry is into an excavated pit, shallower entry angles may be desirable.
 - (ii) The drill exit (pipe entry) angle should be limited to 5° from horizontal, when possible. Increased drill exit (pipe entry) angles up to 10° may be approved by the Engineering Manager on a case-by-case basis.
- (e) **Maximum Permissible Load on Pipeline**
 The Contractor shall carry out engineering calculations for determining the maximum permissible load on pipe, to check that the empty pipeline is safe from collapse at any point along the bored/drilled cross section.
- (f) **Pipeline Axis**
 The plane containing the pipeline route axis shall be perpendicular to the horizontal plane. There shall be no bending of the pipeline route axis at depths lower than 2 meters below the ground level.
- (g) **Pipe Edges**
 The ends of each section of HDPE pipe shall be inspected and cleaned as necessary to be free of debris immediately prior to joining the pipes by means of thermal butt-fusion. The Polyethylene pipe shall be of the same type, grade, and class of the polyethylene compound used in the

process. This process provides joint weld strength equal to or greater than the tensile strength of the pipe.

(h) **Pipe Handling**

The handling of the joined pipeline shall be in such a manner that the pipe is not damaged by dragging it over sharp or jagged objects. Sections of the pipes with cuts and gouges exceeding 10 percent of the pipe wall thickness or kinked sections shall be removed and the ends rejoined.

(i) **Storage**

HDPE Pipes shall be stored on level ground, free of sharp objects, which could damage the pipe. Stacking of the polyethylene pipe shall be limited to a height that will not cause excessive deformation, bending, or warping of the bottom layers of pipes under anticipated temperature condition.

(j) **Pipe String Fabrication**

Sufficient space shall be allocated to fabricate and layout the product pipeline into one continuous pipe length, thus enabling the pull back to be conducted during a single operation. If space considerations are discovered that make this impossible, the permit applicant shall obtain specific alternative instructions from the owner's Engineer.

(k) **Working Space**

Sufficient space is required on the rig side of the machine to safely set up and perform the operation.

(l) **Drill Path Alignment**

The drill path alignment shall be as straight as possible to minimize the frictional resistance during pullback and maximize the length of the pipe that can be installed during a single pull.

(m) **Radius of Curvature**

The Contractor shall carry out engineering computations for determining the minimum allowable radius of curvature for the pipeline.

For plastic pipe, the actual minimum bending radius for boring should be at least 50 times the nominal pipe diameter (although a bending radius of 25 times the nominal diameter is allowed for above-ground pipe coils and pipe strings prior to pullback). However, for HDD installations involving plastic pipe, this bending radius can never be achieved because the minimum bending radius is normally limited to that of the drilling rods (usually 30-35 m, depending on the rod). Although the bending radius of the rods may be used if necessary, a 60 m bending radius should be used for plastic pipe when possible to limit pullback forces, provide room for corrections and a margin for error, and to facilitate a smooth bore.

For steel pipe, the minimum safe bending radius should be such that the bending stress does not exceed 75% of SMYS, calculated as:

Safe Bending Radius (R, m) = $(348,000 \times D) / (2 \times 0.75 \times SMYS \times 12)$, Where D = pipe O.D. in mm.

However, a bending radius of 1.2 m per mm of nominal pipe diameter (1,200 times the nominal diameter of the pipe to be installed) should be used when possible to limit pullback forces, provide room for corrections and a margin for error, and to facilitate a smooth bore.

(n) **Entry angles**

Entry and exit angles are limited by equipment capabilities. However, per most mini-HDD drilling rigs' design, the best entry angle should be between 100 and 120 and for Maxi-HDD

drilling rigs design best Entry angle should be between 60 and 80, subject to the condition that pipeline combined stress shall not exceed 95% of the SMYS of the pipe material.
Contractor shall determine the required pipeline configuration to allow smooth pull in the crossing entry point and admissible stress in the supported pipeline string.

(o) **Exit angles**

Exit angles should generally range from 50 (for large diameter steel pipelines) to 100. However, when high exit angles are encountered or designed, the pipe must be supported in an elevated position during the pullback operation to prohibit the pipe from bending, deforming, kinking, or even breaking.

(p) **Wireline Tracking**

In cases of River Crossings and other such locations where walk over locating is difficult or not feasible due to site constraints, Probe-system (Wire-line-system) should be adopted. The relative drill head position should be recorded with the help of device capable of recording Azimuth and Depth based on calculations without the need for walkover locating to ensure that bore path is confirming to the designed drilling path and future record of product pipeline is available to facilitate future utility laying near the same.

(q) **Sewer Bores**

The stringent grade and line requirements of Gravity Sever Bores require minimum "Pitch Accuracy" of 0.1% for the tracking device to be used for drilling pilot bore for its installation.

(r) **Drill Head Location Recording**

In-order to ensure that the bore path is confirming to the designed drilling path, relative position of cutting head should be recorded at regular interval.

(s) **Void Backfilling**

Should the drilling operation be unsuccessful, the contractor shall ensure the backfill of any void(s) with flowable fill.

(t) **Pipe String Preparation**

The required piping shall be assembled in a manner that does not obstruct adjacent roadways or public activities. The HDD operator shall erect temporary fencing around the entry and exit pipe staging areas.

(u) **Hole Reaming**

Several pre-reams may be employed to gradually enlarge the hole to the desired diameter and reduce road surface heaving potential. Back ream diameter increase shall be gradual to avoid clenching of the reamer in the hole or abrupt rise in the pulling force due to abrupt diameter increase. Furthermore, during the final pullback, the pull-back rate shall not exceed 3 m per minute.

(v) **Pipe End Sealing**

The pipe shall be sealed at both ends with a cap or a plug to prevent water, drilling fluids and other foreign materials from entering the pipe as it is being pulled back.

(w) **Swab Reaming**

To avoid damage by leftover cuttings in HDD bore-path during installation of HDPE pipes, Swab reamer of size slightly higher than product pipe may be passed through the hole ahead of product line installation.

(x) **Hole Stabilization**

In case of Sand formation/lose of the soil conditions, suitable hole stabilizing additives to be used along with the mud so as proper wall cake formation takes place and hole does not collapses on the product pipeline.

(y) **Pipe Protection**

Pipe rollers, skates or other protective devices shall be used to prevent damage to the pipe, eliminate ground drag, reduce pulling force, and reduce the stress on the pipe and joints.

(z) **Annulus Fluid**

The drilling fluid in the annular region outside of the pipe shall not be removed after installation, and remain in place to provide support for the pipe and neighbouring soil.

(aa) **Pressure Measurement**

To avoid damage & buckling of busy highways and roads, mud pressure during pull back of large diameter pipes (Greater than 500mm) need to monitored in real time; "Mud Pressure" monitoring device may be installed. And record of Mud Pressure on time line can be recorded for future inspection on adherence of the same.

(bb) **Proof Length**

In case of HDPE pipe, an extra 1.8 m section of the pipe must be pulled out of the borehole to check for any sign of stress or damage.

3.11 Drilling Site

(a) **Drill Entry & Exit Locations**

A minimum distance of 6 m, from the edge of the paved shoulder or curb to the face of any equipment, and supplies, shall be maintained in areas posted at 50 kmph or less; otherwise, a minimum distance of 9 m shall be maintained. The selection of the drill and exit locations should be based on the following considerations:

- (i) The terrain must be cleared, leveled and suitable for the work (sites with negligible longitudinal or side slopes should be preferred); In case such locations are unavailable, proper restrains should be used.
- (ii) Entry and exit location should be of sufficient size and configuration to undertake the work safely; this should include consideration of:
 - Drill rig entry and exit points (generally the entry point should ideally be at an equal or lower elevation than the exit point);
 - rig size and layout requirements;
 - pipe lay down area or false right-of-way (note that a straight approach to the exit point is preferred to avoid the need for false right-of-way);
 - fabrication area;
 - returns pit; and
 - bulk storage of materials;
- (iii) The resulting drill path must be feasible with a low risk of inadvertent returns; and
- (iv) Existing infrastructure and land use.

(b) **Protection**

At discretion of owner's Engineer/Inspector, traffic barriers shall be installed adjacent to machine site locations according to the owner's plans and current Standard Specifications for such Construction. Temporary beam guardrail shall also be installed according to the current applicable Standard Specifications for Construction.

Fencing barriers shall be installed adjacent to equipment and supplies with suitable fencing and barricading to prohibit pedestrian access to the work site. Equipment shall not be used as fencing to protect access pits.

3.12 Reaming Coefficient

The reaming coefficient should be 1.20 for stable grounds with low frictional coefficients and of 1.50 for unstable grounds, the walls of which tend to give way, and for grounds with high frictional coefficient. A lesser diameter should be preferred to ensure excessive voids are avoided to restrict post installation settlement.

3.13 Watertight Joints

Water tight pipe joints are required to ensure the integrity of the roadbed. Pipe shall be constructed to prevent water leakage or earth infiltration throughout its entire length.

A watertight specification for each type of pipe material should be obtained through each pipe material manufacturer/relevant BIS standards. Necessary reference must be made to the appropriate industry specifications for more detailed information.

3.14 Drilling Fluids

- (a) Drilling fluid shall be used during drilling and back reaming operations. Using water exclusively may cause a collapse of the borehole while in unconsolidated soils, and may also cause soil swelling while in clay soils. Either case may significantly impede the installation of the pipe.
- (b) Excess drilling fluids shall be contained within a lined pit or containment pond, or trailer-mounted portable tank, until removed from the site.
- (c) All drilling fluids shall not enter the streets, manholes, sanitary and storm sewers, and other drainage systems, including streams and rivers.
- (d) Any damage to any highway or non-highway facility caused by escaping drilling fluid, or the directional drilling operation, shall be immediately restored by the HDD operator.

3.15 Pipe Locating and Tracking

- (a) During construction, continuous monitoring and plotting of pilot drill progress shall be undertaken to ensure compliance with the proposed installation alignment and allow for appropriate course corrections to be undertaken. Monitoring shall be accomplished by manual plotting based on location and depth readings provided by the locating/tracking system or by computer generated bore logs which map the bore path based on information provided by the locating/tracking system. Readings or plot points shall be undertaken on every drill rod.
- (b) Pipe installed by the HDD method shall be located in plan as shown on the drawings, and shall be no shallower or deeper than shown on the Drawings unless otherwise approved. The Contractor shall plot the actual horizontal and vertical alignment of the pilot bore at intervals not exceeding 9 m. This "as built" plan and profile shall be updated as the pilot bore is advanced. The HDD operator shall at all times provide and maintain instrumentation that will accurately locate the pilot hole and measure drilling fluid flow and pressure. The HDD operator shall grant the Engineer/ Inspector access to all data and readout pertaining to the position of the bore head, the fluid pressures, and flows.

3.16 Settlement/Heaving Monitoring

- (a) This method shall be performed in a manner that will minimize the movement of the ground in front of, above, and surrounding the boring operation; and will minimize subsidence of the surface above and in the vicinity of the boring.
- (b) Potential settlement shall be monitored at each edge of right of way, each shoulder point, each edge of pavement, the edge of each lane (or centerline for two lane roads), and otherwise at 15 m intervals along the pipe centerline.
- (c) A survey shall be performed 1 day prior to initiating this operation at each required monitoring location. A similar survey shall then be performed at each location, on a daily basis, until the permitted activity has received a final inspection. This survey establishes the pre-existing and post construction conditions, and the amount of settlement. All survey readings shall be recorded to the nearest one-hundredth (0.01) of a meter. Whenever possible, trenchless pipe installations shall not be installed directly under a pavement crack. Digital photographs of the pavement conditions shall also be taken prior and after the pipe installation.
- (d) All operations shall stop immediately whenever monitored points indicate a vertical change in elevation of 12 mm or more, or any surface disruption is observed. The Contractor shall then immediately report the amount of settlement to the owner's Engineer/Inspector.

3.17 Stress Evaluation

- (a) Maximum Longitudinal Stress during Installation
 - (i) Total maximum longitudinal stress in the pipeline due to tension and bending at location shall not exceed 90% of the SMYS of the pipe material. Prior to the pipe installation, stress evaluation must be done as per the following limits:
 $A_s > B_s + H_s$
Where
 A_s Allowable stress or 90% of S.M.Y.S. (Specified Minimum Yield Strength)
 B_s Bending stress or $ED/2R$ where E = Modulus of Elasticity, D = Pipe Diameter, R Radius
 H_s = Hoop Stress or $PD/4T$ where P = Design pressure of pipeline, D = Pipe Diameter, T = Wall thickness.
 - (ii) For the drilling projects exceeding 150 m in length, engineer should calculate the estimated tensile load (pull- back force), to ensure that the estimated pullback forces do not exceed the allowable tensile load of the pipe that is to be installed. The maximum permissible stress calculated should be reduced by an applied design factor of 0.75 (safety factor of 1.33) to conservatively account for If borehole imperfections and geological unknowns. Necessary, HDPE pipe and/or heavier wall pipe may be required or it may be necessary to fill the pipe with water to reduce friction with the top of the borehole.
- (b) Maximum Equivalent Stress during Final Hydrostatic Test during the hydrostatic testing, the combined equivalent stress in the pipeline due to bending and test pressure shall not exceed 90% of the SMYS of pipe material.

3.18 Boring Failure

- (a) Should anything prevent completion of this operation, the remainder of the pipe shall be constructed by methods approved by the owner's Engineer/Inspector.
- (b) Abandonment of any component of the installation shall only be allowed as approved by the owner's Engineer/Inspector.
- (c) While abandoning any failed bore necessary safety precautions like backfilling the entry/exit pits, or other excavations with suitable material, Safe disposal of the drilling fluids etc. must be taken as per the instructions of the Engineer-in-Charge.

3.19 Contamination

When an area of contaminated ground is encountered, all operations shall stop immediately, and shall not proceed until approved by the owner's Engineer/Inspector. Any slurry shall be tested for contamination and disposed of in a manner, which meets locally applicable requirements.

3.20 Bulkhead

Pipe ends shall be temporarily sealed with a cap until the connection is made permanent, to prevent water or earth infiltration.

3.21 Work Site Restoration

- (a) Access pits and excavations shall be backfilled with suitable material, and in a method approved by the owner's Engineer/Inspector.
- (b) The disturbed grass-surface area shall be top soiled, seeded, fertilized, mulched, and anchored according the owner's Specifications for construction. Slopes steeper than 1-on-3, shall be sodded. If a final site restoration is not completed within 5 days after completion of the operation, the installation of temporary soil erosion and sedimentation control measures shall be required.
- (c) Upon completion of the work, the contractor shall remove and properly dispose of all excess materials and equipment from the work site.
- (d) The permit, including the surety requirements, shall remain in effect for a minimum of one year after completing the work to monitor for settlements of the pavement and/or slope.
- (e) After the work has been completed, the surface area should be restored to its original condition.

3.22 Trained Workmen

All the specialized equipment must be operated by properly trained, tested, and certified operators. For long or geologically challenged HDD crossings with steel pipes, presence of skilled drill operator and mud engineer having experience with respect to soil data, pipe size, & length of crossing shall be necessary.

Testing and certification of workmen must have been done on the basis of current competency standards created and revised from time to time by IndSTT or comparable standards created by other Trenchless Technology Societies and approved by IndSTT

3.23 Documentation for As-Built Drawing

When the job is completed, the referenced plan and profile information of the final alignment should be provided to the project owner.

CHAPTER 4 OVERHEAD LINE WORKS

4.1 SCOPE

This chapter covers the requirements for installation, testing and commissioning of overhead lines for power supply upto and including 33KV, service connections, including the materials used therein.

4.2 MATERIALS AND CONSTRUCTION

4.2.1 Supports

4.2.1.1 Types of supports

- (i) Supports for overhead lines shall be any of the following types, as specified in tender documents. (In particular cases, any other type of supports may also be specified)
 - (a) Steel tubular poles,
 - (b) Steel rail poles,
 - (c) Cement concrete (RCC/PCC) poles, and (d) Fabricated poles.
- (ii) Supports shall be of adequate strength and conform to Regulation-57 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010. The sizes of pole sections shall be selected in accordance with relevant IS Specifications to suit the requirements of loading.
- (iii) Length of supports shall be specified in tender papers, so, as to satisfy the relevant functional requirements like the ground clearance of lines when installed, street lighting etc.
- (iv) Pole spacing and clearance between lowest conductors above the ground level across/along the street shall be in accordance with Regulation -58 & 67 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
- (v) The pit should be excavated in the direction of the line. The depth and size of pit shall be such that normally 1/6th of the length of the pole is buried in the ground and suitable for foundation of support.

4.2.1.2 Steel tubular poles

These shall conform to IS: 2713 (Parts 1 to 3)-1980. These shall be of seamless/swaged and welded type in three stepped sections as specified. The pole shall be complete with cap and base plate.

Unless otherwise specified, one sixth of the length of the pole plus 30cm from its base shall be coated with black bituminous paint, both internally and externally. The remaining portion of the pole shall be painted with one coat of red oxide primer on its external surface.

The test requirement as per IS is given below:

Sl.No	Tests used in with clause reference	Test equipment
1.	Tolerances- outside diameter, thickness, length, straightness (Clause 8 of part-1)	a) Vernier caliper b) Micrometer c) Outside micrometer d) Steel tape
2	Mean weight of bulk supplies (Clause 8.4 of Part-1)	Weighing machine
3	Deflection test (Clause-10.1.3 of part-1)	Deflection testing arrangement with loads
4	Drop Test (Clause-10.1.5 of part-1)	Testing arrangement for drop test

4.2.1.3 Steel rail poles

These shall conform to the standard specifications of the Indian Railways.

4.2.1.4 Cement concrete (RCC/PCC) poles

- (i) Reinforced cement concrete (RCC) and pre-stressed cement concrete (PCC) poles shall conform to IS: 785- 1998 and IS: 1678-1998 respectively. These shall carry an earth bond in accordance with Regulation -57 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
- (ii) The dimensions shall be as per designed conforming to local requirements. (iii) Concrete poles shall be treated with suitable chemicals like silicate for the portion to be buried in ground where the subsoil water level is high and/or acidic as in coastal areas.
- (iv) The selection, handling and erection of the poles shall be done in accordance with IS: 7321-1974.
- (v) Method of testing of the poles shall be done in accordance with IS: 2905-1966.
- (vi) The cement used in the manufacturing of the poles shall be ordinary or rapid hardening port land cement conforming to IS: 269-1976 or IS: 8041 E-1978 respectively.
- (vii) The aggregate used for manufacturing of the concrete poles shall conform to IS: 383-1970.
- (viii) The factor of safety for all these poles shall not be less than 2.5.

4.2.1.5 Fabricated poles

These shall be made from Galvanized Iron (GI) pipes or mild steel (MS) pipes (seamless or ERW), or fabricated from structural steel. Details of such supports shall be specified in tender papers.

4.2.2 Line materials

4.2.2.1 Cross arms

- (i)
 - (a) The cross arms for overhead lines shall be made either of MS angle iron of size not less than 50mmx50mmx6mm thick (4.5kg/m) for LV/MV lines and 65mmx65mmx6mm thick (5.8kg/m) for 11 KV lines, or of MS channel iron of size not less than 75mmx40mmx4.8mm thick (7.14Kg/m), (for LV/MV/11KV lines) as specified. The channel iron cross arms may be straight type for LV/MV, and straight or V-cross arms for 11KV, as specified.
 - (b) The cross arms for overhead lines for 22KV/33KV shall be fabricated either from 75mmx40mm (7.14Kg/m), or from 100mmx50mm (9.56 Kg/m) channel iron, as specified, fabricated as V-cross arms.
- (ii)
 - (a) The length of cross arms shall be suitable for accommodating the required number of insulators on them with the spacing of conductors in accordance with clause 4.3.3.1.
 - (b) Where guard wire cradle is specified to be fixed directly to the cross arms (without additional brackets), the length of cross arm supporting the cradle shall be such that the clearances specified in clause 4.3.3.7 is satisfied.
 - (c) The cross arms shall have holes for fixing on to the poles and for taking insulator pins, wire guards etc., as required. A minimum distance of 5cm for LV/MV lines and 10cm for HV lines shall be left from the center of the extreme insulator pin hole to the end of the cross arm.
 - (d) Table XXI indicates the cross-arm lengths based on the above requirements.
- (iii) Unless otherwise specified, a triangular configuration shall be adopted for HV overhead lines: Where specified, the cross arm supporting the lower two conductors over pin insulators shall be provided with 50mmx50mmx6mm thick (4.5kg/m), angle iron bracket duly welded to it so that the cross arms are double clamped to the poles for rigidity. However, such double clamping

arrangement shall necessarily be provided in cases of HV overhead lines, where in-line configuration is adopted, whether or not so specified in tender papers.

- (iv) Details of cross arms are illustrated in figure 5 (A to C) for guidance.
- (v) The cross arms shall be complete with pole clamps made of MS flat of size not less than 50mmx6mm with necessary bolts, nuts and washers.
- (vi) The length of cross arms for carrying guard wires shall always run not less than 300 mm beyond outer most bare conductor of configuration.

4.2.2.2 D-Iron clamp.

- (i) Where vertical configuration is specified in tender documents for MV overhead lines, the conductors shall be supported on shackle insulators which shall be fixed to the poles by means of D-shaped clamps made of MS flat of size not less than 50mmx6mm and galvanized.

Note:- Vertical configuration shall not be adopted for HV lines.

- (ii) The dimensions of D-iron clamp shall be such as to hold a 75mm high and 90mm (nominal) diameter shackle insulator. (As indicated in Figure 6A).
- (iii) The D-iron clamp shall be complete with pole clamp, made of 40mmx6mm flat iron and necessary GI bolts, nuts and washers and holes for fixing insulator pins.



4.2.2.3 GI straps

- (i) Where D-iron clamps are not specified for shackle insulators, a pair of strap plates of hot dip galvanized iron of size 40mmx3mm thick and length 23 cm shall be used. (as indicated in Figure 6B).
- (ii) The fittings shall conform to IS: 7935-1975.

4.2.2.4 Pole top bracket

The pole top bracket for supporting a single pin type insulator shall be made of flat iron 50mm x 8mm as shown in figure - 6C.

4.2.2.5 Cradle guard bracket

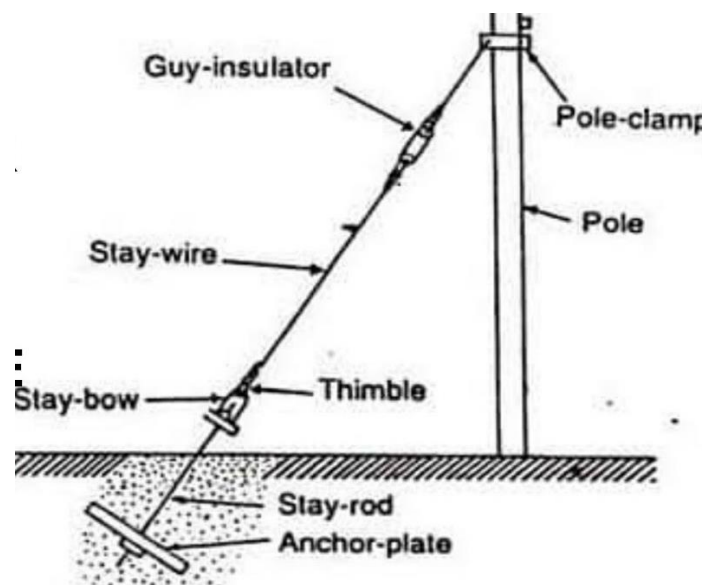
Where brackets are used for supporting cradle guards on cross arms, these shall be made of 50mm x 6mm flat iron, shaped such that clearance with the line conductors as specified in 4.3.3.7 is achieved. These shall be welded to the cross arms or fixed to them with double bolts nuts and washers, thus preventing its movement sideways.

4.2.2.6 The nuts bolts and washers used for fixing of hardware accessories shall be Stainless steel or galvanized.

4.2.3 Stay sets

4.2.3.1 Locations and number

- (i) Stays shall be provided on to the poles at locations where a pull from the conductors on one direction is likely to be experienced such as terminal poles, or at the deviation point(s) of the line from straight run etc. These shall be specified in the drawings.
- (ii) The stays shall be provided on the side of the pole opposite to the likely direction of pull from the line i.e., opposite to where the shackle/disc insulators are provided. In the case of deviations from straight runs, the stays shall be provided opposite to the side of deviation and preferably on the bisection of the angle of deviation.
- (iii) The number of stay sets to be provided shall be decided depending on the likely pull to be experienced on the pole, dictated by the length of span, number and size of conductors etc. Guidelines on the same are given in Appendix-F.
- (iv) Double stay shall be provided at all dead ends in such case, these shall be as far as possible to be set parallel to each other.



4.2.3.2 Construction

- (i) A stay set shall consist of stay rod, anchor plate, bow tightener or turn buckle, thimbles, les, stay wire, strain insulator and stay clamp as per details shown in figs. 7A, 7B & 7C.
- (ii) All components of the stay set assembly shall be of MS and galvanized.
- (iii) The stay rod shall be not less than 1.80m long and 19mm dia. Having tensile strength not less than 42 kgf/sqmm. The stay rod shall be with stay clamp in case turn buckle is used instead of bow tightener. The minimum length buckle shall be 450 mm.
- (iv) The anchor plate shall be not less than 450 mm x 450 mm x 7.5 mm thick.
- (v) The stay wire shall be either 7/4mm dia or 7/3.15mm dia GI as specified having tensile strength of not less than 70 kgf/sq.mm and generally conform to grade 2 of IS:2141-1979 .
- (vi) The strain insulator shall conform to IS:5300 -1969:

The recommended types of strain insulators for use on stay wires of overhead lines of different voltage levels are as follows.

Line voltage	Designation of Insulation
240V/415V	"A"
11KV/33KV	"C"
(2 Insulators to be used/ line for 33KV)	

- (vii) The length of stay rod shall project 450 mm above the ground level. The pit for stay shall be located at a distance not less than 1.8M from the pole.
- (viii) The position of stay pit shall normally be such that stay makes as large an angle as possible with the support and it shall be in the range of 40 to 60 degrees.

4.2.4 Struts

Struts are provided wherever stays cannot be provided due to any obstruction, or where the stays themselves can cause an obstruction. A strut is normally provided opposite to the direction in which a stay would have been provided.

Normally, one strut would be adequate for each pole. However, where the angle of deviation is large, two struts may be required for each pole at the point of deviation.

A strut shall generally consist of a pole of the same section which it supports or slightly lighter as specified in the contract.

4.2.5 Insulators & insulator fittings :

4.2.5.1 Insulators – general

A material that inhibits the flow of electrical current and heat travel through it is known as an insulator. In overhead distribution/transmission lines, insulator prevent the flow of the current from line conductor to the ground/earth/metallic part of the pole or an insulator provides the required protection between the line conductor and the earth.

- (a) Porcelain insulators shall conform to IS:1445-1977 (Reviewed in 2019) for lines below 1000V and to IS:731-1971 for lines with voltage greater than 1000V.
- (b) The insulators shall be vitreous throughout and nonabsorbent. The exposed surface shall be glazed. The glazed shall cover all the porcelain parts of insulators except areas which serve as support during firing are left unglazed for the purpose of assembly.
- (c) These shall have adequate mechanical strength, high degree of resistance to electrical puncture and to climate and atmospheric attack.
- (d) The design and insulators shall be such that stresses due to expansion and contraction in any part of the insulator shall not lead to deterioration.
- (e) The insulators should preferably be manufactured in automatic temperature controlled kilns to obtain uniform baking for better electrical and mechanical properties.
- (f) THE IMPORTANCE OF INSULATORS
 - Help to protect from heat, noise, and electricity
 - Support the overhead conductor
 - Insulate the live parts of equipment or conductor from the earth
 - Help to save switchgear, transformer, and other systems in a substation
- (g) Insulator properties
 - High resistivity
 - Good mechanical strength for the conductor load
 - The high relative permittivity of insulator material
 - Good dielectric strength
 - Waterproof or non-porous

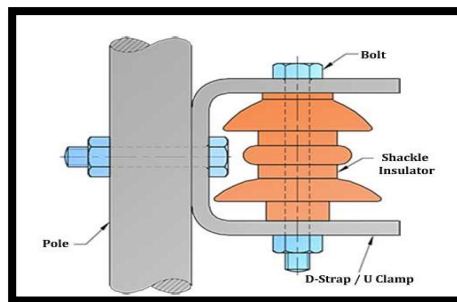
4.2.5.2 Types of Insulators

- (i) The insulator shall be any of the following types, as specified and in compliance with IS: 1445-1977 (Reviewed in 2019) for lines below 1000V and to IS:7311971 for lines with voltage greater than 1000V.

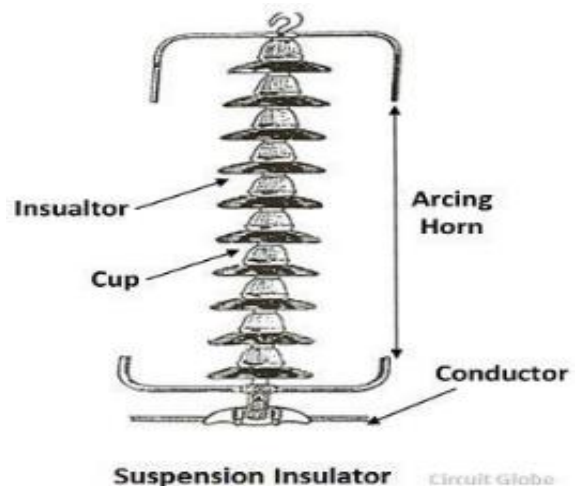
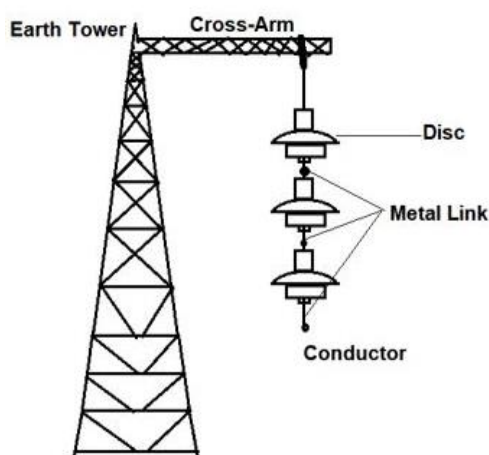
- (a) **Pin Insulator:** - Pin type insulator is used in LT distribution and transmission upto 33KV. Beyond 33 KV operating voltage, the pin type insulators become too bulky and hence uneconomical as in higher voltage system, manufacturing of single piece porcelain pin insulator will require thick insulation to provide sufficient protection which is not practical to manufacture.



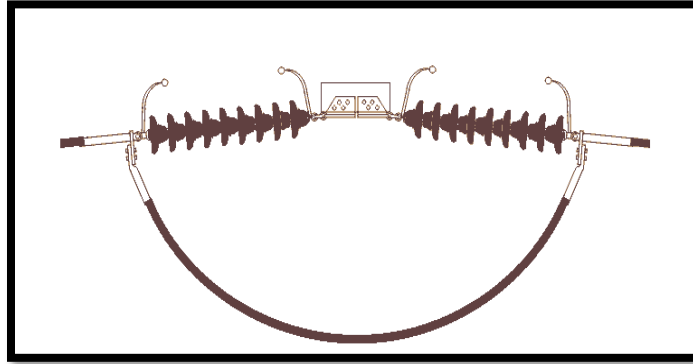
- (b) **Shackle insulator:** - Shackle insulator is used in low and medium voltage overhead distribution system. It can be used both in horizontal or vertical position.



- (c) **Suspension insulator:** - Suspension insulator is used in HV overhead transmission lines. The voltage operating capacity of this insulator is from 11 KV to 765 kV and it overcomes the limitation of pin type insulator. It is constructed by connecting glass or porcelain discs in series by metal links in the form of a string for increasing voltage use. If anyone of the disc of suspension insulator is damaged than it can be replaced easily. By using different number of discs, a suspension string insulator can be made suitable for any voltage level. It is not economical to use below 33kV as it requires more height of supporting structure than pin insulator to maintain the same ground clearance.



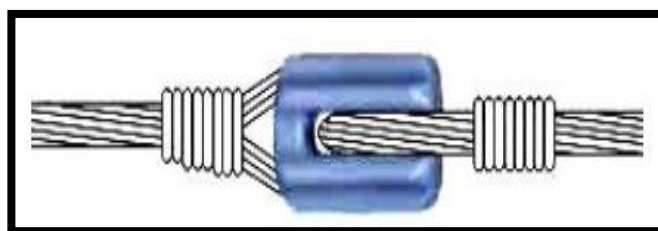
- (d) **Strain insulator:** - When Strain insulator is used in overhead HT transmission lines at dead end or sharp corner to sustain extraordinary tensile load of conductor than the suspension insulator is known as strain insulator. A strain insulator must have considerable mechanical strength as well as the necessary electrical insulating properties.



- (e) **Post insulator:** - Post insulators are similar to pin-type insulators, but they are more suitable for higher voltage applications. Post insulators are commonly used in the substation or generating substation. They are made in ceramic and also composite material for special applications. It can be mounted in horizontal as well as in vertical position.



- (f) **Stay insulator:** - Stay insulator is used in the stay set at locations where a pull from the conductors on one direction is likely to be experienced such as terminal poles, or at the deviation point(s) of the line from straight run etc.



4.2.5.3 Size of insulators

- (i) The minimum size of shackle insulator for LV/MV overhead lines shall be 90mm dia x 75mm high.
- (ii) The minimum size of pin insulator for LV/MV overhead lines shall be 65mm dia x 100mm high.
- (iii) The pin insulator shall be suitable for 12mm cordeau threaded GI pin, nuts and washers.

4.2.5.4 Insulator fittings

- (i) The insulator fittings shall comply with IS:2486(Part1)-1993 for General requirements and tests more than 1000V and IS:2486 (Part-2)-1989 (Reviewed In: 2019) for 11KV and 33KV insulators Dimensional requirements, and with IS:7935-1975 for LV/MV insulators fittings for overhead power lines with a nominal voltage up to and including 1000 V.

(ii) Pin insulator fittings

- (a) The pins suitable for the pin insulator for the LV and MV overhead lines shall have a stalk length 135 mm shank length of 125 mm and minimum failing load of 2 KN.
- (b) The pins suitable for 11 KV pin insulators shall have stalk length of 165 mm and shank length of 150 mm and minimum failing load of 5 KN.
- (c) The pins suitable for the 33 KV, pin insulators shall be large steel head type L 300 N as per IS:2486 Part-II having stalk length of 300 mm and shank length of 150 mm and minimum failing load of 10 KN.
- (d) The pins shall be of single piece MS without joints, obtained by the process of forging. They shall not be made by joining, welding, shrink fittings or any other process using more than one piece material.
- (e) The pins, nuts and washers, shall be galvanized.
- (f) The threads of nuts and tapped holes shall be cut before galvanizing and shall be well oiled or greased.
- (g) The pins shall be of good finish, free from flaws other defects.

(iii) Disc insulator fittings

- (a) The insulator fittings for disc insulators shall be either of ball and socket type or clevis and tongue type depending upon the type of disc insulators specified in the tender documents.
- (b) The strain clamps for string insulators shall be suitable for ACSR conductors 7/3.55mm (50sqmm aluminium area), 7/4.09mm (80 sq mm aluminium area) and 6/4.72mm and 7/1.52mm (100 sq mm aluminium area), as required. The ultimate strength of the clamps shall not be less than 41KN.
- (c) In ball and socket type insulator, the thimble socket shall be attached to the strain insulator with the help of locking pin as per the dimension given in IS: 2486 (Part-2).

4.2.6 Conductors

4.2.6.1 Types of conductors

The conductors shall be any of the following types as per relevant IS as specified below.

- (a) Aluminium stranded conductors, conforming to IS 398(PART-I)-1996 (THIRD REVISION). The details of standard size of AAC Conductor is given in table XXII.
- (b) Aluminium conductors, galvanized steel - Reinforced, conforming to IS: 398 (PART-2)-1996 (THIRD REVISION). The details of standard size of ACSR Conductor is given in table XXIII.
- (c) Aluminium alloy stranded conductors (Aluminium - Magnesium - Silicon Type), conforming to IS:398(Part-4)-1994 is given in table XXIV (Third Revision)
- (d) Aerial Bunched cables for working voltage up to and including 1100 Volts, conforming to IS:14255 (1995)

4.2.6.2 Choice of conductors

- (a) The physical and electrical properties of different conductors shall be in accordance with relevant Indian Standards.
- (b) All conductors shall have a breaking strength of not less than 350Kg. However for LV lines with spans less than 15m, conductors with breaking strength of not less than 140Kg may be used.

- (c) The size of conductors for a line shall be selected considering the power to be transmitted, length of line, line voltage, permissible voltage regulation etc. The size(s) shall be specified in contract.
- (d) No conductor of cross section smaller than the following shall be used for distribution lines.

Voltage of line	All aluminium Stranded	ACSR	Aluminium Alloy Stranded
LV/MV	7/2.21mm	6/1/2.11mm	7/2.09 mm (20 sq mm Al. area)
11KV/33KV		6/1/2.11mm	7/2.56 mm (30 Sq mm Al. area)

- (e) The sizes of conductors are standardized for lines of different voltage as per table XXII to XXIV as per 398(PART-I & 2)-1996 (THIRD REVISION).
- (f) In all aluminium conductors, no joints shall be permitted in any wire.
- (g) In ACSR conductors, no two joints shall occur in the aluminium wires closer together than 15 meters.

4.2.6.3 Joints in conductor

- (a) No conductor of an overhead line shall have more than one joint in a span and joints between conductors of overhead lines shall be mechanically and electrically secure under the conditions of operation.
- (b) The ultimate strength and the electrical conductivity of the joint shall be as per relevant Indian Standards.

4.2.6.4 Testing of Conductor

The Conductor shall be tested for acceptance test and type test as per relevant BIS code as mentioned in the clause 4.2.6.1 the frequency, lot size/sample size etc. shall also be as per relevant BIS code as mentioned above. The acceptance test is a mandatory test.

4.2.7 Binding material

Binding of conductors with the insulators shall be done with 2.6mm (I2SWG) soft aluminium conductors. The binding of conductor to insulator shall be sufficiently firm and tight to ensure that no intermittent contact develops. The end of binding wire shall be tightly twisted in close spaced spiral around the conductor to ensure good electrical contact and strength the conductor.

4.2.8 Guard wire

- (a) Guard wire shall be of GI 4mm dia (8SWG), AAC(7/3.10mm) or ACSR (7/2.59mm). Cross lacings may be of GI wire (minimum 3.15mm dia) or scrap lengths of AAC or ACSR conductors used in the line. It shall have a minimum breaking strength of 635Kg, in accordance with Regulation -70 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
- (b) It shall also be of sufficient current carrying capacity to ensure rendering the line dead without the risk of fusing the guard wire or wires, till the contact of the line wire has been removed.
- (c) Protective guarding of overhead lines shall comply with the requirements of Regulation -70 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.

4.2.9 Earth wire

The size of the continuous earth wire shall not be less than 4 mm (8SWG) GI.

4.2.10 Lightning arresters

4.2.10.1 These shall conform to IS: 3070 (Part-1)-1985.

4.2.10.2 Types of lightning arresters

(a) Horn gap type arrester

This type of arrester shall be used for LV/MV lines as specified.

(b) Surge Diverter Single pole units enclosed in GI case for outdoor mounting shall be used for system not exceeding 650V.

(c) Non-linear Resistor Type Lightning Arrester

(i) This type of arrester shall be used in an effectively earthed system with a nominal line voltage of 11KV/22KV/33KV.

(ii) The rated voltage of the lightning arresters suitable for 33KV lines shall be 30KV (RMS) with nominal discharge current rating of 10KA (Station Class) in lines.

(iii) The rated voltage of lightning arresters suitable for 11KV lines shall be 9KV (RMS) with a nominal discharge current rating of 5KA.

(iv) The system shall be effectively earthed (Coefficient of earth not exceeding 80 percent as per IS:4004- 1985) with the neutrals of all the transformers directly earthed.

4.2.10.3 The lightning arrester system shall conform to Regulation -74 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.

4.2.10.4 The following routine and type test shall be carried out :

(a) Routine test: - Dry power frequency spark over test.

(b) Type Test: -

(i) Voltage withstands test of arrester insulation.

(ii) Power frequency spark over test

(iii) Hundred percent 1.2/550 microsecond impulse spark over test.

(iv) Front-of-wave impulse spark over test.

(v) Residual voltage test

(vi) Impulse current withstand test

(vii) Operating duty test

(viii) Temperature cycle test on porcelain housing

(ix) Porosity test on porcelain component

(x) Galvanizing test on metal parts.

(c) Acceptance test of IS: 3070 (Part-1)-1985 is a mandatory test.

4.2.11 Paint

4.2.11.1 Only paints of approved make and shade conforming to relevant Indian Standards shall be used.

These shall be in original containers of the manufacturers.

4.2.11.2 Primer coats shall be with red oxide paint.

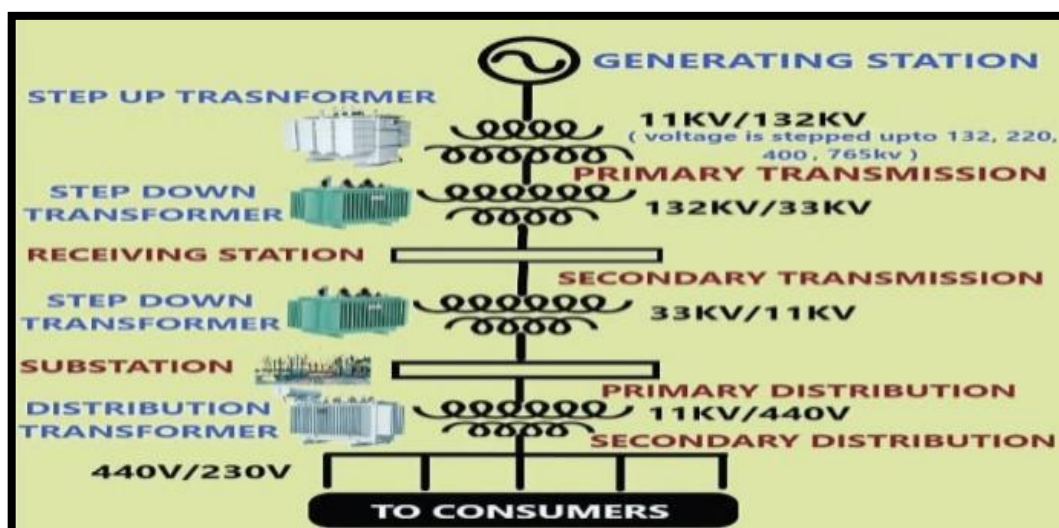
4.3 LINE LAYOUT

4.3.1 Route

4.3.1.1 General

The route of overhead lines shall be adopted considering the following: -

- (a) The route of LV/MV overhead lines shall generally follow the layout of roads except in particular stretches specified. However, HV lines can be routed through cross country also, especially in remote locations.
- (b) As far as possible, the present and future requirements of other agencies and utility services affected shall be considered, both for the line and for stays/struts.
- (c) Overhead lines shall not be erected in the vicinity of Aerodromes until the Aerodrome authorities have approved in writing the route of the proposed lines in accordance with Regulation -66 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
- (d) The minimum factor of safety for stay-wires, guard-wires or bearer-wires shall be 2.5 based on the ultimate tensile strength of the wire.
- (e) The route shall be so chosen as to avoid use of struts and continuous curve in the overhead line as far as possible.

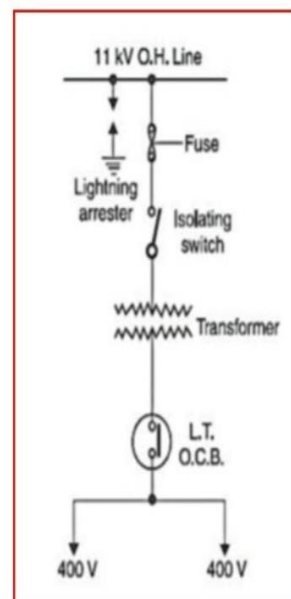
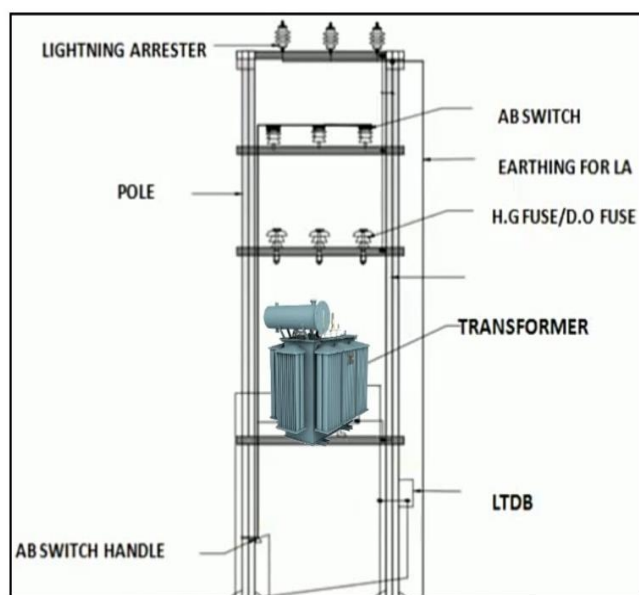


4.3.1.2 LV/MV lines

The following shall be considered for locating the poles of LV/MV overhead lines:-

- (a) Poles shall be located alongside roads, on the road berm, a little away from the road edge and drain.
- (b) There shall be a pole located at each road junction.
- (c) Junction of main road and a service lane shall be preferred for location of pole so that the street light will benefit the service lane as well.
- (d) Front of entrance to building shall be avoided for locating poles. However, in the case of residential colonies, the street lighting poles shall be located such that the entrance to the blocks are lit up, as far as possible.
- (e) There shall be installed a DP structure between primary distribution line of 11KV and secondary distribution line of 440V. There shall be following items at DP structure: -
 - (i) Lightning arrester
 - (ii) Air Break/ Gang operated switch (AB/GO Switch)
 - (iii) Drop Out Fuse (DO Fuse)
 - (iv) Transformer 11KV/0.4KV
 - (v) LT Cable

- (vi) Anti-climbing device
- (vii) LT DB/LT MCCB/LT OCB
- (viii) Earthing for LA
- (ix) AB Switch Handle



4.3.1.3 The route of overhead lines and pole locations shall be indicated in tender drawings. Modifications, if any, required to suit site conditions can be done only with the prior approval of the Engineer-in-charge.

4.3.1.4 Way leave

Way leave for the proposed route of overhead line shall be arranged by the Department from the appropriate authorities, such as State Public Works, Drainage, Public Health and Water Works/ Municipal authorities, Telephone and Telegraph, Gas Works, Railways, Director General of Civil Aviation, other Undertakings, owners of properties etc. as may be required.

4.3.1.5 Cutting of trees etc.

Where the route of overhead lines involves a need to cut branches of trees or clearing of other obstructions that may come in the way of the overhead lines, this may only be done with the prior approval of the Engineer-in-Charge and with the permission of the owners concerned.

4.3.2 Spacing of poles

Spans of overhead lines shall be decided considering the following:

4.3.2.1 Clearances as laid down in clause 4.3.3 shall be satisfied, for the pole length and conductor sizes selected.

4.3.2.2 Requirements of Regulation-67 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010 shall be complied with.

Where street lighting is provided with overhead line system, the span shall be such that lighting is adequate, but the span may not exceed 45 m.

4.3.3 Clearances

4.3.3.1 The spacing of conductors depends on their disposition and is determined by the line voltage, sag, span, swing amplitude and type of structure. It shall comply with the requirements given shall be as Table XXV. Typical formation of conductors are indicated in figure 8 (A) to 8(E).

4.3.3.2 The minimum clearance of the lowest conductor of overhead line including service line above ground level across a street, along a street and elsewhere for different voltage systems shall be in accordance with Regulation -58 of CENTRAL ELECTRICITY AUTHORITY (Measures

Relating to Safety & Electric Supply) Regulations, 2010/NEC-2023 (Part-1- Section-7 at Page-52) which is given below: -

Voltage level	Erected across a street	Erected along any street	Erected elsewhere than along or across any street
For lines of voltage not exceeding 650 volts	5.8 mtr.	5.5 mtr.	4.6 mtr.
For lines of voltage exceeding 650 volts but not exceeding 33 kV	6.1 mtr.	5.8 mtr.	5.2 mtr.

4.3.3.3 An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line. However, if unavoidable where and overhead line of voltage not exceeding 650 volts passes over or adjacent to or terminates on any building, the following minimum clearance of overhead lines and service. lines for different voltage systems from any accessible point of the buildings on the basis of maximum sag shall be observed in accordance with Regulation -60 & 61 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010/ NEC-2023 (Part-1- Section-7):

Voltage level	Line passes above the building (Vertical distance)	Line passes adjacent to the building (Horizontal distance)
For lines of voltage not exceeding 650 volts	2.5 mtr.	1.2 mtr.
For lines of voltage exceeding 650 volts and up to and including 11 kV	3.7 mtr.	1.2 mtr.
For lines of voltage exceeding 11 kV and up to and including 33 kV	3.7 mtr.	2.0 mtr.

Note: -

- (i) Any conductor so situated as to have a clearance less than that specified above shall be adequately insulated and shall be attached at suitable intervals to a bare earthed bearer wire having a breaking strength of not less than 3 50 kg.
- (ii) The horizontal clearance shall be measured when the line is at a maximum deflection from the vertical due to wind pressure.

4.3.3.4 When conductors of different voltages are erected on the same support, then guarding arrangement shall be provided so to guard against the possibility of their coming into contact with each other as per Regulation-62 & 69 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010. The minimum clearance between LV/MV and 11-66kV lines shall be not less than 2.44 mtr.

4.3.3.5 A clearance of not less than the height of the tallest support may be maintained between parallel overhead lines on different supports.

4.3.3.6 When two overhead lines cross, the crossing shall be made at right angles as far as possible. Moreover, guarding arrangement shall be provided so to guard against the possibility of their coming into contact with each other as per Regulation-69 of CENTRAL ELECTRICITY

AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010. The vertical clearance between LV/MV lines and 11-66kV among 11kV-66kV lines shall not be less than 2.44 m.

- 4.3.3.7** The guarding arrangement shall be provided where an overhead line of voltage not exceeding 33kV cross a road or street or in proximity to another overhead line as per Regulation-69 (ii & iii) of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010. The number and position of guard wires shall be provided as per Regulation-70 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
- 4.3.3.8** Regulation-69 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010 shall be followed for clearance between power and telecommunication lines and shall not be less than 2.44m for lines upto 66kV.
- 4.3.3.9** Crossing of Railway lines shall be done as per Specifications for crossings and in consultation with the Railway authorities.
- 4.3.3.10** The minimum factor of safety for stay-wires, guard-wires or bearer-wires shall be 2.5 based on the ultimate tensile strength of the wire as per Regulation-57 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
- 4.3.3.11** The vertical clearance of an overhead line shall be computed as per methodology adopted as per Regulation-65 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010, the maximum sag of any conductor shall be calculated on the basis of the maximum sag in still air and the maximum temperature as specified under regulations 57 and computing any horizontal clearance of an overhead line the maximum deflection of any conductor shall be calculated on the basis of the wind pressure specified under Regulation-57.
- 4.3.3.12** Overhead lines shall not be erected in the vicinity of aerodromes unless the Airport Authorities have approved in writing the route of the proposed lines as per relevant Indian Standards as per Regulation - 66 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
- 4.3.3.13**
 - (a) All conductors shall be attached to supports at intervals not exceeding the safe limits based on the ultimate tensile strength of the conductor and the factor of safety specified under regulations 57 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
 - (b) Maximum interval between supports in the case of overhead lines carrying conductors of voltage not exceeding 650 V when erected in, over, along or across any street, the interval shall not exceed 65 meters as per Regulation-67 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.
- 4.3.3.14** Every overhead telecommunication line erected on supports carrying a power line shall consist of conductors each having a breaking strength of not less than 270 kg and shall be suitably guarded against lightning and shall be protected by cut-outs. Moreover, arrangement shall be made to safeguard any person against injury resulting from contact, leakage or induction between such power and telecommunication lines as per Regulation - 68 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.

4.4 EXCAVATION FOR FOUNDATION

4.4.1 General

- 4.4.1.1** The locations of supports, stays and struts shall be pegged accurately before the excavation work is taken up. The foundation shall include excavation in all type of soil and rocks and back filling, RCC, reinforcement, formwork.

- 4.4.1.2** Care shall be taken to see that the minimum amount of soil is disturbed so as to take advantage of the bearing capacity of the virgin round, (and that the pits are not oversized) after taking into consideration the size of the foundation.
- 4.4.1.3** Pits shall not be left unfilled for unduly long periods so as to avoid accidents. While being kept open, protective measures such as suitable caution signs, caution lights, barricading etc. as necessary should be provided near the pit(s) to warn the pedestrians/ vehicular traffic, till such time the pit is back filled and surface levelled.
- 4.4.1.4** The pit for support/stay/strut shall be filled up or concreted only in the presence of the Engineer-in-Charge of the work.

4.4.2 Excavation for supports

The depth of pit shall be such that normally 1/6th of the length of the pole is buried in the ground. The size of the pit shall be suitable for the foundation of the supports as per clause 4.5.1. The pits should be excavated in the direction of the lines.

4.4.3 Excavation for stays

The position of pit shall normally be such that the stay makes an angle of 30 to 60 degrees with the support. (Higher angle is preferred),

The depth of pit shall be such that normally a length of 45cm of stay rod shall project above the ground level. The size of the pit shall be suitable for the foundation of stay as per clause 4.5.2.

4.4.4 Excavation for struts

The pit for struts shall be located at a distance of not less than 1.8m from the pole. The depth of it shall be such that at least 1.2m of the strut is buried in the ground and the size of the pit shall be suitable for the foundation of the struts.

4.5 ERECTION

4.5.1 Erection of supports

- 4.5.1.1** The supports shall be correctly aligned before concreting or the back filling of the pit, as the case may be.
- 4.5.1.2** All supports including RCC and PCC poles shall be erected over a cement concrete 1:3:6 (1 cement: 3 coarse sand: 6 graded stone aggregate of 40mm nominal size) bed of 15cm thick, either cast in situ or precast and laid in the excavated pit, irrespective of the provision of a base plate. The area of this cement concrete bed shall be 0.35sqm for steel tubular/rail and other steel poles and 0.5sqm for RCC/PCC poles.
- 4.5.1.3** The supports shall be erected in the following manner depending on the type of support.
- Steel tubular /steel rail/ other steel poles shall be fixed in cement concrete 1:3:6 (1 cement: 3 coarse sand: 6 graded stone aggregate of 40mm nominal size) foundation with not less than 20cm thick layer of the cement concrete all-round the support, the foundation being continued upto 15cm above ground level and tapered suitably into a collar.
 - RCC/PCC poles shall be erected with a filling of brick or stone ballast, with excavated earth as binder, well consolidated. The ramming shall be done in layers of 20cm. water as necessary shall be used during this operation. The sectional area of the consolidated ballast foundation shall not be less than 0.5sqm including the area occupied by the support itself and shall be maintained upto the ground level.
RCC/PCC poles except where specified otherwise, do not require any setting in concrete, No cement concrete collar is also necessary for such types of poles.
- 4.5.1.4** After concreting, the excavated earth shall be back filled and well consolidated in layers of 20cm.

4.5.1.5 Watering of concreted foundation above ground level and curing for at least two weeks shall be done by using moist gunny bags etc. before loading the pole.

4.5.2 Erection of stay sets and struts

4.5.2.1 The stay rod will be set in position in the excavated pit as shown in figure 7 (A to C), the rod being straight or bent as laid down in clauses 4.5.2.2(i) or (ii) as the case may be. The correct positioning and setting of stay set is essential.

4.5.2.2

(i) The straight stay rod with the anchor plate shall be embedded in cement concrete 1:3:6 (1 cement :3 coarse sand: 6 graded stone aggregate of 40mm nominal size) not less than 0.28cum in content in such a way that the top of the concrete block is well below the ground level to prevent uprooting of the stay rod.

(ii) Alternatively, the bent stay rod shall be embedded vertically in cement concrete 1:3:6 (1 cement: 3 coarse sand: 6 graded stone aggregate of 40mm nominal size) foundation 42cmx42cm in section, the anchor plate lying over 15cm thick cement concrete. The bend in the stay rod shall be such that the stay wire and the bent portion of stay rod are in correct alignment. Care must be taken to avoid sharp bend or damage to galvanization.

4.5.2.3 After the concrete has set, back filling shall be done with excavated earth and ramming in layers of 20cm using water as required.

4.5.2.4 The top surface of concrete around the stay rod shall be cured by means of moist gunny bags etc. for at least 2 weeks before loading the stays.

4.5.2.5 The stay clamp shall be located just below the lowest cross arm on the pole so that it is as close to the center of gravity of the pull of the overhead conductors as possible. In the case of poles with vertical formation LV/MV line, special bracket as per detail in Figure 7B shall be used for the purpose.

4.5.2.6 One end of the stay wire shall be fixed to the bow tightener or the stay grip of the stay rod and the other end to the clamp fixed to the pole by means of well spliced joints using GI thimbles. A strain insulator shall be provided approximately at the middle of the stay wire. Turn buckle, when used, shall be installed at the top of the stay wire.

4.5.2.7 The stay wire shall be connected and bonded properly to the continuous earth wire.

4.5.2.8 Where Double sets are specified due to large pulling force, these shall as far as possible be set parallel to each other or shall be so placed that one does not reduce the strength of the other by suitable spacing between the two pits.

4.5.2.9 The stay rod, where so specified in tender documents, shall be protected with GI pipe, which shall not be less than 5cm dia and 1.5m long, placed so as to be 0.6m below ground. The length of the stay rod shall accordingly be increased.

4.5.2.10 If the stay rod cannot be erected in accordance with the above clauses due to the existence of a road or an obstruction etc., bow stay, fly stay or strut, whichever is suitable to the location shall be used, as may be directed by the Engineer-in-charge.

4.5.2.11 Bow stay

(a) Bow stay shall consist of a brace made of 50mmx50mmx6mm thick angle iron with a 5cm pulley on the outer end or a hole, to allow for free motion of stay wire, in addition to other accessories required for stay set mentioned in clause 4.2.3.2. The arrangement shall conform to figure 7C.

(b) The stay wire shall be clamped to the pole on the top and the other end to a stay rod, passing over the pulley or through the hole in the brace. The brace shall be clamped at about 2/3rd height of the pole from the ground level. This can be increased if so required, but in no case should the brace be closer to the lowest cross arm/D-Iron clamp by less than 1m. The stay rod shall be embedded in cement concrete foundation in the usual manner, as near as possible to, the pole at about 1m from the pole.

- (c) Where the site condition does not permit allowing a space of 1m as above, the bottom end of the stay wire may be clamped to the pole near ground.

4.5.2.12 Fly stay

- (a) The fly stay shall consist of a fly pole, stay wire running over the obstruction and the usual stay arrangement for the fly pole, as shown in figure 7C.
- (b) The stay wire crossing the obstruction shall be clamped at one end to the top of the fly pole with a turn buckle. The fly stay shall be taken at such a height as may be directed by the Engineer-in-Charge. When a fly stay is taken across a road, it shall conform to traffic regulations.

4.5.2.13 Erection of struts

The strut shall be buried in the ground as mentioned in clause 4.4.4 and erected in the same manner as the pole. It shall rest on the pole squarely and shall be firmly secured by GI clamp. A typical clamping arrangement is shown in figure 9.

4.5.3 Erection of line materials

4.5.3.1 Cross arms

- (a) Cross arms shall be clamped to the support properly taking into consideration the orientation of the lines.
- (b) Double clamping shall be provided where required, as laid down in clause 4.2.2.1.

4.5.3.2 D-Iron clamps

- (i) D-Iron clamps shall be fixed to the support either by a through bolt and nut arrangement, or by a suitable of pole-clamp, bolt and nut.
- (ii) These shall be installed vertically on the supports complying with the required vertical clearance between conductors.

4.5.4 Erection of insulators

4.5.4.1 Pin insulators and shackle/disc insulators shall be erected on cross arms. Where D-iron clamps or GI straps are specified, shackle insulators shall be erected in them.

4.5.4.2 Where so directed by the Engineer-in charge, pin insulators may be provided above in addition to disc/shackle insulators over the cross arm, so that the line conductors are properly routed with adequate clearances.

4.5.4.3 Care shall be taken that insulators are not damaged during handling and erection. Damaged insulators shall not be used for any reason.

4.5.5 STRINGING OF CONDUCTORS

4.5.5.1 Handling

- (a) The precautions during storage and handling shall be taken.
- (b) Particular attention is necessary when handling aluminium (AAC/ACSR) conductors because of their relative softness.
- (c) While laying off, the conductors shall be taken from the top of the drum and the drum shall be rotated in the direction of the arrow marked on it.
Care shall be taken during laying off to avoid damage due to contact with steel works, fence etc. by giving soft wood protection, using wooden rollers etc.

4.5.5.2 When the work is being carried out adjacent to and/or for connecting to an existing system in operation, adequate safety precautions for isolation, discharging earthing etc. shall be taken on the

existing line to ensure that the lines do not inadvertently get charged from live supply. Where “permit to work” system is in vogue, the prescribed safety procedure shall be complied with.

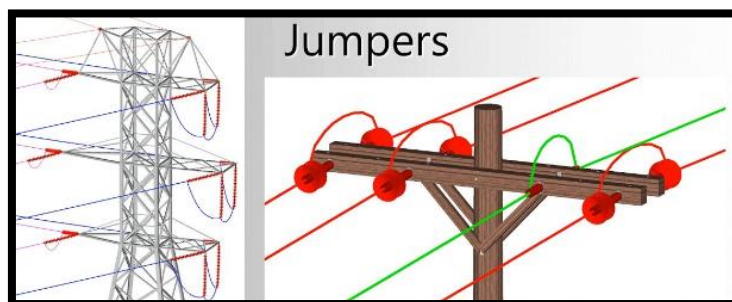
- 4.5.5.3 Appropriate tools only shall be used for the stringing work. All the strands of the conductor must be gripped securely when pulling the conductor.
- 4.5.5.4 During the stringing operation, standard sag tables or charts shall be followed.
- 4.5.5.5 Care shall be taken to see that there are no kinks in the conductors.
- 4.5.5.6 Angle or section points shall be selected while pulling the conductors. While stringing, conductors of sufficient length shall be kept at shackle terminations for making jumpers.
- 4.5.5.7 After stringing the conductor, it shall be clamped permanently with shackle or strain clamps.
- 4.5.5.8 Joints if any in conductors shall be staggered. Mid span joints in conductors shall however be generally avoided.

4.5.6 Binding of conductors

- 4.5.6.1 The binding of conductors to insulators shall be sufficiently firm and tight to ensure that no intermittent contacts develop.
- 4.5.6.2 The ends of the binding wire shall be twisted in a closely spaced spiral around the conductor to ensure good electrical contact and to strengthen the conductor.

4.5.7 Jumpers

- 4.5.7.1 Jumpers shall be neat and as far as possible symmetrical to the run of conductors. These shall be so made as to prevent occurrence of fault due to wind or birds.



- 4.5.7.2 Where necessary, the jumpers shall be with insulated conductors or taken on intermediate pin insulators as specified. For HV lines, the jumpers should be so arranged that there is a minimum clearance of 30cm under maximum deflection condition due to wind between the live jumper and other metallic parts. Erection of intermediate pin insulators may be necessary for fixing these jumpers.
- 4.5.7.3 Parallel groove (PG) Clamps may be preferred to binding of conductors at jumper locations or service taps.
- 4.5.7.4 Jumpers used shall normally be of the same material as the line conductor and they shall be of adequate current carrying capacity. If the material of the jumper wire is different from that of the line conductor, suitable bimetallic clamps should be used. If copper to aluminum bimetallic clamps are to be used, it should be ensured that the aluminium conductor is situated above the copper conductor so that no copper contaminated water comes in contact with aluminium.

4.5.8 Earth wire

- 4.5.8.1 A continuous GI earth wire of size not less than 4mm (8SWG) shall be run all along the LV/MV distribution lines and service lines. This shall be securely fastened to the cross arms by means of cast iron reels or by suitable clamps as directed by the Engineer-in-charge.
- 4.5.8.2 Where a continuous carpet guard wire is provided, the same shall serve the purpose of continuous earth wire run mentioned above.

4.5.9 Erection of guard

4.5.9.1 A guard shall be provided at all road crossings of overhead lines crossings with other lines and between HV and LV/MV lines carried on the same support.

4.5.9.2 A cage guard/cradle guard shall be provided for LV/MV overhead lines of vertical configuration. Cage guard/Cradle guard shall be used for overhead lines of horizontal configuration. Where no guard of any of these types is provided, a ring guard supported from the pole shall be arranged (refer figure 10 & 11). Cage guard/cradle guard shall be made of 6 SWG GI wire confirming to IS: 2633 including netting, stretching and jointing of cage and lacing by 10/12 SWG GI wire, binding by 14/16 SWG GI wire.

4.5.9.3 In case of cradle/cage guards, at least 9 laces shall be provided for each span.

4.5.9.4 The guard wires shall be bonded to the earth wire

4.5.9.5 Reel insulator shall be used to bind the cage guard to the neutral.

4.5.9.6 The guard wire shall always run not less than 30cm beyond the outer most bare conductor of the configuration in the case of 11KV/33KV lines, and 10cm in the case of LV/MV lines.

4.5.10 Earthing

4.5.10.1 Earthing installation shall conform to various clauses of the CPWD General Specifications for Electrical works (part I-Internal) as amended upto date and as per following incompliance with Regulation - 72 of CENTRAL ELECTRICITY AUTHORITY (Measures Relating to Safety & Electric Supply) Regulations, 2010.

- (i) All metal supports and all reinforced and prestressed cement concrete supports of overhead lines and metallic fittings attached thereto, shall be either permanently and efficiently earthed by providing a continuous earth wire and securely fastening to each pole and connecting with earth ordinarily at three points in every km. with the spacing between the points being as neatly equidistant as possible or each support and the metallic fitting attached thereto shall be efficiently earthed.
- (ii) Metallic bearer wire used for supporting insulated wire of overhead service lines of voltage not exceeding 650 V shall be efficiently earthed or insulated.
- (iii) Each stay-wire shall be similarly earthed unless insulator has been placed in it at a height not less than 3.0 metres from the ground.

4.5.10.2 All metal supports and RCC/PCC supports of overhead line, and metallic fittings attached there-to shall be permanently and efficiently earthed, and for this purpose a continuous earth wire shall be run as per clause 4.5.8. In the case of RCC/PCC poles, all insulator pins, cross arms, stays street light brackets and other metallic fittings shall be bonded to the continuous earth wire.

4.5.10.3 The continuous earth wire shall be connected to earth. There shall be not less than 3 connections with the earth per kilometer, spaced at equi-distance as far as possible. Where a continuous earth wire cannot be provided, every pole should be earthed and all the metal parts are to be bonded, with the concurrence of the Engineer-in-charge.

4.5.10.4 Junctions, end terminal locations, and all special structures may be selected of connecting to earth.

4.5.10.5 The lead from the earth electrode shall be suitably protected by a 15mm dia GI pipe upto a height of 3m from the ground level and shall be bonded to the continuous earth wire.

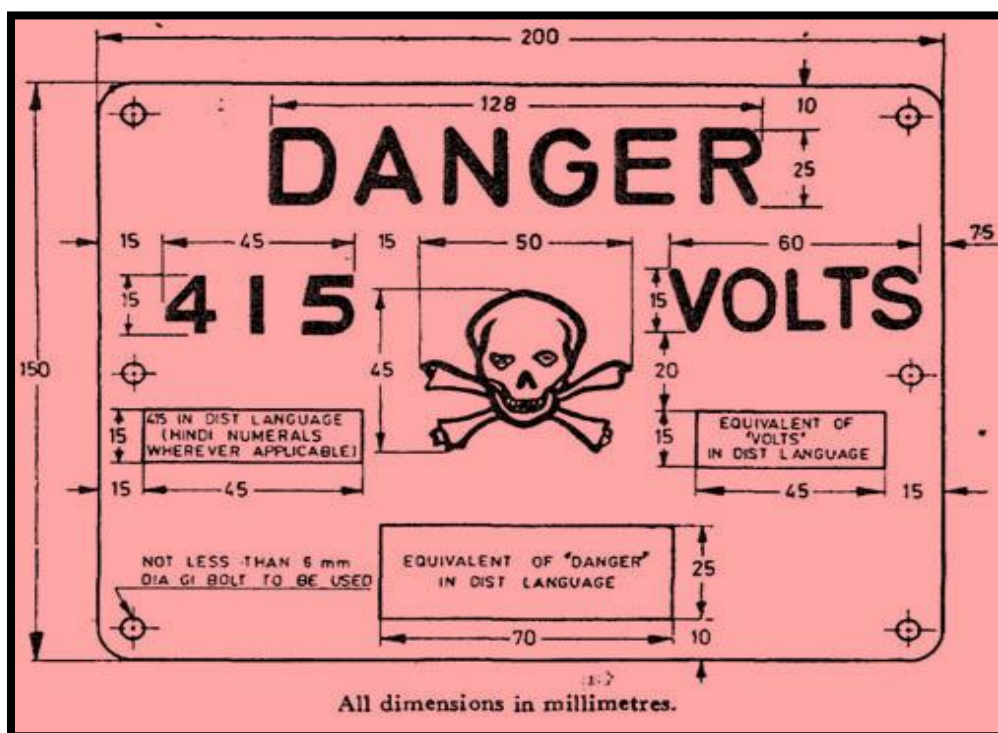
4.5.10.6 The protection pipe and the earth lead shall be suitably clamped to the support.

4.5.11 Safety and protective devices

4.5.11.1 Danger board

- (a) All supports carrying HV lines shall be fitted with danger plate conforming to IS: 2551-1982 at a height of 3m from ground and it shall indicate the voltage of the line.

- (b) The danger plate should be in Hindi or English and in addition in the location language with the sign of skull and bones are required to be provided on power line support and other electrical installation.
- (c) Two sizes of danger notice plate are recommended.
 - (i) For display at 415 V Installation – 200 X 150 mm
 - (ii) For display at 11 KV or higher voltage Installation – 250 X 200 mm.
- (d) The danger plate shall be made from mild steel sheet of at least 1.6 mm thick and vitreous enameled white paint with letters, figures and the conventional skull and cross-bones in signal red color as per IS: 5-1978 on the front side. The rear side of the plate shall also be enameled.

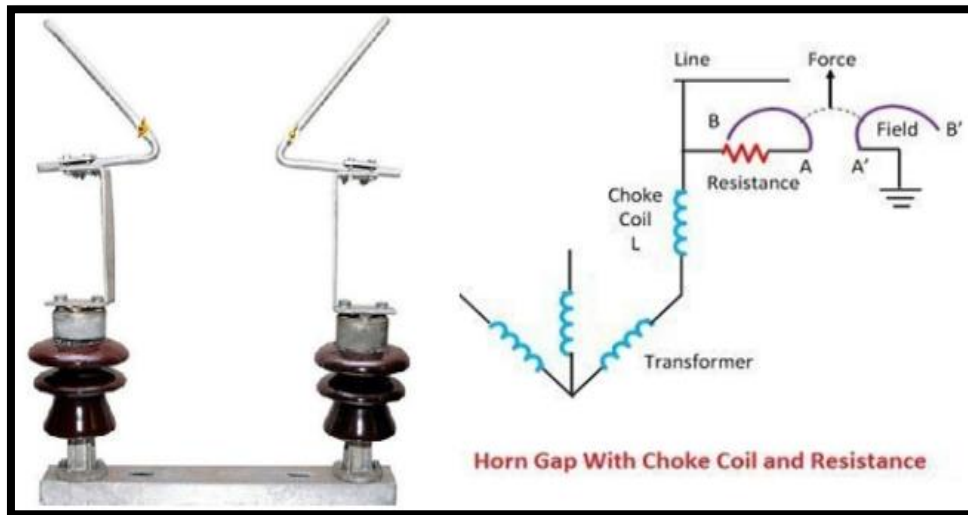


4.5.11.2 Ant climbing devices

Necessary arrangement for preventing unauthorized persons from ascending any of the supports carrying HV lines without the aid of a ladder or special appliances shall be made. Unless otherwise specified, barbed wire conforming to IS:2781978 having 4 point bars, space 75mm $\pm$ 12mm apart and weighing 108/125gm/m, shall be wrapped helically with a pitch of 75mm around the limb of the support and tied firmly commencing from a height of 3.5m and upto a height of 5m or 6m as directed by the Engineer-in- charge.

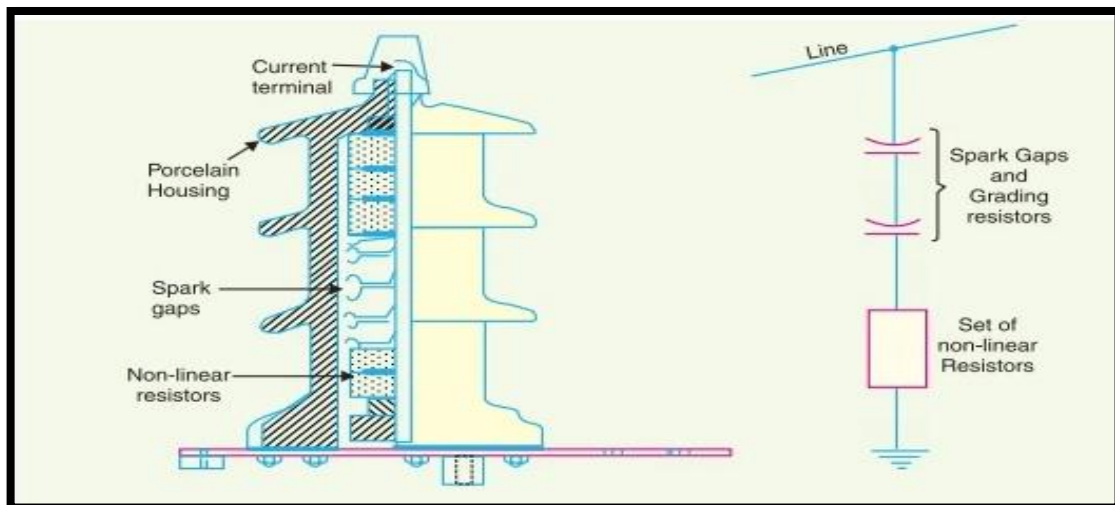
4.5.11.3 Lighting arresters: -Overhead line shall be protected against lightning as per IS/IEC 62305 with following types:-

- (i) **Horn gap type lightning arrester**
 - (a) Horn gap type lightning arrester for LV/MV lines and surge diverter suitable for MV lines shall be employed with each phase at terminals and any other places where specified, depending upon the local climatic conditions. These shall be mounted on cross arms.
 - (b) A short and definite air gap not exceeding 2cm must be maintained between the horns.



(ii) **Non-linear resistor type lightning arrester**

- (a) Non-linear resistor type lightning arrester suitable for HV lines shall be installed, one unit per phase, at the terminals, transformer stations etc. as specified.
- (b) These devices shall be connected ahead of fuses, if any, provided.
- (c) The lightning arrester shall consist of line terminal stud, earth terminal stud, number of spark gaps in series with non-linear resistor. The whole assembly housed inside a hermetically sealed porcelain bushing.
- (d) Neoprene rubber gasket shall be provided between metal caps and porcelain bushing, non-linear resistor shall be silicon carbide blocks metalized at both ends to ensure good electrical contact between terminals, non-linear resistor & spark gaps.



(iii) **Earthing for lightning arresters**

- (a) The earthing lead for any lightning arrester shall not pass through any iron or steel pipe, but shall be taken as directly as possible from the lightning arrester without touching any metal part to a separate-vertical ground electrode or junction of the earth mat already provided for the substation of voltage exceeding 650 V subject to the avoidance of bends wherever practicable.
- (b) The earth lead from the earth electrodes to the lightning arresters shall be continuous, and where specified, it shall be insulated throughout above the earth surface by an alkathene pipe/DMC.
- (c) Mounting bracket shall be hot dip galvanized suitable for mounting lightning arrester on structure.

4.5.12 Double Triple/ Four Pole Structure: -

- 4.5.12.1** Where a HT overhead line deviates by more than 10 degrees, a double pole, triple pole or four pole structure may be erected at the point of deviation depending upon the angle of deviation. Normally, a double pole structure should suffice or an angle of deviation upto about 30 degrees, a triple pole structure upto about 60 degrees, and a four-pole structure beyond that angle of deviation.
- 4.5.12.2** The structure shall comprise of the same type of poles as used for the overhead lines unless otherwise specified.
- 4.5.12.3** A set of double channel iron cross-arms of the same length duly clamped by through bolts and nuts shall be provided for each double pole structure. Similar set of double channel iron length of the double channel iron cross arm shall be likewise clamped to each side of the triple and four pole structures. The two channel iron cross arm shall be stepped to each other by 50mm x 6mm flat iron strips to prevent bulging of the cross arms during stringing of the lines. These flat iron strips shall be bolted to the two channel iron lengths adjacent to each strain/ disc insulator fittings supported by the cross arm.
- 4.5.12.4** A set of double channel iron cross-arms of the same length duly clamped to the poles by through bolts and nuts shall be provided for each double pole structure. Similar set of double channel iron cross-arms shall be likewise clamped to each side of the triple and four pole structures. The two channel iron lengths of the double channel iron cross arm shall be strapped to each other by 50mm x 6 mm flat iron strips to prevent bulging of the cross arms during stringing of the lines. These flat iron strips shall be bolted to the two channel iron lengths adjacent to each strain/ disc insulator fittings supported by the cross arm.
- 4.5.12.5 Cross Bracings**
- (i) A set of cross bracings fabricated out of 50mm x50mmx6mm angle iron for 11 KV lines, and out of 65mmx65mmx6mm for 33 KV lines shall be provided for each double pole structure.
 - (ii) Three such sets of cross bracings shall be provided for each triple pole structure.
 - (iii) Four sets of such cross bracings shall be provided for each four-pole structure.
 - (iv) The horizontal members of the bracings shall be fixed to the poles by means of clamps fabricated out of 50mmx6mm flat iron. The inclined members of the cross bracings shall be fixed to the horizontal members by suitable bolts and nuts, after pressing together by forging the two sides of the angle iron at either end of the inclined members.
 - (v) The cross bracings shall be so fixed as to form a rectangle of minimum size 1.4m width x2.5m height in case of 11KV lines, and of 2.4m width x 2.8m height in case of 33 KV lines.
 - (vi) The cross bracings shall be fixed more or less in the middle of the structure.



4.6 SERVICE CONNECTION LINE

No service connection shall be taken off an overhead line except at a point of support.

The service line shall be either through an overhead service or underground cable, as specified. In case of overhead services, the same may be provided with either

- (a) Bare conductors or,
- (b) Insulated conductors.

4.6.1 Service line with bare conductors:-

4.6.1.1 Any of the following methods shall be adopted as specified: -

(i) First Method

- (a) The bare conductors shall be strung with shackle insulators fixed to cross arms at both ends.

The feeding end cross arm shall be fixed to the support and the one at receiving end shall be mounted on a GI pipe of minimum 5 cm dia. The bare conductors shall be kept at a height of at least 2.5m from the top of the structure In accordance with Rule 79 of the Indian Electricity Rules.

- (b) The GI pipe shall be provided with double bends at the top. The pipe shall be secured by at least 2 clamps made of 50mmx6mm MS flats fixed firmly to the wall in the vertical position. It shall in addition be provided with a GI stay wire of 7/3.15mm size anchored to the building with an eye bolt.
- (c) Service connection shall be given from the overhead service line, with weather proof/XLPE insulated cable through this GI pipe. Well-fitting PVC bushes shall be provided at both ends of this GI pipe.

(ii) Second Method

- (a) The bare conductors shall be strung with shackle insulators as above except at the receiving end, where the insulators shall be fixed to a bracket in a suitable insulator shall be fixed to a bracket in a suitable form made of angle iron of size not less than 50mmx50mmx6mm. The ends of the bracket shall be cut and split and embedded in the wall with cement mortar.
- (b) The bare conductor shall be kept at least 1.2m away from the edge of the structure in accordance with Rule 79 of the Indian Electricity Rules. The service connection shall be given with weather proof/XPLE insulated cable through GI pipe of minimum 4cm dia fixed to the wall. The GI pipe shall be bent downwards near the service entry. Well-fitting PVC bushes shall be provided at both ends of the GI pipe.

4.6.1.2 The service line with bare conductors shall be guarded wherever required in accordance with the Indian Electricity Rules.

4.6.2 Service line with insulated conductors

4.6.2.1 Service connection may be given by weather proof cable/PVC sheathed cable on GI bearer wire.

4.6.2.2 The cable shall be supported by the bearer wire by means of suitable link clips spaced 30cm apart, or by porcelain cleats 50cm apart.

4.6.2.3 The GI bearer wire shall be of minimum 3.2 mm (10SWG) size. One end of the GI bearer wire shall be attached to a clamp which is fastened to the nearest pole carrying the distribution lines from where the service connection is intended to be given. The other end of GI wire shall be fastened to a 5cm dia GI pipe for a span upto 4.5m which shall be fixed to the wall with guy etc.

4.6.2.4 The GI pipe shall be fixed to an angle iron of size 40mmx40mmx6mm thick with suitable guy for high supports and for a span exceeding 4.5m.

Alternatively, when the height of the structure permits minimum ground clearance, the other end of this GI bearer wire may be fixed to a hook, eye-bolt or bracket embedded with cement mortar in the wall. The weather proof/XLPE insulated and PVC sheathed cable shall pass through GI pipe of minimum dia 5cm which is bent downwards. Well-fitting PVC bushes shall be provided at both ends of the GI pipe.

4.6.3 Service line by underground cable

- 4.6.3.1 Service cables shall be XLPE insulated PVC sheathed armoured UG cables to the specified size(s).
- 4.6.3.2 The service cable from an overhead distribution line shall be fixed to the support with 2 Nos. of clamp of MS flat of size 50mmx6mm. This shall be protected upto a height of 3m from ground level by a GI pipe of adequate size, clamped to the support with 2 Nos. of MS flat of size 50mmx6mm. The service cable shall be laid in accordance with chapter 2 of these specifications.
- 4.6.3.3 The service cable shall be terminated to an outdoor cable termination box fixed to the support. The connection with the overhead line shall be given either by the cores of the service cable directly or through separate insulated leads as specified.
- 4.6.3.4 Laying of cable shall be carried out as per chapter-2 “Cable Laying work”.

4.6.4 Service fuses

Unless otherwise specified, an ICDB (with rewire-able fuses) may be provided on the pole for the purpose.

4.7 PAINTING

- 4.7.1 Treatment of all supports and line materials before or at the time of erection shall be done in accordance with the relevant sub clauses of clause 4.2.11. The cross arms and the pole clamps shall be treated with one coat of red oxide primer before erection and finished with two coats of approved paint after erection along with other hardware.
- 4.7.2 After erection, the external surface of metal supports above ground level and all pole fittings shall be given finishing coats of painting.
- 4.7.3 The following procedure for painting/repainting works shall be adopted.
 - (a) In the case of line accessories, all rust and scale shall be removed by scraping, or by brushing with steel wire brush. All dust and dirt shall be carefully and thoroughly wiped away. Painting shall not be done when the surface is wet or covered with dirt/dust.
 - (b) The primer coat shall be applied with red oxide paint. In the case of new supplies, this shall be done before supplying to the site.
 - (c) Application of finishing coat at site shall be done with brushes and the paint shall be spread as evenly and as smooth as possible. The surface shall be given two or more coats as specified.

4.8 NUMBERING OF SUPPORTS

- 4.8.1 All supports shall be numbered after painting, as directed by the Engineer-in-charge.
- 4.8.2 Separate number plates may be used, if so specified.

4.9 TESTING OF OVERHEAD LINE

- 4.9.1 Before connecting the services to transformer, equipment etc. a pressure test of appropriate standard shall be carried out on the line as directed by the Engineer-in-charge.
- 4.9.2 Before charging the MV lines, the same shall be tested with a 500V megger for insulation resistance. Similar testing shall be done for 11KV lines with a 2500V megger and for 33KV lines with a 5000V megger.
- 4.9.3 Where pressure test is not done on a MV line, it shall be tested with a 2500V/5000V megger for insulation before charging.
- 4.9.4 All earth sets associated with the work shall be tested.
- 4.9.5 All test results including earth test results shall be recorded and submitted to the Engineer-in-charge.

4.10 COMMISSIONING

- 4.10.1 The distribution lines shall be charged only if the pressure/ megger test and earth test results are satisfactory.
- 4.10.2 The lines shall be commissioned in the presence of the Engineer-in-charge.

4.11 MEASUREMENT OF OVERHEAD CONDUCTORS AND EARTH WIRE

The weight of overhead conductors and earth wire shall be determined by measuring the distance from center of one pole to the center of another pole and computing the weight of the earth wire or conductor by using a standard table. 3% extra shall be allowed collectively for wastage, sag, jointing, binding, jumpers etc.

CHAPTER 5

AERIAL BUNCHED CABLES WORK

5.1 GENERAL

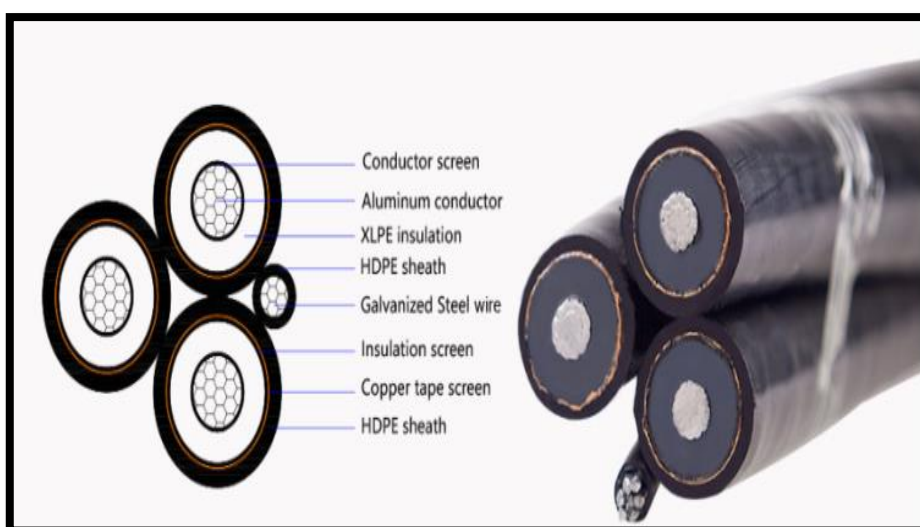
LT Aerial bunched cable shortly known as ABC are available in 1.1 KV volt grade, are used for distributing power to individual consumers by utility service providers to their LT consumers. These cables are suspended overhead using electric poles. The cables can be tapped intermittently from any required position, enabling them to be used in urban as well rural areas. It is used both in single phase or three phase system with or without street light line. ABC should be confirming to IS 14255/1995.

5.2 SIZE AND CONSTRUCTION

In ABC, a messenger cum neutral wire is used as bare or with insulation which support the weight of the cable and keeps the assembly strung under tension. The phase conductor can be of single phase or three phase with XLPE insulation. A lighting conductor can also be incorporated for street lighting. Phase conductors are made of concentrically stranded Aluminium wires having 7 or 19 wires. Messenger conductors are made of either stranded circular or compacted circular type and shall have minimum 7 strands with specially treated Silica, Magnesium and Aluminium Alloy for providing good tensile strength. Generally, the messenger acts as earth and neutral. As the cables remains exposed to environmental elements such as UV rays from sunlight, the insulation is mixed with a small amount of carbon black to prevent the deterioration of polythene insulation.

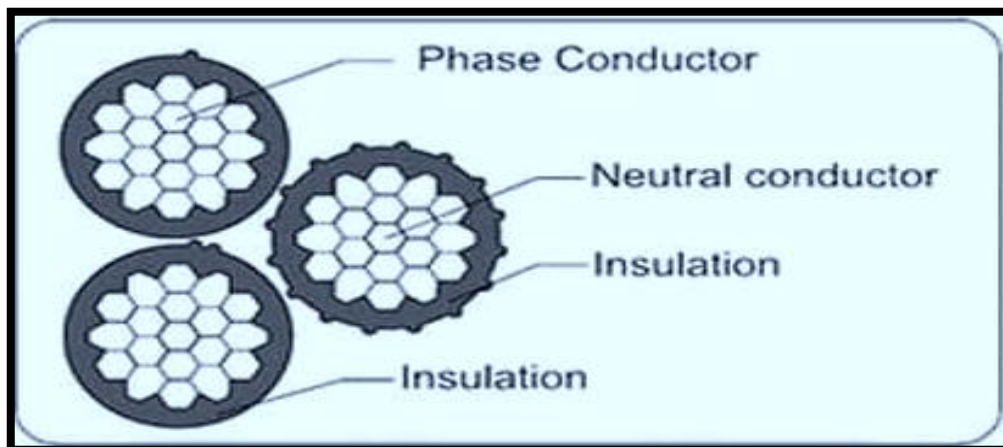
The phase conductor shall be of size 150 mm², 120 mm², 95 mm², 70 mm², 50 mm², 35 mm², 25 mm² and 16 mm² XLPE insulated with H2 or H4 grade Aluminium complying with the requirements of IS 8130: 1984. The messenger-cum-neutral conductor of sizes 70 mm², 50 mm², 35 mm² and 25 mm² shall be bare/insulated, heat treated aluminium-magnesium-silicon alloy wires containing 0.5% magnesium and approximately 0.5% silicon confirming to IS: 398 (Part-IV):1979. The size of street lighting conductor shall be 16 mm². The tensile strength of the aluminium wire used in the conductor shall not be less 90 N/mm².

Three insulated phase conductors, one insulated neutral conductors (if required) and a street lighting conductor (if required) shall be twisted around the bare (or insulated) as required messenger conductor without fillers with a lay not exceeding 35 times the diameter of the insulated phase conductor. The direction of lay shall be right hand.



5.3 CORE IDENTIFICATION

The phase conductors shall be provided with one, two & three 'ridges' and outer insulated conductor, if provided, shall have for ridges for easy identification. The street lighting conductor and messenger conductor (if insulated) shall not have any identification mark.



5.4 ADVANTAGES

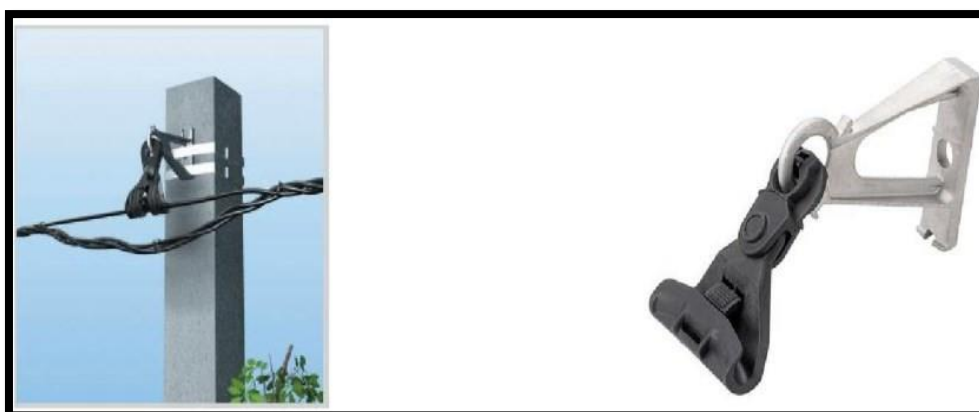
ABC Cables are highly reliable and insulation has been developed to withstand heat, cold and intense sunlight. There is no risk in touching the live cable and the insulation reduces the number of short circuits and over-voltages in overhead cables during thunder-storms.

- (i) Aerial bundled cables do not generate any spark when it touches the trees.
- (ii) Their appearance is good.
- (iii) They can be installed in narrow streets.
- (iv) Insulators and cross arms are not required.
- (v) These cables are very safe as only neutral wire gets damaged in case of any accident.
- (vi) Electricity theft is merely possible.
- (vii) Short circuit is only possible when insulation of a cable is damaged.
- (viii) Insulating bridging wires are needed to connect
- (ix) Non-insulated wires at either side at junction poles.

5.5 VARIOUS ACCESSORIES REQUIRED IN LAYING OF AERIAL BUNCHED CABLE

5.5.1 Suspension Clamps

Suspension Clamps are used along with a bracket or other supporting hardware to suspend and grip in straight run of Aerial Bunched Cable, without damaging, the insulated neutral messenger of a LV-ABC system by means of an adjustable lock that accommodates a range of cable sizes. They can also be used in conjunction with insulation piercing connectors to provide service connections by tapping the mainline.



5.5.2 Anchoring Clamps

Anchoring Clamps are used to terminate at dead end and to provide angle up to 90° along with a bracket or other supporting hardware (as a customized option), and are used to strain the insulated neutral messenger and also to provide angles of a LV-ABC system without damaging the cable's insulation.

- (a) Anchoring Clamp with bracket
- (b) To provide angle up to 90° to the line.
- (c) To terminate in to transformer leads or mains for industrial / residential supply.



5.5.3 Service clamp

Service Clamps are used along with a bracket or other supporting hardware to strain the insulated service conductor of a LV AB Cable system to provide turn angles, anchor service lines on walls, or support tap-offs for the service lines without damaging the cable's insulation.



5.5.4 Insulation Piercing Connectors (IPC)

Insulation Piercing Connectors are used on all AB cable systems (messenger wire as well as self-supporting system) to take a tap connection. This tap can be used to continue the line, distribute the line, used in street lighting or in-service connections to households. The design allows the connection to be completely sealed against any water ingress, thus acting as a waterproof connector. There are four main applications to IPCs, namely street light, tap off, distribution box charging and jumper connections.

(a) Street Light Connections

In this application, IPCs are generally used to tap the main AB Cable and connecting it to a service line that powers streetlights or other applications.



(b) Tap off/Multi Tap Connection to Households

These IPCs are used to tap the main AB Cable and bring power to households. Using the Multiport IPC, one tap connection from the main line can be used to bring connections to multiple households.



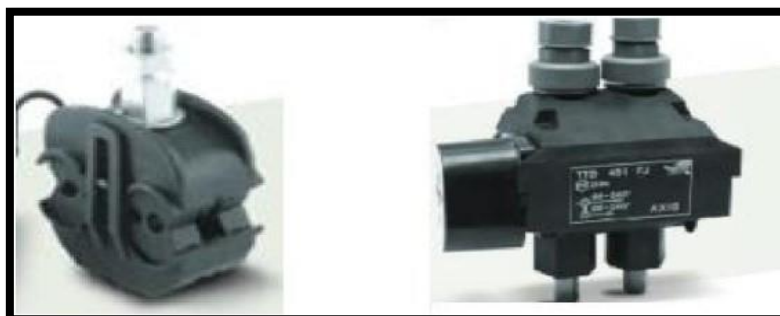
(c) Distribution Box Charging Connection

These IPCs are used to tap the main AB Cable Line to charge Low Voltage Distribution Boxes which are further connected to different applications such residential connections, street lights, etc. Axis Low Voltage Distribution Boxes can provide up to 9 outgoing connections for each incoming line.



(d) Jumper Connection

These IPCs are generally used with the same size of cable on each side to facilitate T-Connections (90° turns) or any other high angle turn.



5.6 Stringing/Laying and Jointing of AB Cables

5.6.1 Stringing/Laying: The stringing of AB cable is easy through conventional methods but care shall be taken that cable insulation does not get damaged during installation. Dragging of cable on the ground can cause damage of insulation. Tension to be applied during stringing shall be 25% of the breaking load of the messenger wires. This will allow line to sag within the specified limit of 1.5% of the span at the lowest ambient temperature.

5.6.2 Jointing: Mid-span jointing is permissible for LT ABC cable through conventional techniques. It is recommended to join the cable in such a way so as to bring the joints at the supports. Mid-span jointing is not recommended in case of HT lines. Line tapping at the support may be allowed under unavoidable circumstances through suitably designed clamp connectors or parallel groove (PG) clamps. Tap off from power conductors in the area where catenaries are under tension is not recommended. The cable may be shorted and earthed through suitable non-liner surge arrestor.

5.7 Sizes and Requirements of Aerial Bunched Cables

The Aerial bunched cables are available in following sizes with specification:-

(A) **With insulated messenger conductor without lighting:-**

Sl.no	Description and type of cables	Numbers of wires		Thickness of XLPE		Approx. overall Diameter	Approx.	Breaking load of messenger	Maximum DC resistance ohms/Km		AC Current rating Amps
		Phase	Messenger	Phase	Messenger						
				mm	mm	mm	kg/km	kn (min)	Phase	Messenger	In air at 40°C
	With insulated messenger conductor without lighting conductor										
1	1 C X 16 mm ² + 25 mm ² (insulated)	7	7	1.20	1.20	20.0	165	7.0	1.910	1.380	72
2	3 C X 16 mm ² + 25 mm ² (insulated)	7	7	1.20	1.20	22.0	301	7.0	1.910	1.380	64
3	1 C X 25 mm ² + 25 mm ² (insulated)	7	7	1.20	1.20	22.4	195	10.1	1.200	1.380	99
4	3 C X 25 mm ² + 25 mm ² (insulated)	7	7	1.20	1.20	25.0	390	10.1	1.200	1.380	84

5	1 C X 35 mm ² + 25 mm ² (insulated)	7	7	1.20	1.20	27.3	227	10.1	0.868	1.380	120
6	3 C X 35 mm ² + 25 mm ² (insulated)	7	7	1.20	1.20	27.4	486	10.1	0.868	1.380	105
7	1 C X 35 mm ² + 35 mm ² (insulated)	7	7	1.20	1.20	28.0	259	10.1	0.868	0.986	120
8	3 C X 35 mm ² + 35 mm ² (insulated)	7	7	1.20	1.20	28.4	518	10.1	0.86	0.986	105
9	1 C X 50 mm ² + 35 mm ² (insulated)	7	7	1.50	1.20	29.0	318	10.1	0.64	0.986	150
10	3 C X 50 mm ² + 35 mm ² (insulated)	7	7	1.50	1.20	32.3	692	10.1	0.64	0.986	130
11	3 C X 70 mm ² + 50 mm ² (insulated)	7	7	1.50	1.50	37.5	939	14.0	0.44	0.689	155
12	3 C X 70 mm ² + 70 mm ² (insulated)	7	7	1.50	1.50	39.0	1002	19.7	0.44	0.492	155
13	3 C X 95 mm ² + 70 mm ² (insulated)	19	7	1.50	1.50	42.7	1237	19.7	0.32	0.492	190
14	3 C X 120 mm ² + 70 mm ² (insulated)	19	7	1.60	1.50	46.0	1482	19.7	0.25	0.492	220
15	3 C X 150 mm ² + 70 mm ² (insulated)	19	7	1.80	1.50	50.0	1791	19.7	0.20	0.492	250

(B) With bare messenger conductor without lighting conductor:-

Sl.no	Description and type of cables	Numbers of wires		Thickness of XLPE		Approx. overall Diameter	Approx. weight of cable	Breaking load of messenger	Maximum DC resistance ohms/Km		AC Current rating Amps
		Phase	Messenger	Phase	Messenger						
				mm	mm	mm	kg/km	kn (min)	Phase	Messenger	In air at 40°C
	With insulated messenger Conductor without lighting conductor										
1	1 C X 16 mm ² + 25 mm ² (bare)	7	7	1.20	N.A	18.5	137	7.0	1.910	1.380	72
2	3 C X 16 mm ² + 25 mm ² (bare)	7	7	1.20	-do-	19.3	272	7.0	1.910	1.380	64
3	1 C X 25 mm ² + 25 mm ² (bare)	7	7	1.20	-do-	19.5	167	10.1	1.200	1.380	99
4	3 C X 25 mm ² + 25 mm ² (bare)	7	7	1.20	-do-	20.5	362	10.1	1.200	1.380	84
5	1 C X 35 mm ² + 25 mm ² (bare)	7	7	1.20	-do-	22.0	199	10.1	0.868	1.380	120
6	3 C X 35 mm ² + 25 mm ² (bare)	7	7	1.20	-do-	23.5	458	10.1	0.868	1.380	105
7	1 C X 35 mm ² + 35 mm ² (bare)	7	7	1.20	-do-	24.6	226	10.1	0.868	0.986	120
8	3 C X 35 mm ² + 35 mm ² (bare)	7	7	1.20	-do-	25.0	485	10.1	0.86	0.986	105
9	1 C X 50 mm ² + 35 mm ² (bare)	7	7	1.50	-do-	26.6	284	10.1	0.64	0.986	150
10	3 C X 50 mm ² + 35 mm ² (bare)	7	7	1.50	-do-	26.8	659	10.1	0.64	0.986	130
11	3 C X 70 mm ² + 50 mm ² (bare)	7	7	1.50	-do-	31.2	890	14.0	0.44	0.689	155
12	3 C X 70 mm ² + 70 mm ² (bare)	7	7	1.50	-do-	34.4	946	19.7	0.44	0.492	155
13	3 C X 95 mm ² + 70 mm ² (bare)	19	7	1.50	-do-	36.0	1179	19.7	0.32	0.492	190
14	3 C X 120 mm ² + 70 mm ² (bare)	19	7	1.60	-do-	38.0	1425	19.7	0.25	0.492	220
15	3 C X 150 mm ² + 70 mm ² (bare)	19	7	1.80	-do-	40.0	1735	19.7	0.20	0.492	250

(C) With insulated messenger conductor with lighting conductor

Sl.no	Description and type of cables	Numbers of wires		Thickness of XLPE		Approx. overall Diameter	Approx. weight of cable	Breaking load of messenger	Maximum DC resistance ohms/Km	AC Current rating Amps	
		Phase	Messenger	Phase	Messenger						
				mm	mm	mm	kg/km	kn (min)	Phase	Messenger	In air at 40°C
	With insulated messenger conductor with lighting conductor										
1	3 C X 16 mm ² + 25 mm ² (insulated) + 16 mm ²	7	7	1.20	1.20	23.5	369	7.0	1.910	1.380	62
2	3 C X 25 mm ² + 25 mm ² (insulated) + 16 mm ²	7	7	1.20	1.20	25.0	457	7.0	1.200	1.380	82
3	3 C X 35 mm ² + 25 mm ² (insulated) + 16 mm ²	7	7	1.20	1.20	27.5	554	7.0	0.868	1.380	103
4	3 C X 35 mm ² + 35 mm ² (insulated) + 16 mm ²	7	7	1.20	1.20	28.4	586	10.1	0.868	0.986	103
5	3 C X 50 mm ² + 35 mm ² (insulated) + 16 mm ²	7	7	1.50	1.20	32.5	760	10.1	0.641	0.986	127
6	3 C X 70 mm ² + 50 mm ² (insulated) + 16 mm ²	7	7	1.50	1.50	37.5	1007	14.0	0.443	0.689	154
7	3 C X 70 mm ² + 70 mm ² (insulated) + 16 mm ²	7	7	1.50	1.50	39.5	1070	19.7	0.443	0.492	154
8	3 C X 95 mm ² + 70 mm ² (insulated) +	19	7	1.50	1.50	42.5	1304	19.7	0.320	0.492	188

	16 mm ²										
9	3 C X 120 mm ² + 70 mm ² (insulated)+ 16 mm ²	19	7	1.60	1.50	46.8	1550	19.7	0.253	0.492	218
10	3 C X 150 mm ² + 70 mm ² (insulated) + 16 mm ²	19	7	1.80	1.50	50.8	1860	19.7	0.206	0.492	248

(D) **With bare messenger conductor with lighting conductor**

Sl.no	Description and type of cables	Numbers of wires		Thickness of XLPE insulation		Approx. overall Diameter	Approx weight of cable	Breaking load of messenger	Maximum DC resistance ohms/Km		AC Current rating Amps
		Phase	Messenger	Phase	Messenger						
				mm	mm	mm	kg/km	kn (min)	Phase	Messenger	In air at 40°C
	With bare messenger conductor with lighting conductor										
1	3 C X 16 mm ² + 25 mm ² (bare) + 16 mm ²	7	7	1.20	-do-	19.5	340	7.0	1.910	1.380	62
2	3 C X 25 mm ² + 25 mm ² (bare) + 16 mm ²	7	7	1.20	-do-	20.5	429	7.0	1.200	1.380	82
3	3 C X 35 mm ² + 25 mm ² (bare) + 16 mm ²	7	7	1.20	-do-	23.5	526	7.0	0.868	1.380	103
4	3 C X 35 mm ² + 35 mm ² (bare) + 16 mm ²	7	7	1.20	-do-	25.0	553	10.1	0.868	0.986	103
5	3 C X 50 mm ² + 35 mm ² (bare) + 16 mm ²	7	7	1.50	-do-	26.8	727	10.1	0.641	0.986	127
6	3 C X 70 mm ² + 50 mm ² (bare) + 16 mm ²	7	7	1.50	-do-	31.5	958	14.0	0.443	0.689	154

7	3 C X 70 mm ² + 70 mm ² (bare) + 16 mm ²	7	7	1.50	-do-	34.5	1013	19.7	0.443	0.492	154
8	3 C X 95 mm ² + 70 mm ² (bare) + 16 mm ²	19	7	1.50	-do-	37.0	1248	19.7	0.320	0.492	188
9	3 C X 120 mm ² + 70 mm ² (bare) + 16 mm ²	19	7	1.60	-do-	39.0	1493	19.7	0.253	0.492	218
10	3 C X 150 mm ² + 70 mm ² (bare) + 16 mm ²	19	7	1.80	-do-	40.0	1803	19.7	0.206	0.492	248

5.8 Applicable Standards

The following Indian standards are necessary adjuncts to this standard: -

- (i) IS – 14255/1995: ABC cables upto 1100 volts.
- (ii) IS – 8130/1984: Conductors for insulated electric cables and flexible cords.
- (iii) IS – 398/Pt.IV/1994: Aluminium conductor for overhead transmission purposes.
- (iv) IS – 10418/1982: Drums for electric cables

5.9 Tests

5.9.1 Type Test

The following type test are required to made before supply on a general commercial basis on a type of cable in order to demonstrate satisfactory performance characteristics to meet the intended application:

Sl. No	Test	Requirement
(1)	(2)	(3)
A.	Tests on phase/street light conductor:	See IS 8130 : 1984
(i)	Tensile test	-do-
(ii)	Wrapping test	-do-
(iii)	Resistance test	-do-
B.	Test on messenger conductor:	-do-
C.	Physical tests for XLPE Insulation:	-do-
(i)	Tensile strength and elongation at break	-do-
(ii)	Ageing in air oven	-do-
(iii)	Hot set test	-do-
(iv)	Shrinkage test	-do-
(v)	Water absorption (Gravimetric)	-do-
(vi)	carbon black:	-do-
(a)	Content	-do-
(b)	Dispersion	-do-

D.	Physical test for PE insulation:	-do-
(i)	Tensile strength and elongation at break	-do-
(ii)	Melt flow index	-do-
(iii)	carbon black	-do-
(a)	Content	-do-
(b)	Dispersion	-do-
E.	Test for thickness insulation	-do-
F.	Insulation resistance (Volume resistivity) test	-do-
G.	High voltage test	-do-

5.9.2 Acceptance Test

The following tests are carried out on samples taken from a lot for the purpose of acceptance of the lot-

- (i) Tensile test (for phase/street light conductor),
- (ii) Wrapping test (for phase/street light conductor),
- (iii) Breaking load test for messenger conductor
- (iv) Elongation test for messenger conductor,
- (v) Conductor resistance test,
- (vi) Test for thickness of insulation
- (vii) Tensile strength and elongation at break test,
- (viii) Hot set test for XLPE insulation
- (ix) Insulation resistance test, and
- (x) High voltage test

5.9.3 Routine Tests

The following Tests made by the manufacturers on all finished cable lengths to demonstrate the integrity of the cables.

- (i) Conductor resistance test, and
- (ii) High Voltage test

5.9.4 Optional Test

The following Special tests to be carried out, when required by agreement between the purchaser and the supplier.

- (i) Bending test

5.10 Details of Test

5.10.1 General

The tests shall be carried out in accordance with appropriate parts of IS 10810 taking into account additional information given in this standard.

5.10.2 High Voltage Test

High voltage test at room temperature (Type, Acceptance and routine test). The cable shall withstand a voltage of 3kV ac (rms) at a frequency of 40 to 60 Hz or a dc voltage of 7.2 kV between conductors for 5 minutes.

5.10.3 Elongation Test for Messenger Conductor

Test shall be carried out on wires before stranding. The specimen shall be straightened by bauld and an original gauge length of 200 mm shall be marked on the wire. A tensile load shall be applied by means of a suitable tensile testing machine. The load shall be applied gradually and the rate of separation of the jaws of the testing machine shall be not less than 25 mm per minute and not greater than 100 mm per minute. The elongation shall be measured after the fractured ends have been fitted together. If the fracture occurs outside the gauge marks, or within 25 mm of either mark and the required elongation is not obtained, the test shall be disregarded and another test made. The elongation of the specimen shall be not less than 4 percent on a gauge length of 200 mm.

5.10.4 Bending Test on the Completed Cable

The test shall be performed on a sample of complete cable. The sample shall be bent around a test mandrel at room temperature for at least one complete turn. It shall then be unwound and the process shall be repeated after turning the sample around its axis by 180 degrees. The cycle of these operations shall then be repeated twice.

The diameter of the mandrel shall be: **10 (D + d)**

D= actual diameter of the cable, for example, minimum circumscribing diameter, mm; and **d**= actual diameter of the phase conductor, mm.

Requirement: No cracks visible to the naked eye are allowed.

5.11 Identification, Packing and Marking

5.11.1 Manufacturers Identification

The manufacturer shall be identified throughout the length of the cable by means of a tape bearing the manufacturer's name or trade-mark and year of manufacture. In case the above method cannot be employed, or if the purchaser so desires, colour identification threads in accordance with a scheme to be approved by the Bureau of India Standards (BIS) shall be employed.

5.11.2 Cable Identification

Cables with cross-linked polyethylene insulation shall be identified throughout the length of the cable by the legend 'XLPE 90' and year of manufacture by a printed tape. No identification is required for polyethylene insulation.

5.11.3 Packing and Marking

- (a) The cable shall be wound in nonreturnable drums conforming to IS-10418/1982 of suitable size and packed. The ends of the cable shall be sealed by non-hygroscopic material.
- (b) The cable shall carry the following information either stenciled on the drum or contained in a label attached to it:
 - (i) Manufacturer's name or Trade-mark
 - (ii) Type of cable and voltage grade
 - (iii) Nominal cross-sectional area of phase conductor

- (iv) Length of cable on the drum
 - (v) Number of lengths on the drum (if more than one)
 - (vi) Direction of rotation of drum (by means of an arrow)
 - (vii) Gross mass
 - (viii) Country of manufacture and
 - (ix) Year of manufacture
- (c) The cable (drum or label) may also be marked with the Standard mark.

NOTE - The use of the Standard Mark is governed by the provisions of Bureau of Indian Standards Act, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the license for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

CHAPTER 6

ROAD LIGHTING AND CONTROL WORK

6.1 Scope

This chapter covers road lighting including normal street lighting, lighting on flyovers, automation of lighting and smart street lighting. This chapter covers the principles governing the lighting of public thoroughfares and makes recommendations on the quantity and quality of lighting to be provided.

6.2 Generic Design Criterion of road lighting

The following aspects are considered while designing the road and street lightings:

- (i) The efficient lamps source for energy saving;
- (ii) Capital cost saving using proper spacing and placement;
- (iii) Maintenance cost saving using lamps with longer life and optimum spacing;
- (iv) Reduced glare and improved visibility by careful selection of luminaries and lamps;
- (v) Improved sense of security by selection of efficient systems and incorporating proper design. This can make an area appear safer and more secure;
- (vi) Improved sense of economic development of communities; and
- (vii) Improved safety of motorist, cyclist and pedestrians, improved traffic guidance and a pleasant environment.

6.3 Terminology

(i) Axial (Conversely Non-axial) Distribution

An asymmetrical distribution in which the directions of maximum luminous intensity lie (do not lie) in vertical planes substantially parallel to the axis of carriageway.

(ii) Arrangement

The pattern according to which luminaries are sited on plan, like staggered, axial, opposite etc.

(iii) Beam

The portion of the light output of the luminaire contained by the solid angle subtended at the effective centre of the luminaire containing the maximum intensity, but no intensity less than 90 percent of the maximum intensity.

(iv) Beam Centre

A direction midway between the directions for which the intensity is 90 percent of the maximum in a vertical plane through the maximum and on a conical surface through the maximum.

(v) Carriageway

That portion of a highway intended primarily for vehicular traffic.

(vi) Central Reserve

A longitudinal space dividing a dual carriageway.

(vii) Coefficient of utilization

The ratio of the luminous flux (lumens) from the lantern received on the surface of the roadway to the lumens emitted by the lantern lamp alone.

(viii) Cycle Track

A way or part of a highway for use by pedal cycles only.

(ix) Cut-off Luminaire

Luminaire employing the technique used for concealing lamps and surfaces of high luminance from direct view in order to reduce glare.

(x) **Dual Carriageway**

A layout of the separated carriageways, each reserved for traffic in one direction only.

(xi) **Footway**

That portion of a road reserved exclusively for pedestrians.

(xii) **Geometry (of a Lighting system)**

The inter related linear dimensions and characteristics of the system, namely the spacing, mounting height, width, overhang and arrangement.

(xiii) **Highway**

A way for the passage of vehicular traffic over which such traffic may lawfully pass.

(xiv) **Illumination**

The luminous flux incident on a surface per unit area. The unit of illumination is lumen per square meter (lux). One lumen of light uniformly distributed over an area of one square meter will be considered to have the illumination of one lux.

(xv) **Isocandela Curve**

A curve traced on an imaginary sphere with a source at its centre and joining all the points corresponding to those directions in which the luminous intensity is the same or a plane projection of this curve.

(xvi) **Intensity Ratio**(in a particular direction)

The ratio of an actual intensity from the luminaire in a particular direction to the mean hemispherical intensity.

(xvii) **Isocandela Diagram**

An array of Isocandela curves.

(xviii) **Integral Luminaire**

Luminaire with all its accessories, such as ballasts, starters, igniters, capacitors, etc. However, integrally with the body of the luminaire.

(xix) **Kerb**

A border of stone, concrete or other rigid material formed at the edge of a carriageway.

(xx) **Luminous Flux**

The light given by a light source or a luminaire or received by a surface irrespective of the direction the luminous flux is lumen (lm).

(xxi) **Lower Hemispherical Flux or Downward Flux**

The luminous flux emitted by a luminaire in all directions below the horizontal.

(xxii) **Luminous Intensity**

The quantity which describes the light giving power of a luminaire in any direction. The unit of luminous intensity is the Candela (cd).

(xxiii) **Luminance** (at a point of surface and in a given direction)

The luminous intensity per unit projected area of a surface. If a very small portion of a surface has an intensity I in a particular direction and its orthogonal projection (that is, its projection on a plane perpendicular to the given direction) has an area D , the luminance in this direction is I/D candelas per unit area. The usual unit is candelas per square meter (cd/m^2).

(xxiv) **Luminosity**

The attribute of visual sensation according to which an area appears to emit more or less light. It is sometimes called 'brightness'.

- (xxv) **Light output**
The Luminous flux emitted by a luminaire.
- (xxvi) **Light Distribution**
The distribution of luminous intensity from a luminaire in various directions in space.
- (xxvii) **Longitudinal Uniformity (UI)**
It is the lowest ratio of the minimum to the maximum road luminance in the middle of each lane.
- (xxviii) **Lighting Installation**
The whole of the equipment provided for lighting the highway comprising the lamps, luminaries, means of support and electrical and other auxiliaries.
- (xxix) **Lighting System**
An array of luminaries having a characteristic light distributed cited in a manner concordant with this distribution.
- (xxx) **Lantern**
The light source or the bulb emitting light is termed 'lamp'.
- (xxxi) **Luminaire**
A housing for one or more lamps, comprising a body and any refractor, reflector, diffuser or enclosure associated with the lamp(s).
- (xxxii) **Layout**
All those physical features of a high way other than the surfacing of the carriage way, which must be considered in planning a lighting installation.
- (xxxiii) **Mean Hemispherical Intensity**
The downward flux divided by $6.28(2\pi)$ (this is the average intensity in the lower hemisphere).
- (xxxiv) **Mounting Height**
The vertical distance between the center of the luminaire and the surface of the carriageway.
- (xxxv) **Overall Uniformity (Uo)**
It is the ratio of the minimum to the average road luminance. A good overall uniformity ensure that all spots on the road are sufficiently visible.
- (xxxvi) **Outreach**
The distance measured horizontally between the center of the column or wall face and the center of a luminaire.
- (xxxvii) **Overhang**
The distance measured horizontally between the centre of a luminaire mounted on a bracket and the adjacent edge of the carriageway.
- (xxxviii) **Peak Intensity Ratio**
The ratio of the maximum intensity to the mean hemispherical intensity of the light emitted below the horizontal.
- (xxxix) **Post Top Luminaire**
Luminaire with arrangement for mounting the same symmetrically on the top of the column.
- (xl) **Polar Curve**
Curve of light distribution using polar co-ordinates.
- (xli) **Refuge**
A raised platform or a guarded area so sited in the carriageway as to divide the streams of traffic and to provide a safety area for pedestrians.

(xlii) **Surround Ratio**

It is the ratio that measures the amount of light falling on the surrounds as a proportion of that falling on the road.

(xliii) **Street Lighting Luminaire**

A housing for a light source or sources, together with any refractor, reflector, dispersive surround, or other enclosure which may be associated with the source in order to modify the light distribution in a desired manner and protect the light from weather conditions and insects and/or for the sake of appearance, brightness and other lighting characteristic the source.

(xliv) **Semi-cut-off Luminaire**

Luminaire employing the technique for concealing lamps and surfaces of high luminance from direct view in order to reduce glare but to a lesser degree than cutoff luminaire.

(xlv) **Spacing**

The distance, measured along the center line of the carriageway, between successive luminaries in an installation.

(xlvi) **Service Road**

A subsidiary road between principle road and buildings or properties facing thereon or a parallel road to the principal road and giving access to the premises and connected only at selected points with the principle road.

(xlvii) **Span**

That part of highway lying between successive luminaires in an installation.

(xlviii) **Symmetrical(Converse Asymmetrical) Distribution**

A distribution of luminous intensity which is substantially symmetrical (conversely asymmetrical) about the vertical axis of the luminaire.

(xlix) **Shoulder**

A strip of highway adjacent to and level with the main carriageway to provide an opportunity for vehicles to leave the carriageway in an emergency.

(l) **Verge**

The unpaved area flanking a carriageway, forming part of the highway and substantially at the same level as the carriageway.

(li) **Width of Carriageway**

The distance between kerblines measured at right angles to the length of the carriageway.

6.4 Design Principles

6.4.1 Requirement of the road users

The following factors shall be considered while designing street lights for the roads:

- (a) Visibility of the road and its surroundings;
- (b) Identification of obstacles;
- (c) Visual guidance of the shape of the road. The motorist should be able to clearly identify bends, lumps & curves, and change in road widths;
- (d) The visual comfort of the driver. The visual field comprises the carriageway, the surrounds to the road, including road signs, the sky and the bright luminaires, the overhanging trees; and
- (e) Lighting of the street should appear continuous and uniform.

6.4.2 Fundamental Quality Criteria

The quality of lighting should meet the requirements of road users such as drivers and pedestrians. It is the 'principles of vision' which are the guiding factors in public lighting.

Hence the function of good road lighting installation is to be provide good visual performance and visual comfort to the road users. The most important criteria to be considered in road lighting are

(a) **Illuminance level**

The illuminance level for road lighting in India is governed by IS-1944 (Parts 1 & 2); the level of luminance should be adequate to provide visibility which guarantee for the user a maximum of safety and sufficient visual comfort. It is obvious that it is the road surface luminance rather than the illumination level which provides for an accurate measure of the effective light in a street lighting installation. Reference may be made to Clause 6.10.2 of this specification regarding illumination values to be provided on the road surfaces.

(b) **Illuminance uniformity**

It ensure that all spots on the road are sufficiently visible. This provides visual comfort to the driver;

(c) **Luminance level**

This is the minimum value to be maintained though out the life of the installation. It is dependent on the light distribution of the luminaries, the luminous flux of the lamps, the geometry of the installation and on the reflection properties of the road surface;

(d) **Visual Guidance**

It shall guide to the driver for the carriage way, surrounding of road etc.; and

(e) **Degree of glare limitation**

Luminaries shall be selected so that road user has no or minimum glare.

6.5 Effect of Road Surface

6.5.1 Contrast in shade or colour holds the key to observation. The recognition of an object springs from the discrimination about the contrast in the luminance of the object and its background. It is said that visual perception mainly occurs by three methods i.e. the observation of Silhouette, reverse -silhouette or surface detail. Silhouette is the formation of a dark area against a bright background, reversesilhouette is that of a bright area against a dark background and observation by surface detail occurs with a high order of direct illumination on the face of the object towards the driver.

Most obstacles on the carriageway are observed by silhouetting. Generally, the objects in areas adjacent to roadways and projections above the pavement surface and vehicles are observed by reverse silhouette. Most road signs are also observed by surface detail.

6.5.2 The basic principles behind road lighting design are therefore to facilitate silhouetting, reverse silhouetting and surface detail observations (along with reduction in glare or visual noise).

6.5.3 The most common form of observation for the vehicle driver is by silhouetting. One of the principal functions of the fixed road light is, therefore, to increase the brightness and uniformity of illumination of the road surface background to improve silhouetting.

In creating the effect of brightness, the surface illumination and the reflection properties of the surface are equally important. The reflecting properties of the road surface depends on the surface texture, the material used in the surfacing and the colour of the road surface. A dark object on a block-topped carriageway needs greater general illumination than that on a newly constructed light colored cement concrete pavement. Age of the road surface is also an important factor since with time and age the light shade of the cement concrete pavement may become darker and a matt surface may gain in polish through ageing with traffic and thereby change its reflective properties.

6.6 CLASSIFICATION

The classification of lighting installations in public thorough fares given in section 6.2 is based on volume, speed and composition of the traffic using them. It is left to the local engineer to decide upon the category of the lighting for the given road.

6.6.1 Types of Roads

For the purposes of this Section, roads are classified as given below:-

Road Classification		
[Taken from IS-1944 (Part-1 & 2)]		
S. No.	Group [as in IS : 1944 (Part 1 and 2)]	Description
(1)	(2)	(3)
i)	A	Main Roads.
	A1	Very important routes with rapid and dense traffic where safety, speed of traffic and comfort to drivers are the only consideration.
	A2	Other main roads with considerable mixed traffic like main city streets, arterial roads, etc.
ii)	B	Secondary roads — Roads which do not require lighting up to Group A standard.
	B1	Secondary roads of considerable traffic such as principal local traffic routes, shopping streets, etc.
	B2	Secondary roads with comparatively light traffic.
iii)	C	Residential and unclassified roads. These are roads not included in Groups A and B.
iv)	D	Grade separated junctions, bridges and elevated roads (<i>see</i> Note 2).
v)	E	Town and city centres and areas of civic importance (<i>see</i> Note 2).
vi)	F	Roads with special requirements (<i>see</i> Note 3).
vii)	G	Tunnels (roads underground).

NOTES

- (1) For the purposes of lighting installations, bridges are classified short or long when their lengths are less than or greater than 60 m.
- (2) Such areas are set apart since their standard of lighting is different from and higher than that described for other groups. Group E also includes important shopping streets, boulevards, promenades, and such other places which are the focus of special activities after dark.
- (3) Group F includes roads in the vicinity of aerodromes, railways, docks, and navigable waterways; where special lighting requirements are to be met in addition to compliance with general principles.

6.7 GENERAL PRINCIPLES

6.7.1 Aims of Public Lighting Installations

(i) Main Roads

The aim of public lighting along main roads, bridges and flyovers (Groups A, B and D) is to permit users of the roads at night to move about with greatest possible safety and comfort so that the traffic capacity of the road at night is as much equal to that planned for daytime as possible. Towards this, end consideration has to be given designing the lighting on road junctions and pedestrian crossings so that these can be easily identified by the driver

(ii) Roads in Residential Areas

The principle aim of public lighting along roads in residential areas (Group C) is to provide light along the stretch of carriage way and footpath for safety and comfort of road users mainly the pedestrians; consideration has to be given to ensure that the lighting is soft and does not cause glare.

(iii) Roads in City Centres

The main consideration while designing the lighting in city centres (Group E) is proper illumination of footpaths for pedestrians, besides the comfort of the drivers. Also, care is required to easily identify flow of traffic and road dividers, islands, roundabouts, etc.

(iv) Roads with Special Requirements (Group F)

Separate considerations are required to be given for each of the following:

(a) Airports

The main consideration in designing lighting of roads in the vicinity of airports is to ensure that under no circumstances, would a pilot mistake the stretch of the road as airport landing strip/runway/taxiway on airport at night time. Also the pattern of luminaire siting and colour of light source should no way resemble/match the runway. Also, the lamps should not cause glare to the pilot either while taking off or more specifically while landing, which may interfere with his/her judgement.

(b) Railways and docks

The driver of the railway is required to observe a number of signal should the tracks in the course of his work. It is necessary that none of the street lamps either glare to the driver or is mistaken by the driver for track signals due to/form/shape/colour of the light. Similar considerations are applicable to navigators in the vicinity of docks.

6.7.2 Principles of Vision in Public Lighting

Though public lighting has to satisfy both drivers and pedestrians, it is in practice requirements of the drivers which are more stringent. The following principles are considered essential [*see also IS 1944 (Parts 1 and 2)*].

(i) Requirements of Drivers

These are as follows:

- (a) Visibility of the whole of the road and its details, such as entry of side-roads, intersecting roads traffic signs, etc.;
- (b) Visual guidance on the alignment of the road;
- (c) Clear visibility of objects in time;
- (d) Good seeing condition by silhouette vision; (e) Continuity and uniformity of lighting; and (f) General or special lighting of signs.

(ii) Visual Field of the Driver

The visual field of the driver comprises, in order of decreasing importance:

- (a) The carriageway;

- (b) The surrounds around /along the road, including signs; and (c) The sky, including the bright luminaires.
- (iii) **Visibility**
The phenomenon of visibility is directly related to contrast. Good contrast should always be produced:
- (a) Between the carriageway and all objects which indicate its boundaries; and
- (b) Between any obstacles which may be present and the background against which it appears; since the characteristics of the obstacles may vary over a very wide range, any factor which tends to increase contrast should be exploited.
- (iv) **Glare and Visual Comfort**
Glare in public lighting is generally caused by the luminaires. Other factors that can lead to glare are presence of undesirable large surface+ of high reflection factor, specular surfaces, excessively bright shop windows, advertisement signs or road direction signs.

6.8 DESIGN

6.8.1 Lighting Layout of Roads

The design, spacing and column heights are governed by the road-width and the classification of the roads. Typical layouts for various road width are given in Figure 16 (A to F).

Classification of Roads and Recommended Arrangement of Columns

Sl No.	Width of Carriage Way (m)	Group	Arrangement as in Fig.	Column Height	Spacing
				$H (m)$	S
(1)	(2)	(3)	(4)	(5)	(6)
i)	24	A1	16 F	9-14	2.5-3.0 H
ii)		A2	16E	9	2.5 H
iii)	18-20	A1	16 D	9-14	2.5-3.0 H
iv)			16C	9	2.5 H
v)		A2	16D	9	2.5 H
vi)			16C	9	2.5 H
vii)		B1	16C	9	3.0 H
viii)	12	A2	16C	9	2.5 H
ix)		B1	16C	9	3.0 H
x)		B2	16B	9	3.0 H
xi)		C	16B	9	2.0 H
xii)	9	B2	16B	9	3.0 H
xiii)		C	16B	7	3.5 H
xiv)	6	C	16A	7	3.5 H

6.8.2 Lighting Layout of Flyovers

The design and column heights on flyovers are governed by the layout of flyovers, height above normal ground level and the width of the low-level roads. The spacing may be governed by the structural design of the flyovers. The layout of typical flyovers is given in Figure 17A to 17D.

- (i) The layout with recommended arrangements, column heights and spacing for various road widths on flyover are given as below:

Recommended Arrangement of Columns on Flyovers

Sl No.	Width of Carriage Way	Group	Arrangement as in Fig.	Column Height	Spacing
	(m)			$H 1)(m)$	S
(1)	(2)	(3)	(4)	(5)	(6)
i)	12	D	17C	9	$2-2.5 H$
ii)		D	17B	9	$2 H$
iii)	9	D	17A	9	$2 H$

Above the flyover road level.

- (ii) The layouts with recommended arrangements, column heights and spacing for various road widths of low-level road are given below:

Recommended Arrangement of Columns on Flyovers (Low Level Roads)

Sl No.	Width of Carriage Way	Group	Arrangement as in Fig.	Column Height	Spacing
	(m)			$H 1)(m)$	S
(1)	(2)	(3)	(4)	(5)	(6)
i)	Over 20	D	17C	9	$2.5 H$
ii)	10-20	D	17D	14	$1.5 H$
iii)	Upto 10	D	17A	9	$2.5 H$

Above low-level road

6.9 Selection of Equipment

6.9.1 Electric Light Sources

The selection of light source is the most important decision and following considerations are required to be made in deciding about the source of light:

- Luminous flux,
- Economy (determined by lumens per watt and light)
- Dimension of the light source
- Colour characteristics

It shall also be borne in mind that the light output of the source undergoes changes during its life period and this effect is to be taken into consideration.

6.9.2 The most used light sources are as below:

(i) **Mercury Lamps**

This type of lamp has been used on various outdoor applications from a long time. Its lamp life is long but as the luminous efficiency is less compared to other new sources in use in the present time, it is almost obsolete in these days.

(ii) **High Pressure Mercury Vapour Lamp (HPMV LAMP)**

Generally, this type of lamp is appropriate for locations where high power is required but rendition of colour is not that important. HPMV lamps with fluorescent bulbs are used where appearance of correct colour is also important. HPMV lamps generally have long life and high luminous efficiency. However, now a day, these are getting obsolete due to introduction of LED lamps which are widely used due to energy saving promoted by Govt.

(iii) **Metal-Halide Lamps**

This is an improved version of HPMV lamp which allows good colour rendition along with high luminous efficiency. However, now a day, these are getting obsolete due to introduction of LED lamps which are widely used due to energy saving promoted by Govt.

(iv) **Tubular Fluorescent Lamps**

These have good colour rendering power as well as high luminous efficiency although capacity for high powers may not be there. These may be used in situations where larger number of luminaire installation may not be aesthetically unsuitable.

(v) **Low Pressure Sodium Vapour Lamp**

Its luminous efficiency is high but does not have a good colour quality. It is widely used where economy in the use of energy outweigh the need of colour rendition.

Use of these lamps are considered good where foggy condition is common.

(vi) **High Pressure Sodium Vapour Lamp**

This version tries to get over the deficiency of the low pressure sodium vapour lamp by improving the colour rendition quality. It has better colour, good luminous efficiency and a reasonably long life. However, the lamp cost is comparatively high. However, now a days, these are getting obsolete due to introduction of LED lamps which are widely used due to energy saving promoted by Govt.

(vii) **LED Lamp**

These lamps are the most improved version of light source with efficacy to the order of 100–150 lm / W with better colour rendering as well as colour changing capability. It is the most energy efficient and operational economy solution available as on date. Also now these days luminaires can be made more compact due to variety in placement of LED formation making dimensions of these lamps flexible. It has also superior light control for specific applications. It has also long burning life(> 50000 hours) and comes in the market with 5 years warranty.

6.9.3 Luminaires

The luminaire has double role of protecting the light source from the weather and redistributing the luminous flux of the source. In the choice of the luminaire the following points should be considered:

- (a) Nature and power of the source or sources;
- (b) Nature of the optical arrangements and the light distribution which they provide;
- (c) Light output ratio;
- (d) Whether the luminaire is open or closed type;
- (e) Resistance to heat, soiling, and corrosion;
- (f) Protection against collection of dust and insects;

- (g) Resistance to atmospheric conditions;
- (h) Ease of installation and maintenance;
- (i) Presence or absence of auxiliaries;
- (j) Fixing arrangements, the weight and area exposed to wind pressure; and (k) Ability to dissipate the heat generated by light source and control gear.

The influence of all these factors varies according to local circumstances and it is difficult to recommend one solution rather than another, but the attention of lighting designers may be drawn to the fact that the most economical installation can be achieved only by the choice of the most suitable luminaire, selected according to the relative importance of the above mentioned factors. There is, however, one essential characteristic of luminaires the choice of which directly influences the quality of the lighting, that is, the general form of its distribution curves of luminous intensity particularly in directions near the usual directions of vision.

- (a) The following general forms of light distribution are considered according to the degree of glare which is acceptable:
 - (i) Post top integral luminaires,
 - (ii) Post top non-integral luminaires,
 - (iii) Cut-off integral luminaires,
 - (iv) Cut-off non-integral luminaires,
 - (v) Semi-cut-off integral.
 - (vi) Non-cut-off tubular luminaires, and
 - (vii) Flood-lighting luminaires

(b) **Cut-off luminaire**

A luminaire whose light distribution is characterized by a rapid reduction of luminous intensity in the region between 80° and the horizontal. The intensity at the horizontal should not exceed 10 cd per 1 000 lm of flux from the light sources and the intensity at 80° is of the order of 30 cd per 1 000 lm. The direction of the maximum intensity may vary but should be below 65° . The principal advantage of the cut-off system is the reduction of glare and its use is favoured under the following conditions:

- (i) Matt carriageway surfaces;
- (ii) Absence of buildings;
- (iii) Presence of large trees;
- (iv) Long straight sections;
- (v) Slight humps, bridges; and
- (vi) Few intersections and obstructions.

(c) **Semi-cut-off luminaire**

A luminaire whose light distribution is characterized by a less severe reduction in the intensity in the region 80° to 90° . The intensity at the horizontal should not exceed 50 cd per 1000 lm of flux from the light sources (*see Note*) and the intensity at 80° is of the order of 1000 cd per 1000 lm. The direction of the maximum intensity may vary but should be below 75° . The principal advantage of the semicut-off system is a greater flexibility in siting, and its use is favoured under the following conditions:

- (i) Smooth carriageway surfaces;
- (ii) Buildings close to carriageway, especially those of architectural interest; and
- (iii) Many intersections and obstructions.

NOTE — Subject to a maximum value of 1000 cd whatever is the luminous flux emitted.

(d) **Non-cut-off luminaire**

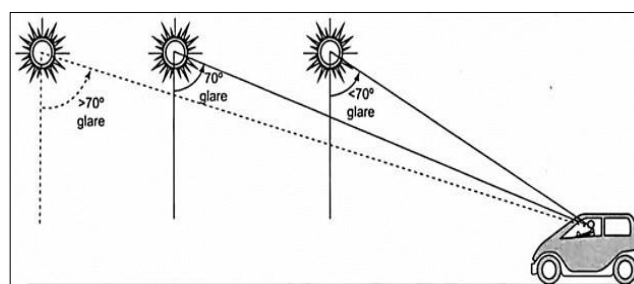
A luminaire whose luminous intensity in directions making an angle equal to or greater than 80° from the downward vertical is not reduced materially and the intensity of which at the horizontal may exceed the values specified for the semicut-off distribution, but should not nevertheless exceed 1 000 cd. Non-cut-off luminaires are permissible only when a certain amount of glare may be accepted and when the luminaires are of large size and of reduced brightness. In certain cases, they have some advantages in increasing the illumination of facades.

(e) **Inclination**

Attention should be given to the inclination of luminaires. An upward inclination which is generally called for reasons of aesthetics should be employed with care. Too great an inclination of the luminaire may modify, particularly in certain directions, the cut off qualities of the luminaires and in certain situations (for example, when there are roads at several levels, bends, roundabouts, etc.) this inclination may lead to unexpected glare.

(f) **Glare Problem**

- (i) To avoid wastage of light energy an ideal light source requires arrangement to reduce or prevent distribution of light in directions in which it is not of any use to the road using traffic (upward to the horizontal) as well as to produce increased luminosity at certain range of higher angles downward of the horizontal so that the light intensity on the pavement at a distance is increased to produce a uniform light patch inspite of its diminishing with distance due to inverse square law.
- (ii) However, incident light on the driver's eye above the horizontal and too intense a light on the whole visual field may create problems of glare for the driver.
- (iii) Two variations of glare have generally been recognized. The first one is termed discomfort glare as it causes a sensation of annoyance and discomfort to the driver and prevents adequate perception. The second one is termed 'disability glare' as it produces a too intense and uniform light on the object area obliterating contrast· discernment and may produce full or partial loss of perception about the presence of objects.
- (iv) In view of these observations, it is obvious that a light-colored pavement with rough texture, which can reflect light from its surface is the best from the point of view of night visibility.
- (v) It is generally observed that light emitted at angles less than 70° from the downward vertical cannot fall on the driver's eye, because the roof of the vehicle acts a shield and thus avoids glare. All rays above this angle cause glare; hence, it is desirable to limit the flux above this angle as shown below.



direction of emitted light rays causing glare

- (vi) The disconcerting effects of glare are sought to be got over by several ingenious ways in modern luminaire design practices, such as
 - (a) By increasing the luminaire height,
 - (b) By increasing the effective luminaire area,
 - (c) By restricting the light at vertical and horizontal angles where it interferes with driver visibility and
 - (d) By increasing the background brightness and contrast.

6.10 GUIDELINES FOR SPECIFIC LOCATIONS

6.10.1 Several factors contribute to good public lighting and those are enumerated for guidance in various parts of IS 1944 series. Recommendations are made on the various components of design of public lighting installations and those require detailed calculations of the level and uniformity of illumination on the road surface and of glare. Several criteria may not be satisfied for want of data, such as characteristics of surface, etc. To get some idea of the extent to which the installation would perform, it may be preferable to make a mock-up installation of a few luminaires on a stretch of road to be lighted.

6.10.2 Tables XVII, XVIII and XIX give a summary of recommendations on the various types of thoroughfares. These shall be used as ready reckoners, though for a detailed guidance reference should be made to the relevant part of IS 1944.

6.10.3 Average illumination, E_a , is given by the formula –

$$E_a = \frac{(lm) \cdot (C_u) \cdot (LMF)}{(S) \cdot (W)}$$

(lux or lumens/m²)

Here,

lm = lamp lumens

C_u = Coefficient of utilisation

LMF = Lamp maintenance factor

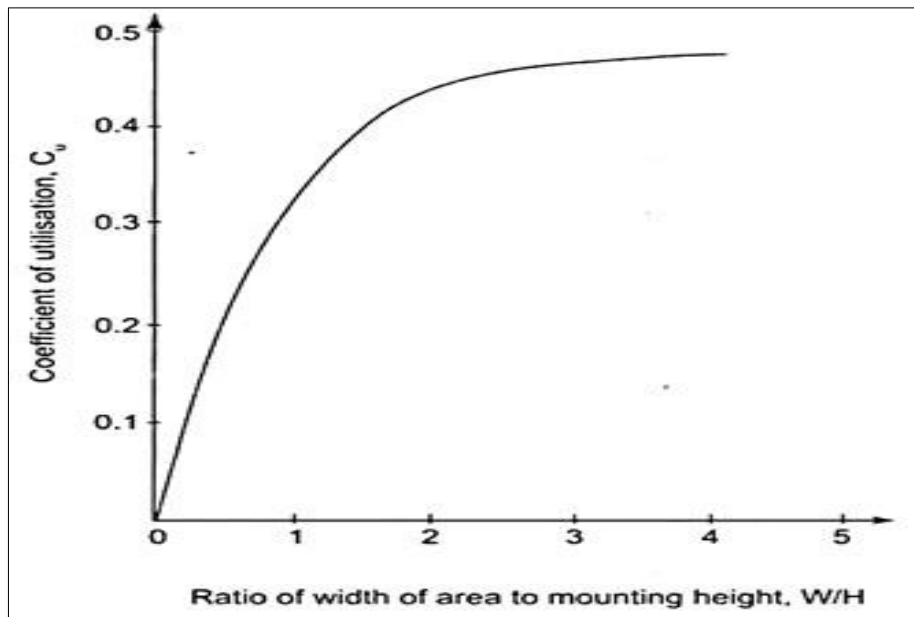
S = Spacing between luminaries (m)

W = Width of roadway/street (m)

LMF takes into account decrease in efficiency with age; it is usually taken as 80% (or 0.8).

The coefficient of utilisation, C_u , is related to the ratio of width of area to mounting height of the luminaire.

The coefficient of utilisation can be determined from the graph shown as shown below.



coefficient of utilization of light from luminaire

6.11 SITING OF LUMINARIES

6.11.1 Types of Arrangement

Arrangement of installation and provision of light support is generally decided on consideration of roadway width, the width of median available, presence or absence of median barriers and the total cost of project. There are four general type of arrangement as described below.

(i) Single side arrangement

All the luminaires are on one side of the carriageway, is recommended only when the width of carriage way is equal to or less than the mounting height. The illumination on the road surface on the side further from the luminaire will be slightly lower.

(ii) Staggered two-side arrangement

The luminaires are situated on either side of the carriageway and in a zig-zag formation. It may be employed when the width of carriageway is greater than the value recommended for single side lighting but not exceed 1.5 times the mounting height. It is superior to single side arrangement in that it provides more nearly uniform illumination and better visibility of the two sides of road.

(iii) Two side arrangement (opposite)

Two side arrangements with mountings opposite to each other is recommended for carriageway widths exceeding 1.5 times the mounting height.

(iv) Axial arrangement

If the road has divided carriageway with adequate median width, axial or central support arrangement may be resorted to. This arrangement cuts down the cost of luminaire support and conductor material, cable cost drastically and gives an overall aesthetic look to the roadway. It is therefore quite common. This arrangement, however, has its limitations for roads with wide carriageway. In the axial or median support arrangement the stress on

illumination reduces towards the shoulder and thus drawing more traffic at night towards the median end creating imbalance in carriageway utilization. Also, the illumination is inadequate on the edge farther from the median where there is more likelihood of objects and obstruction appearing and entry of traffic increasing the potential for accidents.

Other arrangements exist, but they result from the combination of the four preceding fundamental type. A road with dual carriageway can be regarded as a combination of two separate roads.

6.11.2 Mounting Height

The minimum mounting height should be chosen taking into account the power of the source, the light distribution of the luminaires and the geometry of the installation. The mounting height should be greater as the lamps are more powerful to avoid excessive glare; and also greater as the roadway is wider to obtain adequate transverse uniformity. As a general rule heights of 9 of 10 metres are suitable for Group A roads and 7.5 to 9 metres for Group B roads. Heights of less than 7.5 metres are undesirable except in certain special cases, such as the lighting of residential roads or roads bordered by trees.

6.11.3 Spacing

To preserve longitudinal uniformity the spacing should generally be not greater than the values given below:

Type of Luminaire	Maximum Spacing / Height Ratio
Cut-off	3
Semi-cut-off	3.5
Non-cut-off	4

6.11.4 Outreach and Overhang of Luminaire

The outreach is usually determined by architectural or aesthetic considerations. The overhang should not in general exceed one-fourth of the mounting height; excessive overhang leads to reduced visibility of kerbs, obstacles and footways (See Figure 18).

6.12 Photometric Specifications

6.12.1 Key Element

This is a key elements of street light luminaires, as it will determine the lighting quality on the road surface. Photometric specification may include the following:

- (a) Type of distribution;
- (b) Luminaire efficacy; and
- (c) Need for optic or lamp position adjustment.

6.12.2 Lighting Control Elements

- (a) Reflector;
- (b) Refractor;
- (c) Diffuser;
- (d) Filter; and
- (e) Screening Device

These elements (eventually used in combination) define the light distribution.

6.12.3 Luminaire Photometric Principle

- (a) Throw – The extent to which the light from the luminaire is distributed along a road.
- (b) Spread – The amount of sideways spread of the light, across a road; and
- (c) Control – The extent of the facility for controlling glare from the luminaire. The Throw is defined by the angle ($Y_{\max}$) that the beam axis makes with the downward vertical. The beam axis is determined by the direction midway between the 2 directions of 90 percent $I_{\max}$ in the vertical plane of maximum intensity. Figure given below shows the polar intensity curve through the plane of maximum luminous intensity, indicating the angle $Y_{\max}$ used for determination of the Throw.

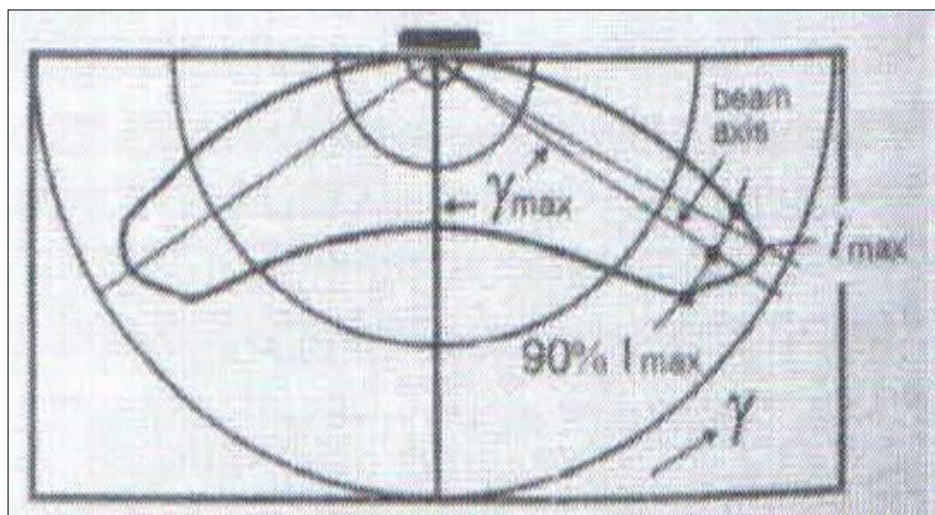
Three degrees of Throw are defined as follows:

$Y_{\max} < 60^\circ$	:	short throw;
$60^\circ < Y_{\max} < 70^\circ$	:	intermediate throw; and
$Y_{\max} > 70^\circ$	:	Long throw

The Spread is defined by the position of the line, running parallel to the road axis that touches the far side of the 90 percent $I_{\max}$ contour on the road. The position of this line is defined by the angle γ .

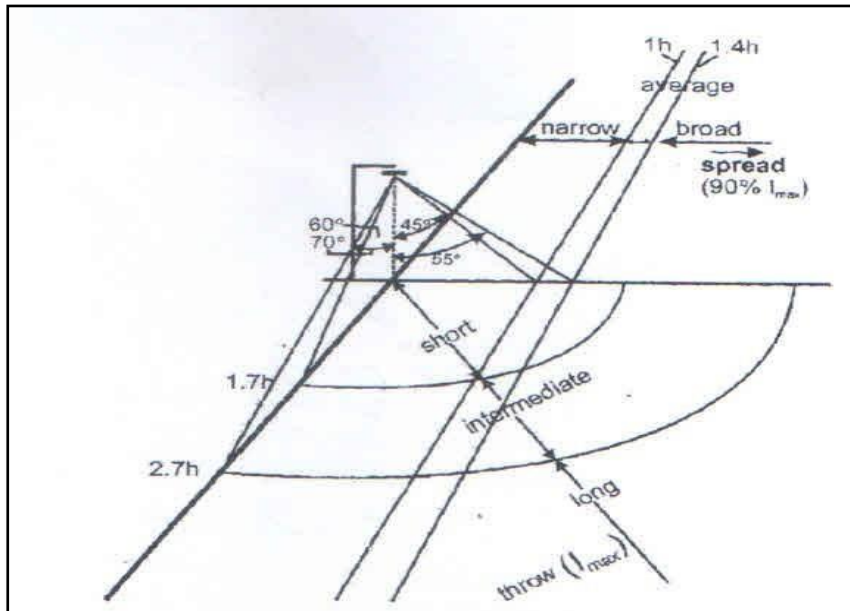
The three degrees of spread are defined as follow and define in diagram as given below:

$\gamma < 45^\circ$	:	narrow spread;
$45^\circ < \gamma < 55^\circ$	:	average spread; and
$\gamma > 55^\circ$	:	broad spread



POLAR INTENSITY CURVE SHOWING THROW

The Control is defined by the Specific Luminaire Index (SLI) of the luminaire. This is the part of the glare control mark G, that is determined by the luminaire properties alone.



DEGREE OF THROW AND SPREAD DEFINED BY CIE

Three degrees of controls are recognized:

SLI < 2	=	limited control;
2 < SLI < 4	=	moderate control; and
SLI > 4	=	tight control.

Summary of CIE Classification for Photometric Properties of Luminaire is as given below:

S.No. (1)	Throw		Spread		Control	
	(2)	(3)	(4)	(5)	(6)	(7)
(i)	Short	$\gamma_{\max} < 60^\circ$	Narrow	$\gamma_{90\max} < 45^\circ$	Limited	SLI < 2
(ii)	Intermediate	$60^\circ \leq \gamma_{\max} \leq 70^\circ$	Average	$45^\circ \leq \gamma_{90} \leq 55^\circ$	Moderate	$2 \leq \text{SLI} \leq 4$
(iii)	Long	$\gamma_{\max} \geq 70^\circ$	Broad	$\gamma_{90} \geq 55^\circ$	Tight	SLI > 4

6.13 Evaluation of Good Road lighting Installation

6.13.1 The most practical method of evaluating a Road Lighting System is by the Cost of Ownership Analysis.

(i) Cost of Ownership

The aim should be to create the most economical solution looking both at the initial investment costs and the running costs.

The total cost is split in two parts:

Preparation Phase: Design, purchase and installation cost = Initial Investment

Running phase: Energy cost and maintenance cost

Total Cost = Investment + Energy + Maintenance

(1) Investment Cost:

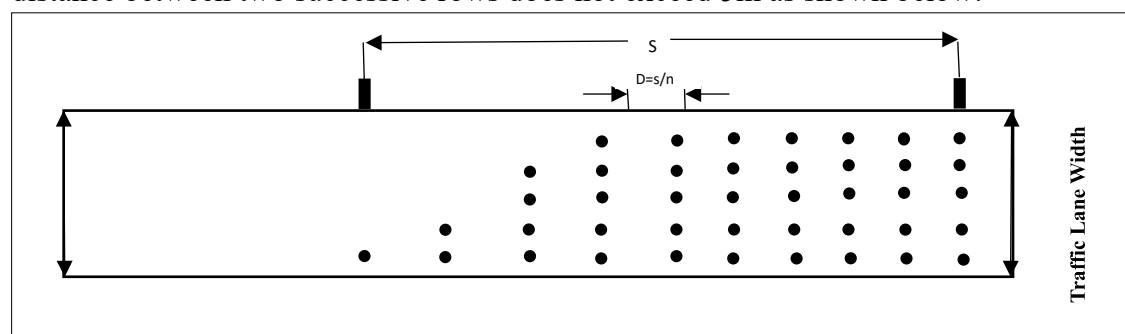
(a) Power supply;

- (b) Cable costs including trunk digging
- (c) Cost of pole, bracket, base and foundation; and
- (d) Luminaire installation.
- (2) Reducing Investment Cost:
 - (a) Use single sided or central arrangement;
 - (b) Use of energy efficient luminaire;
 - (c) Use existing cable and photocell;
 - (d) Increase luminaire spacing and
 - (e) Post top luminaire mounting.
- (3) Energy Cost:
 - (a) Installed power;
 - (b) Number of burning hours; and
 - (c) Energy cost per kWh.
- (4) Reducing Energy Cost:
 - (a) Install the most efficient lighting system = lamp efficacy and luminaire application efficiency (for example using adjustable optics); and
 - (b) Choice of maintenance factor (lamp depreciation and IP classification)
- (5) Maintenance Cost:
 - (a) Maintenance of pole and electrical supply;
 - (b) Scouting;
 - (c) Cleaning;
 - (d) Lamp replacement cost; and
 - (e) Eventual safety check and replacement of electrical components;
- (6) Reducing Maintenance Cost:
 - (a) Lamps with long life and low early failure rate;
 - (b) Reliable, longer life gear;
 - (c) Luminaire which is easy to maintain;
 - (d) Luminaire with high IP rating;
 - (e) Conscious decision for group or spot replacement; and
 - (f) Minimize number of light points by increasing luminaire spacing;

6.14 Lighting Measurements

6.14.1 CIE Method

The CIE (1976a) recommends that where the luminaire spacing does not exceed 50m, there should be 10 evenly spaced transverse rows of calculation points over its length, while for luminaire spacing greater than 50m, the number of transverse rows should be such that the distance between two successive rows does not exceed 5m as shown below:



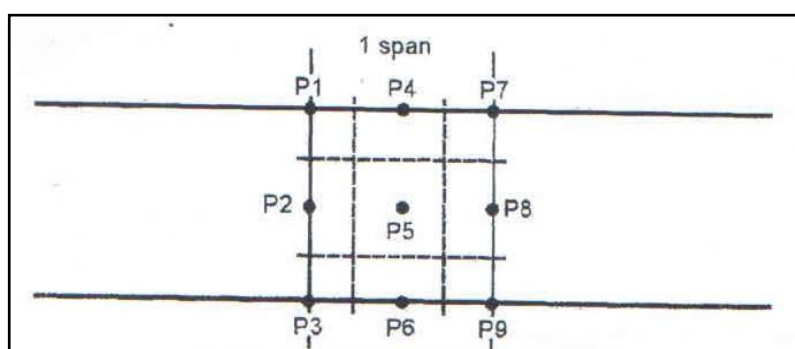
The calculation grid as proposed by CIE (1976a): S = Spacing; d = longitudinal ^{CALCULATION} _{GRID} spacing between calculation points; n = no. of transverse rows. For, $S < 50\text{m}$, $n = 10$; $S > 50\text{m}$, n = smallest integer giving $d, < 5\text{m}$.

Finally the CIE also recommends that there should be 5 points across each traffic lane, with 1 point positioned on the centre line of each lane. It is stated that where the uniformity is good, $U_0 > 0.4$, subsequent calculations may be based on 3 points instead of 5.

6.14.2 Field Measurements

For practical on-site measurement, the 9 points method is most acceptable. Fig. 13 illustrates the layout of a 9 point measuring grid of the kind sometimes used when checking new road lighting installations. The formula above the figure gives the weighting procedure that should be followed when calculating the average lighting level. Where extreme accuracy is required, the CIE recommends using as many measuring points as specified earlier.

Average illuminance E_{av}	=	$P1+P3+P7+P9$	+	$P2+P4+P6+P8$	+	$P5$
		16		8		4



REGULAR NINE POINT MEASURING GRID

6.15 The following Indian Standard Codes of Practice are relevant to “Highway lighting”:

6.15.1 IS: 1944 – (Parts I & II) -1970: ‘Code of Practice for Lighting of Public Through fare’, Bureau of Indian Standards, New Delhi, 1970.

6.15.2 IS: 10322 (Parts-5) – 1987 (Superseding IS: 2149 – 1970); ‘Luminaries for Street lighting’, Bureau of Indian Standard, New Delhi, 1987.

6.16 Control of Street Lighting Installations

6.16.1 The contactors may be provided with additional circuitry for remote/automatic control. Following schemes for remote control are recommended:

Special Relay Control — This may be achieved by special control cables laid up to the street lighting. Special relays operated by normal supply or electronic impulses may be provided.

This scheme enables return indication of the operation by auxiliary contacts of the contactors.

Ripple Control — This may be achieved by injecting audio frequency impulse through suitable power supply network and installing suitable sensors at control points.

6.16.2 Automatic Control

The contactors may, alternately, be controlled automatically by use of auto-control devices. The following controls are recommended:

Photoelectric Control — This may be achieved by installing suitably mounted photoelectric switches near the control points. The photoelectric switch shall be mounted so as to be free the glare caused by headlights of motor vehicles and protected from the weather.

Time Switches — The local electrically operated contractors may be controlled through time switches. The time switches may be manually spring wound or electrically operated. In case of electrical operation, the time switches should be of electrically wound mechanically operated variety so as to prevent their being affected by variations in supply. However, these would need to be regulated from time to time due to seasonal variations.

Electronic Control Switches — It should be possible to install pre-programmed electronic micro-processors, with one-year seasonal variations. These could be installed for each individual lamp. With suitable design these could also be utilised to switch off unwanted lights after peak traffic hours are over.

Intelligent Digital Control Module — This may be achieved by introducing microprocessor based programmable control module wired /unwired (Sim card based) which can provide various control options depending on traffic density (Public/vehicle), natural light, on the spot atmospheric conditions, etc.

- 6.17** Normally the clearance between column and carriageway should be at least 1.5 m. Where there is a footway close to the carriageway, the column should be sited behind the footway. In exceptional cases, a smaller clearance may be used.

If possible, the column itself can be provided with reflective coating up to 1.5-2 m from ground in order to achieve enhanced visual presence in dusty/dim atmosphere.

6.18 Smart Street Lighting

- 6.18.1** The Smart Cities Mission of the Government of India, promotes cities that provide core infrastructure and give a decent quality of life to its citizens, a clean and sustainable environment and application of ‘Smart’ Solutions’, build upon the IoT (internet of things) that allows exchange of data over a network, between objects and devices, allowing analysis of real-time data that can be used to user in a new wave of transformative digital services in cities and enhance citizen services and quality of life.

- 6.18.2** The energy efficient LED lighting with innovative smart IOT technology for street lighting to save further 20-30% energy using smart controlling of street lighting remotely with 40-60% energy saving using LED street Lighting against conventional street lighting vice versa reducing extra inventory of conventional accessories and maintenance spares. The smart Street Lighting Solution is providing smart street lighting solutions in such a way that an user can control and monitor the individual street lights and Group of street lights remotely from any place in the world with respect to the installation site of street lights.

- 6.18.3** The smart street light system communication for controlling and monitoring purpose divided into two categories:

- (i) Individual Street light
- (ii) Group Street light

6.18.4 The various controls can be achieved through smart street light system, are as below

- (i) Remote ON/OFF
- (ii) Dimming
- (iii) Smart Scheduling
- (iv) Parameter Monitoring
- (v) Real-time Alert
- (vi) In-depth Analytics
- (vii) Integration with smart city system

6.18.5 The basic components of smart street lighting solutions are as :

- (i) LED Street Light with inbuilt RF Controller
- (ii) IOT Gateway
- (iii) GCMS/CCMS
- (iv) Cloud based Software
- (v) Operational Features

6.18.6 The various technologies available for smart street lights are as below :

- (i) Traditional Cellular (2G / 3G / 4G / 5G)
- (ii) Short Range Wireless (WiFi-802.11/Zigbee-802.14/900 MHz (Proprietary)/ 6LoPWAN)
- (iii) Wired (Fiber/Ethernet)
- (iv) LPWAN and other WWAN (NB-IoT/LoRA/6LoPWAN)
- (v) Satellite

6.18.7 The various functions of smart street light system are as below :

- (i) Configuration & Control - Options are available to configure and control individual street lights or a group of street lights.
- (ii) Automatic ON/OFF/Dimming of lights of a particular switching point or networked switching points based on preprogrammed time.
- (iii) Manual ON/OFF/Dimming of lights of a particular switching point or networked switching points from Central Control Station through Web based application.
- (iv) All street light can be map with google map with their Geo-coordinates and access its real time status.
- (v) The Street Light Status such as ON hours, voltage, current, power, Energy consumed, Intensity and operating Mode (Auto/Manual) are monitored.
- (vi) Fault monitoring :

The system incorporated with automatic fault detection algorithms to identify various faults and fault details shall be viewed in Web Application.

Faults Includes : Driver failure, RF faults/Power down, Control card failure, RTC failure, NVM failure, Low Load, Over load, Low power factor, Under Voltage, Over voltage and other relevant faults. Email and SMS alerts for at-least 5 mobile numbers per Switching Point.

6.18.8 Luminaire Node

Luminaire shall be integrated with wireless IP66 NEMA node for wireless connectivity with Gateway. Node shall have input voltage range of 230V/ Mains Frequency: 50-60Hz. Node shall have Remote switch on/off option. Node shall have surge protection of 2 Ka / 4 kV combination wave with Internal Controller of ARM Cortex-M4 CPU. Electrical protection shall be of Class II as per EN 610101:2010 Node shall be able to operate between -10°C to +80°C ambient, 10% to 75%, RH non-condensing with internal antenna.

Node shall work on 865-867 MHz self-forming, star of star topology, self-healing wireless network+20 dBm max. transmit power, -98 dBm max. receiver sensitivity upto 1 km open field range and upto 5 kbps microcontroller RF data rate using 32bit microcontroller, 128 kB Flash and 16 kB RAM and 128 AES Multi-layer security with end-to-end encryption.

Node shall have Auto-safe feature, in a case of Gateway failure; all lamps go back to the highest pre-programmed level of brightness. In a case of the controller failure, the lamp goes back to the 100% brightness. Node shall have in built with super-capacitor backed real-time clock; able to switch on/off the lamps at sunset/sunrise and adjust them seasonally.

Compliance Standards: Controller design to be compliant to below:

(a) EMI/EMC: EN 61000-3-2 (2014), EN 61547 (2020),

(b) Environmental Standard: EN 60068-2-1 (2007), & Protection Safety: IEC 60598 / IS 10322

6.18.9 Outdoor Lighting Control CPU (Feeder Pillar):

The CPU module shall be equipped with a 8 Bit microcontroller. The CPU modules shall have internal memory capable of storing data. The CPU shall monitor and control all other modules in the LCS. Direct communication between the panel module and application shall takes place by means of an industrially proven technology. The panel module shall have the ability to automatically switch between different available communications carriers in order to provide stable and reliable communication. It shall support remote Software and configurations updates from the server enabling it to autonomously execute tasks. It shall be light weight compact and easily mountable. It shall have a local Real time clock synchronized with remote time server, to enable functionality even in case of communication network failure. Infrared Photo Sensor shall be provided for Dusk to Dawn operation.

6.18.10 Outdoor Lighting Control CPU Specifications:

(i) Additional System Requirements:

(a) The system shall generate luminaire level failure alarm.

(b) The system should be able to have schedule programming via software.

(ii) Web application Requirements:

(a) Control and Monitoring –

Allows user to communicate with individual and networked controller and allows user to communicate with individual and group street lights under particular controller.

(b) The Street Light Status such as ON /OFF, Alerts state like Mains Fail, High/low voltage, High/Low Current, Load Alerts, Geo-location, LED Wattages, Intensity and operating mode shall be monitored.

(c) Remote configuration – It is possible to configure switching points through web application. Remote configuration includes new ON/OFF timings, RTC time, Real time data, Alert thresholds, of each street light

(d) Fault Alarms/Alerts – Switching point failures, Network Down failures, Group Luminaries failure.

(e) Google Map – Allows user to trace switching points through Google maps.

(f) Lighting Management system should be based on proven technology and deployed in three cities in India with API available to easily integrate it on third party IoT platform.

(g) The cloud-based analytics module should be able to display monthly & weekly heat-maps of excess power drawn across various areas.

(h) The module should analyze the daily & hourly trends of excess power, power factor & base consumption for the selected period.

(i) The analytics engine should simulate the cost impact of excessive energy consumed across the area and should display different scenarios which are affecting the excessive energy consumption & track the auto-generated alerts.

(j) The analytics dashboard should display a map view showcasing geographical spread of power quality performance in the area of Smart Streetlight deployed. It should analyze the

Pole/Device level distribution of power quality performance and display it from best to worse.

- (k) The analytics page should display the monthly & weekly heat-maps of power quality performance. It should be able to generate daily & hourly trends of power quality, voltage trips and frequency.
- (l) A dashboard of power quality analyzer should highlight the summary of all the key insights for decision making like sector-wise voltage & Frequency, Voltage & Frequency distributions, voltage trips and mains fail etc. It should also be able to generate the different scenarios which are affecting the power quality & track the auto-generated alerts.

(iii) Server Requirement:

- (a) Supplier may use suitable server configuration and software/platform as required for the web application. The Server shall be Cloud based Linux server with minimum of seven instances each of 32GB RAM and unlimited bandwidth
- (b) The server should support storage to capture data and shall have real time upgrade of storage facility in case required without any shutdown of system.
- (c) Server provider shall have capable to perform auto backup and restore facilities.
- (d) The cloud infrastructure used by the supplier/OEM should be approved by central government authorities like MeitY or equivalent. Certificate from MeitY or equivalent to be provided on the date of bid submission.
- (e) Server provider shall have regular maintenance of the server and immediately address in case of server down.
- (f) Server shall have minimum uptime of 97%
- (g) Server should able to handle minimum of 1000 concurrent connections.
- (h) The supplier should provide a mobile based application (Mobile App) for installation and commissioning, master data collection from field. It should capture lat./long, phase wise data of the pole to which it is connected, pole no, wattage along with old and new led rated load.
- (i) Mobile App for running day-to-day operation, dash boarding to showcase alert and faults. The App should support both Android and IOS (Apple) platform.
- (j) Bug Tracking tool (Trouble ticketing tool)
 - (i) A web-based bug tracking ticketing / bug reporting tool will be provided.
 - (ii) This would be a ticketing application where designated maintenance team would be allowed to log tickets for our Street light automation / Service issues (device, application), if any.
- (iii) Required logins to be provided
- (k) All historical data shall be stored for the duration as set by the administrator. Supplier should facility for full/incremental data backup on daily basis and data older than 3-months FIFO basis should be archived and handed over to customer project team in a form of secured cloud drive/portable device as provided by customers IT team.
- (l) The cloud should be ISO 20000-1:2018, 27001:2013, 27017:2015 and 27018:2019 certified and in the name of Controller OEM on the date of issue of this tender.
- (m) Web Application OEM should have its own ISO certified cloud. Below Certifications in the name of OEM should be issued by third party reputed certification agency and to be submitted along with supply of materials. Cloud services from any third-party public cloud supplier is not acceptable.

Sr. No.	Certificate / Document
1	ISO/IEC 20000-1:2018 – Cloud hosting, Cloud security, Data centre services, Managed storage & back-up services
2	ISO/IEC 27001:2013 – Cloud hosting, Cloud security, Data centre services, Managed storage & back-up services, Service Delivery & Support Operations
3	ISO/IEC 27017:2015 – Practice for information security controls
4	ISO/IEC 27018:2019 – Information technology, Security Techniques for protection of personal information on cloud
5	SOC Type 2 Report on Control for Cloud Security & Availability

(iv) **Network connectivity:**

The charges for same are inclusive in the offer for the period including Seven year defect liability period. SIM based GPRS connection or LoRaWAN communication technology to be taken in the name of “Customer Name-WCCMS”.

(v) **Reporting:**

- (a) It shall be possible to create reports.
- (b) It shall be possible generate an alarm reports.
- (c) It shall be possible to create an action reports, based on modules, burning ratio, energy consumption report, metering data, alarm, light status.
- (d) It shall be possible to generate reports from Historical data available in Lighting control system server for following:
 - (i) Switching operations
 - (ii) Alarms/ Warning generated
 - (iii) Run Time of Circuits

(vi) **Programming:**

- (a) It shall be possible to create programs based on a fixed time table.
- (b) In the fixed time table program, it shall be possible to define ON and OFF commands.
- (c) It shall be possible to activate the fixed time table on a daily, weekly or monthly or block of months basis
- (d) It shall be possible to activate the fixed time table based on a time period; from day/month to day/month.
- (e) It shall be possible to create programs based on a twilight table,
- (f) It shall be possible to generate a twilight table based on a geographic location.
- (g) In the twilight table program, it shall be possible to include ON and OFF commands at fixed times between sunset and sunrise.

(vii) **Software upgrade:**

It shall be possible to update the settings locally on the CPU using a USB stick or OTA Options.

It shall be possible to update the settings remotely through the web page application.

Customers, permission groups and users

- (a) It shall be possible to define "Customers" to limit access for users within these "Customers" to only a part of the node tree.

- (b) It shall be possible to define permission groups to define different levels of user access.
- (c) It shall be possible to design user accounts within customers, and assign a permission group and a password to each user.

(viii) **Control Room Application:**

- (a) The streetlight automation system shall feature a map in the application that gives an overview of all boxes on top of a street map or a satellite image.
- (b) It shall be possible to visualize the light status of the boxes or the alarm status of the boxes.
- (c) It shall be possible to see the geo-location data of a selected box.
- (d) It shall be possible to see the serial number of a selected box.
- (e) It shall be possible to see the name of a selected box.
- (f) It shall be possible to see the communication status of a selected box.
- (g) It shall be possible to see the alarm status of a selected box.

CHAPTER 7

OTHER EXTERNAL LIGHTING AND CONTROL WORK

7.1 Types of External Luminaires:

- (a) Street light luminaires (please see street lighting chapter in this book)
- (b) Flood light luminaires
- (c) Post Top luminaires.
- (d) Bollard luminaires
- (e) RGB flood light luminaires.
- (f) RGB linear light luminaires

EXTERNAL LIGHT FITTINGS SPECIFICATION

LAMP TYPE	RANGE	LUMINOUS FLUX	EFFICACY (LUMEN S/Watt)	LIFE IN HR	CRI	PF	THD	SD CM	OPERATING VOLTAGE	SPD	IK	COMPLIANCE
GLS	25W-1000 W	225-18000	09-18-	1000	EXCELLENT							
HPMV	80 W-1000W	3520-58000	44-58	5000	Moderate							
HPSV	70W-400W	5800-47600	83-119	12000-15000	FAIR							
LPSV	18W-35W	1800-4500	100-129	10000	POOR							
LED STREET LIGHT	25W-350W	3000-49000	120-140	50000	70-80	0.95-0.98	<10%	<5	120-270W	2-4KV	IP65-66, IK07	IS:10322(Part5/Sec 3)
LED FLOOD LIGHT	35W-1500W	3500-180000	100-120	50000	70-80	0.95-0.98	<10%	<5	120-270W	2-4KV	IP65-66, IK07	IS:10322(Part5/Sec 3)
LED POST TOP	25W-100W	2500-11000	100-110	50000	70-80	0.95-0.98	<10%	<5	120-270W	2-4KV	IP65-66, IK07	IEC 60598-2-3/ IS:10322(Part5/Sec 3)
LED BOLLARD	8W-10W	400-700	50-70	50000	70-80	0.95-0.98	<10%	<5	120-270W	2-4KV	IP65-66, IK07	IEC 60598-1:2008
LED RGB FLOOD LIGHT	21W-600W	NA	NA	50000	NA	0.95-0.98	NA	<4	120-270W	2-4KV	IP65-66, IK07	IS:10322(Part5/Sec 3)
LED RGB LINEAR LIGHT	15W-120W	NA	NA	50000	NA	0.95-0.98	NA	<4	120-277 for AC 24 V for DC	2-4KV	IP65-66, IK07	IS:10322(Part5/Sec 3)
LED NODES	1WNODE	10-30-	10-30-	50000	NA	0.95-.098	<10%	NA	15V DC for Low Wattage, 24 V DC High wattage	2-4KV	IP65-66, IK07	

*IS: 10322(Part5/Sec3)/IEC 60598/: This complies visual examination, Impact Resistance, Endurance test, Thermal test, IP protection test, Insulation resistance, Electric strength test etc.

NOTE : As per CPWD circulars, use of LED is mandatory so use of luminaires other than LED should be avoided.

7.2 Types of Lighting Schemes: (as per applications)

7.2.1 FAÇADE LIGHTING

Facade lighting is a key aspect of architectural design and can bring any building or structure to life. Light influences the perception of its facade and surroundings. Layers of light define the focus and can turn spaces into experiences. When the sun sets, as amazing as the architectural design maybe it fades into the darkness. The exterior facade lighting showcases their unique designs plans during dark hours, communicates emotions, and creates attention. Facade lighting highlights the key features of the building and adds an artistic feel to the exteriors. Architectural design, quality, and styling meet design, aesthetics, economics, to deliver high visual impact.

7.2.1.1 Types of Facade Lighting Techniques

(a) **Direct lighting**

Direct lighting refers to light fixtures that are right on the building facade itself like media facades or spotlights.

(b) **Wash lighting**

Wash lighting technique seeks to highlight the flat facade and illuminate larger objects using a wide beam of lights. Wall washing with uniform illumination helps in better enhancement of the architecture of the wall surface. To achieve an efficient and attractive wall washing effect, you have to place the light fixture a bit far from the facade.

(c) **Wall grazing**

Wall grazing is the method of positioning the light fixture closer to the wall to highlight its texture and architectural character. The light fixtures are gradually placed at the ground, starting illumination from the bottom fading gradually towards the upward direction for mysterious and fascinating effects on surfaces.

(d) **Accentuating**

This type of technique can be used to emphasize vegetation, particular objects, or elements using directed beams of light. This method is perfect for facades, entrances, arcades, and objects, creating a startling effect.

(e) **Floodlights**

Floodlights provide uniform illumination that is casting the same brightness on a vertical surface. It offers optimum beam alignment through lens adjustment and can highlight the details of the building.

One way is to use the silhouette technique by creating shapes through the right angle of the lights without focusing on separate elements. Another way is the contour technique, for that use LED strips that give the structure a certain glow and create a striking effect.

7.2.1.2 RGB LED FLOOD LIGHT:

Application areas: RGB LED lights are used as flood light in architectural lighting of monuments, precious national importance buildings etc. to fill the spaces with desired colors (16 million combinations) like passages, circulation areas and to have wall washing and accent lighting.

7.2.1.3 Fixtures with high lumen output, exterior rated RGBW architectural flood light completes with all accessories, suitable for facade Lighting. The housing material of fixture should be Die-cast

aluminium, powder-coated finish with Anti-aging and anti-UV high optical efficiency PMMA lenses, Clear tempered glass cover.

- (a) Fixture should be available with other architectural optics with choice of beam angles of narrow, medium, and wide (Ranges 3.5°, 8°, 15°, 30°, 40°, 60°) as per application requirement
- (b) SDCM ≤ 4 .
- (c) The fixture should be complete including accessories like leader cable, jumper cable, and male female connector.
- (d) Fixture shall have input voltage for 220 – 240 VAC (e) Operating temperature of -40°C to 50°C.
- (f) Lifetime should be at least 50,000 burning hours L70 at 35°C.
- (g) Fixture shall be at least IP65 protected
- (h) Equipped with mechanical impact of minimum IK07 or better.
- (i) LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters.

7.2.1.4 RGB LED WALL GRAZER LINEAR:

- (a) **Application areas:** It is used for vertical and indirect lighting, placed at certain distance and angle from a wall to create smooth wash or grazing effect.
- (b) Fixtures of Linear wall washer, exterior rated RGB, RGBW, wall washer light completes with all accessories if any; suitable for facade Lighting. The housing material of fixture should be Die-cast aluminium, powder-coated finish with Ant aging and anti-UV high optical efficiency PMMA/PC lenses, Clear tempered glass cover.
 - (i) The fixture should have multiple range of beam angles as per application requirement.
 - (ii) The fixture shall have SDCM ≤ 4 .
 - (iii) Fixture shall be available with 20°X30° ($\pm 5\%$) beam angle with an input voltage of 24 VDC.
 - (iv) Lifetime should be at least 50,000 burning hours L70 at 35°C.
 - (v) Fixture shall be minimum IP65 protected.
 - (vi) Equipped with mechanical impact of minimum IK07 or better.
 - (vii) LM 79 and LM80 reports need to be submitted from a NABL/ to verify above parameters.

7.2.1.5 RGB LED STRIP LIGHTING

Application areas: These fixtures are used for cove lighting, and to have outline effects on certain architecture features on the periphery of building.

7.2.1.6 Flexible RGB LED Strip, system wattage not exceeding 12W and system lumen output of not less than 450 lumens/meter. The fixture shall be complete with aluminum profile with clips and cover.

- (a) Fixture shall operate in a temperature range of -40°C to 50°C,
- (b) Minimum IP65 protection.
- (c) The fixture shall be 24V DC with non-integral driver and complete with all the accessories.

7.2.1.7 LED NODE LIGHTING

LED RGB nodes are flexible strands of large, high-intensity, full-color nodes designed for extraordinary video and graphic effects on expansive installations without the constraints of luminaire size, shape, or space.

- (a) Each Flex RGB strand should consist of at least 50 individually addressable LED nodes
- (b) LED nodes feature dynamic integration of power, communication, and control.

- (c) The flexible form factor accommodates two- and three-dimensional configurations, while high light output affords superior long-distance viewing for large-scale signage and building-covering video displays.

7.2.1.8 Accessories of Façade lighting

(a) **DMX CONTROLLER**

DMX controller allows for greater scalability, makes it easier to program lighting scenes and improves energy efficiency. You can create custom light shows with ease when using a DMX controller. This is Content Program Controller with features of

Single DMX ports control minimum 300 light addresses. This controller should work as a stand-alone or computer-driven controller, Secure Digital (SD) card drive for flexible show storage, USB 2.0 interface for downloading shows and configuring from attached PC, Flexible, adaptable mounting options, Should be provided with light show authoring software, Direct line voltage power should support 100-240VAC. It should provide Minimum Five pre-configurable effects: Using light authoring software to select from five pre-defined effects, adjust effect settings such as color, speed, and brightness. Soft Copy of Catalogues with Link of the official web page of Luminaires, Test Certificates - relevant BIS R Number,

Third Party IEC 60598 test certificates submission should be asked by NIT approving authorities to see proper and authenticity of equipment and as per requirements.

(b) **DMX Amplifier**

DMX Amplifier is used on the standard DMX512 protocol or equivalent for signal shaping and amplification for long distance transmission. The signal amplifier supports magnetic isolation protection, protects the controllers, and ensures that the DMX signal transmitted to various luminaires.

- (i) The temperature range should be -30°C to 50°C.
- (ii) humidity range 10 to 90%.
- (iii) The amplifier should be with minimum IP66 rating with IP67 rating for connectors.
- (iv) NIT approving authority should ask for Soft Copy of Catalogues, Test Certificates - Third party LM79, LM80, BIS R Number.

(c) **Touch Screen**

This is used for addressing device for fast and convenient address writing and testing of DMX512 luminaires. The device should support internal test animation for DMX512 addresses verification. The device should automatically write address to a whole group of interconnecting luminaires within minutes. The device should be compact size and touch screen.

(d) **Leader cable**

Connect from the DMX controller to the first luminaire in the group; one end with 4poles flying wires and the other end is with a female connector. Standard length is 2 m.

(e) **Jumper cable**

Connect the DMX signal between luminaires, one end with a female connector and the other is with a male one. Standard length is 2 m.

(f) **End caps**

IP67 end caps for both male and female connectors that MUST be used at the end of each run.

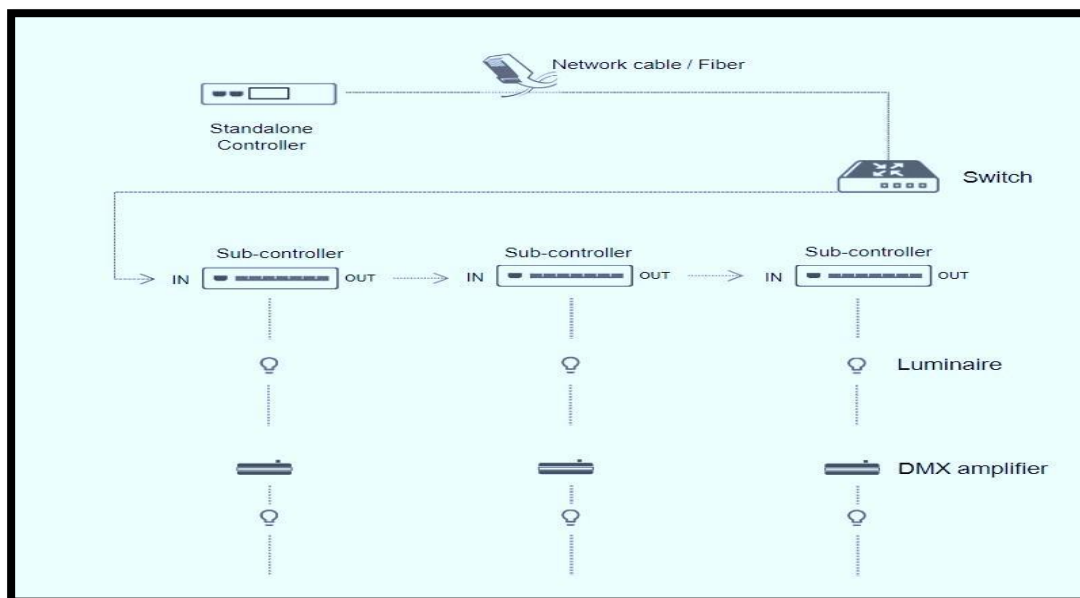
(g) **DMX Splitter & Booster**

Splitter is used to split the signal to different output channels, while booster is used to boost up the signal for long DMX cable after a certain length.

(h) **MX Controller**

DMX controllers (Digital Multiplex) is a universal protocol used in lighting technology and by most manufacturers of lighting control tables and projectors, being used as a means of communication between the equipment and the controllers

(i) **General System Architecture of RGB Façade lighting:**



Note: It is generally seen that to have best quality products, manufacturing facilities should have in-house lab approved by NABL or ministry of science of govt. of India or 3rd party certifications from approved labs which needs to be seen by NIT approving authorities. Also, BIS approval for fittings, drivers or important accessories need to be seen by NIT approving authorities.

7.2.2 LANDSCAPE LIGHTING

Landscape lighting or garden lighting refers to the use of outdoor illumination of private gardens and public landscapes; for the enhancement and purposes of safety, night-time aesthetics, accessibility, security, recreation and sports, and social and event uses.

7.2.2.1 LANDSCAPE LED BOLLARD LIGHT:

Application Areas: to light pathways and driveways.

Fixtures are with Extruded aluminium with polycarbonate diffuser and integral base plate PG gland. Luminaire contains built in LED modules. Operating ambient temperature range is from - 20°C to +40°C. The resistance to heat and force for plastic part should be 850-degree Celsius.

- (a) 3000K 230-240Vac 50-60Hz
- (b) CRI > 70
- (c) Height 800mm/1000mm,
- (d) Electronic driver ON/OFF
- (e) IP rating IP65, impact resistance IK08 or better.

7.2.2.2 LANDSCAPE LED POST TOP LIGHT: (25W-100W)

Application Areas or streetscape

Fixtures are made up of aluminium pressure die-cast aluminium with integrated pole top with polyester powder coating in addition to a specific weathering resistance treatment. On-yellowing transparent and textured Polycarbonate UV-protected round symmetrical diffuser 1.5 mm thick

with circular radial pattern for glare-free and even light distribution. Operating voltage range: 120 V - 277 V Average life L70B50: 50000 hours.

- (a) Powered by Built In electronic. LED driver with THD<10%,
- (b) 4kV inbuilt surge protection,
- (c) High cut off at 325V & auto-restart
- (d) IP 66, IK08 or better, PF>0.95, Luminaire

7.2.3 Solar Lighting

A solar light, is a lighting system composed of an LED lamp, solar panels, battery, charge controller and there may also be an inverter. The lamp operates on electricity from batteries, charged through the use of solar panel (solar photovoltaic panel).

LED lights are used due to their high luminous efficiency and long life. Under the control of a DC charge controller, non-contact control automatically turns on the light at dark and switches off at daytime. It sometimes also combines with time controllers to set a certain time for it to automatically switch light on and off.

7.2.3.1 Application

These lights provide a convenient and cost-effective way to light streets at night without the need of AC electrical grids for pedestrians and drivers. They may have individual panels for each lamp of a system or may have a large central solar panel and battery bank to power multiple lamps.

In rural areas of India where Sun is more exposed, so proper utilization of Sun is expected while we can use solar streetlight and solar flood light using street light pole and high mast.

7.2.3.2 Solar Street light/Flood light:

- (a) Luminaire should have an efficacy of not less than 120lm/watt.
- (b) Life expectancy of 50000 Hr with 70% lumen maintenance, with CRI>70.
- (c) Luminaire should have pressure die cast aluminum housing with minimum IP66 protected with IK08 impact resistance with a working temperature 0 to 50 degree.
- (d) Luminaire should have a factory fitted cable and waterproof connector to terminate in the charge controller.
- (e) LFP Battery of 12.8V/60Ah rating to be provided within the pole. Battery should be IP68 enclosure with life of 2000 cycles.
- (f) Battery should have factory fitted leader cable with waterproof connector matching with charge controller.
- (g) Monocrystalline PV panel 400Wp/23Voc should be integrated into the pole. Waterproof factory fitted connectors should be provided for termination of PV cable into charge controller.
- (h) The charge controller should be housed inside the pole hatch and should have IP67 waterproof connectors to terminate cables from luminaire, PV and battery. Charge controller should be MPPT type with efficiencies > 90%. Charge controller should have a default dimming profile for 15W output which can be and should have protection against Battery Reverse Polarity, Panel Reverse Polarity, Panel Reverse Current, Surge Protection at PV input up to 1kV of Differential Mode, Battery over Voltage, load short circuit, Load Disconnect/Over Voltage, Maximum Battery charging current, Panel over voltage protection.
- (i) Luminaire should conform to IS10322 test standard specification.
- (j) Battery should comply with IS 16046:2015
- (k) Panel should comply with IS14286, IS61730-1&2
- (l) Charge Controller IEC 621 09-1, IEC 62093, IEC 61547, IEC 61347 & EN 50530

7.3 LIGHTING AUTOMATION

A lighting control system is a centralized system that controls the lighting in a building or area. It can be used to control indoor and outdoor lighting and can be programmed to turn lights on and off at specific times or in response to certain conditions. From simple timers and sensors to robust lighting management systems that offer remote controls, real-time tracking, and analysis, and even smartphone apps, there is a perfect lighting control solution for every business.

The manufacturer should directly support the following options within its own range of products.

- (a) User Interfaces (UI) / Keypads
- (b) Color Touchscreens
- (c) Sensors: o Passive Infrared (PIR) o Ultrasonic (US) o Photo Electronic light level (PE) o Infrared Receive (IR)
- (d) Relay Switching controllers
- (e) Double throw relay motor controllers
- (f) Phase-cut (leading and trailing edge) Power Dimming Controllers
- (g) Signal Dimming Controllers supporting 1-10V, DSI, DALI Broadcast, DALI and DALI MultiMaster.
- (h) Multipurpose (modular) controllers
- (i) PWM LED controllers
- (j) Head end user software
- (k) Personal apps

7.3.1 System Architecture

All network devices are to be configured from the same communications port as their operation port. Systems requiring additional device communications ports for configuration will not be accepted. A single communication port for both configuration and operation is required to ensure that configuration changes to the control system can be performed by a commissioning engineer quickly and easily without the need to use multiple protocols, change the network cabling, or having direct access to a network device. This also allows for changes to the system from the user head end software. The control system shall be a network type providing distributed control, with user interfaces communicating directly with load controllers, without requiring the intervention of a central processing unit. This is required so that the system is not dependent on a single device which could create a central point of failure. Control systems that require a central processing unit will not be considered. The control system shall achieve the required functionality by using a distributed intelligence network architecture. In the event of a network cable being severed, the system will automatically continue to operate as two independent lighting control networks. Systems that require operator intervention or where one or both networks cease to operate in the event of a network cable cut shall not be accepted. The control system shall use advanced Windows based graphical programming to configure all network devices and define the behavior and relationship between network devices. The software must be capable of configuring any device within the system from any location on the network. The software shall not be dependent on extra plugins or drivers to enable configuration of devices. The control system shall use a PC running intuitive and user-friendly software for the remote programming of all devices on the control network. It shall be possible to connect the PC to any point on the network, for access to the entire system. The performance of the system shall not be affected when the PC is disconnected. The ability to interrogate the system and save the uploaded configuration information to disk shall be included. The ability to log all network messages to a file for later analysis while the PC is connected to the network shall be included. All load controllers and UIs shall be supplied with a basic configuration that allows the control system to operate in a default way immediately upon installation. This allows for UIs to send on, off and preset selection commands onto the network. All load controllers correctly connected to the network will respond

to these messages appropriately. Achievement of this functionality shall not require any commissioning. All configuration information downloaded from the configuration PC to devices on the network shall be stored in the relevant device, contained within non-volatile EEPROM or FLASH memory. This data shall remain secure for an indefinite period, upon loss of supply. Devices that use battery or super capacitor backed up RAM will not be considered. When supply is restored, the system shall automatically return to the same state as when supply was lost, without requiring any user intervention. Within the system configuration software, the system is able to be viewed in logical groups, so that preset scenes and states can be easily configured for each logical area. This is required so that during project commissioning the control system can be viewed from the same perspective as the physical project layout. Systems that can only be viewed or configured with reference to a physical controller or channel will not be accepted. The control system network power can be supplemented by an external power supply to meet the requirements of devices that are dependent on network power.

7.3.2 Light Controlling

- (a) Switching the control system must be capable of implementing basic switching control to turn lights on and off from a flexible choice of networked user interfaces.
- (b) Dimming the system must be capable of implementing dimming control of lighting from a flexible choice of networked user interfaces. This shall be achieved by a protocol (or signal) to a lamp driver (DALI addressable, DALI Broadcast, DSI, or 110V) or by power control of phase control dimmers.
- (c) Daylight Harvesting/Maintained Illuminance Through sensor based light level detection, it shall be possible to implement level management of artificial lighting to maintain task illuminance at optimum targets. This provision shall minimize energy consumption by preventing over-illumination at any given time, especially in areas that experience high levels of daylight i.e. adjacent to windows.

7.4 Types of Networking

7.4.1 Wired Network:

As we all know, “wired” refers to any physical medium made up of cables. Copper wire, twisted pair, or fiber optic cables are all options.

7.4.1.1 Channel DALI Controller

- (a) Each DALI outputs allowing it to control up to 192 DALI devices.
- (b) The DALI control signals can be programmed to operate automatically and isolate the power circuit when all associated DALI channels are at 0%.
- (c) The device is DIN-rail mountable integral DALI bus power supply, removing the need for an additional external device.
- (d) Supply: 100-240V 50/60Hz Single Phase at 0.1A, Control Outputs: 3 x DALI Control Outputs, each supporting at full DALI universe of 192 channels, including backward channel DALI Bus Power Supply, Internal 24V rated 250mA Control IO :1 x RS485 serial port / DMX, 1 x AUX programmable dry contact input. DC supply 12V @ 200 mA.

7.4.1.2 Channel DALI Controller

- (a) Each DALI output allows it to control up to 64 DALI devices.
- (b) The DALI control signals can be programmed to operate automatically and isolate the power circuit when all associated DALI channels are at 0%.
- (c) The device is DIN-rail mountable integral DALI bus power supply, removing the need for an additional external device.

(d) Supply: 100-240V 50/60Hz Single Phase at 0.1A, Control Outputs: 1 x DALI Control Outputs, each supporting at full DALI universe of 64 channels, including backward channel DALI Bus Power Supply, Internal 24V rated 250mA Control IO :1 x RS485 serial port / DMX, 1 x AUX programmable dry contact input. DC supply 12V @ 200 mA.

7.4.1.3 Channel Relay controller

- (a) Multi types of feed through power circuit design equivalent to 12 pole contactor & each pole should be controllable via system network.
- (b) Each Switched loads should support 20A per channel upto max. of 180A per device with an inrush current of 500A.
- (c) Device should support multiple wiring schemes - single phase & neutral Or three phase & neutral (Star) & should have service override switch can be accessible from the front panel.
- (d) Supply Voltage: 100-240AC, system Output DC current 200mA which can support 10 button panels without any additional power supply.
- (e) The device should also have dry contact Aux input.

7.4.1.4 Channel Relay Controller

- (a) Multi types of feed through power circuit design equivalent to 4 pole contactor & each pole should be controllable via system network.
- (b) Each Switched loads should support 20A per channel upto max. of 80A per device with an inrush current of 500A.
- (c) Device should support multiple wiring schemes - single phase & neutral Or three phase & neutral (Star) & should have service override switch can be accessible from the front panel.
- (d) Supply Voltage: 100-240AC, system Output DC current 200mA which can support 10 button panels without any additional power supply.
- (e) The device should also have dry contact Aux input. Equivalent to Dynalite.

7.4.1.5 Sensor

Surface/Recess mount multifunction Sensor with Motion, PE, IR inbuilt along with Daylight Harvesting Mode & corridor hold on function Maximum Detection Area: 7 m x 5m rectangular (at a height of 2.4m) Detection speed: 1.0m/s Detection object: 700mm x 250mm Detection zones: 64 LED activation indicator Sensor: Quad element pyro- electric R.F.I. Immunity: >15V/m @ 10-1000MHz PE Cell Dynamic range <5 lux to >5000 lux Automatic “Daylight Harvesting” mode Can be used for light measurement for PIR.

7.4.2 Wireless Network

Wireless” means without wire, media that is made up of electromagnetic waves (EM Waves) or infrared waves.

7.4.2.1 Sensor

- (a) Sensor is capable of monitoring occupancy, and daylight harvesting & task tuning.
- (b) The sensor is capable of receiving Bluetooth / IR signals from other devices.
- (c) Data from the sensors is transmitted wirelessly to the gateway and at the same time gives command to supportive bridge which will execute the instruction to DIM and Switching (On/Off) of Luminaires. Sensor (With supportive Driver) compatible with DALI drivers.

7.4.2.2 Wireless gateway

- (a) The Wireless Gateway must create a mesh network with adjacent sensor/wall switch in an open area with a capability to connect to max 150 sensors including wall switch.

- (b) Must be capable of communicating all data from sensors/wall switch/luminaire to the cloud.
- (c) Additionally, connection can be done from this gateway without any additional device or license.

CHAPTER 8

POLES AND HIGH MASTS

8.1 SCOPE

The scope of this specification covers the manufacture, transport, installation, testing and commissioning of the complete lighting system, using Raising and Lowering type of High mast Towers, including the Civil Foundation Works. The purchaser will provide the feeder cable of required size up to the base compartment of the high mast.

8.2 APPLICABLE STANDARDS:

(a)	PLG 07	High Masts for Lighting and CCTV
(b)	SANS 10225	High Mast natural frequency calculation
(c)	IS 875 Part – 3	Wind Loading
(d)	IS 2062/ BS EN 10025	Hot Rolled Steel plates.
(e)	BS EN ISO 1461	Galvanization
(f)	IS 2266	Stainless steel Wire rope
(g)	IS 9968 Part – 1	Trailing Cable
(h)	IS 694	PVC insulated flexible copper cable.
(i)	IS 7098 Part-1	XLPE insulated Aluminium / Copper cable
(j)	IS 12615	Motor
(k)	EN-8 grade, IS 1367 (Part 3)	Foundation Bolt
(l)	AWS	Welding

8.3 HIGHMAST:

8.3.1 Structure

The High mast shall be of continuously tapered, polygonal cross section, 12 / 20 sided, presenting a good and pleasing appearance and shall be based on proven In-Tension design conforming to the standards referred to above to give an assured performance and reliable service. The dimensions of the mast and other details are as per the enclosed data sheet.

8.3.2 Construction

The mast shaft shall be manufactured from Hot Rolled steel plates confirming to IS 2062/ BS EN 10025. Each mast shaft section shall be without any circumferential weld joint except base flange. The mast base flange shall be free from any lamination or incursion and provided with supplementary gussets between the bolt-holes to ensure elimination of helical stress concentration. (Please see attached photo)

The minimum A/F dimension of top shall be 100 mm / 150 mm and bottom as per the design and data sheet enclosed. The minimum section length except for the top section shall be 10 m and top shall depend on the length required to make the specified height. The masts sections shall be joined at site by slip-stress-fit method and minimum overlap distance shall be 1.5 times the diameter at penetration.

A door reinforced with welded steel section, vandal resistant, weather proof with Allen bolts and pad locking facility of dimension not less than 1050 mm x 225 mm shall be provided at a height 2 times the width of door from the base of mast to provide clear access to base compartment equipment's winch, motor, cable, connector etc;

For the environmental protection of the mast, the entire fabricated mast shall be hot dip galvanized internally and externally in single dip having the galvanization thicknesses as per galvanization standard BSEN ISO1461.

8.3.3 Dynamic Loading for the Mast

The mast structure shall be suitable to sustain an assumed maximum reaction arising from a wind speed as per IS 875 part-3-2015 (three second gust) and shall be measured at a height of 10 meters above ground level. The design life of the mast shall be 25 years, Topography factor K3 shall be 1 (Flat terrain) and Importance factor K4 shall be 1 (case 3). The force co-efficient taken for design of the twenty-sided polygonal structure is to be established from the wind tunnel test data.

8.3.4 Luminaries Carriage

Hot dip galvanized Luminaries carriage designed to install luminaries as specified in data sheet or as per illumination design and junction box. The same is to be fabricated from ERW tubes in two halves and flanges joined at site with stainless steel bolts and nyloc nuts. Holes are to be provided in the bottom side of tubes to act as conduit for wiring cable. PVC lining is to be provided in the inner side of carriage to avoid metal contact with mast surface.

Weather-proof junction box shall be provided on the Carriage Assembly for terminating the trailing cable and power cable to luminaries.

8.3.5 Raising and lowering mechanism:

8.3.5.1 Winch

The double drum winch shall be completely self-sustaining without the need for brake shoe, springs or clutches and self-lubricating type by means of an oil bath. The worm gear ratio shall not be less than 53:1 and safe working load shall be as per data sheet. The drums are to be grooved to provide perfect seat for stable and tidy rope lay and arrangement for distortion free rope end termination.

The winch shall have provision to operate manually by a handle or electrically through power tool. The capacity, operating speed, safe working load, recommended lubrication and serial number of the winch shall be marked on each winch.

The winch shall be type tested through reputed institutions like IIT as consultants and the type test report shall be submitted along with offer. A test certificate is to be submitted along with supplies.

8.3.5.2 Head Frame

The hot dip galvanized head frame is to be designed as a capping unit of the mast is of welded steel construction and provided with guides and separators between the ropes and cable. The LM6 Aluminium pulleys with bush mounted through stainless steel shaft shall be suitable to accommodate wire ropes and multi core trailing cable.

The head frame shall be provided with guides and stops with PVC buffer for the docking of luminaires carriage. The pulley assembly shall be covered by a hot dip galvanized canopy.

8.3.5.3 Stainless Steel Wire Ropes

The stainless-steel wire ropes shall be in 7/19 construction with central core in the same material of grade, diameter and breaking load as given in data sheet. The end construction of rope for the winch drum shall be fitted with talurit and for two continuous ropes the end termination in luminary's carriage shall be with stainless steel thimble and copper splicing and for others with stainless steel thimble and bulldog grips.

8.3.6 Electrical System, Cable and Cable Connections

The multi core trailing cable from base compartment to junction box at luminaires carriage shall be 1.1 KV grade and type as specified in the data sheet of copper conductor with male female connectors of shall be as given in data sheet. Wiring from junction box to luminaries is to be done using 3 core 1.5 sq. mm PVC/ XLPE insulated, PVC sheathed, copper conductor flexible cable. Suitable arrangement is to be provided in the base compartment to receive and terminate incoming power cable and MCB in a box for isolation of incoming power supply shall be provided by purchaser.

8.3.7 Power Tool for the Winch

Three phase, single speed, 6 pole high-powered motor of rating suitable to lift the load mounted on adjustable plate to adjust the length of winch motor coupling chain is to be provided in base compartment.

Mechanical torque limiter is to be mounted on motor shaft to stop transmission of motion from motor to winch in case of excess load and thus prevent the damage to winch and breakage of rope.

8.3.8 Lightning Finial

One number heavy duty 1.2 m long hot dip galvanized lightening finial shall be provided for each mast for mast height $\geq 16\text{M}$ on the head frame to get a direct conducting path to the earth through the mast.

8.3.9 Aviation Obstruction Lights

One/Two number Low intensity RED colored LED Aviation Obstruction Lights shall be provided on luminaires carriage as per customer requirement.

8.3.10 Earthing Terminals

Earth terminal using 12 mm diameter hot dip galvanized bolts shall be provided on the door stiffener of the mast for lightning and electrical earthing of the mast.

8.3.11 Feeder Pillar

Each mast shall be provided with a feeder pillar housing Suitable TPN MCB incomer, single dial time switch and outgoing contactor for automatic switching on and off luminaries, 9A two contactors and raise lower push buttons for motor operation, incoming terminals of 50 sq. m and outgoing terminals of 16 sq. mm for power cable and 2.5 sq. mm for motor.

8.3.12 Incoming Power Cable

1.1 KV grade, XLPE insulated, PVC sheathed, Aluminium conductor, armored cable for power supply of suitable size and capacity shall be provided from feeder pillar to the base compartment of the high mast. Cable shall be taken to the base compartment of the high mast through the provision made in the foundation. Power cable of suitable size up to the feeder pillar from supply point shall be provided by purchaser.

Note: Technical Specifications will be upgraded time to time, considering good engineering practices.

Technical Specification of High Masts

	12 Mtr	12.5 Mtr	16 Mtr	20 Mtr	25 Mtr	30 Mtr	40 Mtr
Material Construction	BSEN10025 S355						
Thickness	3, 3 mm	3, 3 mm	3, 4 mm	3, 4 mm	3,4,4 m	3,4,5 mm	4,4,5,5,6,6 mm
Top & Bottom Dia	150/310 mm	150/360 mm	150/360 mm	150/410 mm	150/460 mm	150/500 mm	150/700 mm
Size of base plate	500 mm	540mm	540mm	610 mm	660 mm	700 mm	900mm
Base Plate thickness	16 mm	16 mm	20 mm	25 mm	25 mm	30 m	40 mm
No. of foundation bolts	6 nos.	6 nos	8 nos	8 nos	12 nos	12 nos	16 nos

8.4 Detail specifications of Galvanized Octagonal poles

The Octagonal Poles shall be designed to withstand the maximum wind speed as per IS 875 Part–3-2015. The head loading i.e. windage area and the weight of fixtures are to be considered to calculate maximum deflection of the pole and the same shall meet the requirement of BSEN 40-3-1 & 3.

The pole shaft shall be made from hot rolled steel plate confirming to IS 2062/ BS EN 10025. All octagonal pole shafts shall be provided with the rigid base plate manufactured from MS steel confirming to IS: 2062 of suitable thickness with provision for fixing foundation bolts.

The pole shaft shall have octagonal cross section and shall be continuously tapered with single longitudinal welding. There shall not be any circumferential welding. The welding of pole shaft shall be done by Submerged Arc Welding (SAW) process. The base plate shall be fillet welded to the pole shaft at two locations i.e. from inside and outside.

The octagonal Poles shall have door of approximate 500 mm height at the elevation of 500 mm from the base plate. The pole shall be adequately strengthened at the location of the door to compensate for the loss in section.

The hinged door shall be flush with the exterior surface and shall have suitable locking arrangement. There shall also be suitable arrangement for the purpose of earthing. The welding shall be carried out confirming to approved procedures qualified by third party inspection agency. The welders shall also be qualified for welding the octagonal shafts.

The Octagonal Poles shall be in single section up to 12-meter height and there shall not be any circumferential weld joint except at base flange.

The Manufacturing unit shall have in house galvanizing facility; the poles shall be hot dip galvanized as per BSEN ISO 1461 standard. Zinc used for galvanizing shall have purity of 99.995% as per IS 209. The galvanizing shall be done in single dipping. The galvanized mounting bracket shall be supplied along with the Octagonal Poles for installation of the luminaries.

Provision should be provided in the bottom compartment to mount PVC /Bakelite sheet along with connectors for cable looping and single pole MCB to isolate individual luminaire.

TYPICAL DATASHEET FOR OCTAGONAL POLE

		REQUIREMENTS
1.	OCTAGONAL POLE	
	Material construction of Shaft	Hot Rolled Steel Plate as per IS 2062 / BS EN 10025 Or Equivalent.
	Grade of Steel	Hot Rolled Steel Plate as per IS 2062 / BS EN 10025 Or Equivalent.
	Thickness	To be decided as per height of pole and site condition
	No. of longitudinal welds /section	One
	Cross section of Pole	Octagonal (8-sided polygon)
	Base diameter and top diameter (A/F)	As per structural design calculations.
	Metal protection treatment for Pole Section	Hot Dipped Galvanized, as per BS EN ISO 1461
	Height of door	To be decided as per height of pole and site condition
	Door type	To be decided as per height of pole and site condition
	Dimension of base plate	As per structural design calculations.
	Thickness of base plate	As per structural design calculations.
	Grade of Steel for Base Plate	As per IS 2062
	Max. wind speed	As per IS 875 Part III – 2015
2.	FOUNDATION BOLTS	
	Number of foundation bolts	As per structural design calculations
	PCD of foundation bolts	As per structural design calculations.
	Grade of Foundation Bolts	As per structural design calculations.
	Bolt diameter/Length	As per structural design calculations with J-Bolt

Note: Technical Specifications will be upgraded time to time, considering good engineering practices.

APPENDIX-A
(CLAUSE 1.3)
TERMINOLOGY

- 1. Accessible**
Not obstructed/guarded by locked doors, readily approachable/capable of being physically reached quickly without use of any appliance or special effort for operation.
- 2. Aerial (insulated) Cable**
An insulated cable designed to be suspended overhead and outdoors.
- 3. Air Termination (Lightning Conductor) or Air Termination Network**
Those parts of a lightning protective system that are intended to collect the lightning discharges from the atmosphere.
- 4. Apparatus**
Electrical apparatus and includes all machines, fittings, accessories and appliances in which conductors are used.
- 5. Approach Limit**
A limited approach boundary at a distance from exposed energized electric part from which electric shock hazard exist.
- 6. Appropriate Government**
Central Government or State Government having respective jurisdiction as mentioned under section 2 of *Electricity Act, 2003*.
- 7. Approved**
Permitted by an authority having jurisdiction.
- 8. Arrangement**
The pattern according to which luminaires are sited on plan, for example, staggered, axial, opposite.
- 9. Armour**
A covering consisting of a metal tape(s) or wires, generally used to protect the cable from external mechanical effects.
- 10. Axial (Converse Non-axial) Distribution**
An asymmetrical distribution in which the directions of maximum luminous intensity lie (do not lie) in vertical planes substantially parallel to the axis of the carriageway.
- 11. Bare**
Not covered with insulating material.
- 12. Beam**
The portion of the light output of the luminaire contained by the solid angle subtended at the effective centre of the luminaire containing the maximum intensity, but no intensity less than 90 percent of the maximum intensity.

- 13. Beam Centre**
A direction midway between the directions for which the intensity is 90 percent of the maximum in a vertical plane through the maximum and on a conical surface through the maximum.
- 14. Bonds**
Electrical connection between the lightning protective system and other metal work, and between various portions of the latter.
- 15. Branch Circuit**
A circuit between final circuit breaker protecting the circuit and outlets.
- 16. Braid**
A covering formed from plaited metallic or non-metallic material.
- 17. Busbar**
Low impedance conductor to which several electric circuits can be separately connected.
- 18. Cable Ducting System**
A system of closed enclosure of non-circular sections for insulated conductors, cable and cords in electrical installations, allowing them to be laid in and replaced.
- 19. Cable Trunking System**
A system of closed enclosures comprising a base with a removable cover intended for the complete surrounding of insulated conductors, cables, cords and/or for the accommodation of other electrical equipment.
- 20. Conductor Vibration**
Vibration, generally of small amplitude, of the conductors of overhead lines, which can be suppressed by means of appropriate dampers.
- 21. Conductor insulation**
Insulation applied on a conductor or a conductor screen.
- 22. Conductor Screen**
An electrical screen of non-metallic and/or metallic material covering the conductor.
- 23. Core Insulated Conductor**
An assembly comprising a conductor with its own insulation (and screens, if any).
- 24. Carriageway**
That portion of a highway intended primarily for vehicular traffic.
- 25. Central Reserve**
A longitudinal space dividing a dual carriageway.
- 26. Cycle Track**
A way or part of a highway for use by pedal cycles only.
- 27. CEA**
Central electricity authority, an authority constituted under sub section (1) of section 70, of *Electricity Act*, 2003 having jurisdiction to make regulations as per section 177 of the Act.

- 28. CEA Safety Regulations**
Central electricity authority (measures relating to safety and electric supply) regulations, 2010 as and when amended, as per section 53 of the *Electricity Act*, 2003.
- 29. Conductor**
Any wire, cable, bar, tube, rail, plate used for conducting electricity and so arranged as to be electrically connected to system.
- 30. Cut-off Luminaire**
Luminaire employing the technique used for concealing lamps and surfaces of high luminance from direct view in order to reduce glare.
- 31. Dual Carriageway**
A layout of the separated carriageways, each reserved for traffic in one direction only.
- 32. Down Conductors**
Conductors which connect the air terminations with the earth terminations.
- 33. Distribution (feeder) Pillar**
Totally enclosed structure or cubicle containing busbars connected to incoming and outgoing distribution feeders controlled through links or fuses.
- 34. Downward Flash**
Lightning flash initiated by a downward leader from cloud to earth (A downward flash consists of a first impulse, which can be followed by subsequent impulses. One or more impulses may be followed by a long stroke).
- 35. Designated Person**
A person having certificate of competency or work permit issued by an appropriate government to carryout work on electrical lines and apparatus under Regulation 3 of CEA Safety Regulations and authorised by the owner of the installation as per Regulation 3 of CEA Safety Regulations.
- 36. Distribution Sub-station**
A sub-station used for feeding a distribution network.
- 37. Earthed Neutral System**
A system in which the neutral is connected to earth, either solidly, or through a resistance or reactance of value low enough to reduce materially transient oscillations and to give a current sufficient for selective earth fault protection.
- 38. Electric Line**
A generic term of a set of conductors, with insulation and accessories, used for the transmission or distribution of electrical energy.
- 39. Earth Fault**
Occurrence of an accidental conductive path between a live conductor and the earth or exposed conductive part of a protective conductor.
- 40. Earth Leakage Current**
Current which flows to earth or to extraneous conductive parts in a circuit which is electrically sound.

41. Electric Circuit (or Circuit)

An assembly of electrical equipment supplied from the same origin and protected against over current by the same protective devices.

42. Electrode

Conductive part in electric contact with a medium of lower conductivity and intended to perform one or more of the functions of emitting charge carriers to or receiving charge carriers from that medium or to establish an electric field in that medium.

43. Earth Terminations or Earth Terminations Network

Those part of the lightning protective system which are intended to distribute the lightning discharges into the general mass of the earth. All parts below the testing point in a down conductor are included in this term.

44. Earth Electrode

A conductor or group of conductors in intimate contact with and providing an electrical connection to earth.

NOTE: The word ‘ground’/‘grounding’ used in place of ‘earth’/‘earthing’ has same meaning, purpose, objective in that regard.

45. Earth Electrodes

Conductor or group of conductors in intimate contact with and providing an electrical connection to earth.

46. Earth Fault

An un-intentional electrical connection between live or un-grounded electrical current carrying conductor and normally non-current carrying conductor, metal enclosure/equipment having connectivity to earth.

47. Electrical Inspector

An authority having jurisdiction under section 162 of *Electricity Act*, 2003 and powers conferred thereunder.

48. Electrical Safety

Identifying hazards associated with the working on electrical installation, use of electrical energy and taking precautions to reduce the risk associated with those hazards.

49. Enclosed

A case, housing of apparatus or fence or wall surrounding the installation to prevent person from unintentionally contacting energized parts.

50. Enclosure

Surrounded by case, housing, fence or wall that prevents person from unintentionally contacting energised parts and to protect the equipment from physical damage.

51. Energise

Connecting to electric supply source and commission/charge.

52. Equipment

Any item used for such purposes as generation, conversion, transmission, storage, distribution or utilization of electrical energy, such as machines, transformers, apparatus, measuring instruments, protective devices, equipment for wiring systems, appliances.

53. Exposed

In respect of unguarded conductor or circuit parts capable of being inadvertently touched or approached nearer than a safe distance.

54. Extraneous Conductive Part

A conductive part liable to introduce potential (generally earth potential) and not forming part of electrical installation.

55. Extruded Insulation

Insulation consisting generally of one layer of a thermoplastic or thermosetting material and applied by an extrusion process.

56. Fault Current

B current resulting from fault, which flows through an abnormal unintended path, resulting from insulation failure or bridging of insulation, where impedance between live conductor(s) and exposed extraneous conductive parts at fault position is considered negligible.

57. Fasteners

Devices used to fasten the conductors to the structures.

58. Flashover

Electrical breakdown between conductors in air, a gas or a liquid or in vacuum, at-least partly, along the surface of a solid insulation.

59. Footway

That portion of a road reserved exclusively for pedestrians.

60. Feeder

C line which supplies a point of distribution network without being tapped at any intermediate point.

61. Filler

The material used to fill the interstices between the cores of a multi-conductor cable.

62. Feed Point

D point at which a network or a line receives energy.

63. Fittings

The portions of an insulator provided for making a mechanical connection to it.

64. Geometry (of a Lighting System)

The inter related linear dimensions and characteristics of the system, namely the spacing, mounting height, width, overhang and arrangement.

65. Guarded

Covered, shielded, fenced or protected by means of suitable casings, barrier, rails, metal screen to avoid the possibility of dangerous contact.

- 66. Guarding**
An arrangement provided on overhead electric line to earth the snapped live bare conductor(s) to ensure them rendering dead and also prevent it from becoming accessible from ground.
- 67. Highway**
A way for the passage of vehicular traffic over which such traffic may lawfully pass.
- 68. Insulation Fault**
In the insulation of an equipment, wire or cable which can result either in an abnormal electric current through this insulation or in a disruptive discharge.
- 69. Insulation Screen; Core Screen**
An electrical screen of non-metallic and/or metallic material covering the insulation.
- 70. Isocandela Diagram**
An array of Isocandela curves.
- 71. Integral Luminaire**
Luminaire with all its accessories, such as ballasts, starters, igniters, capacitors, etc. However, integrally with the body of the luminaire.
- 72. Isolated Neutral System**
A system which has no intentional connection to earth except through indicating, measuring, or protective devices of very high impedance.
- 73. Isoceraunic Level**
It is the number of days in a year on which the thunder is heard in the particular region averaged over a number of years.
- 74. Intensity Ratio (in a Particular Direction)**
The ratio of an actual intensity from the luminaire (in a particular direction) to the mean hemispherical intensity.
- 75. Isocandela Curve**
A curve traced on an imaginary sphere with a source at its centre and joining all the points corresponding to those directions in which the luminous intensity is the same or a plane projection of this curve.
- 76. Illumination**
The luminous flux incident on a surface per unit area. The unit of illumination is lumen per square metre (lux).
- 77. Joints**
The mechanical and electrical junctions between two or more portions of the lightning protective system or other metal bonded to the system or both.
- 78. Jumper**
A short length of conductor, not under mechanical tension, making an electrical connection between two sections of a conductor of an electric line.

- 79. Kerb**
A border of stone, concrete or other rigid material formed at the edge of a carriageway.
- 80. Lighting Installation**
The whole of the equipment provided for lighting the highway comprising the lamps, luminaires, means of support and electrical and other auxiliaries.
- 81. Lighting System**
An array of luminaires having a characteristic light distribution sited in a manner concordant with this distribution (lighting systems are commonly designated by the name of the characteristic light distribution, for example, cut-off, semi-cut-off, etc.).
- 82. Luminaire**
A housing for one or more lamps, comprising a body and any refractor, reflector, diffuser, or enclosure associated with the lamp(s).
- 83. Long stroke**
Part of the lightning flash which corresponds to a continuing current (The duration time T_{LONG} {time from the 10 percent value on the front to the 10 percent value on the tail} of this continuing current is typically more than 2 ms and less than 1 s).
- 84. Lightning Protective System (LPS)**
Complete system used to reduce physical damage due to lightning flashes to a structure (It consists of both external and internal lightning protection systems).
- 85. Lightning Equipotential Bonding EB**
Bonding to the LPS of separated conductive parts, by direct connections or via surge protective devices, to reduce potential differences caused by lightning current.
- 86. Lapped Insulation**
Insulation consisting of tapes applied helically in concentric layers.
- 87. Luminaire**
Apparatus which distributes, filters or transforms the light transmitted from one or more lamps/light emitting device(s) and which includes all the parts necessary for supporting, fixing and protecting the lamps, but not the lamps themselves, and where necessary circuit auxiliaries together with the means for connecting them to the supply.
- 88. Lightning Flash to Earth**
Electrical discharge of atmospheric origin between cloud and earth consisting of one or more strokes.
- 89. Luminous Flux**
The light given by a light source or a luminaire or received by a surface irrespective of the directions in which it is distributed. The unit of the luminous flux is lumen (lm).
- 90. Lower Hemispherical Flux or Downward Flux**
The luminous flux emitted by a luminaire in all directions below the horizontal.

91. Luminous Intensity

The quantity which describes the light-giving power of a luminaire in any direction. The unit of luminous intensity is Candela (cd).

92. Limited Approach

An approach limit at a distance from an exposed energised electrical conductor within which shock hazard exists.

93. Luminance (at a Point of Surface and in a Given Direction)

The luminous intensity per unit projected area of a surface. If a very small portion of a surface has an intensity I in a particular direction and its orthogonal projection (that is, its projection on a plane perpendicular to the given direction) has an area D , the luminance in this direction is I/D candelas per unit area. The usual unit is candela per square metre (cd/m^2).

94. Luminosity

The attribute of visual sensation according to which an area appears to emit more or less light. It is sometimes called brightness.

NOTE: Luminosity is the visual sensation which correlates approximately with the photometric quantity 'luminance'.

95. Light Output

The luminous flux emitted by a luminaire.

96. Light Distribution

The distribution of luminous intensity from a luminaire in various directions in space.

97. Length of a Span

The horizontal distance between two consecutive supports of an overhead line.

98. Layout

All those physical features of a high way other than the surfacing of the carriage way, which must be considered in planning a lighting installation.

99. Metal-clad Building

A building with sides made of or covered with sheet metal.

100. Metal-roofed Building

A building with roof made of or covered with sheet metal.

101. Mounting Height

The vertical distance between the centre of the luminaire and the surface of the carriage (see Fig. 13).

102. Mass-impregnated (paper) Insulation

Impregnated paper insulation in which the paper tapes are impregnated after lapping.

103. Mass-impregnated non-draining Insulation

Mass-impregnated paper insulation in which the impregnate is not fluid at the maximum continuous operating temperature.

104. Mineral insulation

Insulation consisting of compressed mineral powder.

105. Mean Hemispherical Intensity

The downward flux divided by $6.28 (2\pi)$ (this is the average intensity in the lower hemisphere).

106. Multiple Stranded Conductor

A stranded conductor consisting of a number of groups of wires assembled together in one or more helical layers, the wires in each group being either bunched or stranded.

107. Overload

Operating conditions in an electrically undamaged circuit which causes an overcurrent.

108. Outreach

The distance measured horizontally between the centre of the column or wall face and the centre of a luminaire (see Fig. 13).

109. Overhang

The distance measured horizontally between the centre of a luminaire mounted on a bracket and the adjacent edge of the carriageway (see Fig. 13).

110. Oversheath

A non-metallic sheath applied over a metallic covering, constituting the outermost sheath of the cable.

111. Outlet

A point on wiring system from which current is taken to supply electrical equipment for its utilization.

112. PPE

Personal Protective equipment for the protection of persons from electrical hazard while working on live electrical installations or work within restricted boundary.

113. Parallel Earth Continuity Conductor

Conductor usually laid along the cable route to provide a continuous low impedance metallic earth connection between the earthing systems at the ends of the cable route.

114. Polar Curve

Curve of light distribution using polar co-ordinates.

115. Peak Intensity Ratio

The ratio of the maximum intensity to the mean hemispherical intensity of the light emitted below the horizontal.

116. Pin Insulator

An insulator consisting of a single piece of porcelain or of two or more porcelain components permanently connected together and intended to be mounted rigidly on a supporting structure by an insulator pin passing up inside the insulator. Unless otherwise stated, this term excludes the insulator pin.

117. Package (Unit) Sub-station

A sub-station in which the switchgear, the busbars and the transformers are all contained in a single robust metal enclosure.

118. PROTECTION FOR SAFETY — PROTECTION AGAINST ELECTRIC SHOCK

(i) Direct Contact

When a person comes in contact with a live conductor, it is called electric shock by direct contact.

(ii) Indirect Contact

When a person comes in contact with an exposed conductive-part, which is normally not live but has become live mainly due to insulation failure, is called electric shock by indirect contact.

119. Radial Circuit

A line emanating from a source of supply and terminating at a point to be supplied which, together with any other points to be supplied by the line, do not have a supply available to them in more than one direction.

120. Rings Circuit

A circuit emanating from a source of supply and terminating at the same or another source of supply, and with points to be supplied along the route of the circuit so connected that the supply is available to all of them from both directions along the circuit.

121. Refuge

A raised platform or a guarded area so sited in the carriageway as to divide the streams of traffic and to provide a safety area for pedestrians.

122. RCD

A mechanical switching device or association of devices intended to cause the opening of the contacts when the residual current attains a given value under specified conditions, intended for protection against electrical shock or over heating due to earth leakage.

123. Shoulder

A strip of highway adjacent to and level with the main carriageway to provide an opportunity for vehicles to leave the carriageway in an emergency.

124. Sub-station

An assemblage of equipment at one place including any necessary housing for the conversion, transformation or control of electrical energy and for connection between two or more circuits.

125. Symmetrical (Converse Asymmetrical) Distribution

A distribution of luminous intensity which is substantially symmetrical (conversely asymmetrical) about the vertical axis of the luminaire.

126. System with Solidly Earthed Neutral

A system with one or more transformers or generators having the neutral point(s) earthed in such a way that the voltage drop along the connection to earth is negligible compared with the nominal voltage of the system under all possible operating conditions.

127. Supply Terminals

The point at which a consumer receives energy.

128. Service Line

A line connecting the consumer's installation to a distributor.

129. Spacing

The distance, measured along the centre line of the carriageway, between successive luminaires in an installation (see Fig. 13).

NOTE: In a staggered arrangement, the distance is measured, along the centre line of the carriageway, between a luminaire on one side of the carriageway and the next luminaire, which is on the other side of the carriage. It is not the distance measured on the diagonal joining them, nor the distance between successive luminaires on the same side of the carriageway.

130. Span

That part of the highway lying between successive luminaires in an installation.

131. Span

The part of an overhead line between two consecutive supports.

132. Sag

Maximum vertical distance, in a span of an overhead line, between a conductor and the straight line passing through the two points of support of the conductor.

133. Screen (of a cable)

Conducting layer(s) having the function of control of the electric field within the insulation. It may also provide smooth surfaces at the boundaries of the insulation and assist in the elimination of spaces at these boundaries.

134. Shield (of a cable)

A surrounding earthed metallic layer to continue the electric field within the cable and or to protect the cable from external electrical influence.

135. Serving

One or more non-extruded layers applied to the exterior of a cable.

136. Shielding Conductor

A separate conductor or single core cable-laid parallel to a cable or cable circuit and itself forming part of a closed circuit in which induced currents may flow whose magnetic field will oppose the field caused by the current in the cable(s).

137. Short-circuit

Accidental or intentional conductive path between two or more conductive parts forcing the electric potential differences between these conductive parts to be equal to or close to zero.

138. Spark

Small luminous electric arc of short duration.

139. Surge Protective Device SPD

Device intended to limit transient over-voltages and divert surge currents; contains at least one non-linear component

140. Side Flash

A spark occurring between nearby metallic objects or between such objects and the lightning protective system or to earth.

141. Short Stroke

Part of the lightning flash which corresponds to an impulse current (This current has a time T_2 to the half peak value on the tail typically less than 2 ms).

142. Service Road

A subsidiary road between principle road and buildings or properties facing thereon or a parallel road to the principal road and giving access to the premises and connected only at selected points with the principle road.

143. Street Lighting Luminaire

A housing for a light source or sources, together with any refractor, reflector, dispersive surround or other enclosure which may be associated with the source in order to modify the light distribution in a desired manner and protect the light from weather conditions and insects and/or for the sake of appearance, brightness and other lighting characteristic the source.

144. Semi-cut-off Luminaire

Luminaire employing the technique for concealing lamps and surfaces of high luminance from direct view in order to reduce glare but to a lesser degree than cut-off luminaire.

145. Testing Points

Joints in down conductors or in bonds or in earth conductors connecting earth electrodes, so designed and situated as to enable resistance measurements to be made.

146. Transmission (Distribution) Line

Electric line forming part of an installation for transmitting electrical energy, normally restricted to overhead construction.

147. Upward Flash

Lightning flash initiated by an upward leader from an earthed structure to cloud (An upward flash consists of a first long stroke with or without multiple superimposed impulses. One or more impulses may be followed by a long stroke).

148. Voltage, Potential Difference

The line of integral from one point to another of an electric field, taken along a given path.

149. Verge

The unpaved area flanking a carriageway, forming part of the highway and substantially at the same level as the carriageway.

150. Voltage Classifications

Nomenclature of commonly used short names of voltages

- a) ELV (extra low voltage) — A system with a nominal voltage $U_n \leq 50 \text{ V a.c. or } \leq 120 \text{ V d.c. ;}$
- b) LV (low voltage) — A system with a nominal voltage $U_n \leq 1\,000 \text{ V a.c. and } \leq 1\,500 \text{ V d.c.}$
- c) MV (medium voltage) — A system with a nominal rms voltage is $1 \text{ kV} < U_n \leq 33 \text{ kV}$
- d) HV (high voltage) — A system with a nominal rms voltage is $33 < U_n \leq 150 \text{ kV.}$
- e) EHV (extra high voltage) — A system with a nominal rms voltage is $> 150 \text{ kV.}$

151. Wiring System

Assembly made up of bare or insulated conductors or cables or busbars and the parts which secure and if necessary, enclose the cables or busbars.

152. Width of Carriageway

The distance between kerb lines measured at right angles to the length of the carriage way (see Fig. 13).

153. Zone of Protection

The space within which the lightning conductor is expected to provide protection against a direct lightning stroke.

APPENDIX-B
(Clause 1.21.2 & 1.22.4.2)
IMPORTANT INDIAN STANDARDS

These are in addition to those given in the General Specifications for Electrical Works (Part I- Internal), 2013 as relevant to External EI works.

S. No.	Description	IS Code	Reaffirmed Year
Poles:			
(1)	Reinforced concrete poles for over- Head power and telecommunication lines	785-1998	2019
(2)	Prestressed concrete poles for overhead power, traction and tele-communication lines	1678-1998	2018
(3)	Precast prestressed concrete street lighting poles	2193-1986	2022
(4)	Precast reinforced concrete street lighting poles	1332-1986	2020
(5)	Methods of test for concrete poles for overhead power and telecommunication lines	2905-1989	2022
(6)	Tubular poles for overhead power and telecommunication lines Conductors	2713-1980 (Part 1 to 3) : 1980	2017
(7)	Design Loads (Other than Earthquake) for Buildings and Structures - Code of Practice Part 3 Wind Loads (Third Revision)	875 : Part 3 : 2015	2020
(8)	Insulator fittings for overhead power lines with a nominal voltage up to and including 1000 V	7935 :1975	2021
Conductors			
(1)	Aluminium conductors for overhead transmission purposes		
(i)	Aluminium stranded conductors	398(Part-1)-1996	2018
(ii)	Aluminium conductors galvanised steel reinforced	398(Part-2)- 1996	2018
(iii)	Aluminium alloy stranded conductors	398(Part-4)-1994	2021
(2)	Aluminised steel core wire for aluminium conductors (ACSR)	3835-1966	1997
(3)	Conductors and earthwire accessories for overhead power lines : Armour rods, binding wires and tapes for conductors	2121(Part-1)-1988	2020
(4)	Conductors and earthwire accessories for overhead power lines: Mid span joints and repair sleeves for conductors	2121(Part-2)-1988	2020
Cables:			

(1)	PVC insulated (heavy duty) electric cables :-		
(i)	For working voltage upto and including 1100V	1554(Part-1)-1988	-
(ii)	For working voltage from 3.3 KV upto and including 11 KV	1554(Part-2)-1988	-
(2)	Paper insulated lead sheathed cables for electricity supply	692-1994	2020
(3)	Cross linked polyethylene insulated PVC sheathed (XLPE) cables:-		
(i)	For working voltage upto and including 1100 Volts	7098(Part-1)-1988	2020
(ii)	For working voltage from 3.3 KV upto and including 33 KV	7098(Part-2)-2011	2021
(4)	Recommended short circuit rating for cables:-		
(i)	Paper insulated and lead sheathed cables	3961(Part-1)-1967	2021
(ii)	PVC insulated and PVC sheathed heavy duty cables	3961(Part-2)-2017	
(5)	Recommended short-circuit ratings of high voltage PVC cables	5819-1980	2021
(6)	Multicore and symmetrical pair/quad cables for digital communication	14493 (all Parts)	2021
(7)	Aerial Bunched Cables for working Voltages upto and including 11000 Volts	14255 : 1995	2005
(8)	Cable accessories for extruded power cables – specification, Part-1 for working voltages from 1.1KV upto and including 3.3kV(E) – Test methods and test requirements.	13573 (Part 1) : 2011	2016
(9)	Cable accessories for extruded power cables – specification, Part-2 for working voltages from 3.3kv (UE) upto and including 33kV(E) – Test requirements.	13573 (Part 2) : 2011	2021
(10)	Cable accessories for extruded power cables – specification, Part-3 for working voltages from 3.3kv (UE) upto and including 33kV(E) – Test methods.	13573 (Part 3) : 2011	2016
(11)	Specification for moulds for cast resin based straight through joints for cables for voltages upto and including 1100 volts.	8438 : 1987	2017
(12)	Moulds suitable for cast resin based joints for cables for voltages upto and including 1100 V - specification.	9646 : 1992	2019
(13)	Conductors for Insulated Electric Cables and Flexible Cords	8130:2013	2018
(14)	Method for testing of cable	10810 (Part 1 to 30) :1984	2021

Insulators:			
(1)	Porcelain insulators for overhead power lines with a nominal voltage upto and including 1000V	1445-1977	2019
(2)	Porcelain insulators for overhead power lines with a nominal voltage greater than 1000V	731-1971	2021
(3)	Porcelain guy strain insulators	5300-1969	2019
(4)	Characteristics of string insulator units	IS/IEC 60471 : 1977	
(5)	Insulator fittings for overhead power lines with a nominal voltage upto and including 1000V	7935-1975	2021
(6)	Insulator fittings for overhead power lines with a nominal voltage greater than 1000V		
(i)	General requirements and tests	2486(Part-1)-1993	2018
(ii)	Dimensional requirements	2486(Part-2)-1989	2019
(iii)	Looking devices	2486(Part-3)-1974	2021
Codes of Practice:			
(1)	Design, installation and maintenance of overhead power lines		
(i)	Upto and including 11KV	5613-(Part-1 sections 1 and 2)-1985)	2018
(ii)	Above 11KV and upto and including 220KV	5613(Part-2 sections 1 and 2)1985	2018
(2)	Selection, handling and erection of concrete poles for overhead power and telecommunication lines	7321-1974	2020
(3)	Installation and maintenance of power cables upto and including 33 KV rating	1255-1983	2016
(4)	Lighting of public thorough-fares for main and secondary roads (Group A & B)	1944- (Part 1 & 2)1970	2018
(6)	Code of practice for lighting of public thorough fares: Part 1 General principles Part 2 Lighting for traffic routes (Groups A1, A2, B1 and B2) (<i>first revision</i>)	1944 (Parts 1 and 2) : 19 '0	2018
(7)	Code of practice for lighting of public thoroughfares: Part 3 Lighting for Secondary roads which do not require lighting up to group A standard(Group B)	1944 (Part 3) : 1981	2018

(8)	Code of practice for lighting of public thoroughfares: Part 4 Lighting for residential and unclassified roads (Group C)	1944 (Part 4) : 1981	2018
(9)	Code of practice for lighting of public thoroughfares: Part 5 Lighting of grade separated junctions, bridges, and elevated road (Group D)	1944 (Part 5) : 1981	2018
(10)	Code of practice for lighting of public thoroughfares: Part 6 Lighting for town and city centres and areas of civic importance (Group E)	1944 (Part 6) : 1981	2018
(11)	Code of practice for lighting of public thoroughfares: Part 8 Lighting in Tunnel	1944 (Part 8) : 1981	
(11)	Code of practice for design, installation, and maintenance of overhead powerlines: Part 1 Lines up to and including 11 kV, Section 1 Design (<i>first revision</i>)	5613 (Part 1/Sec 1) : 1985	2018
(12)	Code of practice for design, installation, and maintenance of overhead power lines Part 1 Lines up to and including 11 kV, Section 2 Installation, and maintenance (<i>first revision</i>)	5613 (Part 1/Sec 2) : 1985	2018
(13)	Code of Practice for earthing	3043:2018	
(14)	Code of Practice for electrical wiring Installation	IS 732 : 2019	
(15)	Protection against lightning – Part 1: General principles	IS/IEC 62305-1 : 2010	
(16)	Protection Against Lightning –. Part 2: Risk Management.	IS/IEC 62305-2 : 2010	
(17)	Protection against lightning – Part 3: Physical damage to structures and life hazard	IS/IEC 62305-3 : 2010	
(18)	Energy Conservation Code of India 2017	ECBC : 2017	
(19)	National Building Code of India 2016(NBC 2016)	SP7 : 2016	
(20)	National Lghting Code of India 2010	SP : 72	
(21)	Code of practice for selection, handling and erection of concrete poles for overhead power and telecommunication lines	7321: 1974	2020

Safety standards:			
(1)	Guide for safety procedures and practices in electrical works:		
(i)	General	5216(Part-1)-1982	2020
(ii)	Life saving techniques	5216(Part-2)-1982	2020
(2)	Excavation work	3764-1992	2022
(3)	Rubber gloves for electrical purposes	4770-1991	2017
(4)	Fire safety of industrial buildings: Generating and Distributing stations	3034 :1993	2017
(5)	Warning symbol for dangerous voltages	8923 : 1978	
(6)	Classification of insulating liquids	13503 : 2013	2018
(9)	Safety of machinery — Electrical equipment of machines: Part 1 General requirements (<i>first revision</i>)	IS 16504 (Part 1) : 2019/IEC 60204-1 : 2016	-
	Safety of machinery — Electrical equipment of machines – Part 11 Requirements for equipment for voltages above 1 000 V AC or 1 500 V DC and not exceeding 36 kV	IS 16504 (Part 11) : 2020/IEC 60204-11 : 2020	-
	Specification for unused natural esters for transformer and similar electrical equipment	16659 : 2017	-
General:			
(1)	Dimensions for hot rolled steel beam, column channel, and angle sections	808-2021	-
(2)	Galvanised stay strand	2141-2000	2021
(3)	Galvanized steel barbed wire for fencing	278-2009	2020
(4)	Cast iron manhole covers and frames:	1726 (Part ii)-1994	2022
(5)	Luminaires for road and street lighting	10322(Part-5/2012)	2017
(6)	Voltage bands for electrical installations including preferred voltage and frequency	12360-1988	2020
(7)	Luminaires: Part 1 General Requirements	10322 (Part 1) : 2014	2019
(8)	Residual current operated circuit breakers for household and similar purposes	12640 (Part 1) : 2016	-
(9)	Residual current operated circuit breakers with integral over current protection (RCBOs)	12640 (Part 2) : 2016	-

(10)	Information technology equipment — Safety — Part 1 General requirements (<i>second revision</i>)	(Part 1) : 2022 13252 60950-1 : 2010/ IEC 2005	
(11)	Effects of current on human beings and livestock: Part 1 General aspects	IS/IEC 60479-1 : 2018	-
(12)	Low - voltage surge protective devices — Part 11 Surge protective devices connected to low — Voltage power systems — Requirements and test methods	16463 (Part 11) : 2016	-
(13)	Low - voltage surge protective devices — Part 12 Surge protective devices connected to low — Voltage power distribution systems — Selection and application principles	16463 (Part 12) : 2017	-
Electro-technical vocabulary			
1)	Overhead transmission and distribution of electrical energy	1885(Part-30)1971	2017
2)	Cables, conductors and accessories for electricity supply (<i>Second Revision</i>)	1885(Part-32)2019	-
3)	Insulators	1885(Part-54)2021	-
4)	Electrotechnical vocabulary — Part 16 Lighting — Section 1 General aspects	1885 (Part 16/Sec 1) : 1968/IEC 60050-45 : 1958	2017
5)	Electrotechnical vocabulary — Part 16 Lighting — Section 2 General illumination, lighting fittings and lighting for traffic and signalling	1885 (Part 16/Sec 2) : 1968/IEC 60050-45 : 1958	2017
6)	Electrotechnical vocabulary — Part 16 Lighting — Section 3 Lamps and auxiliary apparatus	1885 (Part 16/Sec 3) : 1967/IEC 60050-54	2017
7)	Electrotechnical vocabulary — Part 17 Switchgear and control gear (<i>first revision</i>)	1885 (Part 17) : 1979/IEC 60050-441 : 1974	2017
8)	Electrotechnical vocabulary — Part 27 Power electronics (<i>third revision</i>)	1885 (Part 27) : 2008/IEC 60050-551 : 1998	2018
9)	Electrotechnical vocabulary — Part 28 Instrument transformers (<i>first revision</i>)	1885 (Part 28) : 1993/IEC 60050-321 : 1986	2022

10)	Electrotechnical vocabulary — Part 38 Power transformers and reactors (<i>second revision</i>)	1885 (Part 38) : 1993/ IEC 60050421 : 1990	2019
11)	Electrotechnical vocabulary — Part 69 Generation, transmission and distribution of electricity — Generation	1885 (Part 69) : 1993/ IEC 60050602 : 1993	2019
12)	Electrotechnical vocabulary: Part 70 Generation, transmission and distribution of electricity — Operation	1885 (Part 70) : 1993/ IEC 60050604 : 1987	2019
13)	Electrotechnical vocabulary — Part 71 Generation, transmission and distribution of electricity — Substations	1885 (Part 71) : 1993/ IEC 60050605 : 1983	2019
14)	Electrotechnical vocabulary — Part 77 Overhead lines	1885 (Part 77) : 1993/ IEC 60050466 : 1990	2022
15)	Electrotechnical vocabulary — Part 78 Generation, transmission and distribution of electricity — General	1885 (Part 78) : 1993/IEC 60050-601 : 1985	2022
16)	Electrotechnical vocabulary — Part 79 Generation, transmission and distribution of electricity — Power system planning and management	1885 (Part 79) : 1993/IEC 60050-603 : 1986	2022
17)	Electrotechnical vocabulary — Part 80 General terms on measurements in electricity	1885 (Part 80) : 1994/IEC 60050-301 : 1983	2019
18)	Electrotechnical vocabulary — Part 81 Electrical measuring instruments	1885 (Part 81) : 1993/IEC 60050-302 : 1983	2019
19)	Voltage bands for electrical installations including preferred voltages and frequency	12360 : 1988	2020
20)	Distribution system supply voltage quality	17036 : 2018	-
Lighting Arrester			
(1)	Application guide for non-linear resistor type surge arresters for a.c. system	4004-1985	Superseded by : IS 15086 : Part 5
(2)	Lighting arresters for a.c. systems: Non-linear. Resistor type lighting arresters	3070(Part-1)-1985	Superseded by : IS 15086 : Part 1
LIGHTING/ SOLAR PHOTOVOLTAIC			

(1)	Halogen free flame retardant HFFR cables for working voltages up to and including 1100 Volts — Specification	17048 : 2018	-
(2)	Specification for Thermosetting Insulated Fire Survival Cables for Fixed Installation having Low Emission of Smoke and Corrosive Gases when Affected by Fire for Working Voltages upto and including 1100V	17505 (Part 1) : 2021	-
(3)	Code of practice for selection, installation and maintenance of transformers: Part 2 Installation	10028 (Part 2) : 1981	2021
(4)	Testing, evaluation, installation and maintenance of a.c. electricity meters — Code of practice	15707 : 2006	2016
(5)	Low-Voltage Switchgear and Controlgear Assemblies – Part 1 General Rules	IS/IEC 61439-1 : 2011	-
(6)	Electrical safety in low voltage distribution systems up to 1 000 V a.c. and 1500 V d.c.	IS/IEC 61557 series : 2019	-
(7)	Automatic power factor correction (APFC) panels for voltage rating up to and including 1000 V	16636 : 2017	-
(8)	Distribution system supply voltage quality	17036 : 2018	-
(9)	Low-voltage electrical installations — Energy efficiency	16996 : 2018	-
(10)	Low voltage fuses	IS/IEC 60269 (all parts) : 2016	2018
(11)	Electrical accessories — Circuit-breakers for overcurrent protection for household and similar installations	IS/IEC 60898 (all parts) : 2015	2017
(12)	Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code)	IS 17050 : 2018/ IEC 62262 : 2002	-
(13)	Requirements for low-voltage special electrical installations or locations — Solar photovoltaic (PV) power supply systems	61997 : 2018	-
(14)	Crystalline silicon terrestrial PV module design qualification and type approval	14286 : 2010/ IEC 61215 : 2005	2020
(15)	Thin film (a-Si, CIGS and CdTe) terrestrial PV module design qualification and type approval	16077 : 2013/ IEC 61646 : 2008	2018

(16)	Concentrator photovoltaic (CPV) modules and assemblies design qualification and type approval	1228 : 2019/ IEC 62108 : 2016	-
(17)	PV modules safety qualification: Part 1 Requirements for construction	IS/IEC 61730-1 : 2016	2020
(18)	PV modules safety qualification: Part 2 Requirements for testing	IS/IEC 61730 (Part 2) : 2016	
(19)	PV modules salt mist corrosion testing	IS/IEC 61701 : 2011	2019
(20)	PV module performance testing and energy rating: Part 1 Irradiance and temperature performance measurements and power rating	16170 (Part 1) : 2014/ IEC 61853-1 : 2011	2019
(21)	PV modules Test methods for the detection of potential-induced degradation (PID): Part 1 Crystalline silicon PV modules	17210 (Part 1) : 2019/ IEC 62804-1 :2015	-
(22)	PV module ammonia (NH ₃) corrosion testing	16664 : 2018/ IEC 62716 : 2013	-
(23)	Safety of power converters for use in photovoltaic power systems: Part 2 Particular requirements for inverters	16221 (Part 2) : 2015	2020
(24)	Photovoltaic system-power conditioners — Procedure for Measuring efficiency	IS/IEC 61683 : 1999	2020
(25)	Test procedure of islanding prevention measures for utility connected PV inverters	16169 : 2019	-
(26)	Balance-of-system components for photovoltaic systems — Design qualification natural environments	16229 : 2015/ IEC 62093 : 2005	2020
(27)	Secondary cells and batteries for solar PV application-general requirements and methods of test	16270 : 2014	2019
(28)	Cross linked polyethylene insulated thermoplastics sheathed cables: Part 3 For working voltages from 66 kV up to and including 220 kV	7098 (Part 3) : 1993	2019
(29)	Electric cables for photovoltaic systems for rated voltage 1 500 V d.c.	17293 : 2020	-
(30)	Recommended current ratings for cables: Part 5 PVC insulated light duty cables	3961 (Part 5) : 1968	2021
(31)	Recommended current ratings for cables: Part 6 Crosslinked polyethylene insulated PVC sheathed cables	3961 (Part 6) : 2016	-

(32)	Recommended current ratings for cables: Part 7 Crosslinked polyethylene insulated thermoplastic sheathed cables	3961 (Part 7) : 2017	-
(33)	High-voltage switchgear and controlgear: Part 1 Common specifications	IS/IEC 62271-1 : 2007	2018
(34)	High-voltage switchgear and controlgear: Part 100 Alternating — Current circuit — Breakers	IS/IEC 62271-100 : 2008	2017
(35)	High-voltage switchgear and controlgear: Part 102 Alternating current disconnectors and earthing switches	IS/IEC 62271-102 : 2018	-
(36)	High-voltage switchgear and controlgear: Part 105 Alternating current switch — Fuse combinations for rated voltages above 1 kV up to and including 52 kV (<i>first revision</i>)	IS/IEC 62271-105 : 2012	-
(37)	High-voltage switchgear and controlgear: Part 200 a.c. Metal-enclosed switchgear and controlgear for rated voltages above 1 kV up to and including 52 kV (<i>first revision</i>)	IS/IEC 62271-200 : 2011	2018
(38)	High-voltage switchgear and controlgear: Part 202 High-voltage/low — Voltage prefabricated substation	IS/IEC 62271-202 : 2014	-
(39)	Degrees of protection provided by enclosures (IP code)	IS/IEC 60529 : 2001	2019
(40)	Measuring relays and protection equipment — Product safety requirements (First Revision)	IEC 60255-27 : 2013	-
(41)	Connectors for d.c. application in photovoltaic systems, safety requirements and tests	16781 : 2018/ IEC 62852 : 2014	-
(42)	Low-voltage switchgear and control gear: Part 1 General rules	IS/IEC 60947-1 (Part 1) : 2007	2017
(43)	Low-voltage switchgear and control gear: Part 3 Switches disconnectors switch disconnectors and fuse-combination units (<i>first revision</i>)	IS/IEC 60947-3 (part 2) : 2012	2018
(44)	Power transformers: Part 1 General	2026 (Part 1) : 2011	2021
(45)	Power transformers: Part 2 Temperature rise	2026 (Part 2) : 2010	2020
(46)	Power transformers: Part 3 Insulation levels, dielectric tests and external clearances in air	2026 (Part 3) : 2018/ IEC 60076-3 : 2013	-
(47)	Power transformers: Part 5 Ability to withstand short-circuit	2026 (Part 5) : 2011	2016

(48)	Power transformers: Part 10 Determination of sound levels resp	2026 (Part 10) : 2009/ IEC 6007610 : 2001	2019
(49)	Static direct connected smart watt-hour meter Class 1 and 2 specifications (with import and export/net energy measurements)	16444 : 2015	2020
(50)	Guide for general description of PV power generating systems	14153 : 1994	2019
(51)	Photovoltaic PV arrays design requirements	IS/IEC 62548 : 2016	-
(52)	Grid connected photovoltaic systems — Minimum requirements for system documentation, commissioning tests and inspection	16960 (Part 1) : 2018/ IEC 62446-1 : 2016	-
(53)	PV systems performance monitoring; guidelines for measurement, data exchange and analysis (all parts)	IS/IEC 61724 : 2017	-
(54)	Standalone PV systems design qualification	16230 : 2017/ IEC 62124:2004	-
(55)	Protection against lightning (<i>all parts</i>)	IS/IEC 62305 (<i>all parts</i>)	2020

APPENDIX-C
(Clause 1.24 & 1.26.1)
COMPLETION CERTIFICATE

I/We certify that the installation detailed below has been installed by me/us and tested and that to the best of my/our knowledge and belief, it complies with Indian electricity Rules 1956, as amended up-to-date as well as the C.P.W.D. General Specifications for Electrical works (Part II-External) 2023.

Electrical Installation at.....

Voltage and system of supply.....

Test result in the prescribed proforma enclosed.....Yes/No.

Signature of Supervisor

Signature of contractor

Name and address

Name and address

APPENDIX-D
(Clause 1.28)
MAINTENANCE REQUIREMENTS

(1) General

Following details/records shall be maintained. These shall be perused by inspecting officers.

- (a) Drawing (s) showing routes of UG cables and overhead lines.
- (b) Record of tests for Insulation Resistance, earthing, and feeder loading.
- (c) Record of breakdowns/faults encountered, and/or replacements made.

(2) UG Cables

- (a) Inspect every termination once a year. Look for any sign of overheating (visually and/or by unusual burning smell); Identify the cause and rectify.(such as loose connection, bad crimping, weathered soldering, overload etc.)
- (b) Measure with a clip-on ammeter, current in every feeder/ distributor cable once a year and record.(This test may be done at a period when the cable is expected to carry the maximum current).
- (c) Carry out insulation resistance test, section wise, during monsoon.

(3) Overhead lines

- (a) All sections of overhead lines shall be visually inspected by patrolling along the routes. Any extraneous materials clinging to the lines should be removed. Tree branches likely touch the line should be trimmed. Any abnormality noted should be corrected such as slipped conductors, abnormal/ unequal sag, broken/ charred strands at jumpers etc.
- (b) Insulators should be checked for chipping/ cracking, once a year.
- (c) Painting of poles, cross arms etc. should be done once in every year.
- (d) Carry out insulation test section wise, during monsoon and earth test during summer and keep a record of the same.
- (e) Keep a record of the maximum load on each section of the line.

(4) Road/ Street/Compound Lighting

- (a) check weekly for proper working of all luminaires and ensure that the looping boxes are not left open. Check daily for important/ sensitive installations.
- (b) Clean covers of light fittings once every three months and every month for sensitive installations.
- (c) Check looping boxes once every three months for proper termination of cables and ensuring proper condition of components inside.
- (d) Poles to be painted/ marked every year.

(5) Feeder pillars

- (a) Check weekly to ensure doors are locked and not left open.
- (b) Yearly check to ensure proper cable termination, proper conditions of all components including bus bars, insulation, switchgears, fuses. No openings/holes in the enclosure.
- (c) Yearly painting and marking.

APPENDIX E

(Clause 2.7.8)

JOINTING OF POWER CABLE

E-1 General:-

The reliability of a power cable network depends, among other factors, on the quality of joints made (End/Tee/Straight through) in the network. The conductivity and insulation under normal and short circuit conditions should be the same in the joint as in the conductor proper and the breaking load of joint shall be at least 60 percent of that of the conductor. This is achieved by using materials, accessories and tools of approved standards and to a large extent this also depends upon the skill in adhering to standard procedures approved for such work. All materials and accessories shall conform to relevant Indian Standards wherever they exist. For special type of splicing connector kits or epoxy resin spliced joints, makes approved for such applications shall be used.

E-2 Jointing Procedure :-

E-2.1 The joint operation consists briefly of removal of the serving and/or sheathing, armour where provided and core insulation of the cable (s), jointing the conductor and completing the joint either with compound in of *XLPE / PVC cable* or with cold resin epoxy for *XLPE / PVC cables*. Jointing in more than one cable laid in the same trench should be staggered as shown in Figure 12.

E-2.2 Jointing of Conductor :-

Aluminium is now the conductor material in general use. The jointing of aluminium conductors is an important step and the methods generally adopted, irrespective of the type of insulation of the cable are

- (a) Soft soldering
- (b) Welding
- (c) Mechanical compression.

Brief particulars on the above methods of jointing conductors are given in paras F2.2.2., F-2.2.3, and F- 2.2.4 respectively. For complete details reference be made to IS : 1255-1983.

E-2.2.2 Soft Soldering Method:-

E-2.2.2.1 In order to ensure that the wires are firmly embedded solder, the strands at the joint end of the cable are fanned out, after removing the insulation about 10mm more than the length required for the conductor jointing. Individual strands shall be cleaned thoroughly by a Scraper to remove any impregnation compound or oil. The conductor shall be preheated by basting with solder in the interstices of the strands and then applying the aluminium solder flux with a stiff brush; particular care being taken to ensure that the flux and solder reach the underside of the conductor .The flux removes the oxide film and assists the spreading of molten solder. The conductor shall be basted with solder several times and the process repeated till a shining surface is obtained. Basting, which consists of pouring of liquid (solder in this case) with a small ladle, is done usually from the insulated end to the cut end of the conductor to push away any traces of flux left in the interstices of the conductor. The fanned out strands, are closed back together in a circular shape and excess solder wiped out. The ferrule which is generally weak back type and is of copper shall also be tinned with aluminium solder and flux and fitted on to the conductor firmly but not completely. The

ferrule is then basted with solder to allow the solder to fill in between the conductor and ferrule properly. Excess solder is wiped out. As the oxide film which is a bad conductor of electricity forms rapidly over any surface of pure aluminium exposed to atmosphere the whole process should be carried out skilfully and quickly.

In case of termination with a cable lug, a similar procedure is adopted. However in the case of large sockets (say for 50 sq. mm. and above) a hole is drilled at the closed end of the lug before tinning the same. After the conductor is inserted in the lug, the space between the lug and the conductor insulation filled with fiber glass/ asbestos, held in position a flame resistant tape, to save the insulation from being damaged by the molten solder. Molten solder poured through the drilled hole. Gentle tapping of solder point would facilitate the setting of the solder inside the lug. The pouring is continued till the solder flows freely. Excess solder is wiped and the assembly is allowed to cool without disturbance. Particular care is required when jointing PVC cables, ensure that the PVC insulation/sheath are not damaged due to excessive heat.

The most important precaution in this method is maintenance of temperature of solder within Limit, in accordance with the recommendations of the manufactures. The liquids temperature for solder is about 220 degrees C and its solid temperature, 150 degrees C, Care should be taken that the pouring temperature not exceeds 300 degrees C, as the flux tends to char at temperature exceeding this limit. This normally achieved by maintaining the pot temperature at 360 degrees C.

As the temperature of solder plays an important role in the jointing of conductors, any crude method of checking the adequacy of heating the strands and the solder should not be resorted to: A thermometer should invariably be used for the measurement of temperature of solder.

E-2.2.2.2 Friction soldering method is also followed for soldering conductors. Special solder sticks with cadmium crystals are used in this method. No flux is used. After heating the strands and the solder the shrinks and strands are wiped with solder stick when the cadmium scratches out the oxide film and the solder adhesives to the strands.

E-2.2.3 Welding Method

The strands of the conductor end are first welded solid. For this, the strands are cleaned with petrol or kerosene after the insulation is stripped and the strands have been fanned out. The strands are smeared with aluminium flux and then brought back to position. With the conductor end kept vertical a pair of cooling tongs, an asbestos flame shield disc, a top part sheet steel tube, or a carbon tube (mould) are placed along the conductor in the above order from the insulation end. The tube should project about 5 to 10 mm above the conductor end. The conductor is heated and welded by puddling with an iron wire and adding aluminium metal. The welded conductor strands and the solid aluminium at the end should be about 10 to 20 mm depending on the cross sectional area of the conductor.

The two conductor ends to be jointed after being welded solid are placed in a mould whose interior is coated with a mould paint. A thin coat of flux is applied to the ends and the conductors are welded by heating an aluminium wire, coated with a thin layer of flux. Care is taken to ensure that only part of the solid ends is melted in this operation. The flame shield and cooling tongs are also used as above, one set for each cable.

After welding, the mould is allowed to cool and then removed. The joint is then filed smooth. Instead of an open mould, sometimes a closed mould is used, where initial welding of individual conductor end is not necessary after cleaning of the strands and smearing them with flux. The weld should be puddled with continued heating to ensure that it fills the mould fully without any cavity etc.

In the case of end terminations, the solid welded conductor end is welded to the cable lug, using the flame shield, cooling tongs etc.

E-2.2.4 Mechanical Compression Method

A special sleeve is pressed on to the conductor ends to be jointed with some gap between the ends (or the conductor ends and the inner edge of lug as the case may be) to allow for elongation by a hydraulic/ mechanical compressor with a special die holder or compression tool. The compression is done in stages the centre to the sleeve end with an overlap of 1/3 the die width. The recommendations of manufacturers the tool in this regard should be followed.

E-2.4. Jointing of PVC Cables

E-2.4.1. Jointing Using Cable Compound

The procedure outlined above shall mutandis be applicable for outdoor jointing of PVC cables also, but the operations concerning lead sleeve etc. are not applicable as the sheathing is also of PVC. Since too hot compound could damage the PVC insulation, adequate care is to be exercised to maintain the temperature compound within recommended limits. Typical PVC Cable jointing shown in Figure 15.

E-2.4.2. Epoxy jointing

Cold pouring casting resin system for PVC cable jointing has been developed for application upto 11 KV grade cables. The compound consists of a resin base and a poly amino hardener. The two component liquids are mixed at site in accordance with the recommendation of manufacturers. In this system of jointing, the insulation etc. are removed and conductors jointed as explained in clause E-2.2. The jointed cores in case of LV/MV cables should be kept apart to avoid any flash over between them. Spacers are provided between them. Spaces are provided between cores for HV cables. No insulation is applied over the jointed cores. A cover ring is placed tight over the two cut ends of armour and soldered to the armour wires. The two rings are then jointed by a copper wire and the cut ends of armouring are bent over the rings.

Sand paper is applied to the inner sheath surface and cleaned using methyl chloride. The joint is enclosed by plastic mould which is in two parts whose ends are duly cut to match the size of cables. PVC tape is wrapped at the two places where the mould will touch the cables. The two halves are pasted together and kept clamped to avoid any air gap. The mould ends are enclosed with putty which is supplied in the joint kit.

Expiry date of resin is checked and hardener added to resin. The mixture is churned thoroughly for about 15 to 20 minutes till the colour of the mixed compound is grey. The mix is poured slowly into the mould taking care to avoid formation of air bubbles till the mould is filled and it comes out at the risers.

Allow the joint to set for minimum three hours till becomes a solid mass before changing the cable. The mould may be removed, if desired.

Normally all the components required for joints are supplied as a kit for various sizes of cables.

E-2.4.3 Termination Using Compression Glands

PVC cables upto 1.1 KV grade shall preferably be terminated in the indoor electrical switchgear using compression glands.

The nipple of the gland is first screwed to the switch gear to which the cable is to be terminated and locked with check nut from inside the housing. Compression ring, washer, rubber ring and again another washer slipped in succession over the cable. The cable sheath is removed to desired length and the armour strands splayed out. The armored wire is then cut to the overall diameter of the second washer. Sharp edges are removed and armour cleaned. A third washer is now slipped on to trap the trimmed strands and armour between the second and the third washer. The cable end is pushed through the gland nipple (body). The compression ring is then tightened when the rubber ring will expand and hold the cable tightly by the sheath. The cores are terminated in the usual manner. Typical termination by this method is shown in figure 14 illustrates a typical outdoor termination of PVC cable using compression gland with a protective M.S. box.

E-2.4.4 Jointing using Heat shrinkable insulating material for MV cables

Heat shrinkable tubes of different diameters are available, for cable jointing applications. Tube of the appropriate size and of good quality (going by the successful test certificates from Independent bodies) should only be used. After preparing the cable end for jointing, the tube is inserted over the cable end held back. The lugging/straight through conductor jointing operation is done. The surface is thoroughly cleaned. The tube is moved over the joint and heated from one end towards the other by a blow lamp, so that air bubble gets entrapped. The tube shrinks on to insulation and lug. (Cases are reported of moisture travel through the space close to the lugs and hence the covering the same is desirable.) The joint be energized, immediately after the surface has cooled down.

E-2.4.5 Jointing using Cold shrinkable insulating material for MV cables

Cold shrinkable cable jointing was done with heat shrink technology which involved applying heat to artificially expanded plastic tubes. Upon the application of heat, these plastic tubes shrunk to provide electrical insulation or jointing cables. However, heat shrink technology could not be utilized safely in all kinds of environments. For example, it was quite risky to use heat-producing tools in areas with flammable gasses, such as gas pipeline projects. Cold shrinking was developing as a safer alternative for terminating & jointing in areas with such issues.

In cold shrink technology, accessories such as joint body, terminal body and tubes are made with materials such as rubbers and silicones that offer a high level of elasticity at ambient temperatures and are excellent at connecting and sealing power cables. The cold shrink tubes have a removable inner support system, which, when pulled out, causes the tube to shrink to its pre-stretched size. The tube wraps the cable perfectly and constantly

exerts radial pressure for the lifetime of the joint or termination. For correct utilization of cold shrink tubes, technician must only remove the inner core during the installation stage and not before. The cold shrink tubes must remain in their expanded state before they are actually being used for installation.

Cold shrink technology is optimized for field use and can withstand environmental conditions such as UV, pollution and chemicals, and salt fog. It can be utilized easily in onshore, offshore, indoor and outdoor environments. Furthermore, it is more suitable than heat shrink technology in places with space constraints where electrical tools and equipment cannot be used easily. Owing to these excellent practical features, cold shrink technology is experiencing widespread adoption in the electrical industry. Cold Shrink Cable Joints, Cable Terminations, Flexible Bushing Boots, Trifurcation Kit for 3 Core MV Cables, Silicone Rubber Cold Shrink Tubing, EPDM Cold Shrink Tubing, Breakout Boots and End Caps, all materials are made with only good quality raw materials and going by the successful test certificates from Independent bodies.

E-2.5 Use of heat shrinkable joints on High Voltage joints.

This type of joint can be adopted indoor/outdoor, for HV upto and including 33 KV (at present). The size of cable, as well as voltage of system should be indicated while ordering. In the case of HV cables, the electrostatic stresses tend to concentrate near the screen which is earthed, at the location where the cable is cut (for any type of joint). This stress has to be relieved, in order to avoid failure of insulation at that location. Semiconducting tapes can be used for the purpose, but if the workmanship is not satisfactory, there can be entrapped air pockets where partial discharge can occur. Heat shrinkable tube with semiconducting properties can be provided there and shrunk. A further insulating tube is heat shrunk over the same. In HV joints, particular precautions are taken to (i) stress relief at cut ends, (ii) prevention of moisture entry at lug position and (iii) build up of the required insulation with the outer covering having antitracking property. Heat shrinkable joint can be used for PVC and XLPE cables, for end terminations as well as for straight through/tee joints. The procedure, though apparently simple, needs to be carefully and skilfully followed to avoid failures later. Assistance may be taken from Manufacturer's representative.

E-2.6. Jointing of XLPE (Cross Linked Polythene insulated) Cables

E-2.6.1 XLPE cable is highly resistant to moisture, hence jointing and terminations of XLPE cable is similar than PILC cables. Upto 3.3 KV grade, cable can be terminated straight away with compression glands similar to other PVC cables.

E-2.6.2 Jointing of Conductors

Since the maximum permissible short circuit temperature is 250 degrees C in XLPE cables, soldered type joints become unsuitable as solder melts at much lower temperature. Welded or crimped/ compression type conductor joints are therefore adopted for XLPE cables.

E-2.6.3 Insulation Build Up

The conductor joint is insulated lapping of EPR self-amalgamated (SA) tape to required thickness. S.A. tape is universally accepted as most reliable jointing materials because of its resistance to water and ozone, capacity to operate at higher temperature, high dielectric strength and void amalgamating properties.

For cables above 3.3 KV grade semiconducting S.A. tape is to be applied on the conductor before starting insulation build up to equalize distribution of electrical stress on conductor surface.

The insulated jointed cores are further lapped with semi conducting SA tape to serve as core screening. For cable rated at 3.3 KV and below use of semiconducting SA tape is not necessary.

E-2.6.4 Straight Through Joints

The jointed assembly (with insulated cores) is encapsulated in special cast resin compound.

E-2.6.5 Stress Relieved Termination

The termination of cables above 3.3 KV is provided with stress cone to relieve electrical stresses formed on the insulation at the point of termination of cable. SA tape is lapped over the insulation to the required thickness to build up the stress cone.

E-2.6.6. Indoor Type Termination

Indoor type termination for cables above 3.3 K is provided with stress cone but it does not require any compound for encapsulation. However, it must be protected by lapping of adhesive PVC for protection against external abuses.

E-2.6.7. Outdoor Termination

Outdoor type termination for cable above 3.3 KV is provided with stress cone but this termination is encapsulated in special cast compound to protect the joint from atmosphere abuses.

Note:- The above procedure is only indicative of the general requirements. However, the procedure detailed by the manufacturer as suitable for the type of cables, and for the type of joints shall be strictly followed.

APPENDIX-F

(Clause 4.2.3.1 (iii))

Selection of stay sets

F-1 General

Stay sets are required to be provided in overhead line work at locations where there is likely to be a horizontal pull from the line, on a pole, as for example, terminal poles, poles at locations of deviations of a line etc. Stay sets are provided at Tee-off positions and steep gradient locations also. The horizontal component of tension in the stay wire provides the balancing force, so that the stability of the pole is not adversely affected.

F-2 Size of stay wire

Stay wire is a multistranded galvanized steel wire. The sizes adopted in this specification are 7/3.15 mm and 7/4 mm dia. In certain other standards, a size of 7/2.5mm dia is also adopted. The strength of these wires is as under:

Size	Ultimate strength	Permissible strength
.....		
7/2.5mm	2300KG	920KG
7/3.15 mm	3625KG	1450KG
7/4mm	5500KG	2200KG

F-3 Force from the Line

The pull experienced by a pole is governed by the governed by the number of conductors in the line, their sizes, span, and wind force on the conductor and deviation angle of the line. The force can be calculated from these parameters in individual cases. The tension calculation is elaborated in IS: 5613 (Pat 1/sec1)-1985.

F-5 Stay selection

- (i) The stay angle is selected from 30 to 60 degrees to vertical, 45 degrees being common. Higher the angle, higher will be the horizontal component, but beyond a limit the anchoring becomes less effective. Depending on the soil condition, the angle may be around 45 degrees.
- (ii) The tension needed to counter the pull can be calculated. After applying factor of safety, the size may be selected for stay wire. There may be a need to go for 2 stay sets at the same support, if the calculated tension is such that the permissible tension will be exceeded in case a single stay set is proposed.
- (iii) Similarly, the expected tension for the stay wire at angular deviations can be evaluated, from principles of Mechanics, taking into consideration whether the stay set will be located at the angular bisector of the deviation or individually for each arm of the deviation. The size and number are then decided.
- (iv) Standardized charts are also available in the standards like REC standard.

F-6 Two/ Three/ Four Pole structure locations:

- (i) For 2/3 pole structure locations, minimum 2 stays in each direction of the line or in the angular bisector should be provided.
- (ii) Normally no stays may be required at 4 pole structure locations, as the forces are generally balanced. The requirement may however be examined in individual cases.

Note: If cradle guard is purposed below the 11 KV line, additional Z clamps should be provided from the ends of cross arm, with horizontal conductors of 30 cm from the outer conductors.

Appendix G

LIST OF TABLES

TABLE 1 (A)

(Clause 2.4, 2.6, 2.7.16, 2.8.2)

CHART SHOWING THE DISTANCE UPTO WHICH DIFFERENT SIZES OF U.G. ALUMINIUM CONDUCTOR CABLES CAN BE USED FOR DIFFERENT CURRENT RATINGS FOR 8 VOLTS DROP WHEN LAID 'IN GROUND (XLPE INSULATED, PVC SHEATHED, 3 CORE OR 4 CORE) WHEN CABLE GRADING IS 1.1 KV)

(MAXIMUM CONDUCTOR TEMPERATURE 90 DEGREE C)

S. No	Current Amp.	Distance in Meters for the following cable sizes in Sq.mm.															
		6	10	16	25	35	50	70	95	120	150	185	240	300	400	500	630
1	5	156	233	377	599	832	1117	1609	2209	2765	3346	4123	5189	6158	7390	8475	9623
2	10	78	116	188	299	416	558	804	1104	1382	1673	2061	2594	3079	3695	4238	4811
3	15	52	77	125	199	277	372	536	736	921	1115	1374	1729	2052	2463	2825	3208
4	20	39	58	94	149	208	279	402	552	691	836	1030	1297	1539	1848	2119	2406
5	25	31	46	75	119	166	223	321	441	553	669	824	1037	1231	1478	1695	1925
6	30	26	38	62	99	138	186	268	368	460	557	687	864	1026	1232	1413	1604
7	40	19	29	47	74	104	139	201	276	345	418	515	648	769	924	1059	1203
8	50	-	23	37	59	83	111	160	220	276	334	412	518	615	739	848	962
9	60	-	-	31	49	69	93	134	184	230	278	343	432	513	616	706	802
10	70	-	-	26	42	59	79	114	157	197	239	294	370	439	528	605	687
11	80	-	-	-	37	52	69	100	138	172	209	257	324	384	462	530	601
12	90	-	-	-	33	46	62	89	122	153	185	229	288	342	411	471	535

13	100	-	-	-	-	41	55	80	110	138	167	206	259	307	370	424	481
14	110	-	-	-	-	37	50	73	100	125	152	187	235	279	336	385	437
15	120	-	-	-	-	-	46	67	92	115	139	171	216	256	308	353	401
16	130	-	-	-	-	-	42	61	84	106	128	158	199	236	284	326	370
17	140	-	-	-	-	-	-	57	78	98	119	147	185	219	264	303	344
18	150	-	-	-	-	-	-	53	73	92	111	137	172	205	246	283	321
19	160	-	-	-	-	-	-	50	69	86	104	128	162	192	231	265	301
20	170	-	-	-	-	-	-	-	64	81	98	121	152	181	217	249	283
21	180	-	-	-	-	-	-	-	61	76	92	114	144	171	205	235	267
22	190	-	-	-	-	-	-	-	58	72	88	108	136	162	194	223	253
23	200	-	-	-	-	-	-	-	-	69	83	103	129	153	185	212	241
24	225	-	-	-	-	-	-	-	-	-	74	91	115	136	164	188	214
25	250	-	-	-	-	-	-	-	-	-	-	82	103	123	148	170	192
26	275	-	-	-	-	-	-	-	-	-	-	74	94	111	134	154	175
27	300	-	-	-	-	-	-	-	-	-	-	-	86	102	123	141	160
28	350	-	-	-	-	-	-	-	-	-	-	-	-	-	106	121	137
29	400	-	-	-	-	-	-	-	-	-	-	-	-	-	92	106	120
30	450	-	-	-	-	-	-	-	-	-	-	-	-	-	-	94	107
31	500	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	96

Note 1: XLPE Insulated electrical cable for voltage grade upto 1.1 KV is based on 8 volts drop consideration.

1. This table based on technical data sheet by M/s Havells India Ltd.

2. The distances are given in meters and after rounding.

3. The conditions of installation of cable is ground temp. 30 degree C and ambient temperature 40 degree

Note 2: Rating factors for Variation in ground Temperature of XLPE cable.

Ground:	20° C	25° C	30° C	35° C	40° C	45° C	50° C
Rating Factors:	1.08	1.04	1.00	0.96	0.91	0.87	0.82

TABLE 1 (B)
(CLAUSE 2.4.3)
**PERMISSIBLE MAXIMUM SHORT CIRCUIT
CURRENT RATINGS FOR XLPE CABLES**

CONDUCTOR AREA	SHORT CIRCUIT RATINGS FOR ONE SECOND DURATION	
	COPPER CONDUCTOR	ALUMINIUM CONDUCTORS
Sq.mm.	A	B
16	2570	1730
25	3970	2670
35	5500	3690
50	7800	5220
70	10850	7400
95	14600	9740
120	18400	12200
150	23000	15200
185	28200	18700
240	36400	24200
300	45300	30100
400	60200	39900
500	74800	49800
630	92700	62000
800	--	78800
1000	--	97800

Initial conductor temperature - 90 Deg. C.

Final conductor temperature - 250 Deg. C.

For durations other than one second the short circuit current may be calculated from the following formula –

$$I_{sc} = \frac{I}{\sqrt{t}}$$

Where I_{sc} - Short circuit current during time t, amperes

I - Short circuit current during the time one second as given in above table.

t - Short circuit current duration, seconds.

Note : For Large currents the force between the conductors must be considered especially when single core cable are used.

TABLE 1 (C)
(CLAUSE 2.4)
**SHORT-CIRCUIT RATINGS OF ALUMINIUM CONDUCTOR FOR HIGH
VOLTAGE PVC CABLES**

(Applicable to single-core and three-core cables)
[AS PER IS STANDARD: 5819-1970]

S.N.	Cross Section of Conductor (mm ²)	Short-Circuit Rating for 1 Second Duration in kA (rms)			
		3.3 KV (T = 150° C) T ₀ = 70°C)	6.6 KV (Earthed) (T = 150° C) T ₀ = 70° C)	6.6 KV (Unearthed) (T = 150° C) T ₀ = 70° C)	11 KV (Earthed) (T = 140° C) T ₀ = 60° C)
(1)	(2)	(3)	(4)	(5)	(6)
1	25	1.80	1.80	1.80	1.83
2	35	2.52	2.52	2.52	2.56
3	50	3.60	3.60	3.60	3.65
4	70	5.04	5.04	5.04	5.11
5	95	6.84	6.84	6.84	6.94
6	120	8.64	8.64	8.64	8.76
7	150	10.80	10.80	10.80	11.00
8	185	13.30	13.30	13.30	13.50
9	225	16.20	16.20	16.20	16.40
10	240	17.30	17.30	17.30	17.50
11	300	21.60	21.60	21.60	21.90
12	400	28.80	28.80	28.80	29.20
13	500	36.00	36.00	36.00	36.50
14	625	45.00	45.00	45.00	45.60
15	800	57.60	57.60	57.60	58.40
16	1000	72.00	72.00	72.00	73.00

T – Maximum short-circuit temperature of conductor.

T₀ – Temperature of conductor prior to short-circuit.

For any duration t seconds, multiply the above ratings by 1/√t

TABLE 1 (D)
AC RESISTANCE, REACTANCE AND IMPEDANCE PARAMETERS OF
ALUMINIUM CONDUCTOR FOR MV XLPE AND PVC CABLES

Size of Conductor in Sq.mm	XLPE insulated			PVC Insulated		
	AC Resistance (R)	Reactance (X)	Impedance (Z)	AC Resistance (R)	Reactance (X)	Impedance (Z)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
6	5.910	0.089	5.910	5.540	0.097	5.540
10	3.950	0.084	3.950	3.700	0.091	3.700
16	2.450	0.081	2.450	2.300	0.086	2.300
25	1.540	0.081	1.540	1.440	0.085	1.440
35	1.110	0.079	1.110	1.040	0.083	1.040
50	0.823	0.078	0.827	0.771	0.083	0.775
70	0.569	0.074	0.574	0.533	0.077	0.539
95	0.412	0.072	0.418	0.386	0.077	0.394
120	0.326	0.072	0.334	0.306	0.075	0.315
150	0.266	0.073	0.276	0.249	0.075	0.260
185	0.212	0.072	0.224	0.199	0.074	0.212
240	0.163	0.071	0.178	0.153	0.074	0.170
300	0.132	0.071	0.150	0.123	0.074	0.144
400	0.104	0.070	0.125	0.098	0.073	0.122
500	0.083	0.070	0.109	0.078	0.073	0.107
630	0.066	0.070	0.096	0.062	0.073	0.096

TABLE II
[CLAUSE 2.4, 2.6.1. (b)]
Minimum Permissible Bending Radius for cables.
[AS PER IS: 1255 STANDARD-1983]

Voltage Rating (in KV)	PVC and XLPE Cables	
	Single Core	Multi Core
(1)	(2)	(3)
Up to 1.1	15 D	12 D
Above 1.1 to 11	15 D	15 D
Above 11	20 D	15 D

“D” Is the overall diameter of the cable.

TABLE III
(2.4, 2.6.11.3)
Clamping of cables on surface

Type of cables	Size	Clamping by	Fixing Intervals
MV	Upto and including 25sq.mm.	Saddles 1mm thick	45cm
MV & HV	35 sq.mm to 120 sq.mm.	Clamps 3mm thick 25mm wide	60cm
MV & HV	150 sq.mm and above	Clamps 3mm thick 40mm wide	60cm

Note: The fixing intervals specified apply to straight runs. In the case of bends, additional clamping shall be provided at 30cm from the center of the bend on both sides.

TABLE IV
[CLAUSE 2.4, 2.6.12.2 (d)]
Schedule of permissible loads for cable trays (M.S)

The figures given below represent the maximum permissible uniformly distributed load in Kgs. Per running meter for different unsupported free spans.

Channel Section size			Unsupported Free Span			
Width (mm)	Depth (mm)	Thickness (mm)	800 mm	1200 mm	1800 mm	2500 mm
100	50	1.6	1430	355	156	87
150	50	1.6	1458	362	159	88
225	50	1.6	1498	371	160	89.4
300	50	1.6	1540	380	162	91.3
375	50	2.0	1955	483	210	116.7
450	50	2.0	1958	483	210	110.7
600	50	2.0	1964	481	208	110.4
300	62.5	2.0	2680	664	290	161.3
375	62.5	2.0	2685	664	290	158.9
450	62.5	2.0	2689	664	289	161.3
600	62.5	2.0	2698	664	289	157.1
750	62.5	2.0	2707	666	287	155.7
900	62.5	2.0	2716	667	287	153.7
600	75.0	2.0	3491	861	377	206.5
750	75.0	2.0	3513	868	377	205.02
900	75.0	2.0	3535	874	378	205.02

Note 1 – (i) The maximum permissible load at mid-span may be obtained from the formula.

$$P = (\frac{1}{2}) \times U \times S$$

Where, P = Maximum permissible point load in Kgs. at mid-span.

U = Maximum permissible uniformly distributed load in Kg. per running meter,
and,

S = The particular free span in meter.

(ii) In case of a span not listed above, the corresponding maximum permissible uniformly distributed load may be found from the formula.

$$U = 4 \times (U \text{ at } 2 \text{ m. span}) / S^2$$

(iii) Free spans exceeding 16 ft. are not recommended.

TABLE V
(CLAUSE 2.4, 2.7.7)
TYPES OF CABLE JOINTS

S.No.	Type of Joint	PVC/ XLPE cables up to 1.1KV	PVC/ XLPE cables up to 11KV & above
(1)	(2)	(3)	(4)
1.	Epoxy resin	Yes	Yes
2.	Heat shrinkable	Yes	Yes
3.	Cold shrinkable	Yes	Yes
4.	Cast resin based	Yes	Yes

TABLE VI
(CLAUSE 2.4, 2.8.2.1)
Test pressure in KV

Working Volts in KV	AC 15 minutes test		DC 15 minutes test	
	Between conductors	Conductors to earth	Between conductors	Conductors to earth
Upto 1.1	2.0	2.0	3.0	3.0
3.3	6.0	3.5	9.0	5.0
6.6	12.0	7.0	18.0	10.5
11	20.0	11.5	30.0	17.5
22	40.0	23.0	60.0	35.0
33	-	-	-	60.0

TABLE VII
(CLAUSE 2.4)
Current ratings (a.c.) for Three, Four and Five core general purpose 70°C
PVC insulated cables, 1100 V [AS PER IS: 3961 (PART 2) : 2017]

Sl. No.	Nominal area of Conductor mm ²	Buried direct in the ground		Buried Duct		In air	
		Copper	Aluminium	Copper	Aluminium	Copper	Aluminium
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	1.5	21	18	18	15	17	14
ii)	2.5	28	22	24	19	23	18
iii)	4	37	29	31	25	30	24
iv)	6	46	38	39	32	39	32
v)	10	61	47	51	40	53	40
vi)	16	78	61	66	51	67	52
vii)	25	101	78	85	66	90	70
viii)	35	121	94	102	79	110	85
ix)	50	143	111	120	93	134	104
x)	70	175	136	147	115	169	131
xi)	95	210	163	177	138	209	162
xii)	120	237	185	200	156	238	186
xiii)	150	265	206	224	175	272	212
xiv)	185	300	234	254	198	314	245
xv)	240	345	271	293	230	371	291
xvi)	300	387	305	330	260	425	335
xvii)	400	436	348	372	297	490	390
xviii)	500	488	395	424	343	560	452
xix)	630	544	447	473	389	640	525

TABLE VIII
(CLAUSE 2.4)
Current ratings (a.c.) for Three, Four and Five core Heat Resisting 85°C PVC insulated cables, 1100 V [AS PER IS: 3961 (PART 2) : 2017]

Sl. No.	Nominal area of Conductor mm ²	Buried direct in the ground		Buried Duct		In air	
		Copper	Aluminium	Copper	Aluminium	Copper	Aluminium
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	1.5	25	20	21	17	21	17
ii)	2.5	32	25	27	21	28	22
iii)	4	42	33	36	28	38	30
iv)	6	52	43	44	36	48	39
v)	10	70	53	59	45	65	50
vi)	16	90	69	75	58	83	65
vii)	25	116	90	97	75	111	86
viii)	35	139	108	116	90	137	106
ix)	50	164	127	138	107	166	129
x)	70	201	156	169	131	209	163
xi)	95	241	187	203	157	259	201
xii)	120	272	212	229	178	296	230
xiii)	150	304	236	257	200	337	262
xiv)	185	343	268	291	227	389	304
xv)	240	396	310	336	264	459	360
xvi)	300	445	350	379	298	527	415
xvii)	400	502	399	428	340	608	483
xviii)	500	563	453	489	394	696	560
xix)	630	629	514	547	447	796	650

TABLE IX
(CLAUSE 2.4)
Current ratings (a.c.) for Three, Four and Five core cables with XLPE Insulation and
Rated voltage 1100 V [AS PER IS: 3961 (PART 6) : 2016]

Sl. No.	Nominal area of Conductor mm ²	Buried direct in the ground		In single way Ducts		In air	
		Copper	Aluminium	Copper	Aluminium	Copper	Aluminium
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	1.5	26	22	22	18	23	19
ii)	2.5	34	27	29	23	30	24
iii)	4	45	35	38	30	41	32
iv)	6	56	46	47	38	52	42
v)	10	74	57	62	48	70	54
vi)	16	95	74	79	61	89	69
vii)	25	122	95	102	79	119	93
viii)	35	146	114	122	94	147	114
ix)	50	173	134	144	112	179	138
x)	70	212	164	177	137	226	175
xi)	95	254	197	212	164	279	216
xii)	120	287	223	240	187	320	249
xiii)	150	321	249	269	209	365	284
xiv)	185	362	282	304	238	422	329
xv)	240	418	327	352	276	500	392
xvi)	300	469	369	396	312	574	452
xvii)	400	528	420	447	356	662	526
xviii)	500	593	478	511	412	760	612
xix)	630	661	542	571	468	870	712

TABLE X**(CLAUSE 2.4)****Current ratings for Three core cables with XLPE Insulation, Aluminium Conductor
and Rated voltage 6.6/6.6 KV to 11/11 KV [AS PER IS: 3961 (PART 7) : 2017]**

Sl. No.	Nominal area of Conductor mm²	Buried direct in the ground	In a Buried Duct	In air
(1)	(2)	(3)	(4)	(5)
i)	25	94	81	103
ii)	35	112	97	124
iii)	50	131	114	148
iv)	70	161	139	184
v)	95	190	165	222
vi)	120	216	188	256
vii)	150	242	209	288
viii)	185	273	240	330
ix)	240	315	278	387
x)	300	354	312	441
xi)	400	404	356	512
xii)	500	457	403	590

TABLE XI
(CLAUSE 2.4)

Rating factors for Variations in Ambient Air Temperature for cables in Free Air
[AS PER IS: 3961 (PART 6) : 2016]

Maximum contractor tempature	Ambient air temperature							
°C	25	30	35	40	45	50	55	60
90	1.14	1.10	1.05	1.00	0.95	0.89	0.84	0.77

TABLE XII
(CLAUSE 2.4)

Rating factors for Variations in Ground Temperature for Direct Buried cables
[AS PER IS: 3961 (PART 6) : 2016]

Maximum conductor Temperature	Ground Temperature							
°C	15	20	25	30	35	40	45	50
90	1.12	1.08	1.04	1.00	0.96	0.91	0.87	0.82

TABLE XIII
(CLAUSE 2.4)
Rating factors for Depths of Laying for direct buried cables
[AS PER IS: 3961 (PART 6) : 2016]

Sl. No.	Depth of Laying in mm	Upto 25sq.mm.		Above 25sq.mm. and upto 300sq.mm.		Above 300sq.mm.	
		Single core	Multi core	Single core	Multi core	Single core	Multi core
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	750	1.00	1.00	1.00	1.00	1.00	1.00
ii)	900	0.98	0.98	0.98	0.98	0.98	0.98
iii)	1050	0.97	0.98	0.96	0.97	0.96	0.96
iv)	1200	0.96	0.97	0.95	0.95	0.94	0.95
v)	1500	0.94	0.95	0.93	0.93	0.92	0.93
vi)	1800	0.93	0.93	0.91	0.92	0.90	0.91
vii)	2000	0.92	0.93	0.90	0.91	0.89	0.90
viii)	2500	0.90	0.92	0.89	0.89	0.87	0.88
ix)	3000	0.90	0.90	0.87	0.88	0.86	0.87

TABLE XIV
(CLAUSE 2.4)

Rating factors for Variations in Soil Thermal Resistivity for multi-core cables Laid Direct in Ground [AS PER IS: 3961 (PART 6) : 2016]

Sl. No.	Nominal area of Conductor mm ²	Values of Soil Thermal Resistivity Km/W					
		1.0	1.2	1.5	2.0	2.5	3.0
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	1.5	1.14	1.08	1.0	0.90	0.83	0.77
ii)	2.5	1.15	1.08	1.0	0.90	0.82	0.76
iii)	4	1.15	1.08	1.0	0.89	0.82	0.76
iv)	6	1.16	1.09	1.0	0.89	0.81	0.75
v)	10	1.16	1.09	1.0	0.89	0.81	0.75
vi)	16	1.17	1.09	1.0	0.89	0.80	0.74
vii)	25	1.17	1.09	1.0	0.89	0.80	0.74
viii)	35	1.18	1.10	1.0	0.88	0.80	0.74
ix)	50	1.18	1.10	1.0	0.88	0.80	0.74
x)	70	1.18	1.10	1.0	0.88	0.80	0.74
xi)	95	1.18	1.10	1.0	0.88	0.80	0.73
xii)	120	1.18	1.10	1.0	0.88	0.80	0.73
xiii)	150	1.18	1.10	1.0	0.88	0.80	0.73
xiv)	185	1.18	1.10	1.0	0.88	0.80	0.73
xv)	240	1.19	1.10	1.0	0.88	0.80	0.73
xvi)	300	1.19	1.10	1.0	0.88	0.80	0.73
xvii)	400	1.19	1.10	1.0	0.88	0.80	0.73
xviii)	500	1.19	1.10	1.0	0.88	0.80	0.73
xix)	630	1.19	1.10	1.0	0.88	0.80	0.73

TABLE XV
(CLAUSE 2.4)
Group Rating factors for multi-core cables Laid Direct in the Ground, in Tier
Formation [AS PER IS: 3961 (PART 6) : 2016]

Sl. No.	Number of Cables	Number of Tiers	Spacing Between Group Centre mm				
			Touching	150	300	450	600
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	2	1	0.80	0.84	0.87	0.90	0.91
ii)	3	1	0.68	0.74	0.79	0.83	0.86
iii)	4	2	0.60	0.66	0.73	0.77	0.79
iv)	5	2	0.55	0.61	0.68	0.71	0.73
v)	6	2	0.51	0.57	0.63	0.67	0.69
vi)	7	3	0.48	0.54	0.59	0.63	0.64
vii)	8	3	0.46	0.51	0.56	0.60	0.61
viii)	9	3	0.44	0.48	0.53	0.57	0.58
ix)	10	4	0.42	0.47	0.52	0.55	0.56
x)	11	4	0.41	0.46	0.50	0.54	0.55
xi)	12	4	0.40	0.45	0.49	0.53	0.54

TABLE XVI

(CLAUSE 2.4)

Group Rating factors for multi-core cables Laid Direct in the Ground, in Horizontal Formation [AS PER IS: 3961 (PART 6) : 2016]

Sl. No.	Number of Circuits	Spacing Between Duct Group Centre mm				
		Touching	150	300	450	600
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	2	0.80	0.84	0.87	0.90	0.91
ii)	3	0.68	0.74	0.79	0.83	0.86
iii)	4	0.62	0.69	0.75	0.80	0.83
iv)	5	0.58	0.65	0.72	0.77	0.80
v)	6	0.55	0.62	0.69	0.75	0.78
vi)	7	0.42	0.59	0.67	0.73	0.77
vii)	8	0.50	0.57	0.66	0.72	0.75
viii)	9	0.48	0.55	0.65	0.71	0.75
ix)	10	0.46	0.54	0.64	0.70	0.74
x)	11	0.45	0.53	0.63	0.70	0.74
xi)	12	0.44	0.52	0.62	0.69	0.73

TABLE XVII
(CLAUSE 6.10.2)
Lighting Installation in Group A and B Roads

Sl. No	Description	Group A1	Group A2	Group B2	Group B2
(1)	(2)	(3)	(4)	(5)	(6)
i)	Average level of illumination on road surface, lux	30	15	8	4
ii)	Ratio minimum/average illumination ratio	0.4	0.4	0.3	0.3
iii)	Transverse variation of illumination, percent	33	33	20	20
iv)	Preferred Type of luminaire:	Cut-off	Cut-off	Cut-off or semi cut-off	Cut-off or semi cut-off
	Permitted	Semi-cut-off	Semi cut-off	Non cut-off	Non cut-off
v)	Mounting height, m	9-10	9-10	7.5-9	7.5-9
vi)	Maximum spacing of luminaire/height ratio	Cut-off = 3 Semi-cut-off = 3.5		Semi-cut-off = 3.5 Non-cut-off = 4	

NOTES

1. In Group A lighting, the level and uniformity of illumination shall be as high as possible and the glare strictly reduced.
2. In Group B lighting, greater tolerances on uniformity and glare are admitted, which may be justified by the character of the roads and by the presence of facades.
3. Mounting heights less than 7.5 m are undesirable except in special cases, such as lighting of residential roads or roads bordered by trees.

TABLE – XVIII
(CLAUSE 6.10.2)
Lighting Installations in Group D Roads

Sl. No.	Description	Remark
	1. Grade Separated Junctions	
	i) Lighting by conventional street lighting technique Complex junctions:	
	a) General principles	See IS 1944 (Parts 1 and 2)
	b) Mounting height, m	10-12
	c) Luminaire	cut-off
	d) Light sources	LED
	2. Lighting by high-mast lighting	
	a) Minimum service level value (lux)	30
	b) Uniformity ratio (avg)	0.4
	c) Height of masts, m	Not less than 20
3. Short bridges (<60 m)		Normal street lighting techniques with minor adjustments
		[see IS 1944 (Part 5)]
Long bridges (> 60 m)		See IS 1944 (Part 5)
Bridges of historical importance		Special considerations apply
Parapet lighting		
a)	Mounting height, m	Not greater than 1
b)	Separation between rows of lighting, m	Not greater than 12
c)	Choice of lamps	LED, HPSV lamps or other linear sources of luminance
		weatherproof, dustproof, vermin proof, robust
Footbridges		
a)	Illumination, lux	Not less than 6
<i>Elevated Roads</i>		
a)	General lighting	Like heavy traffic roads
b)	Choice of luminaire	Cut-off

TABLE – XIX
(CLAUSE 6.10.2)
Lighting Installation in Group E Road

Sl. No. (1)	Description (2)	Remarks (3)
i)	<i>Main squares [see IS 1944 (Part 6)]:</i>	
	1) Average level of illumination, lux 2) Mounting height, m	20 10-15
ii)	<i>Shopping streets/promenades</i>	Special considerations
iii)	<i>Pedestrian precincts</i> 1) Average horizontal illuminance (footway level), lux 2) Average horizontal illuminance (under canopies), lux 3) Mounting height, m 4) Luminaires	20 50 5-6 Special considerations
iv)	<i>Public car parks (above ground)</i> 1) Average horizontal illuminance, lux 2) Luminaires preferred	10 Floodlight luminaries on high masts
v)	<i>Pedestrian stairways/footbridges</i> 1) Average level of illuminance (surface of footbridge), lux 2) Average level of illuminance (sub-ways), lux	6 10

Table XX
(CLAUSE 2.4.4)
Voltage Drop Limitation

The ECBC specifies the provisions for the power cabling for the Power Distribution Systems:

The power cabling shall be sized so that the distribution losses do not exceed

- (a) 3% of the total power usage in ECBC Buildings
- (b) 2% of the total power usage in ECBC+ Buildings
- (c) 1% of total power usage in Super ECBC Buildings

Record of design calculation for the losses shall be maintained. Load calculation shall be calculated up to the panel level.

In the design / selection of the cables, the consideration of the power distribution losses as defined in the ECBC shall be complied with, subject to the condition that the voltage drop permissible shall not be more than 8 volts.

TABLE XXI
(CLAUSE 4.2.2.1 (ii))
CROSS ARM LENGTHS

Line voltage	No. of Horizontal conductors	Length of cross arm
LV/MV	2	55cm
LV/MV	4	115cm
LV/MV	4 + Guard	175cm
11KV	3	230cm

Note:- If Cradle guard is proposed below the 11 KV line, additional Z clamps should be provided from the ends of cross arm, with horizontal conductors of 30 cm from the outer conductors.

TABLE-XXII
(CLAUSE 4.2.6.1)
OVERHEAD CONDUCTOR DETAILS

ALL ALUMINUM STRANDED CONDUCTORS (A.A.C.) (AS PER IS 398(PART-I)1996 (THIRD REVISION))

Nominal Aluminium conductor cross section area	Number of Strands and Diameter of each strand	Sectional Area	Approximate Overall Dia	Approximate Mass	Calculated Resistance at 20°C Max	Approximate Calculated Breaking Load
(1)	(2)	(3)	(4)	(5)	(6)	(7)
mm ²	mm	mm ²	mm	kg/km	ohms/km	kN
25	7/2.21	26.85	6.63	74	1.096	4.52
50	7/3.10	52.83	9.30	145	0.5525	8.25
100	7/4.39	106.0	13.17	290	0.2752	15.96
150	19/3.18	150.9	15.90	415	0.1942	23.28
240	19/3.99	237.6	19.95	654	0.1235	35.74
300	19/4.65	322.7	23.25	888	0.09107	48.74

Note: The sectional area of a stranded conductor has been taken as the sum of the cross-sectional areas of the individual wires.

TABLE-XXIII
(CLAUSE 4.2.6.1)
ALUMINUM CONDUCTOR, STEEL-REINFORCED (A.C.S.R.) (AS PER IS:398
(PART-2)-1996 (THIRD REVISION)

Nominal cross section area of Aluminium conductors	Number of Strands and Diameter of each strand		Cross Sectional Area of aluminium conductors	Total Cross Sectional area	Approximate diameter	Approximate Mass	Calculated Resistance at 20°C Max	Approximate Calculated Breaking Load
	Aluminium	Steel						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
mm ²	mm	mm	mm ²	mm ²	mm	kg/km	ohms/k m	kN
10	6/1.50	1/1.5 0	10.60	12.37	4.50	43	2.780	3.97
18	6/1.96	1/1.9 6	18.10	21.12	5.88	73	1.618	6.74
20	6/2.11	1/2.1 1	20.98	24.48	6.33	85	1.394	7.61
30	6/2.59	1/2.5 9	31.61	36.88	7.77	128	0.9289	11.12
50	6/3.35	1/3.3 5	52.88	61.70	10.05	214	0.5524	18.25
80	6/4.09	1/4.0 9	78.83	91.97	12.27	319	0.2792	26.91
100	6/4.72	7/1.5 7	105.0	118.5	14.15	394	0.2792	32.41
150	30/2.59	7/2.5 9	158.1	194.9	18.13	726	0.1871	67.34
200	30/3.00	7/3.0 0	212.1	261.5	21.00	974	0.1390	89.67
400	42/3.50	7/1.9 6	404.1	425.2	26.88	1281	0.0731 1	88.79
420	54/3.18	7/3.1 8	428.9	484.5	28.62	1621	0.0686 8	130.32
520	54/4.13	7/3.5 3	528.5	597.0	31.77	1998	0.0559 5	159.60
560	42/4.13	7/2.3 0	562.7	591.7	31.68	1781	0.0523 1	120.16

Note:

1. The sectional area is the sum of the cross-sectional areas of the relevant individual wires.

TABLE-XXIV
(CLAUSE 4.2.6.1)

Aluminium alloy Stranded Conductors

Sl NO	Actual Area	Standing and wire Dia	Approx. Overall Dia	Approx. Mass	Calculated Maximum Resistance at 20° C	Approx. Calculated Breaking Load
	Mm ²	mm	mm	Kg/km	Ohms/km	kN
(1)	(2)	(3)	(4)	(5)	(6)	(7)
1	15	3/2.50	5.39	40.15	2.3040	4.33
2	22	7/2.00	6.00	60.16	1.5410	6.45
3	34	7/2.50	7.50	94.00	0.9900	10.11
4	55	7/3.15	9.45	149.20	0.6210	16.03
5	80	7/3.81	11.43	218.26	0.4250	23.41
6	100	7/4.26	12.78	272.86	0.3390	29.26
7	125	19/2.89	14.45	342.51	0.2735	36.64
8	148	19/3.15	15.75	406.91	0.2290	43.50
9	173	19/3.40	17.00	474.02	0.1969	50.54
10	200	19/3.66	18.30	549.40	0.1710	58.66
11	232	19/3.94	19.70	636.67	0.1471	68.05
12	288	37/3.15	22.05	794.05	0.1182	84.71
13	346	37/3.45	24.15	952.56	0.0984	101.58
14	400	37/3.71	25.97	1101.63	0.0829	117.40
15	465	37/4.00	28.00	1280.50	0.0734	136.38
16	525	61/3.31	29.79	1448.39	0.0651	146.03
17	570	61/3.45	31.05	1573.71	0.0598	158.66
18	604	61/3.55	31.95	1666.00	0.0568	167.99
19	642	61/3.66	32.94	1771.36	0.0534	178.43
20	695	61/3.81	34.29	1919.13	0.0492	193.25
21	767	61/4.00	36.00	2115.54	0.0446	213.01

Table XXV
(CLAUSE 4.3.3.1)
MINIMUM CLEARANCE BETWEEN CONDUCTORS ON THE SAME
SUPPORTS

(a) L. V. Lines

(i) Vertical configuration of conductors

Minimum Clearance Between earth and live conductors	:	30 cm.
Minimum Clearance Between live conductors	:	20 cm.

(ii) Hoarizontal configuration of conductors

Minimum Clearance Between live wires on either side of a support	:	45 cm.
Minimum Clearance Between live wires on same side of a support	:	30 cm.
Minimum distance Between the centre of insulator pin hole and end of cross arm: 5 cm.		

(b)H.V.Lines

Triangular configuration:-

(i) 11 KV LINES

The conductors shall be erected in such a way that they form an equilateral triangular pattern of side of 1 meter minimum.

(ii) 33 KV LINES

The conductors shall be erected in such a way that they form an equilateral triangular pattern of side of 1.5 meter minimum.

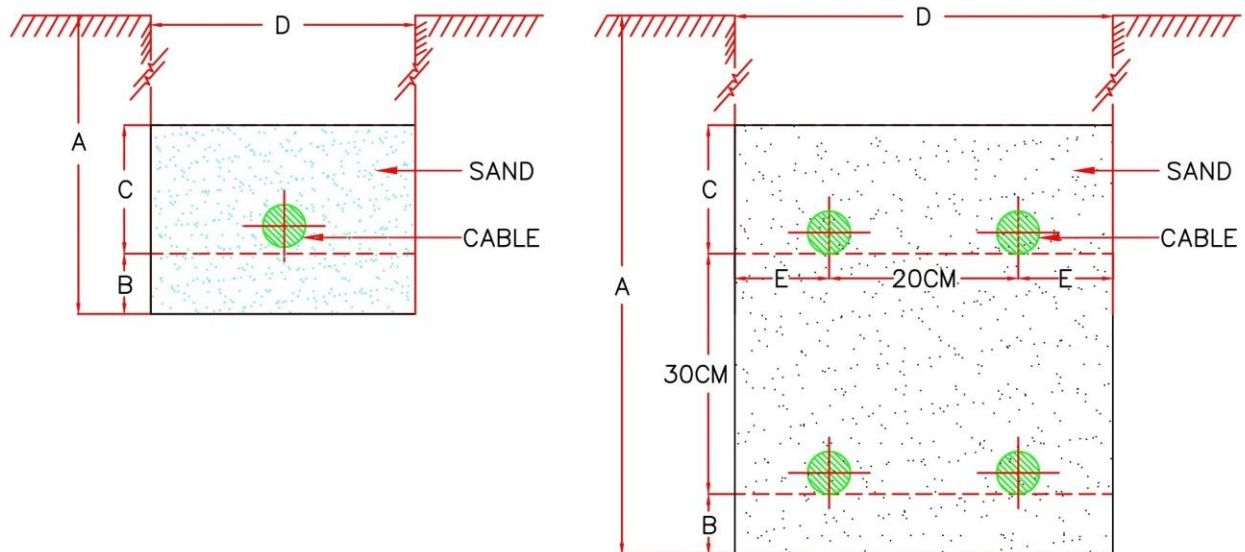
APPENDIX-H

LIST OF FIGURES

NOT TO SCALE

FIGURE-1
(Clause 2.6.8)

MINIMUM DIMENSION OF CABLE TRENCH



FOR CABLE OF RATING		
REF.	UP TO 1.1KV	EXCEEDING 1.1KV
A	75 CM	120 CM
B	8 CM	8 CM
C	17 CM	17 CM
D	35 CM	35 CM

FOR CABLE OF Rating		
REF.	UP TO 1.1 KV	EXCEEDING 1.1KV
A	$(75+n_1 \times 30)$ CM	$(120+n_1 \times 30)$ CM
B	8 CM	8CM
C	17 CM	17 CM
D	$(30+n_2 \times 20)$ CM	$(30+n_2 \times 20)$ CM
E	15 CM	15 CM

n1—Number of Additional Cables in
Vertical Formation

n2—Number of Additional Cables in
Horizontal Formation

FIGURE-2
{Clause 2.6.8.7(c)}
(CONCRETE ROUTE MARKER)

NOT TO SCALE

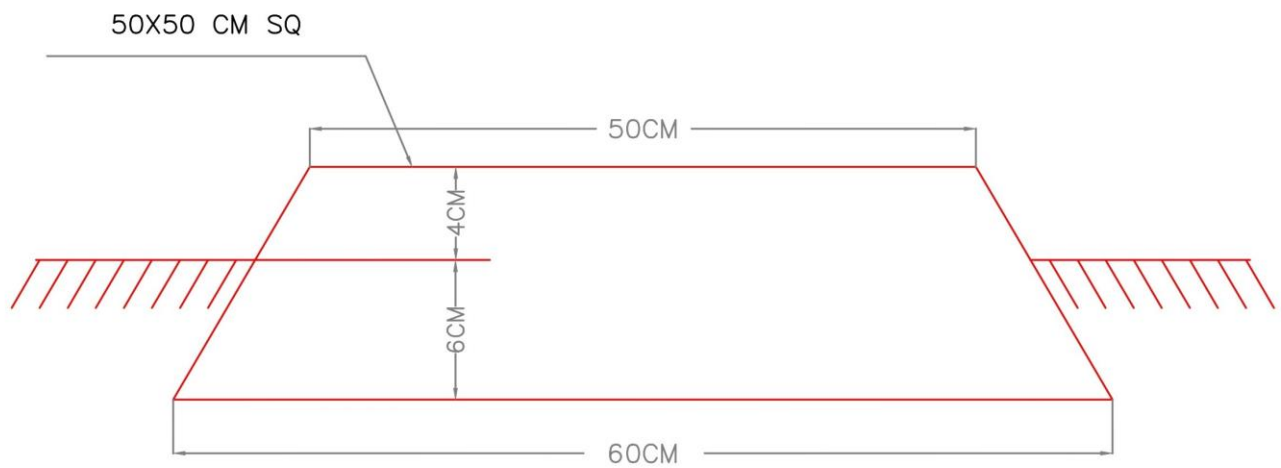
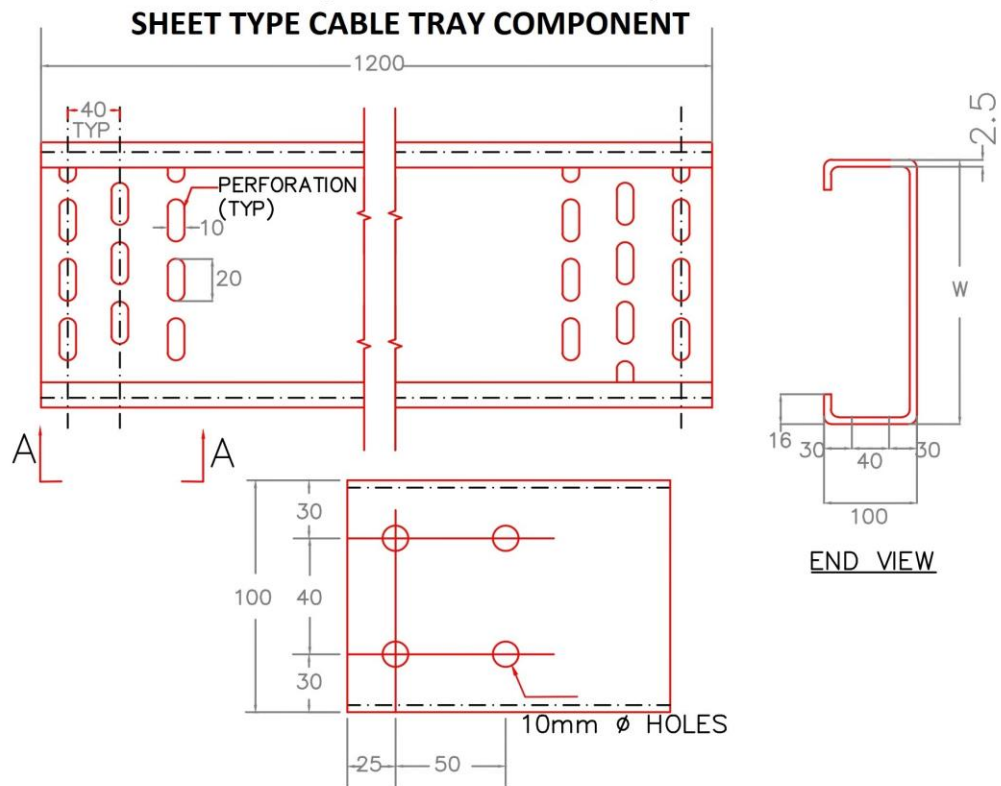


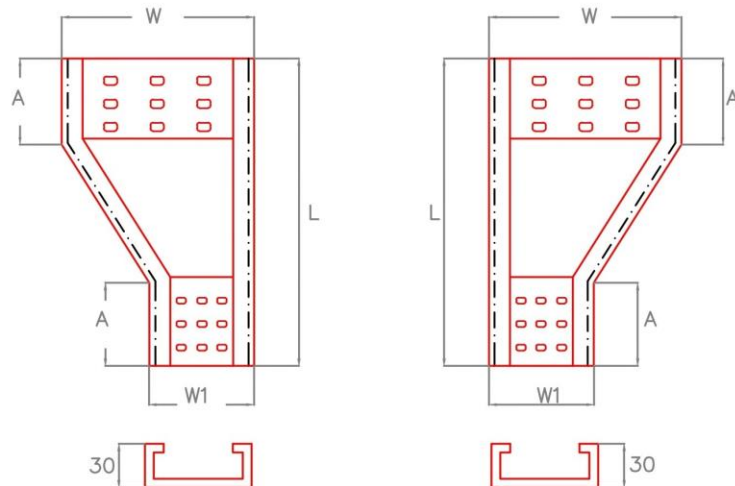
FIGURE-3A
(Clause 2.6.12.2)

NOT TO SCALE



INSIDE W(WIDTH)–150,300,450,600mm
OR SPECIFIED MATERIAL–2.5mm M.S. SHEET

SECTION ON –AA



W	W1	L	A	TH
600	450	350	100	3
450	300	350	100	3
600	300	500	100	3

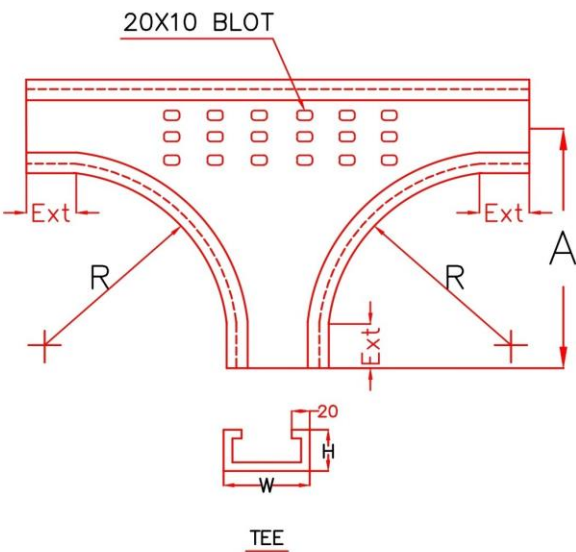
NOTE: –
DETAILS SHOWN ARE ILLUSTRATIVE
DEVIATIONS AS PER MANUFACTURER'S
STANDARD DESIGN'S
PERMISSIBLE

RIGHT HAND (REDUCER) LEFT HAND

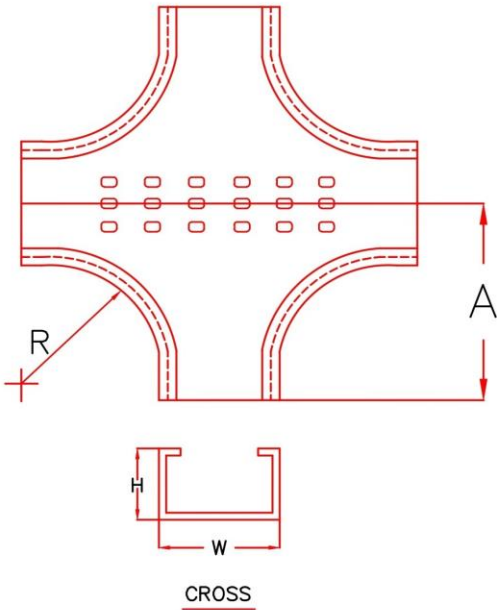
NOT TO SCALE

FIGURE-3B
(Clause 2.6.12.2)

SHEET TYPE CABLE TRAY COMPONENT



W	H	R	EXT.	TH.
600	50	600	100	3
450	-	-	-	-
300	-	-	-	-
150	-	-	-	-
-	-	1000	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-



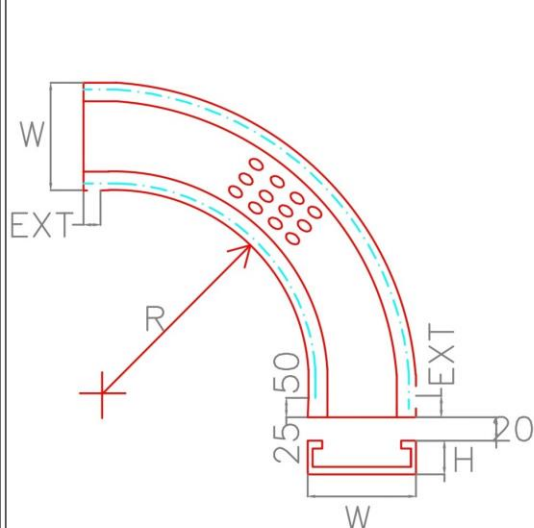
W	H	R	EXT.	TH.	A
600	50	600	100	3	1000
450	-	-	-	-	925
300	-	-	-	-	850
150	-	-	-	-	~

NOTE: –
DETAILS SHOWN ARE ILLUSTRATIVE
DEVIATIONS AS PER MANUFACTURE’S
STANDARD DESIGN IS PERMISSIBLE

NOT TO SCALE

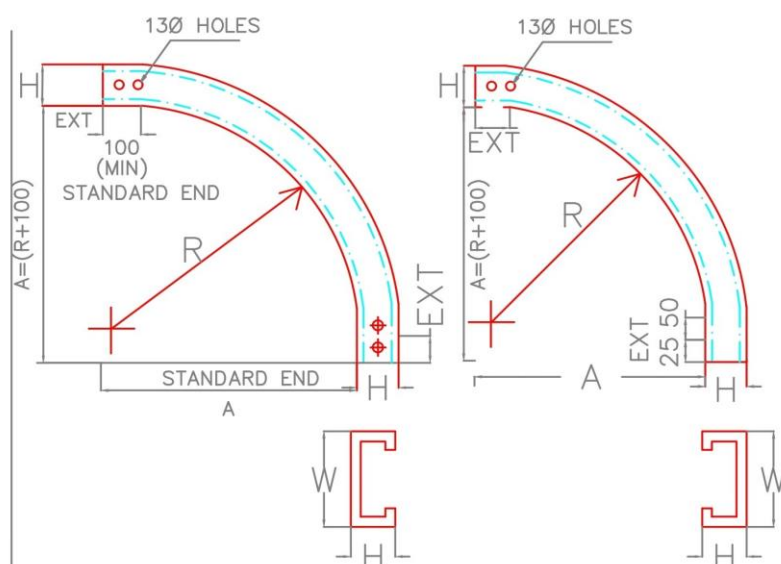
FIGURE-3C
(Clause 2.6.12.2)

SHEET TYPE CABLE TRAY COMPONENT



W	H	R	EXT	TH
600	50	600	100	3
450	"	"	"	"
300	"	"	"	"
150	"	"	"	"

90° HORIZONTAL BED OUTSIDE



W	H	R	EXT	TH	A (R+100)
600	50	1000	100	3	1100
450	"	"	"	"	"
300	"	"	"	"	"
150	"	"	"	"	"
600	"	600	"	"	700
450	"	"	"	"	"
300	"	"	"	"	"
150	"	"	"	"	"

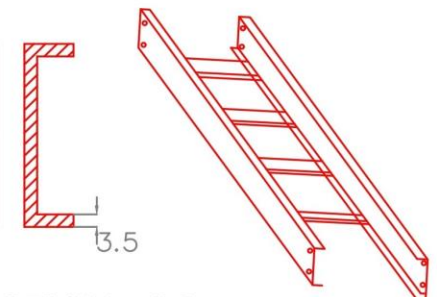
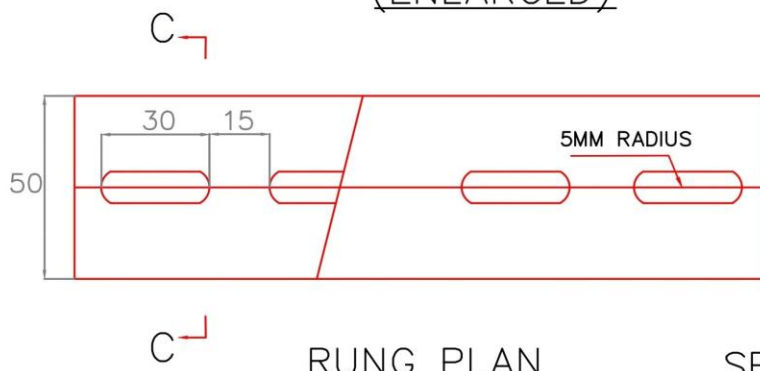
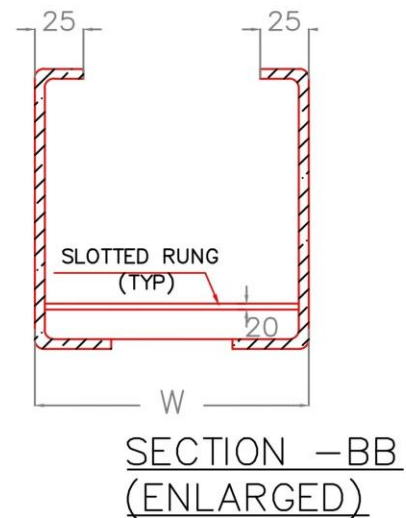
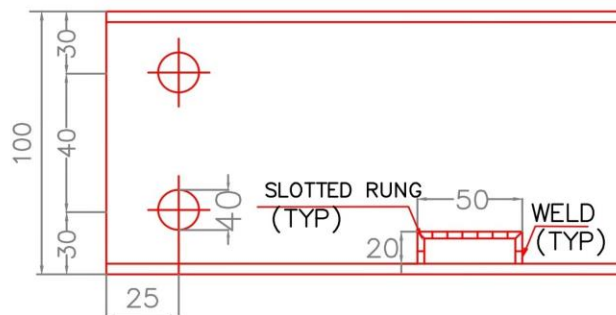
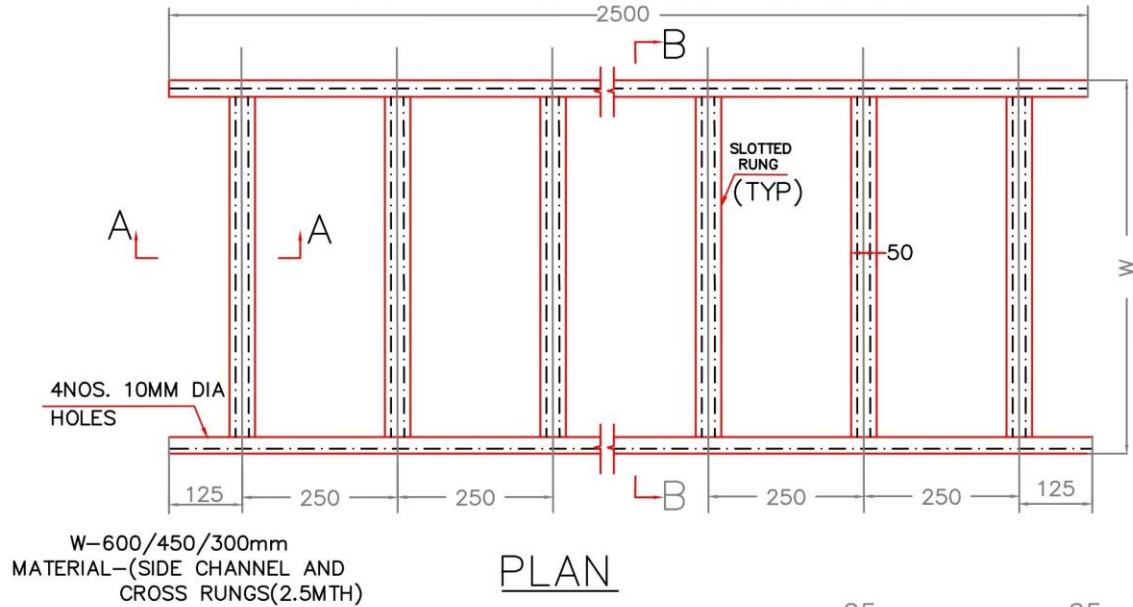
90° HORIZONTAL BED OUTSIDE AND INSIDE

NOTE: –
DETAILS SHOWN ARE ILLUSTRATIVE
DEVIATIONS AS PER MANUFACTURER'S
STANDARD DESIGN'S
PERMISSIBLE

NOT TO SCALE

FIGURE-4A
{Clause 2.6.12.3(c)}

LADDER TYPE CABLE TRAY COMPONENT

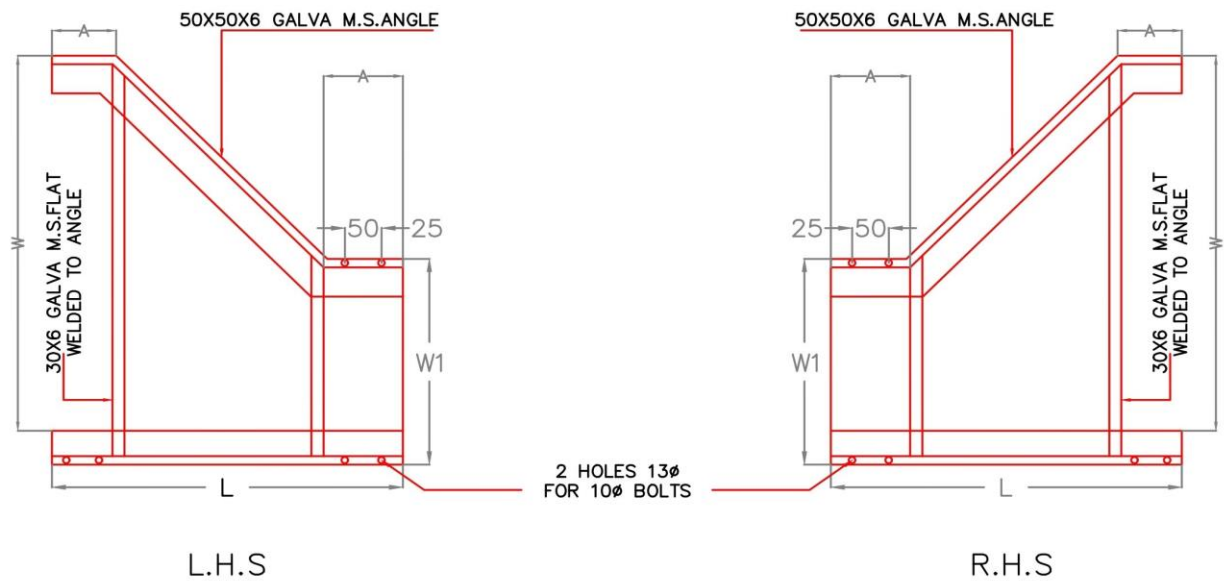


NOTE: -
DETAILS SHOWN ARE ILLUSTRATIVE
DEVIATIONS AS PER MANUFACTURER'S
STANDARD DESIGN'S
PERMISSIBLE

NOT TO SCALE

FIGURE-4B {Clause 2.6.12.3(c)}

LADDER TYPE CABLE TRAY COMPONENT



W	W1	L	A
600	300	500	100
600	450	350	100
450	300	350	100

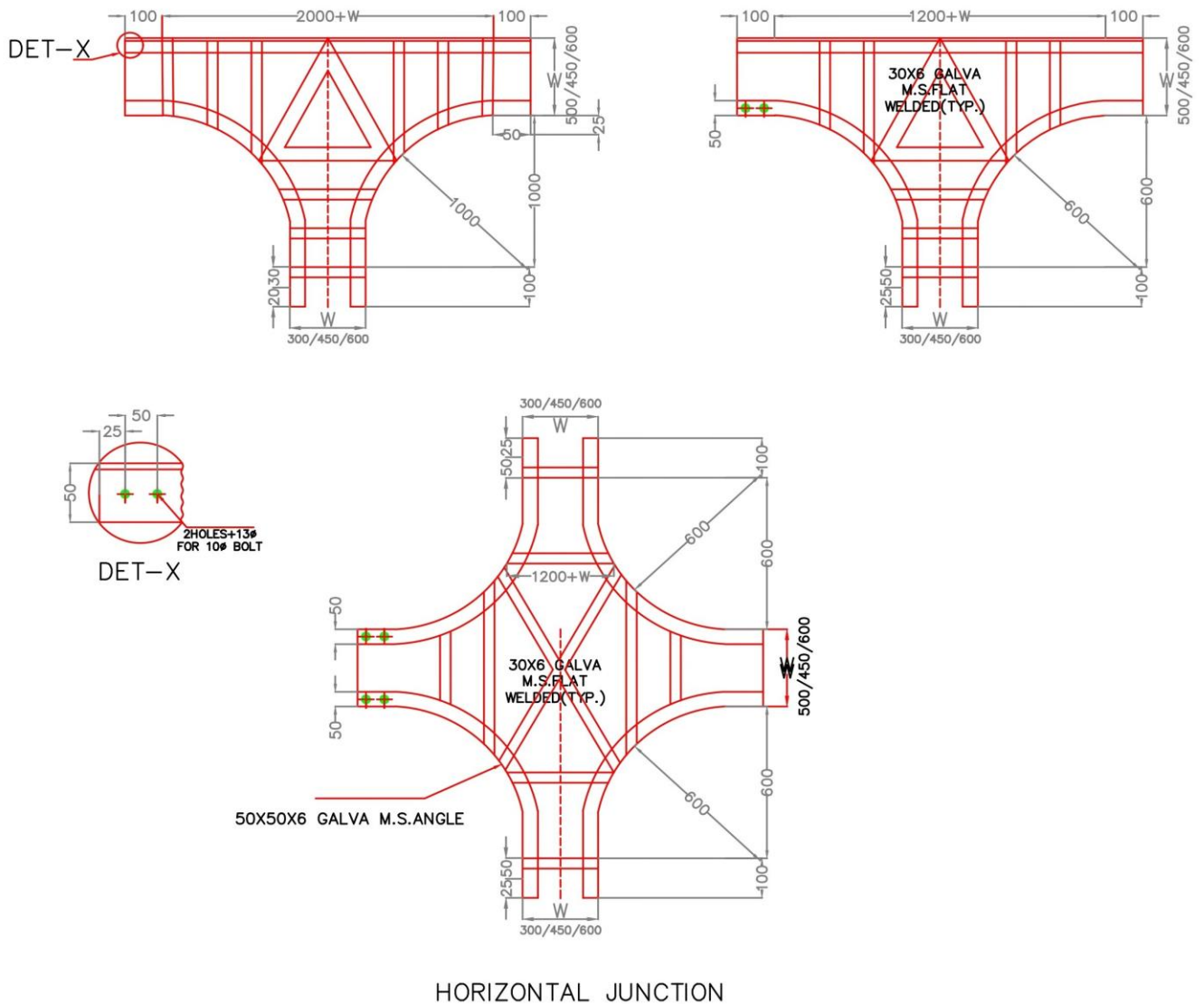
VERTICAL REDUCER

NOTE: –
DETAILS SHOWN ARE ILLUSTRATIVE
DEVIATIONS AS PER MANUFACTURE'S
STANDARD DESIGN IS PERMISSIBLE

NOT TO SCALE

FIGURE-4C
{Clause 2.6.12.3(c)}

LADDER TYPE CABLE TRAY COMPONENT

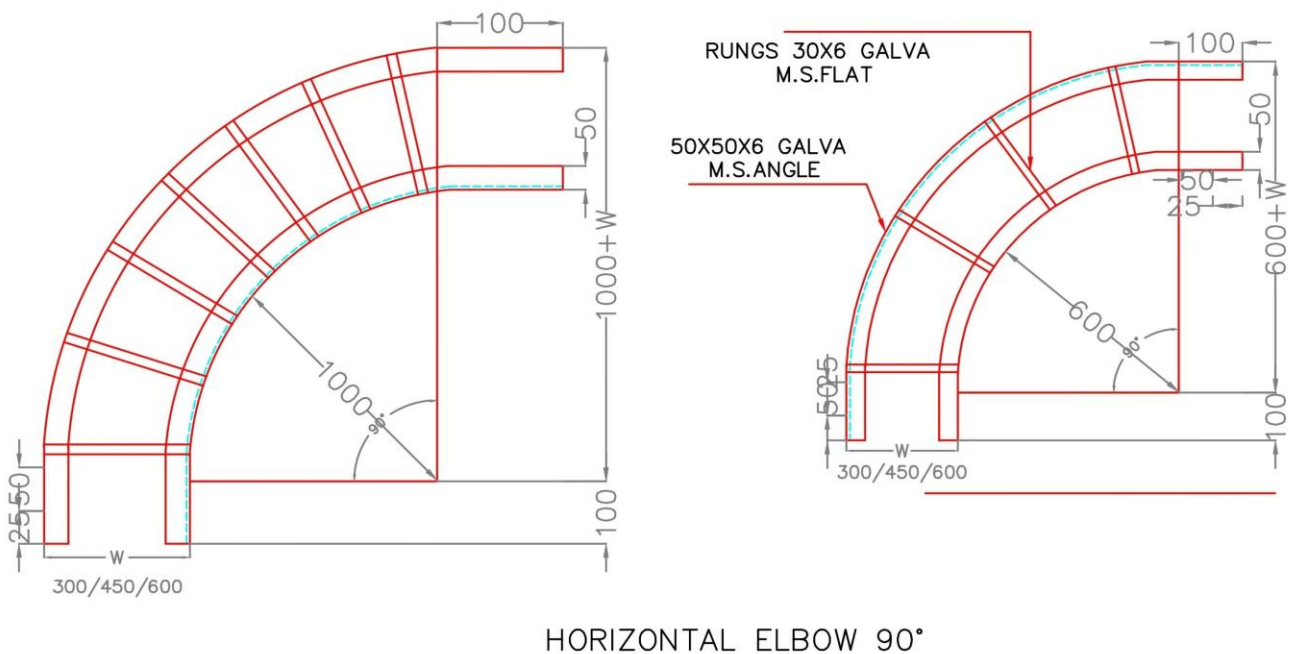
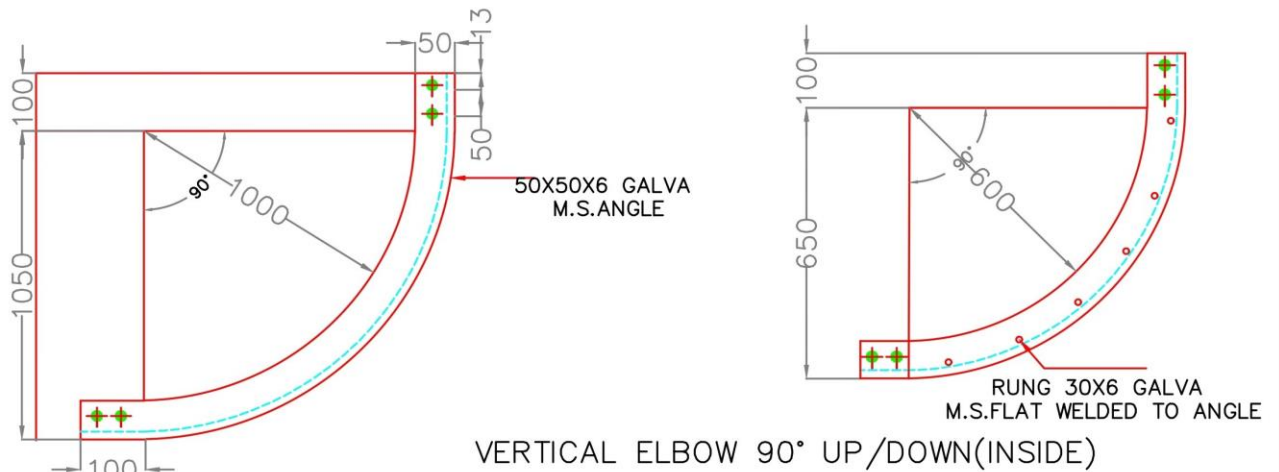


NOTE: —
DETAILS SHOWN ARE ILLUSTRATIVE
DEVIATIONS AS PER MANUFACTURE'S
STANDARD DESIGN IS PERMISSIBLE

NOT TO SCALE

FIGURE-4D
{Clause 2.6.12.3(c)}

LADDER TYPE CABLE TRAY COMPONENT

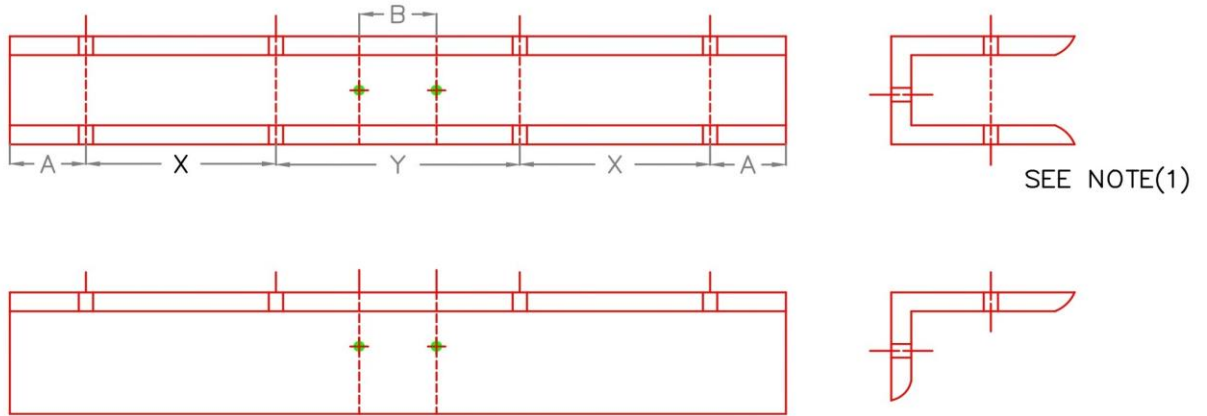


NOTE: —
DETAILS SHOWN ARE ILLUSTRATIVE
DEVIATIONS AS PER MANUFACTURE'S
STANDARD DESIGN IS PERMISSIBLE

NOT TO SCALE

FIGURE-5A {Clause 4.2.2.1(iv)}

STRAIGHT CROSS ARM (LV/MV/11KV LINES)



DIMENSION: –

A: –50MM FOR LV/MV LINES
100MM FOR 11KV LINES

B: –TO SUIT THE POLES

X: –50MM }

Y: –100MM} FOR LV/MV LINES(SEE NOTE 2&3)

X+Y=1050MM FOR HV LINES(SEE NOTE 3&4)

CHANNEL SIZE: –75X40X4.8MM(7.14KG/M)

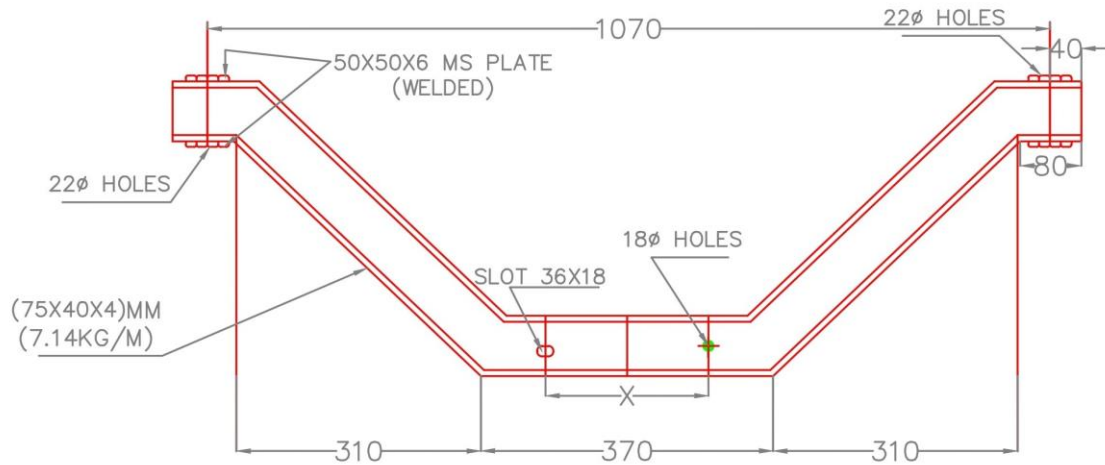
ANGLE SIZE: –50X50X6MM(4.5KG/M) FOR LV/MV LINES
&65X65X6MM(5.8KG/M) FOR LV/MV LINES

NOTES: –

1. HOLES TO BE DRILLED (SHOWN TYPICALLY) SHALL SUIT THE REQUIREMENT OF FIXING HARDWARE, AS PER NUMBER AND TYPE OF INSULATOR
2. X&Y DIMENSION ARE BASED ON 750MM(MAX.) SAG
3. FOR TAKING ONLY & CONDUCTORS , 'X' DIMENSIONS WILL BE ZERO
4. FOR 11KV LINES THE SPACING BETWEEN HOLES(3 NOS 3 CONDUCTORS) SHOULD BE 1050MM(MIN)
5. FOR HV LINES, DOUBLE CLAMPING WITH WELDED ADDITIONAL ANGLE IRON BRACKET SHALL BE ADOPTED AS SHOWN IN FIGURE –5C

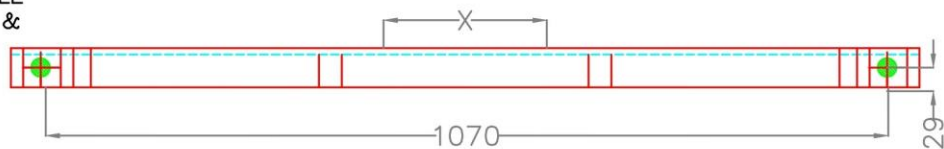
NOT TO SCALE

FIGURE-5B
{Clause 4.2.2.1(iv)}
V-CROSS ARM FOR HV LINES

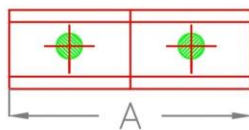


ELEVATION

X= TO SUIT THE POLE
Y= 1050 FOR 11KV &
1500 FOR 33KV



PLAN



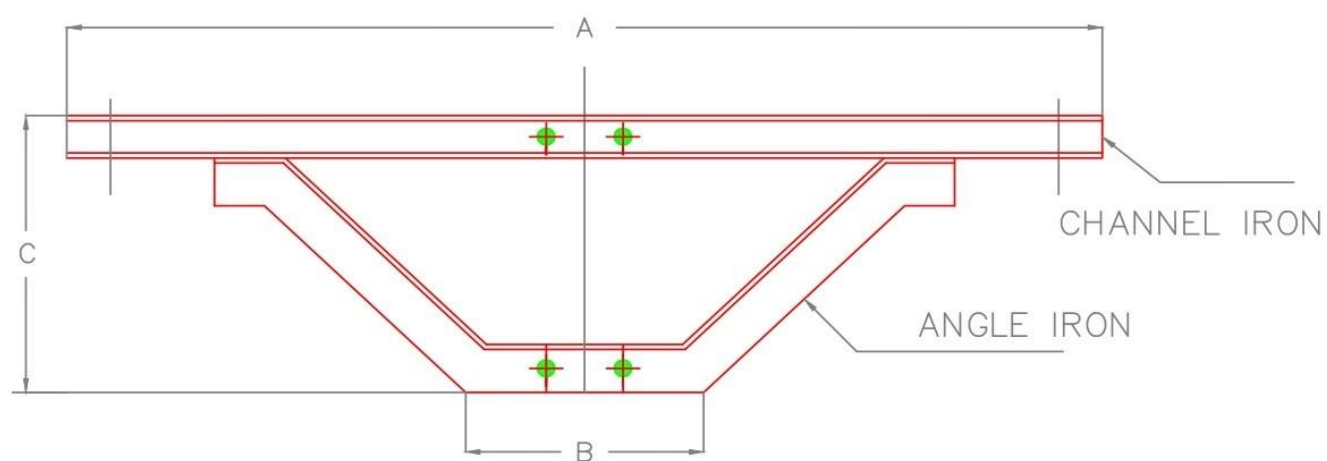
CHANNEL IRON

	A
11KV	0.25M
33KV	0.30M

NOTE: –
DETAILS SHOWN ARE ILLUSTRATIVE
DEVIATIONS AS PER MANUFACTURER'S
STANDARD DESIGN'S
PERMISSIBLE

NOT TO SCALE

FIGURE-5C
{Clause 4.2.2.1(iv)}
V-CROSS ARM FOR HV LINES

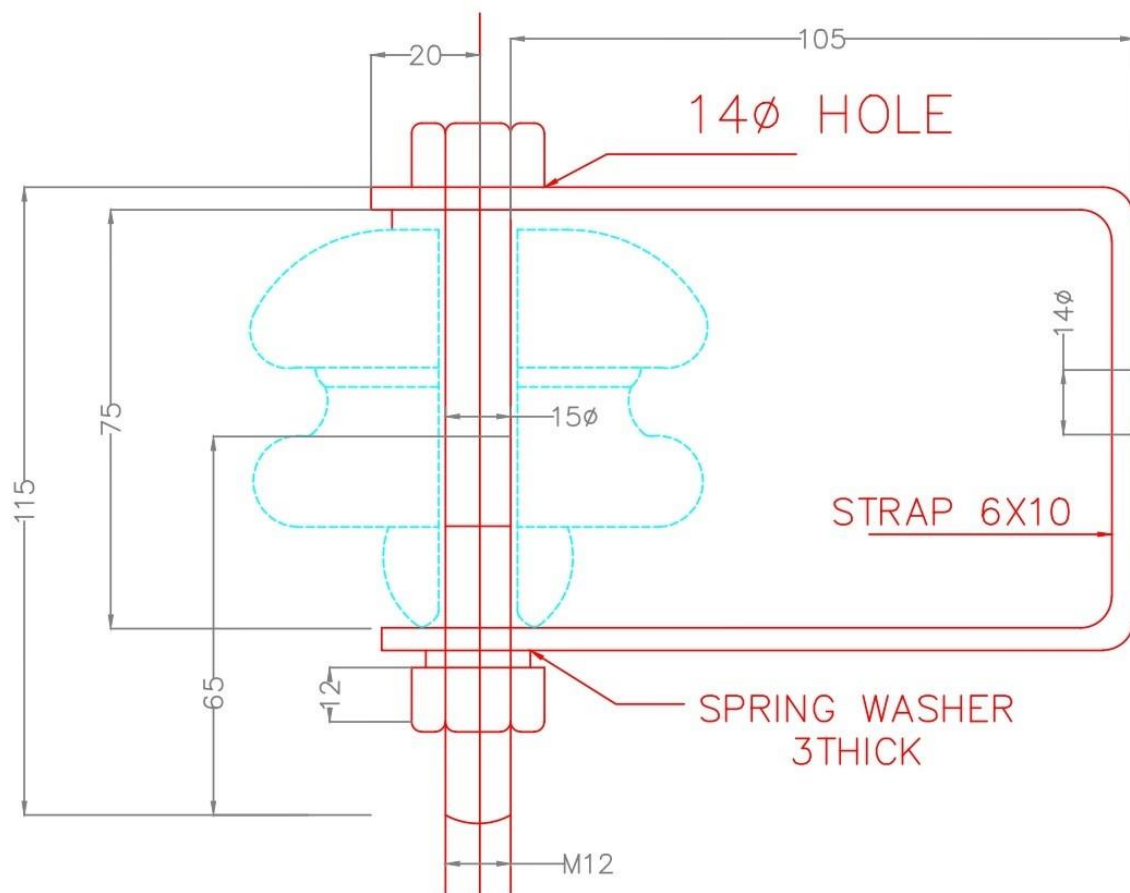


	A	B	C
11KV	1.1M	0.45M	0.6M
33KV	1.6M	0.6M	0.73M

NOT TO SCALE

FIGURE-6A
{Clause 4.2.2.2(ii)}

415/240V INSULATORS & FITTINGS HARDWARE FOR SHACKLE INSULATOR

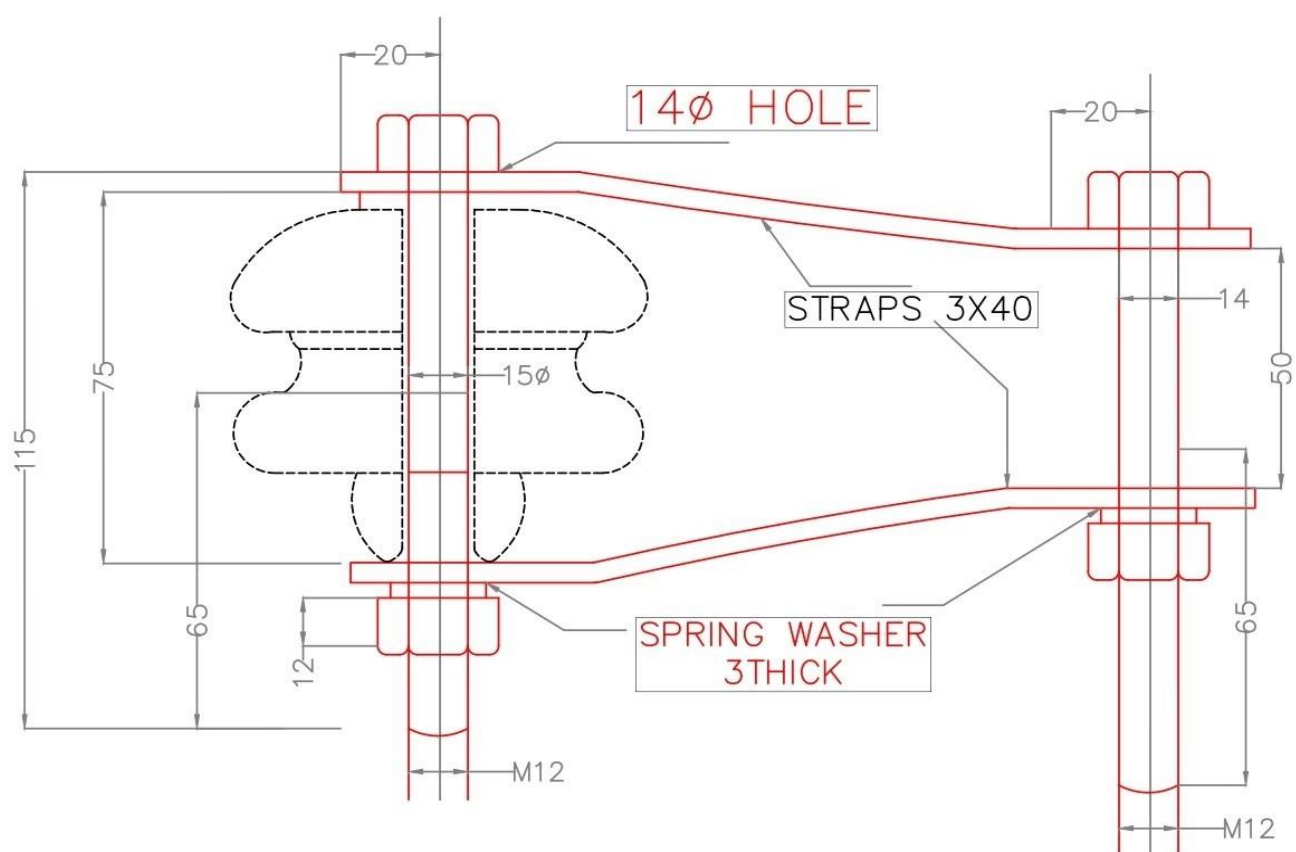


NOT TO SCALE

FIGURE-6B

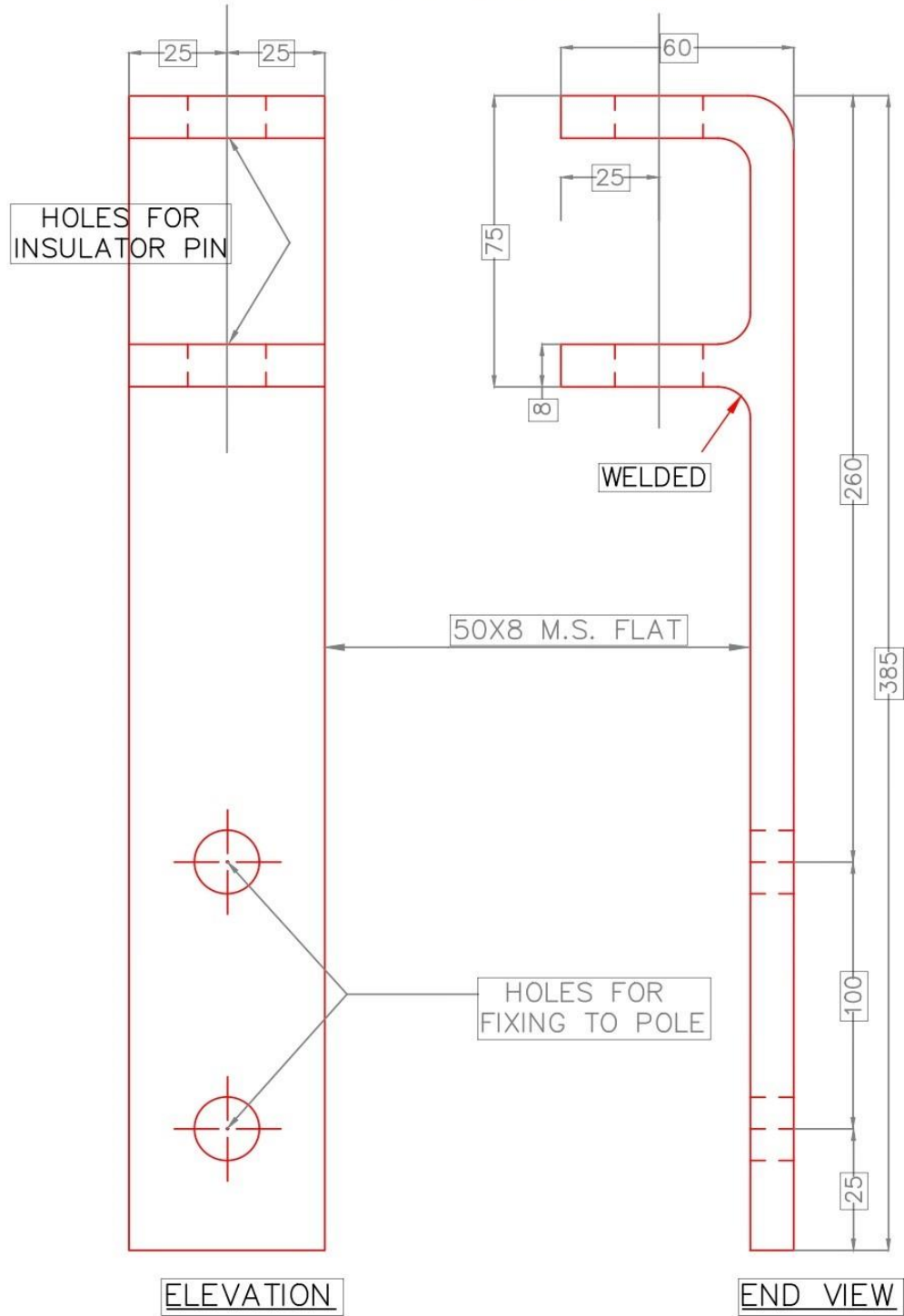
Clause 4.2.2.3

G.I. STRAPS



NOT TO SCALE

FIGURE-6C
{Clause 4.2.2.2(ii)}
POLE FOR BRACKET

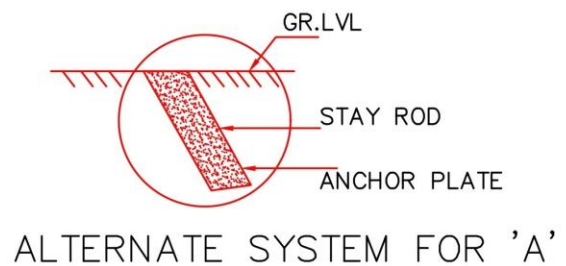
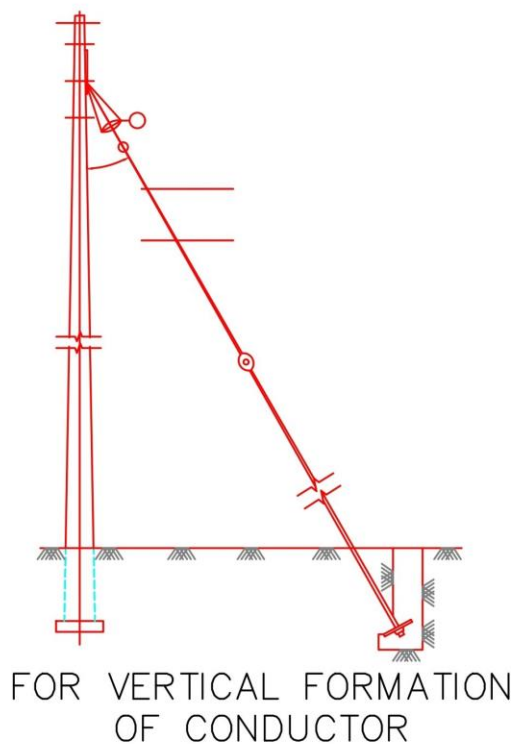
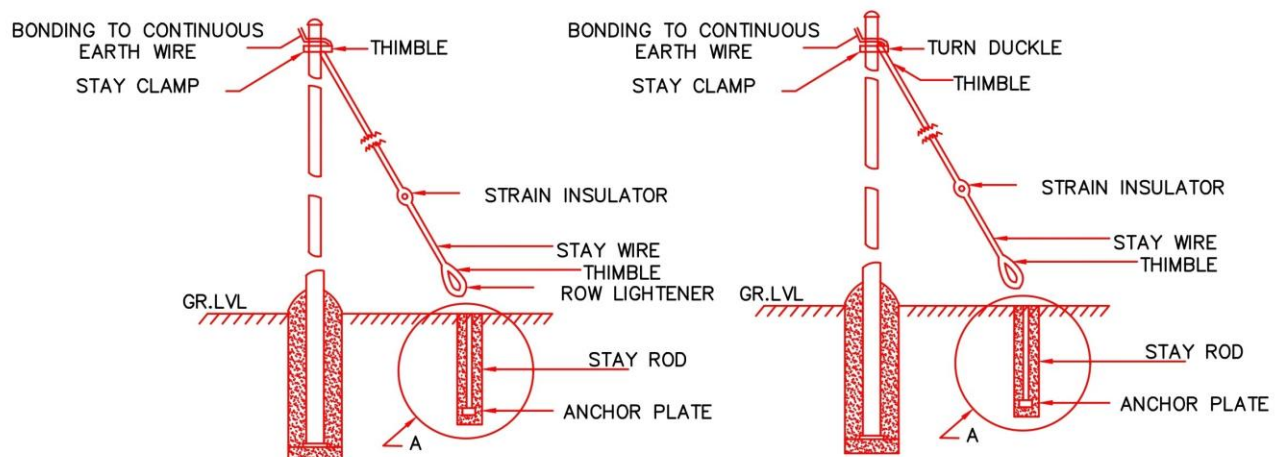


ALL DIMENSIONS IN MM

NOT TO SCALE

FIGURE-7A
(Clause 4.5.2.1)

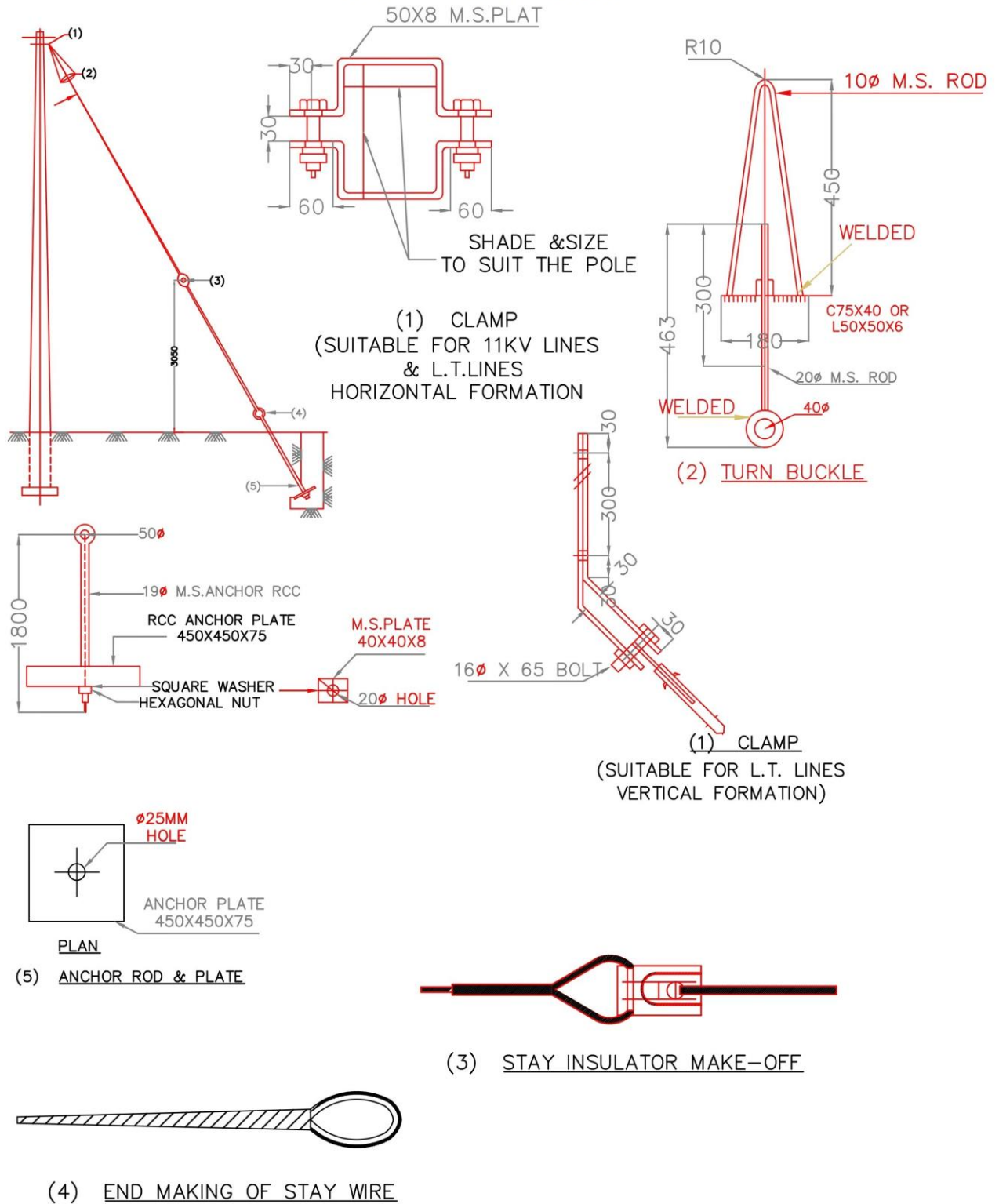
TYPICAL ARRANGEMENT OF STAY SET



NOT TO SCALE

FIGURE-7B
(Clause 4.5.2.1)

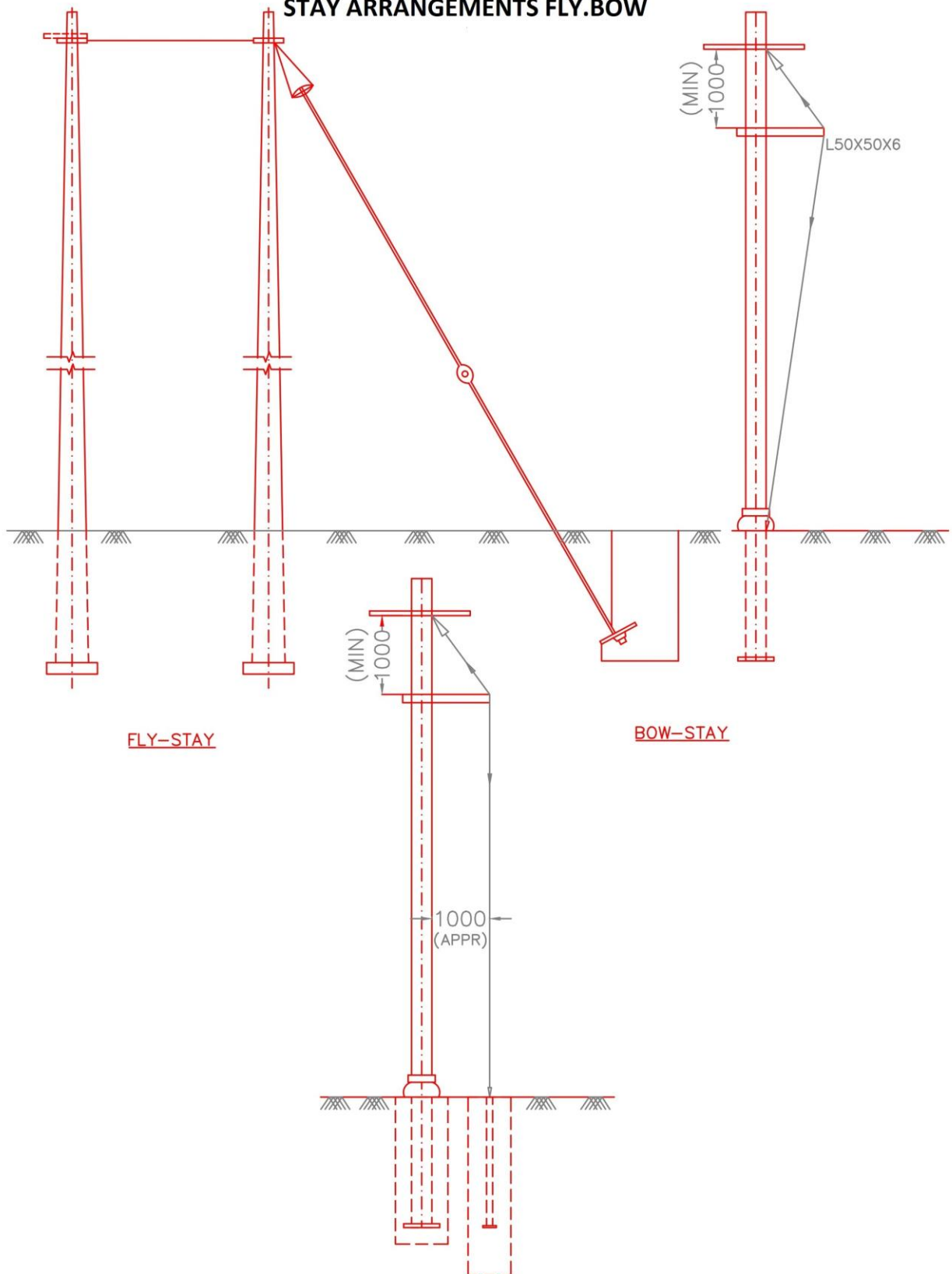
STAY ASSEMBLY DETAILS



NOT TO SCALE

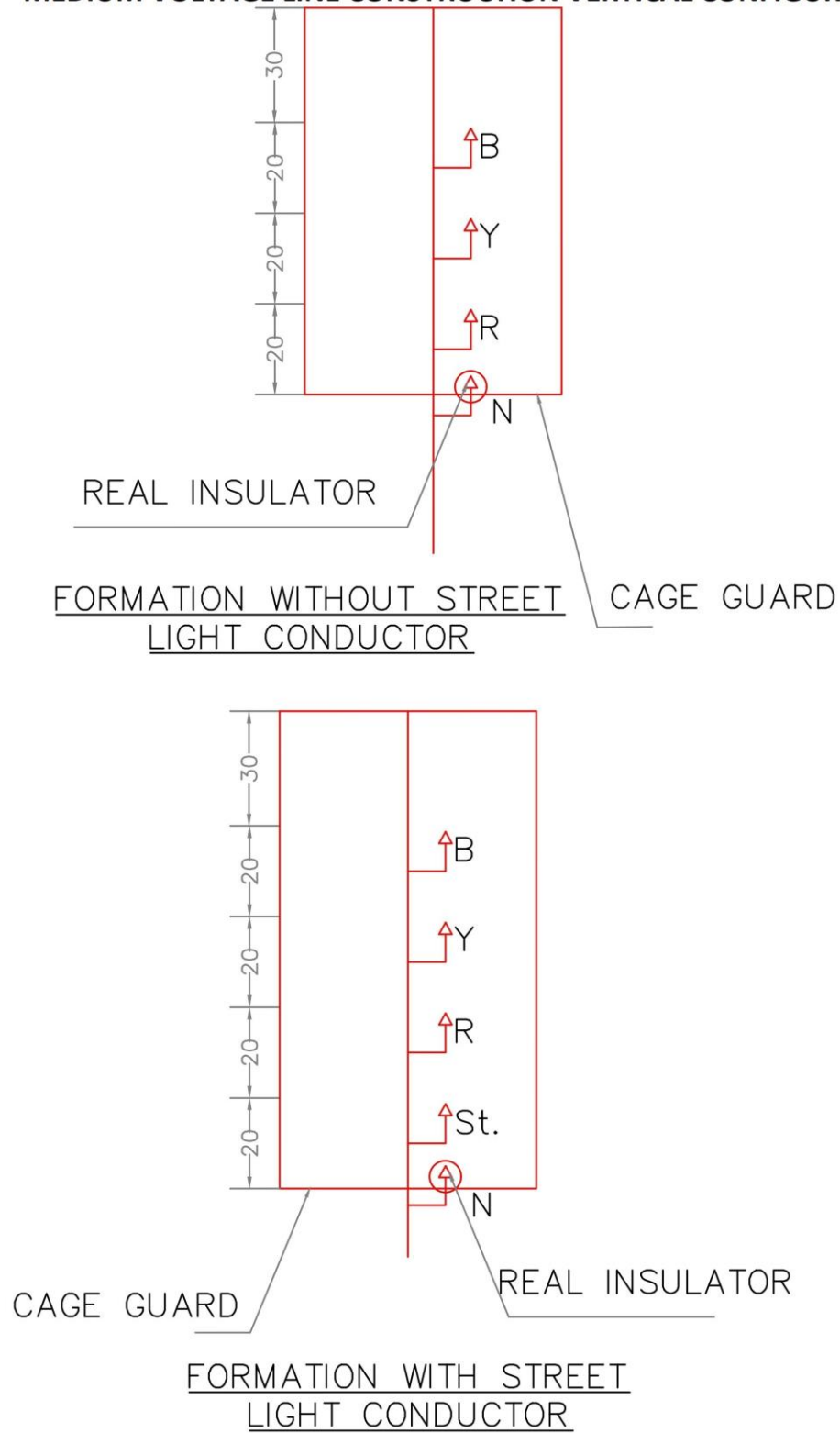
FIGURE-7C
(Clause 4.5.2.1)

STAY ARRANGEMENTS FLY.BOW



NOT TO SCALE

FIGURE-8A
(Clause 4.3.3.1)
MEDIUM VOLTAGE LINE CONSTRUCTION VERTICAL CONFIGURATION



NOT TO SCALE

FIGURE-8B
(Clause 4.3.3.1)

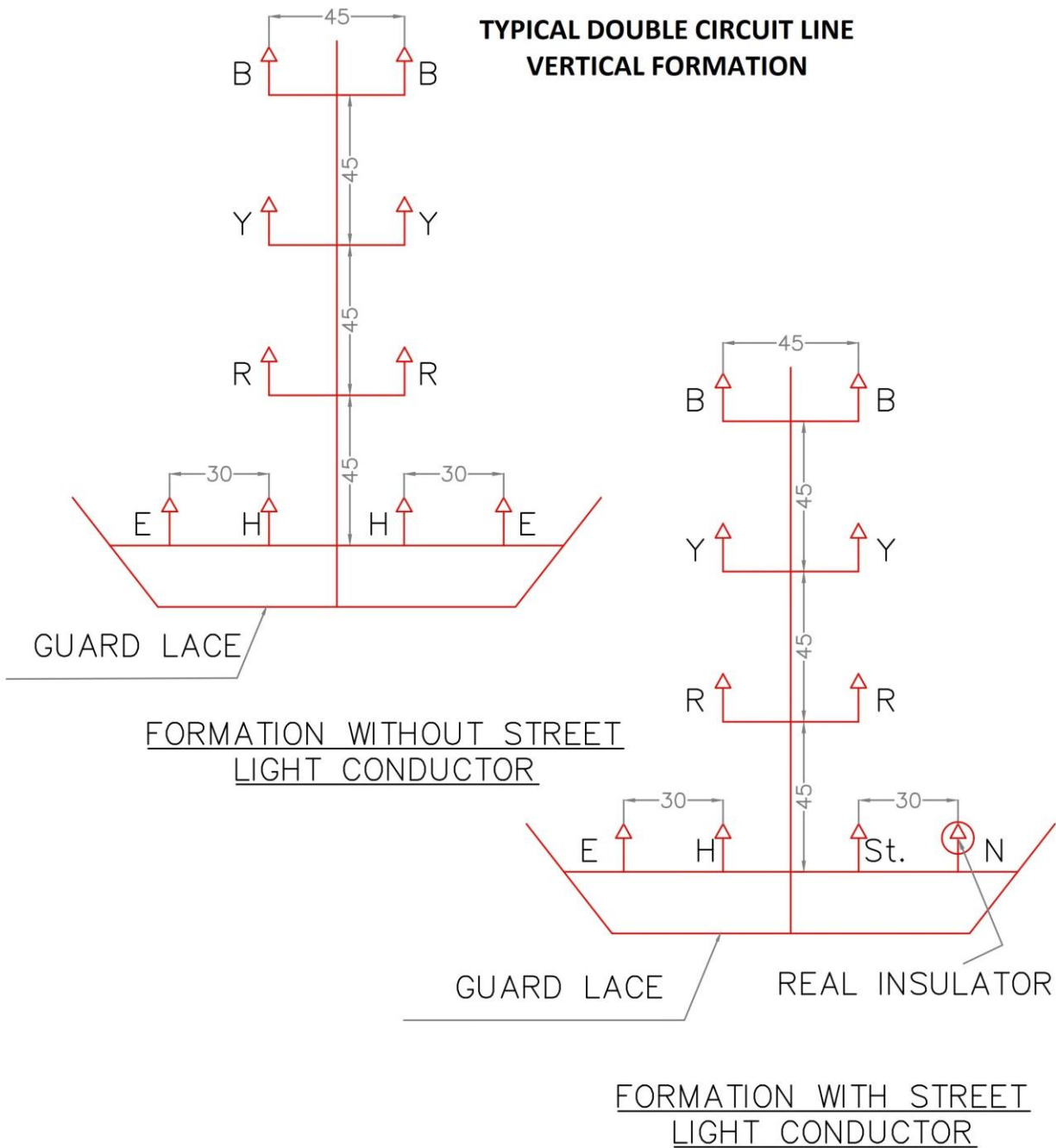
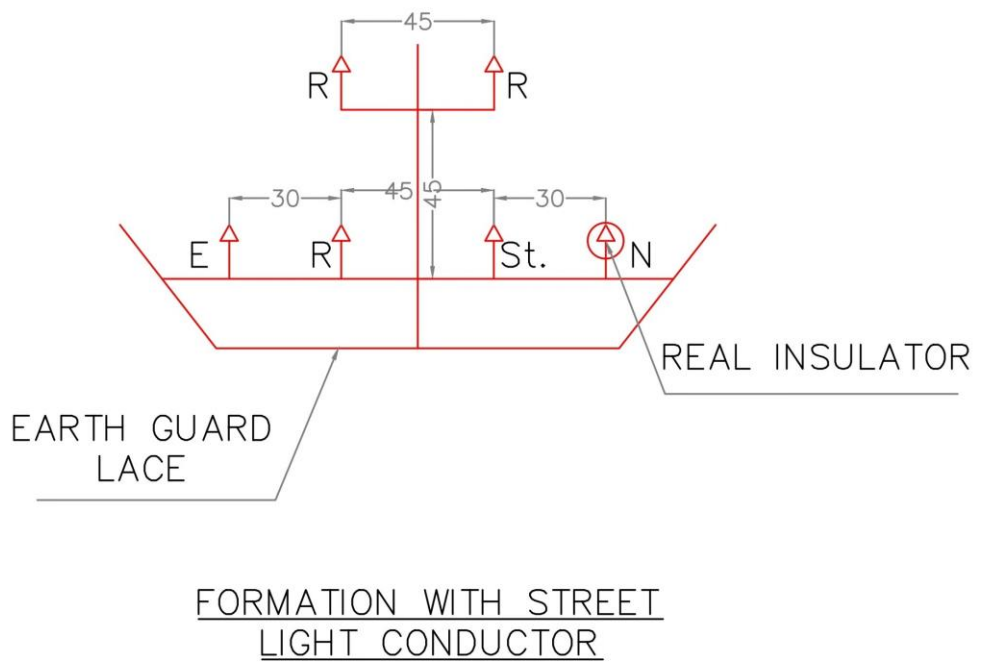
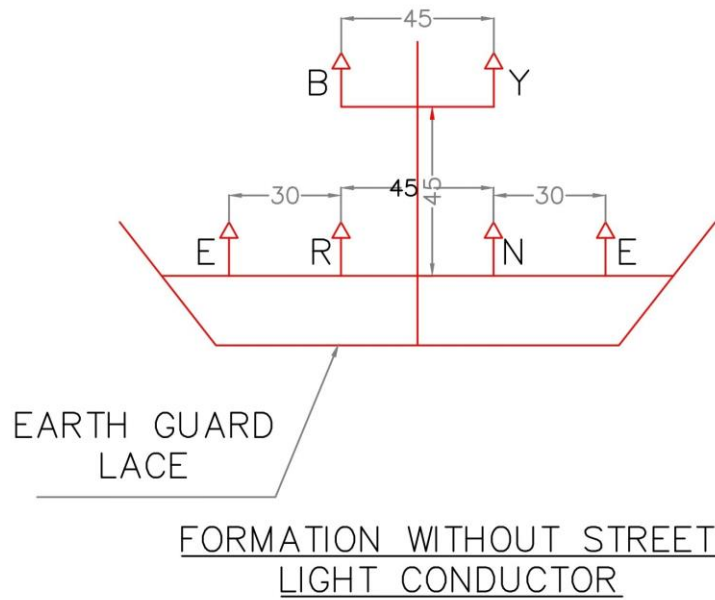


FIGURE-8C
(Clause 4.3.3.1)

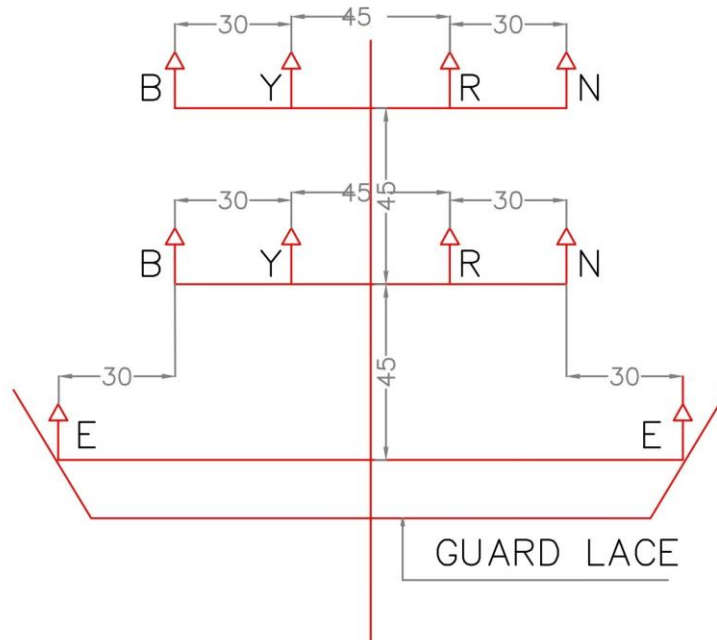
TYPICAL SINGLE CIRCUIT M. V. HORIZONTAL FORMATION



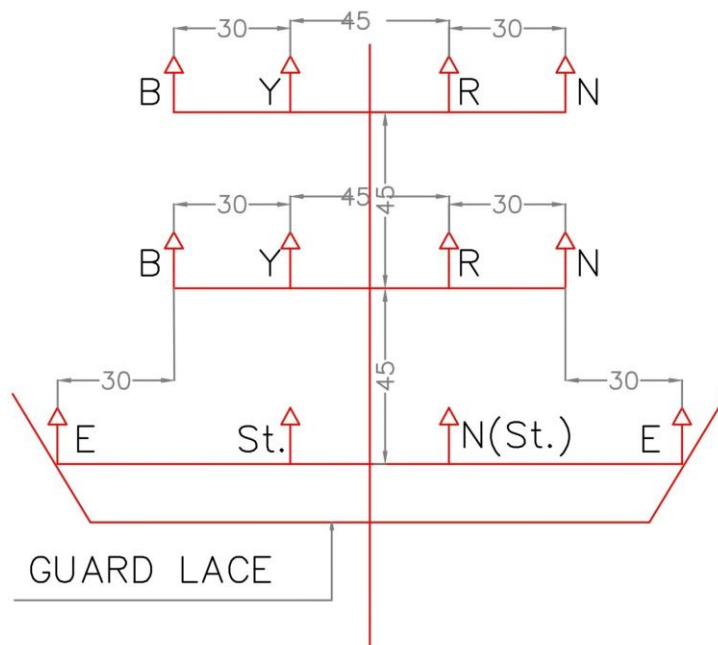
NOT TO SCALE

FIGURE-8D
(Clause 4.3.3.1)

TYPICAL DOUBLE CIRCUIT M. V. HORIZONTAL FORMATION



FORMATION WITHOUT STREET
LIGHT CONDUCTOR

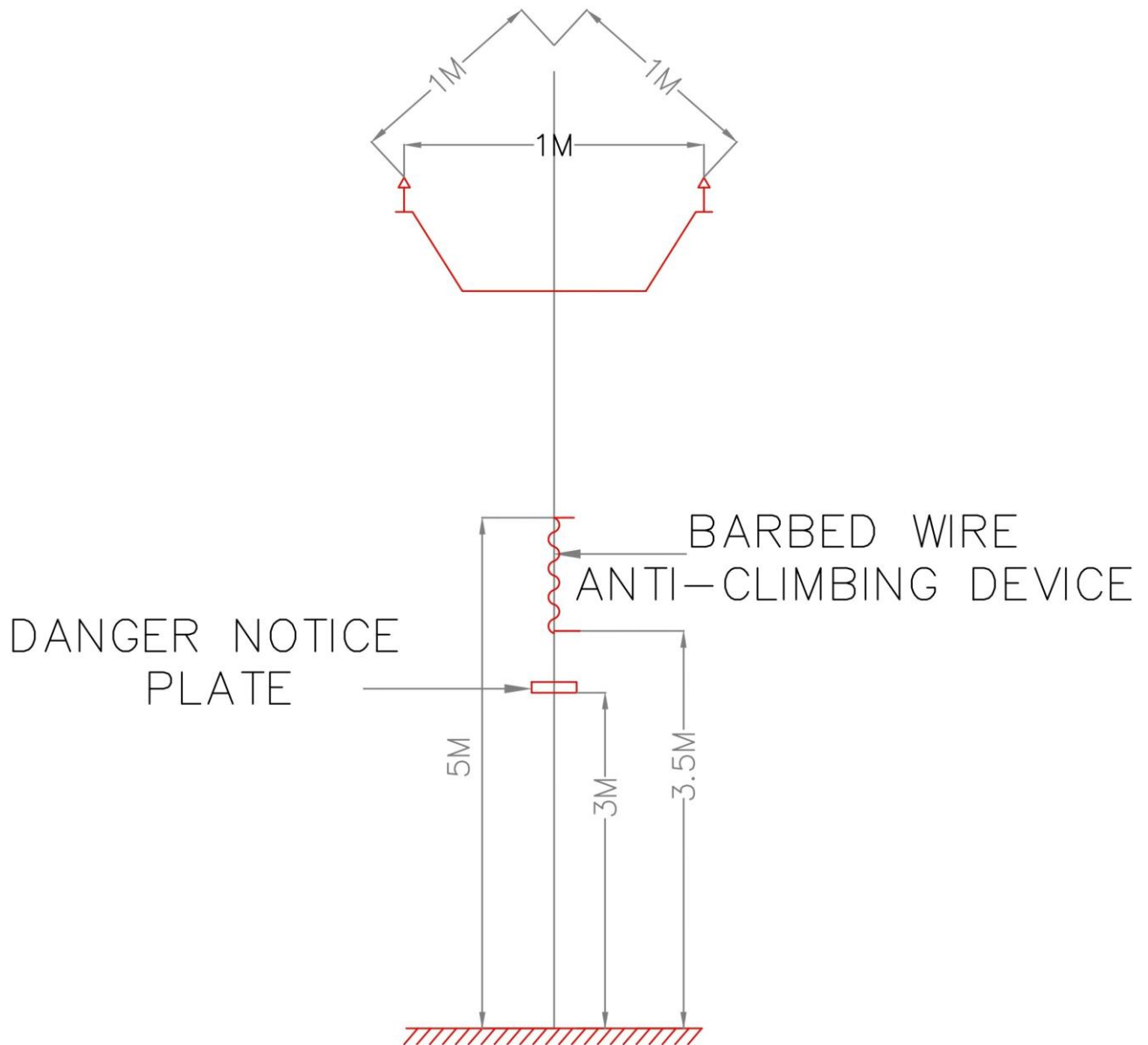


FORMATION WITH STREET
LIGHT CONDUCTOR

NOT TO SCALE

FIGURE-8E
(Clause 4.3.3.1)

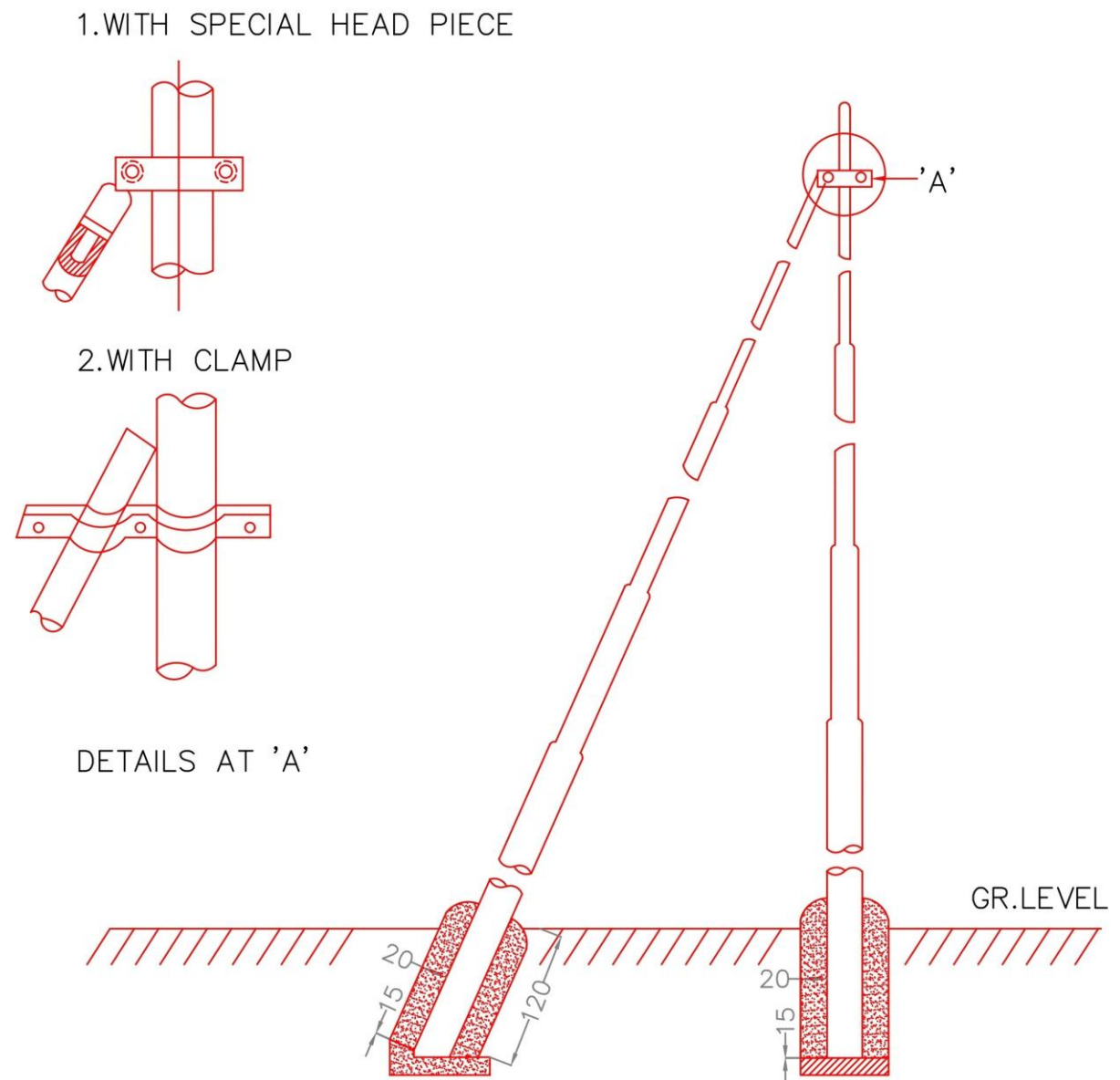
TYPICAL H. V. LINE TRIANGULAR FORMATION



NOT TO SCALE

FIGURE-9
(Clause 4.5.2.13)

STRAIGHT CROSS ARM (LV/MV/11 kV LINES)



ALL DIMENSIONS IN CENTIMETERS

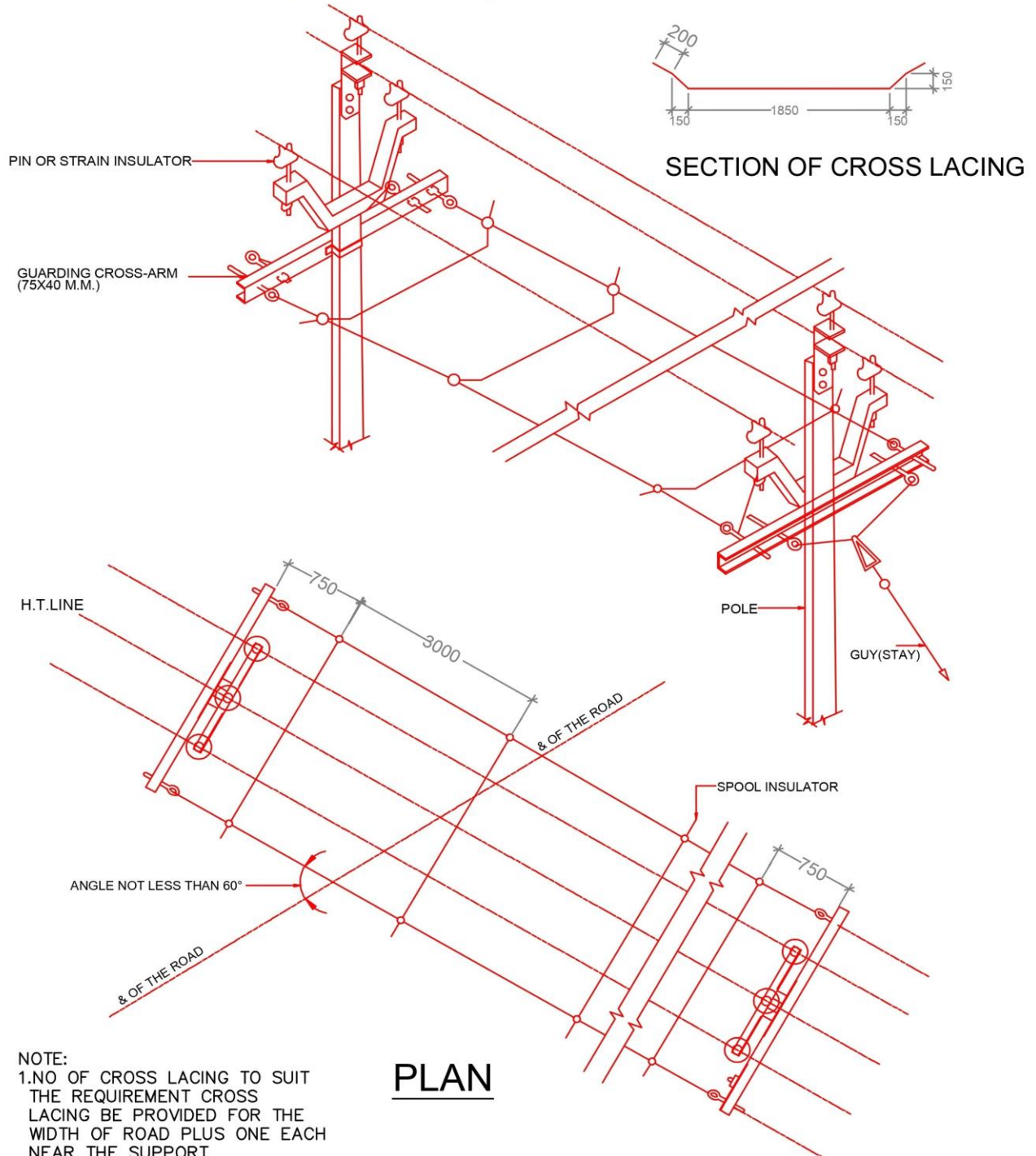
NOT TO SCALE



NOT TO SCALE

FIGURE-10B
(Clause 4.5.9.2)

PROTECTIVE GUARDING ACROSS/RC ROAD HORIZONTAL FORMATION



PLAN

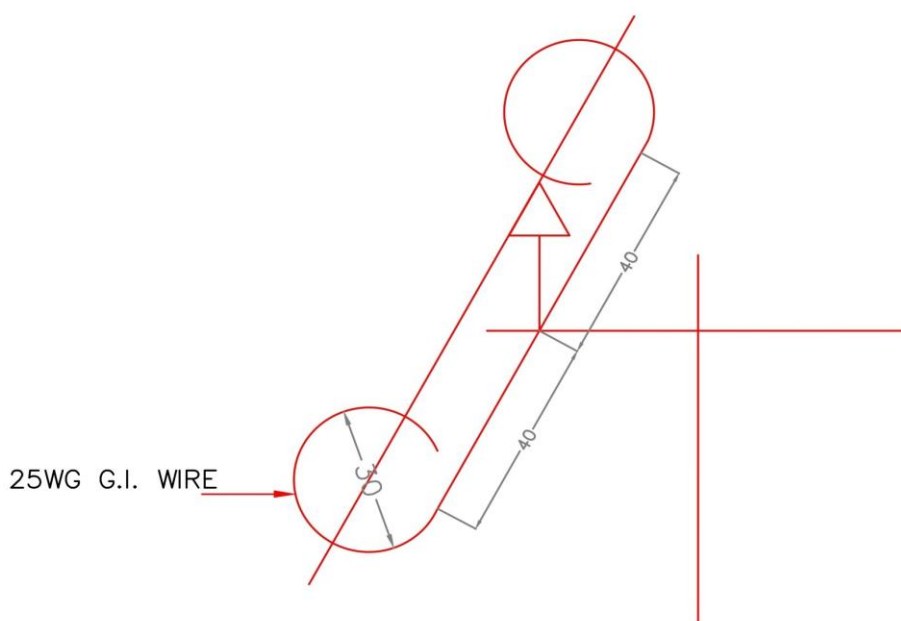
NOTE:

1. NO. OF CROSS LACING TO SUIT THE REQUIREMENT CROSS LACING BE PROVIDED FOR THE WIDTH OF ROAD PLUS ONE EACH NEAR THE SUPPORT
2. STRUCTURES ON EITHER SIDE OF THE ROAD TO BE EARTHED.
3. CROSSING ANGLE SHOULD NOT BE LESS THEN 60°
4. SPECIAL SUPPORT MAY BE NEEDED TO ALLOW MINIMUM CLEARANCE FROM GROUND AS PER I.E. RULE NO 77 i.e & 5.8 METERS.

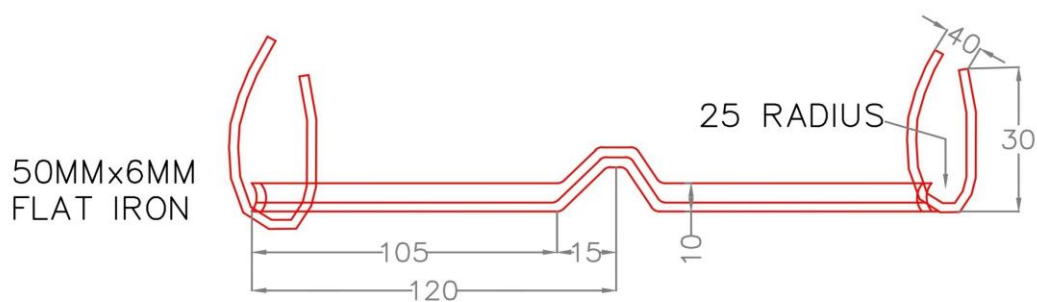
NOT TO SCALE

FIGURE-11
(Clause 4.5.9.2)
TYPICAL ARRANGEMENT OF RING GUARD

TYPICAL ARRANGEMENT OF RING GUARD

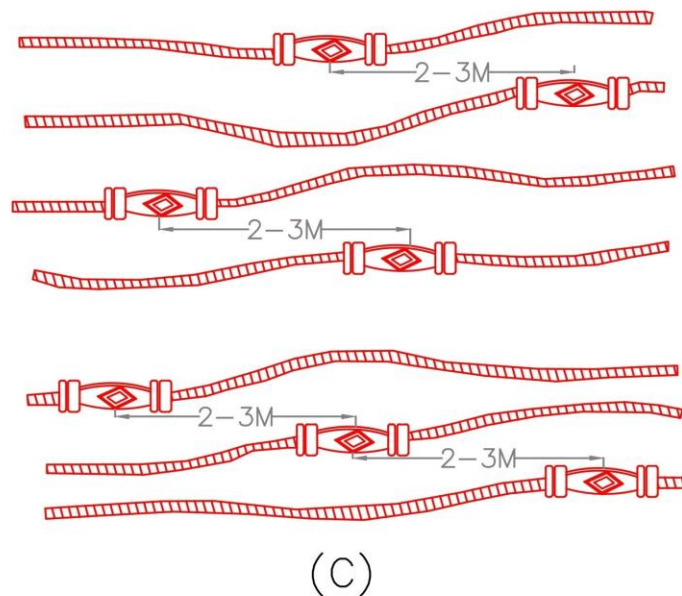
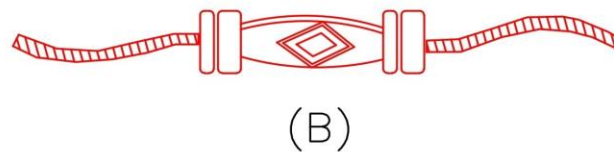
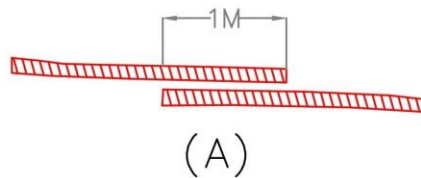


DOUBLE ARM ANTI-BUCKLING TYPE FOR FIXING
UNDER THE INSULATOR PIN & GUARDING ONE CONDUCTOR



ALL DIMENSIONS IN CENTIMETERS

FIGURE-12
(Appendix E-2.1)
STAGGERING OF JOINTS



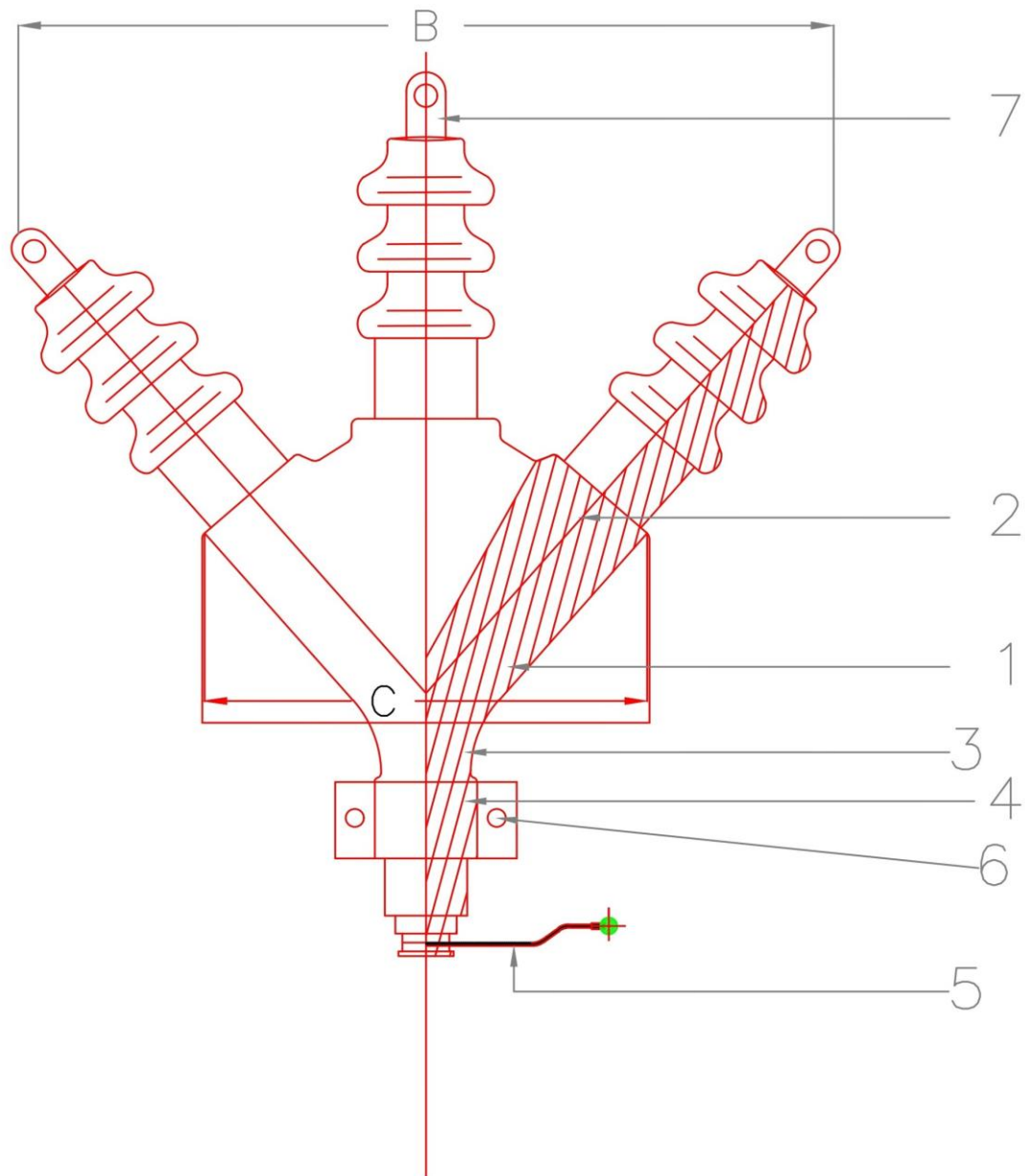
Arrangement of Cable and Joint Boxes

- Determining center position of joint box cable end.
- Wave of slack left near joint box.
- Staggering of joint boxex.

NOT TO SCALE

FIGURE-13
(Appendix E-2.4.2)

TYPICAL OUTDOOR TERMINATION (PVC CABLE WITH EPOXY)

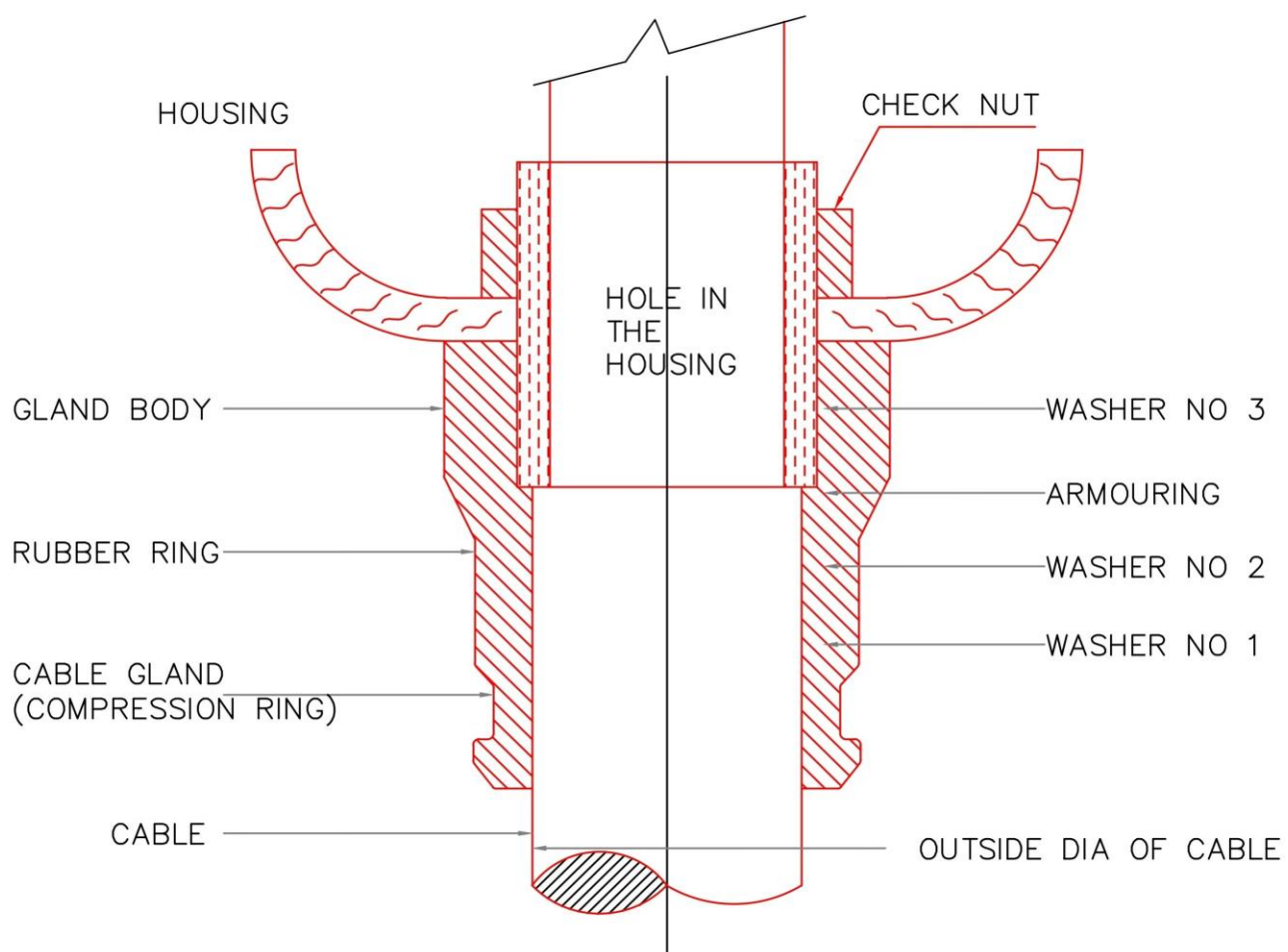


- 1.Seeling End Body. 2. Pearling Bushing 3. Tapered Serving
I/C Screen to Distribute Electric Stressed More Uniformly
4.Neck of Sealing End 5. Earthing Conductor

NOT TO SCALE

FIGURE-14
(Appendix E-2.4.3)

COMPRESSION TYPE CABLE GLAND ASSAMBLY



NOT TO SCALE

FIGURE-15
(Appendix E-2.4)
TYPICAL PVC CABLE JOINTING

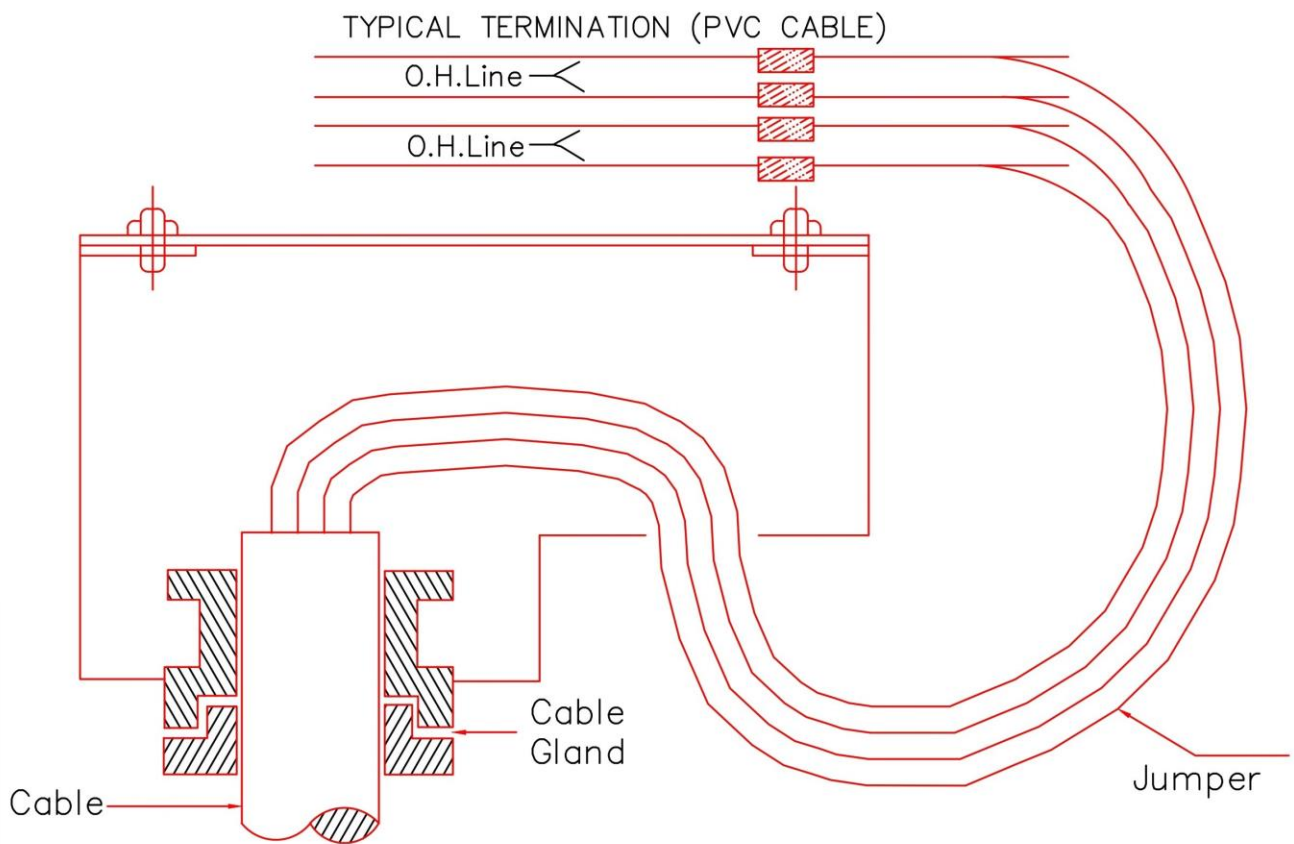


FIG 16A to 16F
(Clause 6.8.1)
STANDARD LAYOUTS FOR ROAD
LIGHTING

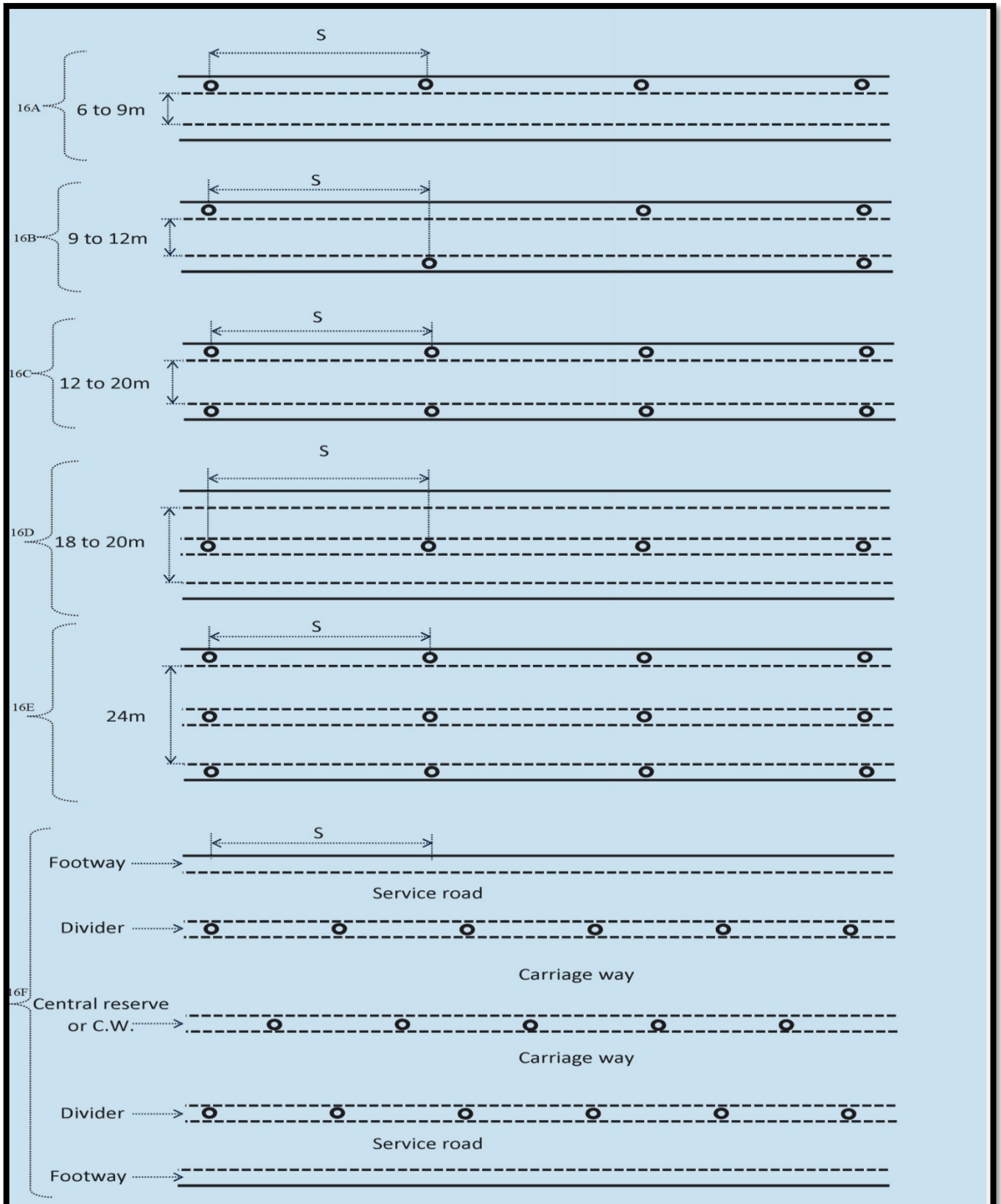


FIGURE 17A
(Clause 6.8.2)

NOT TO SCALE

LIGHTING LAYOUTS OF FLYOVERS

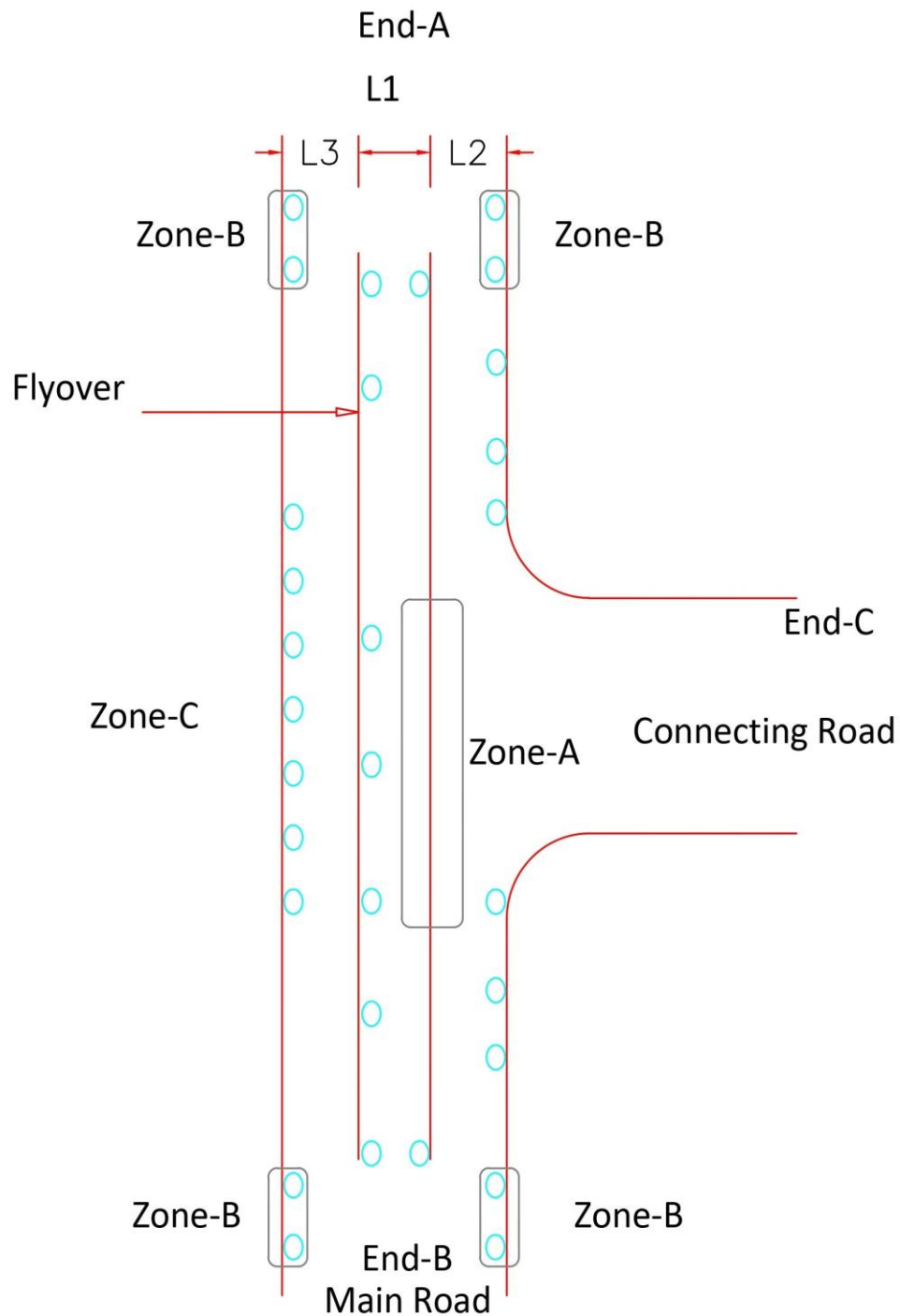


FIGURE 17B
(Clause 6.8.2)
LIGHTING LAYOUTS OF FLYOVERS

NOT TO SCALE

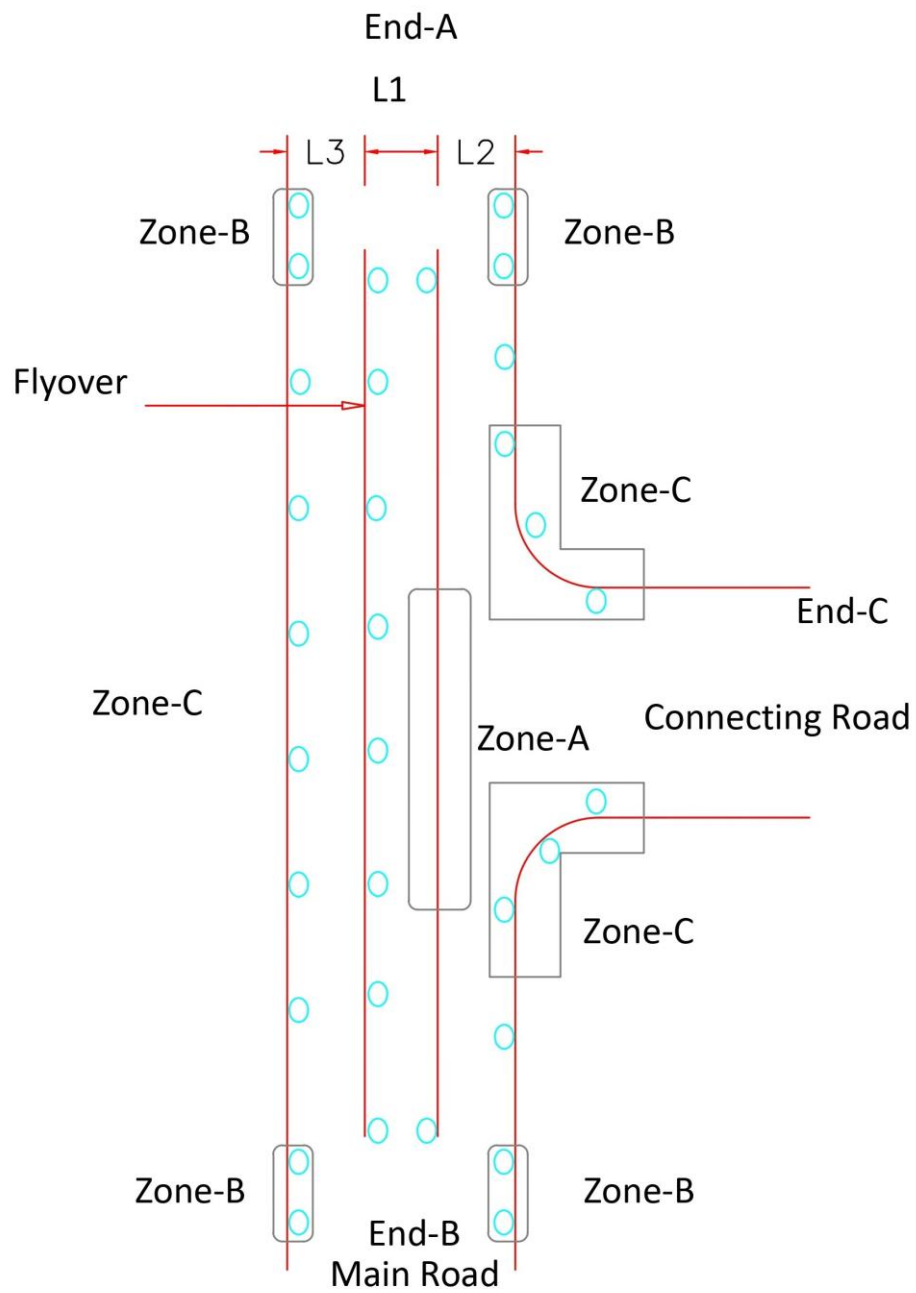


FIGURE 17C
(Clause 6.8.2)

NOT TO SCALE

LIGHTING LAYOUTS OF FLYOVERS

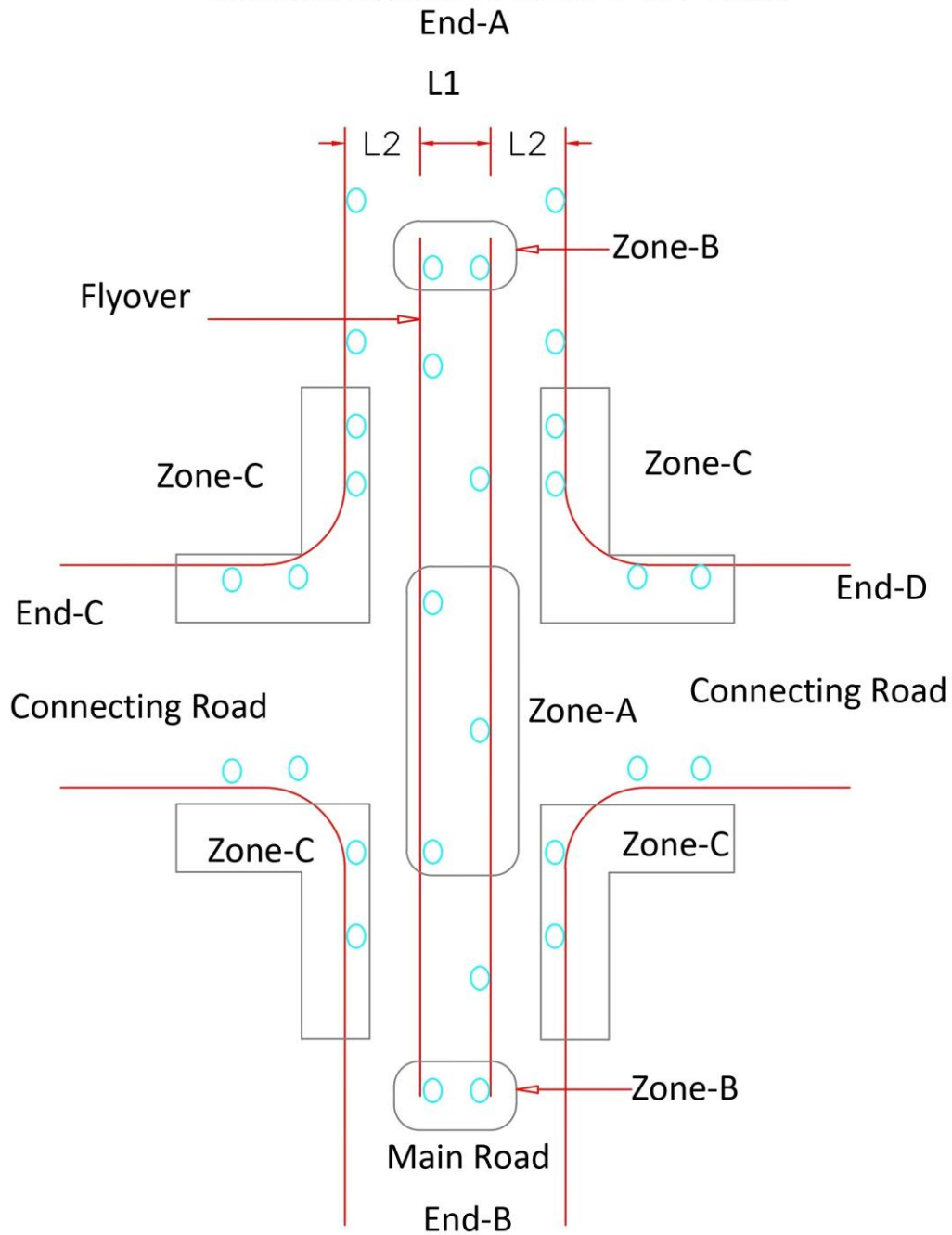


FIGURE 17D
(Clause 6.8.2)

NOT TO SCALE

STANDARD LAYOUTS FOR FLYOVERS (FOUR ALTERNATIVES)

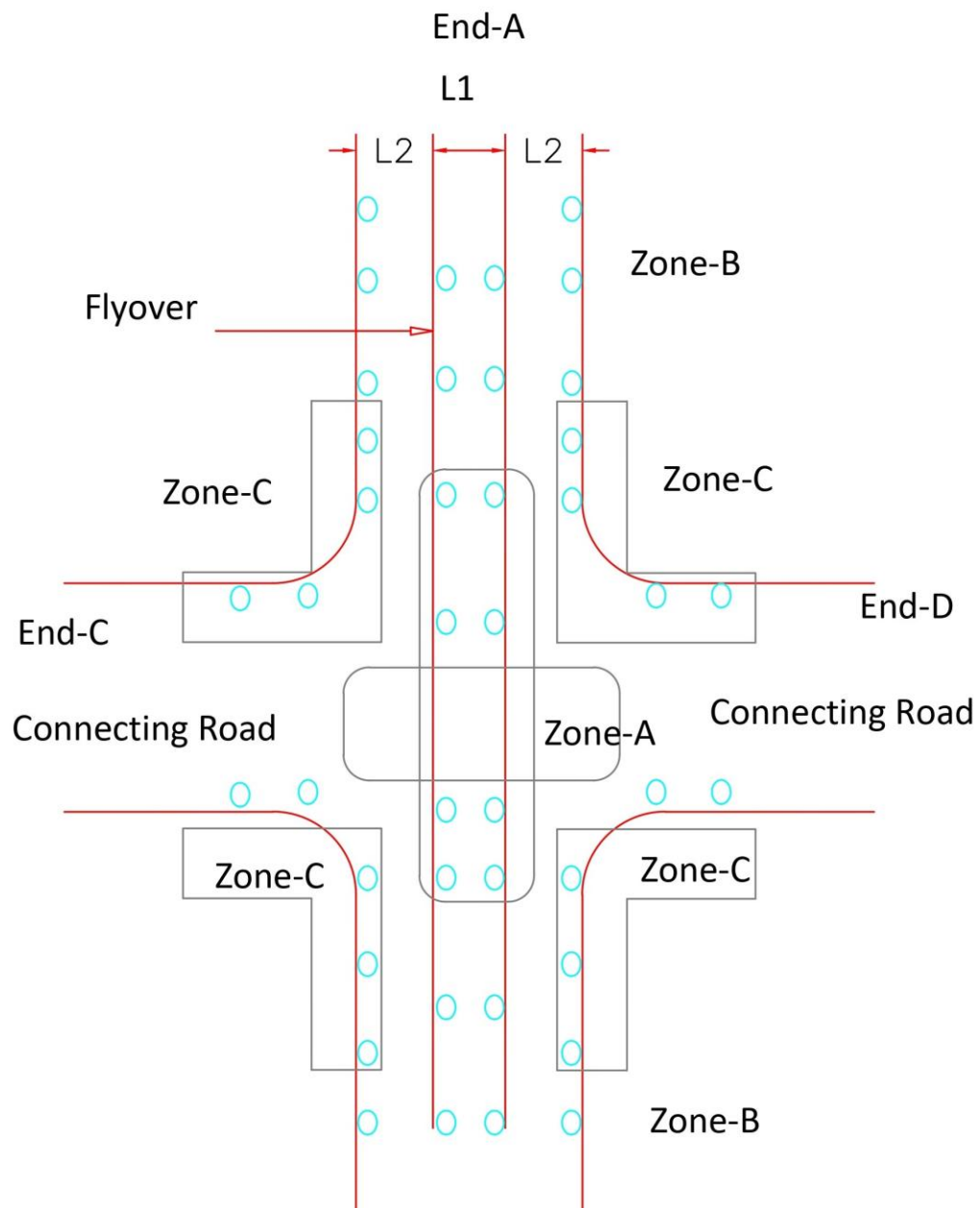
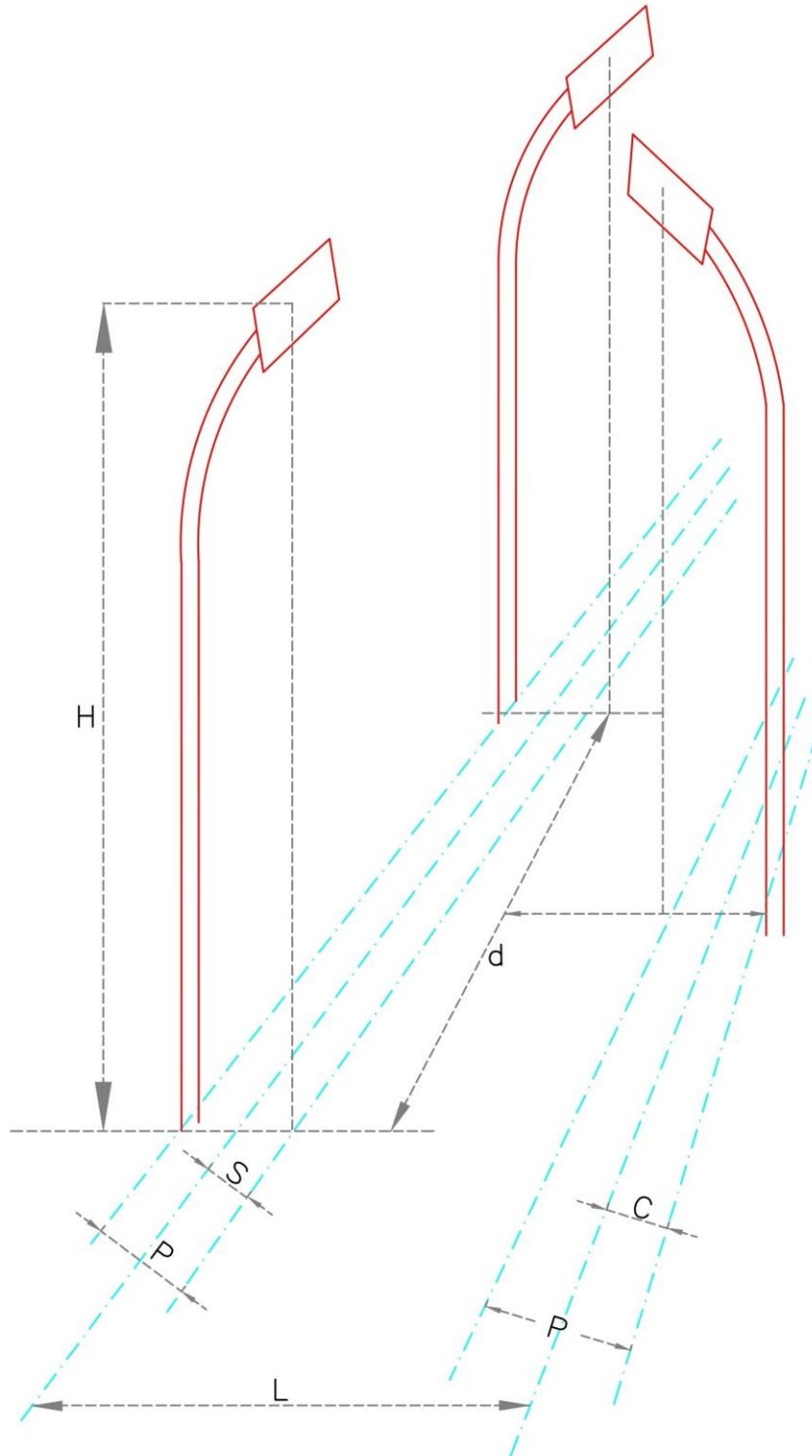


FIGURE 18
(Clause 6.11.4)

NOT TO SCALE

SITTING OF LUMINARIES CHARACTERISTIC DIMENSIONS



Key:-

p = outreach, h =mounting height of luminaires, s = overhang, L =width of carriage way, d = spacing, c = clearance

